Unabridged recorded council responses, including unsuccessful truncated responses. These reports are historical model outputs, not independently validated findings or laboratory instructions. Full prompts, machine-readable records and original source hashes accompany this document. See the interpretation notes in the revised ESI, Section 7.3. No new model calls were made.

F5-S1: Figure 5, Set 1
Source: outputs/figure5_pure_oxygen_physics_20260918/figure5_set1/attempt_01. Selected Candidate 1; 12-candidate pool; downstream model claude-sonnet-4-6. The objective policy is balanced.

Call 1: DrChemistry, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry / Stream a composition: Substrate 1 (0.1 M, 1.0 equiv), acrylonitrile (0.2 M, 2.0 equiv), [Ir(dF(CF3)ppy)2(dtbpy)]PF6 (0.5 mol%, 0.0005 M) in EtOH:pH9 buffer (5:1 v/v), Ar-sparged offline. No inline degasser available; offline preparation is the only deoxygenation route.

Shared assessment / Chemistry / Stream b: Pure O2, 0.09 sccm (inlet/STP), introduced at Stage 2 only. Gas identity is chemist-confirmed (Q-GAS-001). Switching from air to pure oxygen relative to the batch protocol is a change requiring chemistry, inventory, and oxygen-service review; this change is not yet approved.

Shared assessment / Chemistry / Wavelength note: Protocol specifies 452 nm. UV-150 450 nm (C1–C2, C4–C12, Stage 1) and Manual 2 448 nm (all candidates, Stage 2) are the closest available; no proof of photon-dose adequacy at these wavelengths.

Shared assessment / Chemistry / Acrylonitrile pump note: Acrylonitrile is excluded from RED pump tubing. All candidates correctly assign stream A to pump_e_series_blue (BLUE). Acceptable.

Shared assessment / Gas equivalents: All candidates deliver 2.0078 equiv O2 at inlet STP (0.09 sccm MFC grid step). This meets the ≥2.0 equiv minimum. Equiv are proposed delivered molar ratios at inlet/STP; uptake and consumption in-channel are not modeled.

Shared assessment / Residence time basis: Stage 1: liquid-only V/Q (no gas). Stage 2: inlet/STP apparent index V/(Q_liq + Q_gas_STP). These are not measured liquid exposure or photon-dose equivalents. Batch times (240 min Stage 1, 360 min Stage 2) are screening anchors only.

Shared assessment / Backflow unresolved / Finding bf-reverse-path: Applies to all 12 candidates. The liquid path from the Stage 2 gas junction back through Stage 1 reactor carries no forward-oriented non-return valve. This is an unresolved credible risk of O2 reaching Stage 1 under startup, flow interruption, shutdown, or pressure transients. A check valve on the MFC branch (CV-3301) protects only that branch; it does not protect the liquid upstream path. CV-3301 pressure rating is unconfirmed (cracking pressure 1.0 bar ≠ pressure rating). Reverse connectivity cannot be closed by reducing gas flow alone, as confirmed by the KHU observation and stated in the backflow findings.

Shared assessment / Backflow unresolved / Finding bf-verify-protection: CV-3301 direction, leakage, service compatibility, pressure rating, and low-flow suitability unverified for all candidates. This is a shared unresolved requirement before laboratory execution.

Shared assessment / Backflow unresolved / Consequence: If O2 reaches Stage 1, the oxygen-exclusion hard constraint is violated and Stage 1 photocatalysis may be disrupted. This risk must be addressed by engineering review (e.g., branch-specific NRV on the liquid interstage line, or pressure-decoupled arrangement) before execution of any candidate.

Shared assessment / Backflow unresolved / Operability status: backflow_resolved: false for all candidates

Shared assessment / Stage1 light: All candidates use UV-150 450 nm for Stage 1. UV-150 is compatible with E-series platform (confirmed). Photon-dose adequacy at 450 nm vs. 452 nm protocol is unproven.

Shared assessment / Stage2 light: All candidates use Manual 2 448 nm Blue Strip LED. No intensity specification listed for manual2; light-dose adequacy for Stage 2 is unproven.

Shared assessment / Stage2 temperature: Stage 2 batch temperature is not stage-resolved in the protocol. All candidates screen at 25 °C; no validated rate correction exists.

Shared assessment / Cv3301 pressure rating: Maximum pressure rating of CV-3301 is not supplied. Cracking pressure (1.0 bar) is not a pressure rating. Laboratory verification required before execution.

Shared assessment / Oxygen service: Pure O2 at Stage 2 requires chemistry, inventory, and oxygen-service review. Not yet approved. The gas-source change from air (protocol) to pure O2 (chemist answer Q-GAS-001) is unapproved and requires laboratory review before use.

Shared assessment / No yield prediction: Kinetics are uncharacterized. No yield, conversion, or intensification factor is predicted for any candidate. Hold ratios are experimental exposure indices only; physical exposure equivalence is not established.

Shared assessment / Etfe uv150 compatibility: coil_row11 and coil_row12 (ETFE, 2.4 mm ID) are listed with photoreactor_module_ids including uv150. UV-150 max reactor volume is 10 mL; coil_row11 is 5 mL and coil_row12 is 10 mL, both within limit. However, the large bore (2.4 mm) creates very low Re (<0.18) and may affect gas-liquid distribution in Stage 2 ETFE coils; this is an unresolved experimental uncertainty.

Shared assessment / Stage1 100min note: Candidates 1, 2, 4, 5, 7, 8, 10, 12 all achieve Stage 1 liquid tau = 100 min (2 mL coil at 0.02 mL/min), which is 0.42× the 240 min batch anchor. Whether this exposure is adequate for the Giese reaction is uncharacterized. Candidates 3, 6, 9, 11 achieve 250 min (0.04 or 1.04× batch) using 5 mL coils.

Shared assessment / Stage2 apparent tau: Candidates using FEP 20 mL coil (C1, C2, C3, C4, C6) achieve inlet/STP apparent tau ≈ 181.8 min (0.51× 360 min batch). Candidates using 10 mL coils (C5, C7, C8, C9, C10, C11, C12) achieve ≈ 90.9 min (0.25× batch). The apparent index understates actual liquid time due to gas volume compression at elevated BPR.

Shared assessment / Bpr note: BPR setpoint affects in-channel gas volume and gas-liquid contact; higher BPR compresses gas more. Candidates with 7 bar BPR have lower in-channel gas volume fraction than 1.5 or 2.5 bar candidates at identical inlet/STP flow. Neither regime is proven superior for mass transfer.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Baseline: Stage 1 tau=100 min (0.42× batch) in 2 mL PFA/UV-150 450 nm; Stage 2 tau_inlet=181.8 min (0.51× batch) in 20 mL FEP/Manual2, BPR 7 bar. Balanced stage ratios. Interpretable anchor for comparison. Backflow unresolved (shared); all other checks pass.

Reviews 1 / Uncertainty 1: Stage 1 at 100 min (0.42× batch) may be inadequate for Giese reaction; uncharacterized.

Reviews 1 / Uncertainty 2: Stage 2 apparent tau is inlet/STP index only; liquid exposure much longer but unverified.

Reviews 1 / Uncertainty 3: Backflow: unprotected reverse path to Stage 1 unresolved.

Reviews 1 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 1 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1 identical to C1 (100 min, 0.42×). Stage 2 same reactor (20 mL FEP) and inlet tau (181.8 min) but BPR 2.5 bar; in-channel gas volume larger (0.028 mL/min vs 0.012 mL/min), potentially different gas-liquid contact. Useful BPR comparison to C1. Backflow unresolved (shared).

Reviews 2 / Uncertainty 1: Lower BPR may increase gas void fraction; gas-liquid mass transfer and slug stability uncharacterized.

Reviews 2 / Uncertainty 2: Stage 1 adequacy at 100 min uncharacterized.

Reviews 2 / Uncertainty 3: Backflow: unprotected reverse path unresolved.

Reviews 2 / Uncertainty 4: CV-3301 rating unconfirmed; O2 service unapproved.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 tau=250 min (1.04× batch) in 5 mL ETFE/UV-150; provides near-batch Stage 1 exposure. Stage 2 tau_inlet=181.8 min (0.51×) in 20 mL FEP, BPR 7 bar. Large-bore ETFE in Stage 1 (2.4 mm ID, Re≈0.18) is laminar; photon penetration depth at 2.4 mm bore unverified. Backflow unresolved.

Reviews 3 / Uncertainty 1: ETFE 2.4 mm bore: photon penetration and mixing at Re≈0.18 uncharacterized in UV-150.

Reviews 3 / Uncertainty 2: Stage 2 apparent tau 0.51× batch; adequacy uncharacterized.

Reviews 3 / Uncertainty 3: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1 tau=100 min (0.42×) in 2 mL PFA; Stage 2 tau_inlet=181.8 min in 20 mL FEP, BPR 1.5 bar. Lowest BPR of the 20 mL FEP group; highest in-channel gas volume fraction (0.040 mL/min). Useful low-BPR comparison point. Stage 1 adequacy and backflow unresolved.

Reviews 4 / Uncertainty 1: BPR 1.5 bar: largest gas void fraction among FEP 20 mL candidates; slug flow stability uncharacterized.

Reviews 4 / Uncertainty 2: Stage 1 at 100 min (0.42×) may be insufficient.

Reviews 4 / Uncertainty 3: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1 tau=100 min (0.42×) in 2 mL PFA; Stage 2 tau_inlet=90.9 min (0.25×) in 10 mL ETFE 2.4 mm, BPR 7 bar. Shortest apparent Stage 2 index in the set. Stage 2 exposure substantially below batch anchor; adequacy is an open question. ETFE large bore in Stage 2 may affect gas distribution.

Reviews 5 / Uncertainty 1: Stage 2 at 0.25× batch may be insufficient for sulfide oxidation; uncharacterized.

Reviews 5 / Uncertainty 2: ETFE 2.4 mm bore Stage 2: gas-liquid distribution uncharacterized.

Reviews 5 / Uncertainty 3: Stage 1 at 0.42× may be insufficient.

Reviews 5 / Uncertainty 4: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 tau=250 min (1.04×) in 5 mL PFA 1 mm/UV-150; near-batch Stage 1 in a narrow-bore coil with better photon penetration than ETFE 2.4 mm. Stage 2 tau_inlet=181.8 min (0.51×) in 20 mL FEP, BPR 7 bar. Preferred Stage 1 geometry among near-batch Stage 1 candidates. Backflow unresolved.

Reviews 6 / Uncertainty 1: Stage 2 at 0.51× batch; oxidation adequacy uncharacterized.

Reviews 6 / Uncertainty 2: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 6 / Uncertainty 3: 5 mL PFA quantity: 2 available; one consumed here, leaving one for other use.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1 tau=100 min (0.42×) in 2 mL PFA; Stage 2 tau_inlet=90.9 min (0.25×) in 10 mL PFA 1 mm, BPR 7 bar. Short Stage 2 in narrow-bore PFA; no large-bore concern. Both stages substantially below batch. Provides short-time comparison point against C1 and C6. Backflow unresolved.

Reviews 7 / Uncertainty 1: Both stages significantly below batch anchors; adequacy uncharacterized.

Reviews 7 / Uncertainty 2: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1 tau=100 min (0.42×) in 2 mL PFA; Stage 2 tau_inlet=90.9 min (0.25×) in 10 mL FEP 1 mm (max 50 °C), BPR 7 bar. FEP temperature ceiling 50 °C is not limiting at 25 °C screen. Functionally identical to C7 in times; differs only in Stage 2 material (FEP vs PFA). Redundant with C7 unless material comparison is intended.

Reviews 8 / Uncertainty 1: FEP coil max 50 °C limits future temperature exploration in Stage 2.

Reviews 8 / Uncertainty 2: Both stages below batch; adequacy uncharacterized.

Reviews 8 / Uncertainty 3: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 tau=250 min (1.04×) in 5 mL ETFE 2.4 mm; Stage 2 tau_inlet=90.9 min (0.25×) in 10 mL ETFE 2.4 mm, BPR 7 bar. Near-batch Stage 1 but large-bore throughout; photon penetration concern in both stages. Short Stage 2. Backflow unresolved. All-ETFE design limits photon penetration assessment.

Reviews 9 / Uncertainty 1: Large bore (2.4 mm) in both stages: photon penetration and gas-liquid distribution uncharacterized.

Reviews 9 / Uncertainty 2: Stage 2 at 0.25× batch may be insufficient.

Reviews 9 / Uncertainty 3: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1 tau=100 min (0.42×) in 2 mL PFA; Stage 2 tau_inlet=90.9 min (0.25×) in 10 mL ETFE 2.4 mm, BPR 2.5 bar. Lower BPR increases in-channel gas volume vs C5; may affect slug stability. ETFE large bore Stage 2 concern retained. Provides BPR vs C5 comparison for large-bore Stage 2.

Reviews 10 / Uncertainty 1: ETFE 2.4 mm Stage 2 with BPR 2.5 bar: larger gas void fraction; slug stability uncharacterized.

Reviews 10 / Uncertainty 2: Stage 1 and Stage 2 both below batch.

Reviews 10 / Uncertainty 3: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1 tau=250 min (1.04×) in 5 mL ETFE 2.4 mm; Stage 2 tau_inlet=90.9 min (0.25×) in 10 mL PFA 1 mm, BPR 7 bar. Near-batch Stage 1 (large bore, photon concern) with short narrow-bore Stage 2. Mixes bore geometries across stages; provides near-batch Stage 1 with short Stage 2 in PFA. Backflow unresolved.

Reviews 11 / Uncertainty 1: ETFE 2.4 mm Stage 1: photon penetration at Re≈0.18 uncharacterized.

Reviews 11 / Uncertainty 2: Stage 2 at 0.25× batch; oxidation adequacy uncharacterized.

Reviews 11 / Uncertainty 3: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1 tau=100 min (0.42×) in 2 mL PFA; Stage 2 tau_inlet=90.9 min (0.25×) in 10 mL PFA 1 mm, BPR 2.5 bar. Narrow bore both stages; lower BPR than C7. Useful BPR comparison to C7. Both stages well below batch. Functionally overlaps C7 and C10 in screen space.

Reviews 12 / Uncertainty 1: Both stages below batch; adequacy for both reactions uncharacterized.

Reviews 12 / Uncertainty 2: Lower BPR increases gas void fraction vs C7.

Reviews 12 / Uncertainty 3: Backflow unresolved; CV-3301 unconfirmed; O2 service unapproved.

Cross cutting findings / Backflow requirement: The reverse path from the Stage 2 gas junction through the liquid line back to Stage 1 reactor is unprotected in all 12 candidates. A forward-oriented non-return valve on the interstage liquid line between Stage 1 outlet and Stage 2 junction, or an equivalent pressure-decoupled arrangement, must be evaluated and verified before any candidate is executed. The CV-3301 on the MFC branch does not address this path. This is the highest-priority unresolved engineering requirement.

Cross cutting findings / Gas source change: Use of pure O2 instead of the batch protocol's air is unapproved. Chemistry, inventory, and oxygen-service review is required before use of pure O2 in any candidate.

Cross cutting findings / Preferred for screening rationale: Under the balanced objective, C1 and C6 together provide the most interpretable baseline: C1 offers a short Stage 1 / mid Stage 2 starting point; C6 offers a near-batch Stage 1 / mid Stage 2 point with narrow-bore PFA Stage 1 (better photon penetration than ETFE). Neither is proven superior. No yield or kinetic prediction is made.

Cross cutting findings / Candidates with redundancy: C7 and C8 are functionally identical in residence times; they differ only in Stage 2 material (PFA vs FEP). C8 adds a 50 °C temperature ceiling constraint. If material comparison is not the goal, running both is low priority. Similarly C5 and C10 share identical inlet tau but differ in BPR and Stage 2 material (ETFE with 7 vs 2.5 bar).

Cross cutting findings / No hard vetoes: No candidate has a confirmed hard constraint violation. All pressure ratings pass computed checks. All carry the shared unresolved backflow risk and the unapproved gas-source change, which are engineering and safety review requirements, not confirmed violations of a specific design rule.

Call 2: DrKinetics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Kinetics: No flow kinetic data exist. Batch times (S1: 240 min, S2: 360 min) are screening anchors only. Stage 1 liquid V/Q ratios range 0.42–1.04×; Stage 2 inlet/STP apparent index ratios range 0.25–0.51×. Inlet/STP apparent index is not liquid exposure time. No yield, conversion, or degradation ranking can be inferred from these ratios.

Shared assessment / Gas source: Pure O2 is confirmed by chemist answer (Q-GAS-001). This supersedes the original batch air atmosphere. O2 service compatibility, inventory review, and oxygen-service hardware review are required before laboratory execution; this is not an approved gas supply absent those reviews.

Shared assessment / Backflow unresolved: All 12 candidates share identical unresolved backflow findings: (1) BF-REVERSE-PATH: the liquid branch from Stage 2 junction back to Stage 1 reactor has no declared forward-oriented non-return valve; O2 gas can propagate upstream under startup, pump interruption, shutdown, or pressure transient scenarios, violating the hard constraint that Stage 1 must remain oxygen-free. (2) BF-VERIFY-PROTECTION: the CV-3301 on the MFC gas branch requires direction, leakage, service compatibility, pressure rating (unspecified maximum), and low-flow cracking verification. The CV-3301 cracking pressure (1.0 bar) concerns the differential pressure across that valve, not BPR minus cracking pressure. A valve on the MFC branch does not protect the liquid branch. Reducing gas flow does not close the unprotected liquid reverse path. KHU observation of reverse flow at 0.02 mL/min liquid / 0.43 mL/min O2 is qualitative evidence consistent with this unresolved risk. No tested configuration has demonstrated absence of reverse flow.

Shared assessment / Light sources: Stage 1 uses UV-150 / 450 nm (2 nm from protocol 452 nm); adequacy of this small offset is unproven but not a hard violation. Stage 2 uses Manual 2 / 448 nm strip LED; wavelength match is similar but photon dose adequacy for oxidation is unproven. No optical intensification is claimed. Light-dose adequacy is not established for any candidate.

Shared assessment / Wavelength note: UV-150 is assigned to Stage 1 (E-series platform, compatible). Manual 2 is assigned to Stage 2 (strip LED, compatible with coil). No equipment boundary violations detected.

Shared assessment / Temperature: Stage 2 batch temperature is not stage-resolved. All candidates apply 25 °C to Stage 2. This is an unresolved deviation; no validated rate correction exists.

Shared assessment / Cv3301 rating: CV-3301 maximum pressure rating is not supplied. Cracking pressure (1.0 bar) is not a pressure rating. Verification required before execution for all candidates.

Shared assessment / Pure o2 requires review: Pure O2 use requires chemist confirmation, chemistry inventory review, and oxygen-service hardware review. This is not an approved gas supply. Recommend mitigations (e.g., branch-specific liquid-side check valve before Stage 1 reactor, verified O2-service fittings) are unapproved proposals requiring full review.

Shared assessment / Analogy records: All three retrieved analogy records are from different reaction classes or light regimes and are not usable as kinetic evidence for this Giese/oxidation sequence.

Shared assessment / Stage1 etfe photon note: ETFE coil (ID 2.4 mm) in candidates 3, 9, 11 is listed as compatible with UV-150 module. However, the larger bore provides lower surface-to-volume ratio; photon penetration depth relative to the batch geometry is uncharacterized and represents an additional uncertainty, not a hard violation.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: S1: 2 mL PFA, τ_liq=100 min (0.42× batch). S2: 20 mL FEP, τ_inlet=181.8 min (0.50× batch), BPR 7 bar, O2 0.09 sccm, 2.01 equiv. Provides a mid-range S1/S2 combination as a baseline. Backflow and O2-service issues unresolved but not a hard violation absent confirmed occurrence.

Reviews 1 / Uncertainty 1: S1 liquid τ 0.42× batch; adequacy for Giese completion unproven.

Reviews 1 / Uncertainty 2: S2 inlet/STP index 0.50× batch; liquid exposure ~1000 min (unconstrained by gas index).

Reviews 1 / Uncertainty 3: Unprotected liquid reverse path to S1 unresolved across all candidates.

Reviews 1 / Uncertainty 4: CV-3301 pressure rating unconfirmed; O2-service compatibility unreviewed.

Reviews 1 / Uncertainty 5: Stage 2 temperature not batch-resolved.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical reactor geometry to C1 but BPR 2.5 bar; lower system pressure increases gas volumetric fraction in S2 (0.028 mL/min vs 0.012 mL/min in-channel), potentially aiding gas-liquid contact. Same S1 τ limitation. BPR difference is a useful screen variable alongside C1.

Reviews 2 / Uncertainty 1: Lower BPR reduces O2 solubility in S2; net effect on oxidation unproven.

Reviews 2 / Uncertainty 2: Same unprotected liquid reverse path as all candidates.

Reviews 2 / Uncertainty 3: S1 τ 0.42× batch; Giese completion unproven.

Reviews 2 / Uncertainty 4: CV-3301 and O2-service unreviewed.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: S1: 5 mL ETFE (ID 2.4 mm), τ_liq=250 min (1.04× batch), BPR 7 bar. S2: 20 mL FEP, τ_inlet=181.8 min. Exceeds batch S1 time; provides near-batch S1 anchor. ETFE large bore reduces surface/volume for photon delivery vs protocol geometry; uncharacterized impact.

Reviews 3 / Uncertainty 1: ETFE 2.4 mm bore: photon penetration depth uncharacterized vs batch.

Reviews 3 / Uncertainty 2: S2 inlet/STP index 0.50× batch; same as C1.

Reviews 3 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 3 / Uncertainty 4: O2-service and CV-3301 unreviewed.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: S1: 2 mL PFA, τ_liq=100 min (0.42× batch), BPR 1.5 bar. S2: 20 mL FEP, τ_inlet=181.8 min. Lowest BPR in 20 mL FEP group; highest in-channel gas volumetric fraction (0.040 mL/min). Useful low-pressure screen point. Stage 1 τ limitation same as C1.

Reviews 4 / Uncertainty 1: BPR 1.5 bar: lowest system pressure; O2 solubility in S2 lowest of this group.

Reviews 4 / Uncertainty 2: S1 τ 0.42× batch; Giese completion unproven.

Reviews 4 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 4 / Uncertainty 4: CV-3301 and O2-service unreviewed.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: S1: 2 mL PFA, τ_liq=100 min (0.42× batch). S2: 10 mL ETFE (ID 2.4 mm), τ_inlet=90.9 min (0.25× batch), BPR 7 bar. Shortest S2 inlet-index; large bore may improve gas dispersion but reduces liquid τ index further. Useful short-S2 reference with ETFE geometry.

Reviews 5 / Uncertainty 1: S2 τ_inlet 0.25× batch; oxidation completeness most uncertain in this set.

Reviews 5 / Uncertainty 2: ETFE 2.4 mm bore in S2: gas-liquid contact pattern uncharacterized.

Reviews 5 / Uncertainty 3: S1 τ 0.42× batch unproven for Giese.

Reviews 5 / Uncertainty 4: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: S1: 5 mL PFA (ID 1 mm), τ_liq=250 min (1.04× batch), BPR 7 bar. S2: 20 mL FEP, τ_inlet=181.8 min. Near-batch S1 with narrower bore than C3; better photon geometry for S1. Paired with C3 allows bore effect assessment in S1.

Reviews 6 / Uncertainty 1: S2 inlet/STP index 0.50× batch same as C1/C3.

Reviews 6 / Uncertainty 2: S1 near-batch τ does not guarantee equivalent conversion; no kinetics.

Reviews 6 / Uncertainty 3: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Reviews 6 / Uncertainty 4: Stage 2 temperature not batch-resolved.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: S1: 2 mL PFA, τ_liq=100 min. S2: 10 mL PFA (ID 1 mm), τ_inlet=90.9 min (0.25× batch), BPR 7 bar. Same inlet-index as C5 but narrow-bore PFA in S2; allows bore comparison for gas-liquid contact vs ETFE in C5. Short S2 τ risk shared with C5.

Reviews 7 / Uncertainty 1: S2 τ_inlet 0.25× batch; oxidation adequacy unproven.

Reviews 7 / Uncertainty 2: S1 τ 0.42× batch; Giese completion unproven.

Reviews 7 / Uncertainty 3: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: S1: 2 mL PFA, τ_liq=100 min. S2: 10 mL FEP (ID 1 mm, max 50 °C), τ_inlet=90.9 min, BPR 7 bar. FEP coil in S2; temperature limit (50 °C) is not a concern at 25 °C. Same inlet-index as C7; material difference FEP vs PFA is the distinguishing variable.

Reviews 8 / Uncertainty 1: FEP S2 coil max 50 °C; adequate at 25 °C but confirmed temperature margin needed.

Reviews 8 / Uncertainty 2: S2 τ_inlet 0.25× batch; adequacy unproven.

Reviews 8 / Uncertainty 3: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: S1: 5 mL ETFE (ID 2.4 mm), τ_liq=250 min (1.04× batch). S2: 10 mL ETFE (ID 2.4 mm), τ_inlet=90.9 min, BPR 7 bar. Both stages large bore; photon penetration in S1 and gas-liquid contact in S2 both uncharacterized. Useful all-ETFE reference point.

Reviews 9 / Uncertainty 1: Both stages 2.4 mm bore; photon delivery in S1 and gas-liquid contact in S2 uncharacterized.

Reviews 9 / Uncertainty 2: S2 τ_inlet 0.25× batch; oxidation adequacy unproven.

Reviews 9 / Uncertainty 3: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: S1: 2 mL PFA, τ_liq=100 min. S2: 10 mL ETFE (ID 2.4 mm), τ_inlet=90.9 min, BPR 2.5 bar. Lower BPR with large-bore S2; highest in-channel gas volumetric fraction among 10 mL S2 candidates (0.028 mL/min). Pairs usefully with C5 (same geometry, different BPR).

Reviews 10 / Uncertainty 1: Lower BPR reduces O2 solubility; higher gas volumetric fraction effect on contact unproven.

Reviews 10 / Uncertainty 2: S1 τ 0.42× batch; S2 τ_inlet 0.25× batch; both below batch.

Reviews 10 / Uncertainty 3: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: S1: 5 mL ETFE (ID 2.4 mm), τ_liq=250 min (1.04× batch). S2: 10 mL PFA (ID 1 mm), τ_inlet=90.9 min, BPR 7 bar. Near-batch S1 with large bore paired with narrow-bore S2; allows mixed-geometry assessment across stages.

Reviews 11 / Uncertainty 1: S1 ETFE 2.4 mm bore: photon penetration uncharacterized.

Reviews 11 / Uncertainty 2: S2 τ_inlet 0.25× batch; adequacy unproven.

Reviews 11 / Uncertainty 3: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: S1: 2 mL PFA, τ_liq=100 min. S2: 10 mL PFA (ID 1 mm), τ_inlet=90.9 min, BPR 2.5 bar. Narrow-bore throughout, lowest BPR with PFA S2; pairs with C7 (same geometry, 7 bar) and C2 (same BPR, larger S2). Useful low-pressure, narrow-bore reference.

Reviews 12 / Uncertainty 1: S1 τ 0.42× batch; S2 τ_inlet 0.25× batch; both below batch.

Reviews 12 / Uncertainty 2: Lower BPR; O2 solubility consequence in S2 unproven.

Reviews 12 / Uncertainty 3: Unprotected liquid reverse path; CV-3301 and O2-service unreviewed.

Global unresolved before execution 1: Unprotected liquid reverse path from Stage 2 junction to Stage 1 reactor must be resolved (branch-specific forward-oriented non-return valve on liquid inlet to Stage 2 junction, or pressure-decoupled arrangement) before any candidate is executed. This is required to satisfy the hard constraint that Stage 1 remains oxygen-free.

Global unresolved before execution 2: CV-3301 on MFC gas branch: maximum pressure rating unconfirmed; direction, leakage, O2-service compatibility, and low-flow cracking behavior unverified.

Global unresolved before execution 3: Pure O2 gas supply requires chemist confirmation, chemistry and inventory review, and oxygen-service hardware review; not yet approved.

Global unresolved before execution 4: Stage 2 batch temperature is not stage-resolved; 25 °C assignment is unvalidated.

Global unresolved before execution 5: No inline degasser available; Ar-sparge deoxygenation protocol for feed must be verified as adequate for Stage 1 oxygen exclusion.

Global unresolved before execution 6: Photon dose adequacy at 450/448 nm vs protocol 452 nm is unproven for both stages.

Global unresolved before execution 7: Gas uptake and O2 consumption in Stage 2 are not modeled; actual dissolved O2 available to the reaction is unknown.

Global unresolved before execution 8: Candidates 3, 9, 11 (ETFE 2.4 mm S1): photon penetration adequacy relative to batch geometry is uncharacterized and requires assessment before execution.

Call 3: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Tool name: simulate_flow_transients

Candidate ids 1: 1

Candidate ids 2: 2

Candidate ids 3: 3

Candidate ids 4: 4

Candidate ids 5: 5

Candidate ids 6: 6

Candidate ids 7: 7

Candidate ids 8: 8

Candidate ids 9: 9

Candidate ids 10: 10

Candidate ids 11: 11

Candidate ids 12: 12

Call 4: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Common issues 1: ALL candidates: The liquid branch (st2_mixer → st1_reactor) carries NO forward-oriented non-return protection. This unprotected reverse path is an unresolved credible risk for oxygen intrusion into Stage 1 across all candidates. The CV-3301 check valve on the MFC gas branch does not protect the liquid branch. This finding is unresolved and requires engineering review before any execution.

Shared assessment / Common issues 2: ALL candidates: Pure O2 was introduced in the prior experiment without approval (air was specified); this gas-source change is unapproved and requires laboratory review before use.

Shared assessment / Common issues 3: ALL candidates: CV-3301 maximum pressure rating is unconfirmed. Cracking pressure (1.0 bar) is not a pressure rating.

Shared assessment / Common issues 4: ALL candidates: Physics tool results are conditional on unmeasured parameters (viscosity, compliance, gas plenum, MFC ΔP, pump dynamics). No reversal predicted in these finite-window simulations does not establish safe operation or exclude backflow. Liquid displacement is the reported metric, not oxygen reaching Stage 1.

Shared assessment / Common issues 5: ALL candidates: Kinetics are uncharacterized; stage hold ratios are experimental exposure indices only—not yield predictors. Inlet/STP apparent residence time for Stage 2 is V/(Q_liquid + Q_gas_STP), not measured liquid residence or photon exposure time.

Shared assessment / Common issues 6: ALL candidates: Stage 2 batch temperature is not stage-resolved; 25 °C is applied by analogy with Stage 1. No validated rate correction exists.

Shared assessment / Common issues 7: ALL candidates: No inline degasser available; offline Ar sparge of feed is the only deoxygenation method; effectiveness is unconfirmed.

Shared assessment / Common issues 8: ALL candidates: Gas equivalents (2.0078 equiv) are proposed delivered molar ratios at inlet/STP, not established stoichiometric uptake.

Shared assessment / Common issues 9: ALL candidates: Light wavelength—UV-150/450 nm (actual 450 nm) vs. protocol 452 nm; Manual 2/448 nm vs. 452 nm. Neither deviation is proven adequate; photon-dose adequacy is unproven.

Shared assessment / Common issues 10: ALL candidates: Stage 1 ETFE coil (coil_row11, ID 2.4 mm) used in candidates 3, 9, 11 is listed as compatible with UV-150 (uv150 module). However, the UV-150 max reactor volume is 10 mL and the ETFE 5 mL coil is within that limit; no hard inventory violation identified, but photon penetration through ETFE 2.4 mm ID at this scale is uncharacterized.

Shared assessment / Common issues 11: ALL candidates: Feed: Stream A — (((4-methoxyphenyl)thio)methyl)trimethylsilane 0.1 M (1.0 equiv), acrylonitrile 0.2 M (2.0 equiv), [Ir(dF(CF3)ppy)2(dtbpy)]PF6 0.0005 M (0.5 mol%), in EtOH:pH 9 buffer (5:1 v/v), Ar-sparged, 0.02 mL/min via pump_e_series_blue; molar feed rate of limiting substrate = 0.002 mmol/min. Stream B — O2 (pure), 0.09 mL/min at inlet/STP via mfc_ffc00, 2.0078 equiv.

Shared assessment / Common issues 12: ALL candidates: Acrylonitrile is on the excluded_chemicals list for RED pump tubing; all candidates correctly use BLUE pump tubing (pump_e_series_blue) — no violation.

Shared assessment / Common issues 13: Candidates 3, 6, 9, 11 (ETFE coil_row11 Stage 1): Large-bore 2.4 mm ID coil gives very low Re (~0.18), laminar, potentially poor radial mixing and light distribution. This is an unresolved experimental uncertainty, not a hard violation.

Shared assessment / Common issues 14: Candidates 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12: Mixer type at Stage 2 junction is not explicitly identified in designs; must be a T- or Y-mixer from inventory (P-632/P-713/P-513), not a union. Confirm before assembly.

Shared assessment / Common issues 15: All gas stages: Q_gas_sccm and Q_gas_STP mean mL/min at STP (273.15 K, 1.01325 bar). In-channel actual mL/min differs by pressure ratio and is not the liquid residence basis.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA, τ_liq = 100 min (ratio 0.42 vs. 240 min batch). Stage 2: 20 mL FEP, τ_inlet = 181.8 min (ratio 0.51 vs. 360 min batch), BPR 7 bar. Balanced sub-batch screen for both stages. No BPR violations; pressure headroom confirmed within model assumptions. Unprotected liquid reverse path unresolved (shared). Adequate balanced exploratory baseline.

Reviews 1 / Uncertainty 1: Unprotected liquid branch reverse path to Stage 1 unresolved.

Reviews 1 / Uncertainty 2: Stage 1 τ_liq 100 min is 42% of batch; photocatalytic conversion unknown.

Reviews 1 / Uncertainty 3: Stage 2 photon adequacy at 448 nm (vs. 452 nm) unconfirmed.

Reviews 1 / Uncertainty 4: BPR opening pressure reached in simulation but under illustrative, not measured, parameters.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical reactor geometry to Candidate 1 but BPR 2.5 bar. Stage 2 in-channel gas volume is larger (0.028 mL/min vs. 0.012 mL/min), increasing gas fraction and potential slug flow. τ_inlet identical (181.8 min). Lower BPR reduces dissolved O2 driving force—this is an unconfirmed trade-off, not a hard violation. Acceptable screening point at lower pressure.

Reviews 2 / Uncertainty 1: Lower BPR (2.5 bar) reduces dissolved O2 potential; impact on oxidation yield unknown.

Reviews 2 / Uncertainty 2: Larger in-channel gas volume fraction at lower pressure; slug/flow regime uncharacterized.

Reviews 2 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 2 / Uncertainty 4: Stage 1 τ_liq 100 min may be insufficient; unknown without kinetic data.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 5 mL ETFE (ID 2.4 mm), τ_liq = 250 min (ratio 1.04, at-batch). Stage 2: 20 mL FEP, τ_inlet = 181.8 min, BPR 7 bar. Near-batch Stage 1 exposure with the 2.4 mm ID coil; photon penetration and mixing at this bore are uncharacterized uncertainties. Startup simulation only reached early transient (BPR not opened in that window); steady-state behaviour with liquid check valve showed no predicted reversal. Operability at startup warrants care.

Reviews 3 / Uncertainty 1: ETFE 2.4 mm ID: photon penetration and lateral mixing in Stage 1 uncharacterized.

Reviews 3 / Uncertainty 2: Startup simulation did not reach BPR opening pressure; full steady-state behaviour conditional.

Reviews 3 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 3 / Uncertainty 4: Stage 2 τ_inlet still sub-batch (0.51 ratio); adequacy for oxidation unknown.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 2 mL PFA, τ_liq = 100 min (ratio 0.42). Stage 2: 20 mL FEP, τ_inlet = 181.8 min, BPR 1.5 bar. Lowest BPR of the 20 mL Stage 2 group; further reduces dissolved O2 driving force. At 1.5 bar, gas compression is minimal; largest in-channel gas volume fraction (0.040 mL/min). Acceptable as a low-pressure exploratory point but lowest dissolved O2 potential.

Reviews 4 / Uncertainty 1: BPR 1.5 bar: lowest dissolved O2 driving force of all candidates; oxidation adequacy most uncertain here.

Reviews 4 / Uncertainty 2: Highest in-channel gas volume fraction; two-phase flow regime uncharacterized.

Reviews 4 / Uncertainty 3: Stage 1 τ_liq 100 min may be insufficient.

Reviews 4 / Uncertainty 4: Unprotected liquid branch reverse path unresolved.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA, τ_liq = 100 min (ratio 0.42). Stage 2: 10 mL ETFE (ID 2.4 mm), τ_inlet = 90.9 min (ratio 0.25, well below batch). BPR 7 bar. Sub-batch both stages; shortest Stage 2 exploration among 7-bar candidates. ETFE 2.4 mm in Stage 2 with gas introduces uncharacterized two-phase flow and photon distribution. Acceptable as a short-exposure exploratory point.

Reviews 5 / Uncertainty 1: Stage 2 τ_inlet 90.9 min is 25% of batch; oxidation completeness most uncertain.

Reviews 5 / Uncertainty 2: ETFE 2.4 mm ID Stage 2 two-phase photon distribution uncharacterized.

Reviews 5 / Uncertainty 3: Stage 1 τ_liq 100 min may be insufficient.

Reviews 5 / Uncertainty 4: Unprotected liquid branch reverse path unresolved.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA (ID 1 mm), τ_liq = 250 min (ratio 1.04, at-batch). Stage 2: 20 mL FEP, τ_inlet = 181.8 min, BPR 7 bar. Near-batch Stage 1 in narrow-bore PFA is better-characterised optically than ETFE 2.4 mm. Startup simulation reached only early transient (BPR not opened); steady-state with liquid check showed no predicted reversal. Good balanced candidate for near-batch Stage 1 screen.

Reviews 6 / Uncertainty 1: Startup simulation did not reach BPR opening pressure in finite window.

Reviews 6 / Uncertainty 2: Stage 2 τ_inlet still sub-batch (0.51); oxidation adequacy unknown.

Reviews 6 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 6 / Uncertainty 4: Photon dose adequacy at 448 nm unconfirmed.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA, τ_liq = 100 min (ratio 0.42). Stage 2: 10 mL PFA (ID 1 mm), τ_inlet = 90.9 min (ratio 0.25), BPR 7 bar. Both stages sub-batch. PFA narrow-bore in both stages is well-defined optically. Short Stage 2 is a legitimate short-exposure screen point. Acceptable exploratory candidate.

Reviews 7 / Uncertainty 1: Stage 2 τ_inlet 90.9 min is 25% of batch; oxidation likely incomplete.

Reviews 7 / Uncertainty 2: Stage 1 τ_liq 100 min sub-batch; Giese step completeness unknown.

Reviews 7 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 7 / Uncertainty 4: No kinetic data; cannot predict whether short Stage 2 is adequate.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 2 mL PFA, τ_liq = 100 min (ratio 0.42). Stage 2: 10 mL FEP (ID 1 mm), τ_inlet = 90.9 min (ratio 0.25), BPR 7 bar. Functionally identical exposure profile to Candidate 7; FEP vs. PFA for Stage 2 are both 1 mm ID, max 50 °C vs. 100 °C for FEP. At 25 °C operation, FEP temperature limit is not a constraint. Acceptable; minor material difference from Candidate 7.

Reviews 8 / Uncertainty 1: FEP max temperature 50 °C; adequate at 25 °C but limits future temperature exploration.

Reviews 8 / Uncertainty 2: Stage 2 τ_inlet 90.9 min is 25% of batch; oxidation likely incomplete.

Reviews 8 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 8 / Uncertainty 4: Stage 1 τ_liq sub-batch; Giese completeness unknown.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 5 mL ETFE (ID 2.4 mm), τ_liq = 250 min (ratio 1.04). Stage 2: 10 mL ETFE (ID 2.4 mm), τ_inlet = 90.9 min (ratio 0.25), BPR 7 bar. Near-batch Stage 1 with large-bore ETFE in both stages. Photon penetration and two-phase mixing are doubly uncharacterized. Pump interruption scenario predicted negligible displacement in simulation (conditional). Acceptable but carries the most geometric uncertainty of the long-Stage-1 group.

Reviews 9 / Uncertainty 1: Both stages use ETFE 2.4 mm ID; photon penetration and mixing uncharacterized in both.

Reviews 9 / Uncertainty 2: Stage 2 τ_inlet 90.9 min is 25% of batch; oxidation adequacy most uncertain.

Reviews 9 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 9 / Uncertainty 4: Pump interruption simulation conditional on illustrative parameters only.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA, τ_liq = 100 min (ratio 0.42). Stage 2: 10 mL ETFE (ID 2.4 mm), τ_inlet = 90.9 min (ratio 0.25), BPR 2.5 bar. Lower BPR reduces dissolved O2 driving force; ETFE 2.4 mm Stage 2 adds two-phase flow uncertainty. Short Stage 2 with reduced pressure is the most limiting O2-delivery condition of the short-Stage-2 group. Acceptable as a bracketing low end.

Reviews 10 / Uncertainty 1: BPR 2.5 bar reduces dissolved O2; ETFE 2.4 mm two-phase flow uncharacterized.

Reviews 10 / Uncertainty 2: Stage 2 τ_inlet 90.9 min at lowest O2 driving force: oxidation likely most limited.

Reviews 10 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 10 / Uncertainty 4: Stage 1 τ_liq sub-batch; Giese completeness unknown.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 5 mL ETFE (ID 2.4 mm), τ_liq = 250 min (ratio 1.04). Stage 2: 10 mL PFA (ID 1 mm), τ_inlet = 90.9 min (ratio 0.25), BPR 7 bar. Near-batch Stage 1 in large-bore ETFE; Stage 2 in

narrow-bore PFA is better-characterized optically. Startup simulation did not reach BPR opening pressure (finite window); steady-state conditional. Acceptable near-batch Stage 1 / short Stage 2 screen.

Reviews 11 / Uncertainty 1: Stage 1 ETFE 2.4 mm ID photon penetration and mixing uncharacterized.

Reviews 11 / Uncertainty 2: Startup simulation did not reach steady state in finite window.

Reviews 11 / Uncertainty 3: Stage 2 τ_inlet 90.9 min is 25% of batch; oxidation adequacy unknown.

Reviews 11 / Uncertainty 4: Unprotected liquid branch reverse path unresolved.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 2 mL PFA, τ_liq = 100 min (ratio 0.42). Stage 2: 10 mL PFA (ID 1 mm), τ_inlet = 90.9 min (ratio 0.25), BPR 2.5 bar. Short sub-batch both stages with lower BPR. Functionally similar to Candidate 7 but with reduced dissolved O2 driving force. Acceptable as the low-pressure, short-both-stages bracketing candidate.

Reviews 12 / Uncertainty 1: BPR 2.5 bar reduces dissolved O2 driving force; adequacy for oxidation unknown.

Reviews 12 / Uncertainty 2: Both stages sub-batch; Giese and oxidation completeness both unknown.

Reviews 12 / Uncertainty 3: Unprotected liquid branch reverse path unresolved.

Reviews 12 / Uncertainty 4: Stage 2 τ_inlet 90.9 min is 25% of batch.

Physics assessment / Tool result id:
528a4b483ec4da68740988da21792a042541bcba6304dd1e73e7bea6e98e0510

Physics assessment / Evidence ids 1: C1/as_designed/entered_profile/steady

Physics assessment / Evidence ids 2: C1/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 3: C2/as_designed/entered_profile/steady

Physics assessment / Evidence ids 4: C2/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 5: C3/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 6: C3/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 7: C4/as_designed/entered_profile/steady

Physics assessment / Evidence ids 8: C4/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 9: C5/as_designed/entered_profile/steady

Physics assessment / Evidence ids 10: C5/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 11: C6/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 12: C6/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 13: C7/as_designed/entered_profile/steady

Physics assessment / Evidence ids 14: C7/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 15: C8/as_designed/entered_profile/steady

Physics assessment / Evidence ids 16: C8/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 17: C9/as_designed/entered_profile/pump_interruption

Physics assessment / Evidence ids 18: C9/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 19: C10/as_designed/entered_profile/steady

Physics assessment / Evidence ids 20: C10/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 21: C11/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 22: C11/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 23: C12/as_designed/entered_profile/steady

Physics assessment / Evidence ids 24: C12/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Proposed alternatives 1: Unapproved mitigation proposal (requires chemistry, inventory, and O2-service review before adoption): Install a forward-oriented non-return valve (e.g., CV-3301, service-verified) on the liquid branch between Stage 1 outlet and the Stage 2 gas-introduction junction, protecting Stage 1 from reverse gas intrusion. This valve's cracking pressure concerns the pressure difference across it, not BPR minus cracking pressure. Its maximum pressure rating, O2 compatibility, and low-flow behaviour must be confirmed before use.

Physics assessment / Proposed alternatives 2: Unapproved mitigation proposal: A pressure-decoupled arrangement (e.g., intermediate collection and re-pressurisation) could isolate Stage 1 from Stage 2 gas pressure; this is not a designed or approved element and requires full engineering review.

Physics assessment / Proposed alternatives 3: Reducing gas flow alone does not resolve the unprotected reverse path on the liquid branch, as confirmed by the measured backflow event in the prior experiment and the topology finding.

Physics assessment / Limitation 1: Tool results are conditional on unmeasured parameters: illustrative viscosity (1.0 mPa·s), gas plenum (0.2 mL), MFC required ΔP (1.0 bar), pump response (1.0 s), liquid compliance (0.01 mL/bar), and multiphase resistance multiplier (3.0). These are not laboratory setpoints.

Physics assessment / Limitation 2: No reversal predicted in the finite simulation windows does not establish safe operation or exclude backflow under real startup, transient, or off-normal conditions.

Physics assessment / Limitation 3: Reported metric is predicted liquid displacement from the liquid branch at the junction, not oxygen reaching Stage 1 Reactor.

Physics assessment / Limitation 4: Candidates 3, 6, and 11 startup simulations did not reach BPR opening pressure; steady-state behaviour was only assessed with the hypothetical liquid check valve configuration, which is unassigned and service-unverified.

Physics assessment / Limitation 5: The CV-3301 check valve cracking pressure (1.0 bar) concerns the pressure difference across that valve; its maximum pressure rating is unconfirmed.

Physics assessment / Limitation 6: Isothermal ideal gas model; slug flow, gas dissolution, capillary effects, reaction, and gas-front dynamics are not modelled.

Physics assessment / Limitation 7: All alternative equipment and controls remain unapproved; no inventory or design is changed by the tool.

Physics assessment / Limitation 8: The prior experimental gas reverse flow observation (KHU, 15 Sep 2026) used unverified reference conditions and remains unresolved by these simulations.

Physics assessment / Limitation 9: BPR hysteresis, valve closing dynamics, and pressure-wave propagation are not resolved.

Physics assessment / Limitation 10: Pure O2 use is a gas-source change from the protocol (air); it is unapproved and requires laboratory review before use.

Call 5: DrSafety, attempt 1
Status: recorded response, parsed successfully.

Tool name: simulate_flow_transients

Candidate ids 1: 1

Candidate ids 2: 2

Candidate ids 3: 3

Candidate ids 4: 4

Candidate ids 5: 5

Candidate ids 6: 6

Candidate ids 7: 7

Candidate ids 8: 8

Candidate ids 9: 9

Candidate ids 10: 10

Candidate ids 11: 11

Candidate ids 12: 12

Call 6: DrSafety, attempt 1
Status: unsuccessful response; retained for provenance. Unterminated string starting at: line 269 column 7 (char 19558)


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{
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    {
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      "hard_violation": false,
      "justification": "Stage 1: 2 mL PFA, τ_liq=100 min (ratio 0.42 vs batch 240 min), UV-150/450 nm, no gas, Ar-sparged feed. Stage 2: 20 mL FEP, τ_inlet=181.8 min (ratio 0.51 vs batch 360 min), O2 0.09 sccm STP, 2.008 equiv, BPR 7 bar. Pure O2 use is confirmed by chemist (Q-GAS-001) and requires oxygen-service and inventory review before execution. Unprotected reverse path st2_mixer→st1_reactor persists; CV-3301 on MFC branch does not protect the liquid branch. Tool predicted zero liquid displacement in entered-profile scenarios only—this is not safety confirmation. Acrylonitrile compatibility with blue pump tubing confirmed. Stage 2 temperature not stage-resolved; no validated rate correction.",
      "uncertainties": [
        "Unprotected reverse path on liquid branch to Stage 1 reactor unresolved; engineering review required before execution.",
        "CV-3301 pressure rating unconfirmed; cracking pressure ≠ pressure rating.",
        "Pure O2 requires oxygen-service, materials, and inventory review; not approved.",
        "Stage 2 batch temperature not stage-resolved; yield and adequate exposure time unknown.",
        "Tool results conditional on illustrative, uncalibrated assumptions; not measured evidence of safety.",
        "Gas-liquid uptake, slug flow, and photon adequacy in Stage 2 not modeled.",
        "Stage 1 τ_liq = 100 min vs batch 240 min; adequacy of photochemical exposure unconfirmed.",
        "Inline degasser unavailable; offline Ar sparging must be verified as sufficient."
      ]
```

```
  },
  {
    "candidate_id": 2,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Identical reactor geometry to C1 (2 mL PFA S1, 20 mL FEP S2, τ_inlet_S2=181.8 min) but BPR 2.5 bar. Lower system pressure expands gas volume in channel (in-channel 0.028 mL/min vs 0.012 mL/min at 7 bar); O2 STP delivery identical at 0.09 sccm, 2.008 equiv. Same unprotected reverse-path finding applies. Lower BPR reduces MFC back-pressure margin against junction pressure (tool: min MFC margin 7.29 bar vs 2.80 bar for C1), which is conditional model output only. Pure O2 requires oxygen-service review.",
    "uncertainties": [
      "Unprotected reverse path on liquid branch unresolved.",
      "Lower BPR increases gas-phase volume fraction in S2 channel; two-phase flow behavior uncharacterized.",
      "CV-3301 pressure rating unconfirmed.",
      "Pure O2 oxygen-service and inventory review required.",
      "Stage 2 temperature not stage-resolved.",
      "Tool results conditional on illustrative assumptions only."
    ]
  },
  {
    "candidate_id": 3,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 5 mL ETFE coil, ID 2.4 mm, τ_liq=250 min (ratio 1.04 vs batch 240 min), UV-150/450 nm. Wider bore reduces linear velocity and may affect photon penetration depth—not an established equivalence. Stage 2: 20 mL FEP, τ_inlet=181.8 min, BPR 7 bar. Tool startup simulation (gas_first_start) did not reach BPR operating pressure in the simulated window; full-pressure steady-state behavior not captured in as_designed scenario. Unprotected reverse path persists. Pure O2 requires oxygen-service review.",
    "uncertainties": [
      "Unprotected reverse path on liquid branch to Stage 1 unresolved.",
      "Tool as_designed startup simulation did not reach BPR; steady-state pressure behavior unconfirmed.",
      "ETFE 2.4 mm ID: photon penetration depth and mixing vs 1 mm PFA uncharacterized for this photocatalyst.",
      "CV-3301 pressure rating unconfirmed.",
      "Pure O2 oxygen-service review required.",
      "Stage 2 temperature not stage-resolved."
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  },
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{
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 "uncertainties": [
  "Unprotected reverse path on liquid branch unresolved.",
  "Lowest BPR (1.5 bar) yields highest gas-phase fraction in channel; slug/annular flow regime uncharacterized.",
  "CV-3301 pressure rating unconfirmed.",
  "Pure O2 oxygen-service review required.",
  "Stage 1 τ_liq=100 min; photochemical adequacy unconfirmed.",
  "Stage 2 temperature not stage-resolved."
 ]
},
{
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 "uncertainties": [
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  "Stage 2 τ_inlet=90.9 min is 25% of batch 360 min; oxidation adequacy unconfirmed.",
  "ETFE 2.4 mm ID in S2: gas-liquid contact and photon penetration uncharacterized.",
  "CV-3301 pressure rating unconfirmed.",
  "Pure O2 oxygen-service review required.",
  "Stage 2 temperature not stage-resolved."
 ]
},
{
 "candidate_id": 6,
 "recommendation": "acceptable",
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    "hard_violation": false,
    "justification": "Stage 1: 5 mL PFA ID 1 mm (coil_row4), τ_liq=250 min (ratio 1.04), UV-150/450 nm. Stage 2: 20 mL FEP, τ_inlet=181.8 min, BPR 7 bar. S1 coil_row4 occupies one of two available PFA 5 mL coils; quantity constraint confirmed. Tool startup simulation did not reach BPR operating pressure in as_designed scenario—same limitation as C3. Unprotected reverse path persists. Pure O2 requires oxygen-service review.",
    "uncertainties": [
     "Unprotected reverse path on liquid branch unresolved.",
     "Tool as_designed startup did not reach BPR; steady-state behavior unconfirmed for this candidate.",
     "CV-3301 pressure rating unconfirmed.",
     "Pure O2 oxygen-service review required.",
     "Stage 2 temperature not stage-resolved.",
     "PFA 5 mL coil (coil_row4) quantity: 2 available; concurrent use in other candidates must be tracked."
    ]
   },
   {
    "candidate_id": 7,
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    "hard_violation": false,
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    "uncertainties": [
     "Unprotected reverse path on liquid branch unresolved.",
     "Stage 2 τ_inlet=90.9 min; oxidation completeness unconfirmed.",
     "CV-3301 pressure rating unconfirmed.",
     "Pure O2 oxygen-service review required.",
     "Stage 2 temperature not stage-resolved.",
     "Photon delivery adequacy in S2 (Manual 2, 448 nm) not established."
    ]
   },
   {
    "candidate_id": 8,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA, τ_liq=100 min, UV-150/450 nm. Stage 2: 10 mL FEP ID 1 mm (coil_row6), τ_inlet=90.9 min (ratio 0.25), BPR 7 bar. FEP max temperature 50 °C; operation at 25 °C is within limit. Identical τ_inlet and hydraulic profile to C7; differentiator is FEP vs PFA in S2. Unprotected reverse path persists. Tool: no liquid displacement in entered-profile scenarios.",
```

```
    "uncertainties": [
      "Unprotected reverse path on liquid branch unresolved.",
      "Stage 2 τ_inlet=90.9 min; oxidation adequacy unconfirmed.",
      "FEP coil_row6 max 50 °C; temperature excursion risk if system heats above ambient—verify thermal control.",
      "CV-3301 pressure rating unconfirmed.",
      "Pure O2 oxygen-service review required.",
      "Stage 2 temperature not stage-resolved."
    ]
  },
  {
    "candidate_id": 9,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 5 mL ETFE ID 2.4 mm (coil_row11), τ_liq=250 min, UV-150/450 nm. Stage 2: 10 mL ETFE ID 2.4 mm (coil_row12), τ_inlet=90.9 min (ratio 0.25), BPR 7 bar. Both stages use wide-bore ETFE; photon penetration and gas-liquid contact in S2 uncharacterized at 2.4 mm. Tool pump-interruption scenario showed negligible predicted displacement (4.9×10⁻¹³ μL). Unprotected reverse path persists.",
    "uncertainties": [
      "Unprotected reverse path on liquid branch unresolved.",
      "Both stages use ETFE 2.4 mm ID; photon penetration and mixing uncharacterized.",
      "Stage 2 τ_inlet=90.9 min; oxidation adequacy unconfirmed.",
      "CV-3301 pressure rating unconfirmed.",
      "Pure O2 oxygen-service review required.",
      "Stage 2 temperature not stage-resolved.",
      "ETFE coil_row11 and coil_row12 quantities (2 each): concurrent allocation across candidates must be tracked."
    ]
  },
  {
    "candidate_id": 10,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA, τ_liq=100 min, UV-150/450 nm. Stage 2: 10 mL ETFE ID 2.4 mm (coil_row12), τ_inlet=90.9 min, BPR 2.5 bar. Lower BPR increases in-channel gas volume (0.028 mL/min). Tool: no liquid displacement. Unprotected reverse path persists. Wide-bore S2 gas-liquid contact uncharacterized.",
    "uncertainties": [
      "Unprotected reverse path on liquid branch unresolved.",
```

```
      "Stage 2 τ_inlet=90.9 min at BPR 2.5 bar; oxidation adequacy and flow regime unconfirmed.",
      "ETFE 2.4 mm ID in S2: gas-liquid contact uncharacterized.",
      "CV-3301 pressure rating unconfirmed.",
      "Pure O2 oxygen-service review required.",
      "Stage 2 temperature not stage-resolved."
    ]
  },
  {
    "candidate_id": 11,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 5 mL ETFE ID 2.4 mm (coil_row11), τ_liq=250 min, UV-150/450 nm. Stage 2: 10 mL PFA ID 1 mm (coil_row5), τ_inlet=90.9 min, BPR 7 bar. Tool as_designed startup did not reach BPR in the simulated window (same limitation as C3 and C6); steady-state simulation available only for hypothetical_liquid_check alternative. Unprotected reverse path persists.",
    "uncertainties": [
      "Unprotected reverse path on liquid branch unresolved.",
      "Tool as_designed startup did not reach BPR; steady-state as_designed behavior unconfirmed.",
      "ETFE 2.4 mm in S1: photon penetration depth uncharacterized.",
      "Stage 2 τ_inlet=90.9 min; oxidation adequacy unconfirmed.",
      "CV-3301 pressure rating unconfirmed.",
      "Pure O2 oxygen-service review required.",
      "Stage 2 temperature not stage-resolved."
    ]
  },
  {
    "candidate_id": 12,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA, τ_liq=100 min, UV-150/450 nm. Stage 2: 10 mL PFA ID 1 mm (coil_row5), τ_inlet=90.9 min, BPR 2.5 bar. Tool: no liquid displacement in entered-profile scenarios. Lower BPR increases in-channel gas volume. Unprotected reverse path persists. All other shared limitations apply.",
    "uncertainties": [
      "Unprotected reverse path on liquid branch unresolved.",
      "Stage 2 τ_inlet=90.9 min at BPR 2.5 bar; oxidation adequacy unconfirmed.",
      "CV-3301 pressure rating unconfirmed.",
      "Pure O2 oxygen-service review required.",
      "Stage 2 temperature not stage-resolved.",
```

```
      "Stage 1 τ_liq=100 min; photochemical adequacy unconfirmed."
    ]
   }
  ],
  "shared_safety_assessment": {
   "applies_to_all_candidates": true,
   "issues": [
    {
     "id": "SA-001",
     "title": "Unprotected reverse path: Stage 2 mixer to Stage 1 reactor (liquid branch)",
     "severity": "requires_engineering_review_before_execution",
     "finding": "In all twelve candidates, the O2 gas junction connects via the liquid branch back to Stage 1 with no declared forward-oriented non-return device on that branch. The CV-3301 check valve is on the MFC/gas branch only; it does not protect the liquid branch. This topology is structurally identical across all candidates and is consistent with the KHU observation of gas reverse flow reported 2026-09-15.",
     "tool_note": "The deterministic pressure tool predicted zero liquid displacement in entered-profile scenarios for most candidates and negligible values (sub-femtolitre to sub-picometre) in startup/interruption scenarios. These are conditional model outputs based on illustrative, uncalibrated assumptions with a finite scenario window. They do not establish that reverse flow cannot occur under actual laboratory conditions, startup sequences, or transients not captured in the model. No reversal in these tests does not establish safety.",
     "consequence": "Potential O2 intrusion into Stage 1 would violate the hard constraint that Stage 1 remain oxygen-free and could disrupt the photocatalytic Giese reaction and introduce flammability/explosion risk with the EtOH-containing feed under illumination.",
     "required_action": "Install and verify a forward-oriented non-return device (e.g., a check valve of confirmed service compatibility, pressure rating, and low-flow cracking characteristics) on the liquid branch between Stage 1 outlet and the Stage 2 gas junction, or adopt a pressure-decoupled interstage arrangement. Laboratory review of startup sequence, shutdown protocol, and pressure control is required before execution.",
     "resolved": false
    },
    {
     "id": "SA-002",
     "title": "CV-3301 check valve on MFC branch: unconfirmed pressure rating and service",
     "severity": "requires_laboratory_verification",
     "finding": "CV-3301 has a declared cracking pressure of 1.0 bar. Its maximum pressure rating is not supplied. The cracking pressure is a differential pressure across the valve, not the system BPR setpoint. Service compatibility with pure O2 and with the EtOH/buffer/acrylonitrile stream has not been verified.",
     "resolved": false
    },
    {
     "id": "SA-003",
     "title": "Pure O2 gas source: unapproved change from batch protocol",
```

```
"severity": "requires_chemist_and_safety_confirmation",
"finding": "The batch protocol used ambient air as oxidant. The chemist has answered Q-GAS-001 specifying pure O2. The gas source change to 100 mol% O2 is confirmed by chemist intent but has not been approved through oxygen-service, materials compatibility, flammability/enrichment, or inventory review. Pure O2 with EtOH solvent and photocatalyst under LED irradiation requires explicit safety assessment. This is an unapproved proposal requiring laboratory review before use.",
"resolved": false
},
{
"id": "SA-004",
"title": "Acrylonitrile exclusion from red pump tubing",
"severity": "informational_confirmed",
"finding": "All candidates assign acrylonitrile-containing Stream A to pump_e_series_blue (blue tubing). Acrylonitrile is explicitly excluded from red tubing. Assignment is correct across all candidates.",
"resolved": true
},
{
"id": "SA-005",
"title": "Stage 2 temperature not stage-resolved in batch protocol",
"severity": "unresolved_experimental_uncertainty",
"finding": "The batch protocol does not explicitly state the temperature for the 6 h oxidation stage. All candidates assign 25 °C to Stage 2. No validated rate correction or degradation risk assessment is available at this temperature. Reduced-temperature compensation and comparative degradation risk remain untested hypotheses.",
"resolved": false
},
{
"id": "SA-006",
"title": "FEP coil (coil_row6 and coil_row7) maximum temperature 50 °C",
"severity": "informational_monitor",
"finding": "FEP coils used in Stage 2 of C1, C2, C3, C4, C6, C8 are rated to 50 °C maximum. Operation at 25 °C is within limits, but LED heating effects on coil temperature in manual modules must be monitored.",
"resolved": false
},
{
"id": "SA-007",
"title": "Inline degasser unavailable; Ar sparging adequacy unconfirmed",
"severity": "unresolved_experimental_uncertainty",
```

```
      "finding": "No inline degasser is available. Offline Ar pre-sparging of the EtOH:buffer feed is the only declared deoxygenation method. Residual dissolved O2 level in Stage 1 feed is not confirmed and must be assessed before execution.",
      "resolved": false
    },
    {
      "id": "SA-008",
      "title": "Candidates 3, 6, 11: as_designed startup simulation did not reach BPR operating pressure",
      "severity": "unresolved_model_limitation",
      "finding": "For C3, C6, and C11, the tool's as_designed gas_first_start scenario terminated before the BPR opening pressure was reached. Steady-state as_designed pressure behaviour was not captured in these runs. The hypothetical_liquid_check steady-state simulation did reach operating pressure but uses an unapproved hypothetical valve. Actual startup behaviour for these candidates at 7 bar BPR is unconfirmed.",
      "resolved": false
    }
  ]
 },
 "physics_assessment": {
  "tool_result_id": "528a4b483ec4da68740988da21792a042541bcba6304dd1e73e7bea6e98e0510",
  "evidence_ids": [
    "C1/as_designed/entered_profile/steady",
    "C1/hypothetical_liquid_check/entered_profile/steady",
    "C2/as_designed/entered_profile/steady",
    "C2/hypothetical_liquid_check/entered_profile/steady",
    "C3/as_designed/entered_profile/gas_first_start",
    "C3/hypothetical_liquid_check/entered_profile/steady",
    "C4/as_designed/entered_profile/steady",
    "C4/hypothetical_liquid_check/entered_profile/steady",
    "C5/as_designed/entered_profile/steady",
    "C5/hypothetical_liquid_check/entered_profile/steady",
    "C6/as_designed/entered_profile/gas_first_start",
    "C6/hypothetical_liquid_check/entered_profile/steady",
    "C7/as_designed/entered_profile/steady",
    "C7/hypothetical_liquid_check/entered_profile/steady",
    "C8/as_designed/entered_profile/steady",
    "C8/hypothetical_liquid_check/entered_profile/steady",
    "C9/as_designed/entered_profile/pump_interruption",
    "C9/hypothetical_liquid_check/entered_profile/steady",
```

"C10/as_designed/entered_profile/steady",
"C10/hypothetical_liquid_check/entered_profile/steady",
"C11/as_designed/entered_profile/gas_first_start",
"C11/hypothetical_liquid_check/entered_profile/steady",
"C12/as_designed/entered_profile/steady",
"C12/hypothetical_liquid_check/entered_profile/steady"
],
"proposed_alternatives": [
"Install a verified forward-oriented check valve on the

Call 7: DrSafety, attempt 2
Status: recorded response, parsed successfully.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA at 100 min liquid tau (0.42× batch). Stage 2: 20 mL FEP at 181.8 min inlet/STP tau (0.51× batch), 7 bar BPR, 2.01 equiv O2 at 0.09 sccm. Unprotected reverse path to Stage 1 unresolved; pure O2 use requires chemist and oxygen-service confirmation. Tool predicts no liquid displacement in entered-profile scenarios (unvalidated). CV-3301 pressure rating unconfirmed.

Reviews 1 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 reactor unresolved; engineering review required before execution.

Reviews 1 / Uncertainty 2: CV-3301 on MFC branch does not protect liquid branch; max pressure rating unconfirmed.

Reviews 1 / Uncertainty 3: Pure O2 substitution for air is unapproved; requires chemist, inventory, and oxygen-service review.

Reviews 1 / Uncertainty 4: Gas check-valve cracking pressure (1.0 bar) concerns differential pressure across that valve, not BPR minus cracking pressure; low-flow seating unverified.

Reviews 1 / Uncertainty 5: Tool results conditional on illustrative, unmeasured parameters; no reversal predicted is not proof of safe operation.

Reviews 1 / Uncertainty 6: Stage 2 batch temperature not stage-resolved; rate and yield implications unknown.

Reviews 1 / Uncertainty 7: Acrylonitrile excluded from RED pump tubing; BLUE pump assignment confirmed.

Reviews 1 / Uncertainty 8: Photon adequacy at 450 nm vs. 452 nm target not established.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical reactor geometry to C1 but BPR reduced to 2.5 bar, yielding larger in-channel gas volume (0.028 mL/min) and wider MFC margin (7.29 bar above required). Stage 1 tau and Stage 2 inlet/STP tau unchanged at 100 and 181.8 min. Unprotected liquid-branch reverse path unresolved. Same O2 confirmation requirement applies.

Reviews 2 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 2 / Uncertainty 2: Lower BPR increases in-channel gas volume; gas-liquid flow pattern unverified.

Reviews 2 / Uncertainty 3: Pure O2 unapproved; requires chemist and oxygen-service confirmation.

Reviews 2 / Uncertainty 4: CV-3301 max pressure rating and low-flow seating unconfirmed.

Reviews 2 / Uncertainty 5: Tool results conditional on illustrative assumptions only.

Reviews 2 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 uses 5 mL ETFE coil (ID 2.4 mm) at 250 min liquid tau (1.04× batch), the only candidate exceeding the batch hold at Stage 1. Stage 2: 20 mL FEP at 181.8 min inlet/STP tau, 7 bar BPR. Gas-first startup simulation reached only pre-operating pressure; BPR_opening_pressure_reached false for as-designed scenario—full startup behaviour unconfirmed. Large ID coil raises photon penetration uncertainty for Stage 1.

Reviews 3 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 3 / Uncertainty 2: Gas-first startup simulation did not reach operating pressure; steady-state behaviour unconfirmed for as-designed case.

Reviews 3 / Uncertainty 3: ETFE 2.4 mm ID coil: photon penetration depth to reaction mixture not established.

Reviews 3 / Uncertainty 4: Pure O2 unapproved; requires chemist and oxygen-service confirmation.

Reviews 3 / Uncertainty 5: CV-3301 pressure rating unconfirmed.

Reviews 3 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 2 mL PFA at 100 min (0.42× batch). Stage 2: 20 mL FEP at 181.8 min inlet/STP tau, 1.5 bar BPR. Lowest system pressure; in-channel gas volume largest among 20 mL FEP candidates (0.040 mL/min). MFC margin 8.28 bar. Tool predicts no liquid displacement in entered-profile scenarios. Unprotected liquid-branch path unresolved.

Reviews 4 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 4 / Uncertainty 2: 1.5 bar BPR yields highest in-channel gas fraction; two-phase flow pattern unverified.

Reviews 4 / Uncertainty 3: Pure O2 unapproved; requires chemist and oxygen-service confirmation.

Reviews 4 / Uncertainty 4: CV-3301 pressure rating and low-flow seating unconfirmed.

Reviews 4 / Uncertainty 5: Tool results conditional on illustrative assumptions.

Reviews 4 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA at 100 min (0.42× batch). Stage 2: 10 mL ETFE (ID 2.4 mm) at 90.9 min inlet/STP tau (0.25× batch), 7 bar BPR. Shortest Stage 2 apparent index across all candidates. Whether

90.9 min is adequate for oxidation is unknown; shorter exposure is a screening point, not a defect. Large ID raises photon penetration uncertainty at Stage 2.

Reviews 5 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 5 / Uncertainty 2: Stage 2 inlet/STP tau 90.9 min is 0.25× batch; adequacy for oxidation unknown.

Reviews 5 / Uncertainty 3: ETFE 2.4 mm ID at Stage 2: photon penetration to liquid not established.

Reviews 5 / Uncertainty 4: Pure O2 unapproved.

Reviews 5 / Uncertainty 5: CV-3301 pressure rating unconfirmed.

Reviews 5 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA (ID 1 mm) at 250 min liquid tau (1.04× batch), 7 bar BPR. Stage 2: 20 mL FEP at 181.8 min inlet/STP tau. Gas-first startup simulation did not reach operating pressure (BPR_opening_pressure_reached false for as-designed); steady-state unconfirmed. Photon penetration in 1 mm ID coil is better than 2.4 mm ID but adequacy not established.

Reviews 6 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 6 / Uncertainty 2: As-designed gas-first startup did not reach operating pressure; full steady-state behaviour unconfirmed.

Reviews 6 / Uncertainty 3: Pure O2 unapproved.

Reviews 6 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 6 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 6 / Uncertainty 6: Photon adequacy at 450 nm in Stage 1 5 mL PFA coil not established.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA at 100 min (0.42× batch). Stage 2: 10 mL PFA (ID 1 mm) at 90.9 min inlet/STP tau (0.25× batch), 7 bar BPR. Shorter Stage 2 exposure is a screening coordinate. Narrower ID than ETFE alternatives may improve gas-liquid mixing. Tool predicts no liquid displacement in entered-profile scenarios.

Reviews 7 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 7 / Uncertainty 2: Stage 2 inlet/STP tau 90.9 min (0.25× batch); oxidation adequacy unknown.

Reviews 7 / Uncertainty 3: Pure O2 unapproved.

Reviews 7 / Uncertainty 4: CV-3301 pressure rating and low-flow seating unconfirmed.

Reviews 7 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 2 mL PFA at 100 min (0.42× batch). Stage 2: 10 mL FEP (ID 1 mm, max 50°C) at 90.9 min inlet/STP tau (0.25× batch), 7 bar BPR. FEP max temperature 50°C limits future elevated-

temperature screening at Stage 2. Hydraulics and tool results identical to C7. Unprotected reverse path unresolved.

Reviews 8 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 8 / Uncertainty 2: FEP coil Stage 2 max 50°C limits temperature flexibility.

Reviews 8 / Uncertainty 3: Stage 2 inlet/STP tau 90.9 min (0.25× batch); adequacy unknown.

Reviews 8 / Uncertainty 4: Pure O2 unapproved.

Reviews 8 / Uncertainty 5: CV-3301 pressure rating unconfirmed.

Reviews 8 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 5 mL ETFE (ID 2.4 mm) at 250 min liquid tau (1.04× batch). Stage 2: 10 mL ETFE (ID 2.4 mm) at 90.9 min inlet/STP tau (0.25× batch), 7 bar BPR. Both stages use large-bore coils; photon penetration in both stages unestablished. Pump-interruption scenario produced negligible predicted displacement (~0.5 pL). Unprotected liquid-branch path unresolved.

Reviews 9 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 9 / Uncertainty 2: Both stages use 2.4 mm ID ETFE; photon penetration to mixture unestablished at both stages.

Reviews 9 / Uncertainty 3: Stage 2 inlet/STP tau 90.9 min (0.25× batch); adequacy unknown.

Reviews 9 / Uncertainty 4: Pure O2 unapproved.

Reviews 9 / Uncertainty 5: CV-3301 pressure rating unconfirmed.

Reviews 9 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA at 100 min (0.42× batch). Stage 2: 10 mL ETFE (ID 2.4 mm) at 90.9 min inlet/STP tau (0.25× batch), 2.5 bar BPR. Lower BPR yields higher in-channel gas volume (0.028 mL/min). MFC margin 7.32 bar. Large ID at Stage 2 raises photon penetration uncertainty. Unprotected reverse path unresolved.

Reviews 10 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 10 / Uncertainty 2: ETFE 2.4 mm ID at Stage 2; photon penetration unestablished.

Reviews 10 / Uncertainty 3: Lower BPR increases in-channel gas fraction; two-phase flow pattern unverified.

Reviews 10 / Uncertainty 4: Pure O2 unapproved.

Reviews 10 / Uncertainty 5: CV-3301 pressure rating unconfirmed.

Reviews 10 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 5 mL ETFE (ID 2.4 mm) at 250 min liquid tau (1.04× batch). Stage 2: 10 mL PFA (ID 1 mm) at 90.9 min inlet/STP tau (0.25× batch), 7 bar BPR. Gas-first startup simulation did not reach operating pressure for as-designed case; steady-state unconfirmed. Large ID at Stage 1 raises photon penetration uncertainty. Narrower ID at Stage 2 is preferable for gas-liquid contact.

Reviews 11 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 11 / Uncertainty 2: As-designed gas-first startup did not reach operating pressure; steady-state unconfirmed.

Reviews 11 / Uncertainty 3: ETFE 2.4 mm ID at Stage 1; photon penetration unestablished.

Reviews 11 / Uncertainty 4: Pure O2 unapproved.

Reviews 11 / Uncertainty 5: CV-3301 pressure rating unconfirmed.

Reviews 11 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 2 mL PFA at 100 min (0.42× batch). Stage 2: 10 mL PFA (ID 1 mm) at 90.9 min inlet/STP tau (0.25× batch), 2.5 bar BPR. Lower BPR increases in-channel gas volume (0.028 mL/min). MFC margin 7.30 bar. All-PFA, narrow-bore arrangement throughout is compatible with photon delivery. Unprotected reverse path unresolved.

Reviews 12 / Uncertainty 1: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 12 / Uncertainty 2: Lower BPR increases in-channel gas fraction; two-phase flow pattern unverified.

Reviews 12 / Uncertainty 3: Stage 2 inlet/STP tau 90.9 min (0.25× batch); adequacy unknown.

Reviews 12 / Uncertainty 4: Pure O2 unapproved.

Reviews 12 / Uncertainty 5: CV-3301 pressure rating unconfirmed.

Reviews 12 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Physics assessment / Tool result id:
528a4b483ec4da68740988da21792a042541bcba6304dd1e73e7bea6e98e0510

Physics assessment / Evidence ids 1: C1/as_designed/entered_profile/steady

Physics assessment / Evidence ids 2: C1/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 3: C2/as_designed/entered_profile/steady

Physics assessment / Evidence ids 4: C2/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 5: C3/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 6: C3/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 7: C4/as_designed/entered_profile/steady

Physics assessment / Evidence ids 8: C4/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 9: C5/as_designed/entered_profile/steady

Physics assessment / Evidence ids 10: C5/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 11: C6/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 12: C6/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 13: C7/as_designed/entered_profile/steady

Physics assessment / Evidence ids 14: C7/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 15: C8/as_designed/entered_profile/steady

Physics assessment / Evidence ids 16: C8/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 17: C9/as_designed/entered_profile/pump_interruption

Physics assessment / Evidence ids 18: C9/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 19: C10/as_designed/entered_profile/steady

Physics assessment / Evidence ids 20: C10/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 21: C11/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 22: C11/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 23: C12/as_designed/entered_profile/steady

Physics assessment / Evidence ids 24: C12/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Proposed alternatives 1: A dedicated check valve (CV-3301 or equivalent) on the liquid branch between Stage 1 reactor outlet and the Stage 2 gas-introduction junction is an unapproved proposal: requires engineering review, inventory confirmation, service verification, and oxygen-compatibility assessment before use.

Physics assessment / Proposed alternatives 2: A pressure-decoupled interstage arrangement (e.g., gravity or pressure-controlled transfer) is an unapproved hypothesis requiring separate engineering and chemistry review.

Physics assessment / Proposed alternatives 3: Pure O2 gas supply in place of the batch air atmosphere is an unapproved gas-source change: requires chemist confirmation, oxygen-service review, and materials compatibility assessment before laboratory use.

Physics assessment / Proposed alternatives 4: CV-3301 cracking pressure (1.0 bar) concerns the differential pressure across that valve only; the valve on the MFC branch does not protect the liquid branch. Branch-specific protection on the liquid line is a separate unapproved item.

Physics assessment / Limitation 1: Tool uses illustrative, unmeasured parameters (viscosity, compliance, gas plenum, conductance, pump and MFC response times); results are conditional model outputs, not measured evidence or safety certification.

Physics assessment / Limitation 2: No liquid displacement predicted in entered-profile scenarios does not establish that backflow cannot occur; the unprotected liquid-branch reverse path (st2_mixer → st1_reactor) is an unresolved credible risk confirmed by the KHU qualitative observation.

Physics assessment / Limitation 3: Candidates 3, 6, and 11 (as-designed, gas-first startup): BPR opening pressure was not reached within the simulated window; steady-state behaviour during full pressurisation is unconfirmed for those as-designed cases.

Physics assessment / Limitation 4: Predicted quantities are liquid displacement at the junction, not confirmation that oxygen reaches Stage 1 Reactor; actual gas transport through the liquid column is not modelled.

Physics assessment / Limitation 5: Gas check-valve cracking pressure (1.0 bar) governs pressure difference across that valve; it does not set a minimum system pressure margin above which backflow is precluded.

Physics assessment / Limitation 6: Isothermal ideal-gas model; no slug, dissolution, capillary, multiphase front, or reaction model included.

Physics assessment / Limitation 7: Finite scenario window; no lifetime, frequency, or reliability assessment is provided.

Physics assessment / Limitation 8: Hypothetical liquid-branch check valve is not assigned, not service-verified, and not in the approved inventory; zero leakage assumed is an idealisation.

Physics assessment / Limitation 9: CV-3301 maximum pressure rating not supplied; suitability at 7 bar or 8 bar system pressures requires laboratory verification before use.

Physics assessment / Limitation 10: All candidates share the same unresolved reverse-path finding (BF-REVERSE-PATH:st2_mixer:s3:st1_reactor); reducing gas flow does not close this path.

Physics assessment / Limitation 11: Acrylonitrile is listed as excluded from RED pump tubing; all candidates correctly assign liquid feed to BLUE pump, but solvent-wetted component compatibility beyond pump tubing should be confirmed.

Call 8: Skeptic, attempt 1
Status: recorded response, parsed successfully.

Vetoes: []

Shared assessment / Gas source change: Pure O2 replacing batch air is a gas-source change. source_context.gas_delivery.requires_chemist_confirmation is not explicitly set to true in the schema field, but the intake record (Q-GAS-001) records this as a user-confirmed requirement and the measured-evidence context notes the prior experiment used O2 without design approval, resulting in observed reverse flow. Pure O2 use remains unapproved for oxygen-service purposes across all candidates and must undergo chemist, inventory, and oxygen-service review before laboratory execution.

Shared assessment / Reverse path finding: All twelve candidates share an unresolved credible risk: BF-REVERSE-PATH:st2_mixer:s3:st1_reactor. The liquid branch between Stage 1 outlet and the Stage 2 gas-introduction junction carries no forward-oriented non-return protection. The CV-3301 check valve is declared only on the MFC (gas) branch; it does not protect the liquid branch. A valve on one branch does not protect a different branch. The KHU qualitative observation (15 Sep 2026) of gas reverse flow is unresolved measured evidence. Reducing gas flow does not close this unprotected reverse path. No simulation result — including the tool's prediction of zero liquid displacement — establishes that backflow cannot occur: the tool uses illustrative unmeasured parameters and a finite scenario window, and its output is predicted liquid displacement at the junction, not confirmation that oxygen reaches or does not reach Stage 1. This finding requires branch-specific engineering review before any candidate is executed.

Shared assessment / Physics tool status: DrFluidics and DrSafety results are conditional on unmeasured illustrative parameters (viscosity 1.0 mPa·s, gas plenum 0.2 mL, MFC ΔP 1.0 bar, liquid compliance 0.01 mL/bar, multiphase resistance multiplier 3.0). No reversal predicted in these finite scenario windows does not establish safety or exclude backflow under real startup, off-normal, or shutdown conditions. Candidates 3, 6, and 11 gas-first startup simulations did not reach BPR opening pressure within the simulated window; steady-state behaviour for those as-designed cases is unconfirmed.

Shared assessment / Cv3301 pressure rating: CV-3301 cracking pressure is 1.0 bar; this concerns the pressure difference across that valve. The maximum pressure rating of CV-3301 is not supplied and must be verified before use at 7 bar or 8 bar system pressures. Cracking pressure is not a pressure rating.

Shared assessment / Stage2 temperature: Stage 2 batch temperature is not stage-resolved in the protocol. All candidates assign 25°C; this is unverified. No validated rate correction or yield consequence can be stated.

Shared assessment / Stage2 gas residence time: Stage 2 Q_gas_sccm and Q_gas_STP values are gas volume at inlet/STP (273.15 K, 1.01325 bar). The apparent residence-time index V/(Q_liquid + Q_gas_STP) is not a measured liquid residence time or photon exposure index. Gas equivalents are proposed delivered molar ratios, not established stoichiometric consumption or uptake.

Shared assessment / Wavelength note: Protocol specifies 452 nm. Assigned light sources are UV-150/450 nm (Stage 1) and Manual 2/448 nm (Stage 2). These are nearest available; adequacy for catalyst excitation is not established.

Shared assessment / Acrylonitrile pump: Acrylonitrile is excluded from RED pump tubing. All candidates assign liquid feed to BLUE pump; this is consistent with the exclusion constraint.

Shared assessment / No kinetic data: Kinetics are uncharacterized for both stages. Stage-hold ratios are exposure indices only. No yield, conversion, or rate superiority can be inferred from any ratio or residence-time comparison. Near-batch times are not inherently the best choice; short times are not inherently failures.

Shared assessment / Large bore etfe photon: Candidates using ETFE 5 mL or 10 mL coils (ID 2.4 mm) in Stage 1 or Stage 2 carry an uncharacterized photon penetration uncertainty. The batch protocol used a narrow-bore tube irradiated at close range; 2.4 mm ID reduces surface-to-volume ratio and increases path length. This is an unresolved experimental uncertainty, not a hard violation.

Shared assessment / Fep temperature: FEP coils (coil_row6, coil_row7) have max temperature 50°C. At the 25°C screen temperature this is not a constraint, but it limits future elevated-temperature exploration at Stage 2 for candidates using those reactors.

Candidate justifications / C1: Baseline: S1 2 mL PFA τ_liq=100 min (0.42×), S2 20 mL FEP τ_inlet=181.8 min (0.51×), BPR 7 bar, 2.01 equiv O2. Balanced sub-batch screen; interpretable anchor. No hard violations. All shared uncertainties apply.

Candidate justifications / C2: Identical reactors to C1; BPR reduced to 2.5 bar. In-channel gas volume 0.028 mL/min (vs 0.012 mL/min in C1); higher gas fraction may alter slug behaviour. Lower pressure reduces O2 solubility driving force — effect uncharacterized. Useful BPR comparison. No hard violations.

Candidate justifications / C3: S1 5 mL ETFE ID 2.4 mm τ_liq=250 min (1.04×); S2 20 mL FEP τ_inlet=181.8 min, BPR 7 bar. Only candidate with S1 at or above batch. Gas-first as-designed startup simulation did not reach BPR opening pressure (finite window); steady-state unconfirmed for that case. ETFE 2.4 mm photon penetration uncharacterized. No hard violations; startup operability warrants chemist attention.

Candidate justifications / C4: S1 2 mL PFA τ_liq=100 min (0.42×); S2 20 mL FEP τ_inlet=181.8 min, BPR 1.5 bar. Lowest system pressure of all candidates; in-channel gas volume 0.040 mL/min is highest in 20 mL FEP group. O2 solubility driving force lowest here. Useful low-pressure bracket. No hard violations.

Candidate justifications / C5: S1 2 mL PFA τ_liq=100 min (0.42×); S2 10 mL ETFE ID 2.4 mm τ_inlet=90.9 min (0.25×), BPR 7 bar. Both stages well below batch. ETFE 2.4 mm in S2: photon penetration and gas-liquid distribution uncharacterized. Shortest S2 apparent index. No hard violations; S2 adequacy most uncertain here.

Candidate justifications / C6: S1 5 mL PFA ID 1 mm τ_liq=250 min (1.04×); S2 20 mL FEP τ_inlet=181.8 min, BPR 7 bar. Near-batch S1 in narrow-bore PFA is better-characterized optically than ETFE 2.4 mm. Gas-first as-designed startup simulation did not reach BPR opening pressure (finite window). No hard violations; startup confirmation needed.

Candidate justifications / C7: S1 2 mL PFA τ_liq=100 min (0.42×); S2 10 mL PFA ID 1 mm τ_inlet=90.9 min (0.25×), BPR 7 bar. Both stages sub-batch, narrow-bore throughout. Provides short-S2 reference with favourable photon geometry. No hard violations.

Candidate justifications / C8: S1 2 mL PFA τ_liq=100 min (0.42×); S2 10 mL FEP ID 1 mm τ_inlet=90.9 min (0.25×), BPR 7 bar. Functionally identical exposure profile to C7; FEP Stage 2 max 50°C limits future elevated-temperature use but is not a constraint at 25°C. No hard violations; distinguishing variable vs C7 is Stage 2 material only.

Candidate justifications / C9: S1 5 mL ETFE ID 2.4 mm τ_liq=250 min (1.04×); S2 10 mL ETFE ID 2.4 mm τ_inlet=90.9 min (0.25×), BPR 7 bar. Both stages large-bore ETFE; photon penetration and gas-liquid contact uncharacterized in both. Carries the greatest geometric uncertainty of the near-batch-S1 group. No hard violations.

Candidate justifications / C10: S1 2 mL PFA τ_liq=100 min (0.42×); S2 10 mL ETFE ID 2.4 mm τ_inlet=90.9 min (0.25×), BPR 2.5 bar. Lower BPR with large-bore S2; highest in-channel gas fraction among 10 mL S2

candidates (0.028 mL/min). O2 solubility and slug stability least favourable in this sub-group. No hard violations.

Candidate justifications / C11: S1 5 mL ETFE ID 2.4 mm τ_liq=250 min (1.04×); S2 10 mL PFA ID 1 mm τ_inlet=90.9 min (0.25×), BPR 7 bar. Gas-first as-designed startup did not reach BPR opening pressure (finite window); steady-state unconfirmed. S1 photon penetration uncharacterized (2.4 mm). Mixed-bore design provides a cross-stage geometry comparison. No hard violations.

Candidate justifications / C12: S1 2 mL PFA τ_liq=100 min (0.42×); S2 10 mL PFA ID 1 mm τ_inlet=90.9 min (0.25×), BPR 2.5 bar. All-PFA narrow-bore throughout; lower BPR than C7. Both stages well below batch. Low-pressure, short-both-stages bracketing candidate. No hard violations.

Requested measurement 1: Branch-specific non-return protection on the liquid line between Stage 1 outlet and the Stage 2 gas-introduction junction: verify device identity, service direction, cracking pressure differential, maximum pressure rating, O2 compatibility, and low-flow seating behaviour before execution of any candidate.

Requested measurement 2: CV-3301 maximum pressure rating: must be confirmed independently from cracking pressure specification before use at system pressures up to 7–8 bar.

Requested measurement 3: Chemist and oxygen-service review of pure O2 substitution for the batch air atmosphere: O2 concentration, materials compatibility, MFC service range, and laboratory safety approval required before any candidate is executed.

Requested measurement 4: Gas supply pressure and MFC inlet differential pressure at operating conditions: required to validate model margin estimates; illustrative values (12 bar supply, 1.0 bar ΔP required) are not measured setpoints.

Requested measurement 5: Stage 2 batch temperature: protocol does not resolve a temperature for Stage 2; confirm operating temperature before designing rate or yield expectations.

Requested measurement 6: Startup sequence and pressure traces for Stage 1–Stage 2 connected operation: required to validate or falsify the unresolved backflow risk and simulation predictions.

Requested measurement 7: Photon penetration depth for 2.4 mm ID ETFE coils (Stage 1 in C3, C6 startup comparison; C9, C11): verify irradiance at reaction locus relative to 452 nm protocol lamp at 3 cm distance.

Requested measurement 8: Stage 1 Giese conversion as a function of flow residence time: minimum data point needed to select a Stage 1 reactor for yield-optimised operation.

Requested measurement 9: Stage 2 oxidation completion as a function of flow residence time and O2 equivalent at 7 bar and 2.5 bar: required to distinguish among Stage 2 reactor and BPR options.

Requested measurement 10: Acrylonitrile compatibility and solubility in EtOH:pH 9 buffer at 0.2 M stock concentration: confirm no precipitation or phase separation in the pump and inlet tubing at 0.02 mL/min.

Call 9: Chief, attempt 1
Status: recorded response, parsed successfully.

Candidate: 1

Justification: Candidate 1 (S1: 2 mL PFA, τ_liq=100 min, 0.42× batch; S2: 20 mL FEP, τ_inlet=181.8 min, 0.51× batch; BPR 7 bar, 2.01 equiv O2, 0.09 sccm) is selected as the interpretable balanced baseline. Under balanced priority with no kinetic data, neither stage should be assumed adequate or inadequate from time ratios alone. C1 provides mid-range exposure in both stages, uses narrow-bore (1 mm) PFA and FEP giving defined photon geometry, and sets the highest system pressure among the 20 mL FEP group (7 bar), maximising dissolved O2 driving force for Stage 2. All shared unresolved issues apply; no hard violations found.

Objective alignment: Balanced priority requires an interpretable starting point from which shorter and longer stage holds can be compared. C1 anchors the screen at 0.42× batch (S1) and 0.51× batch (S2), covering sub-batch exposure for both stages simultaneously. This is not the shortest or longest design; it is the mid-range

combination that allows yield data from a single run to direct subsequent experiments toward longer S1 (C3, C6) or longer S2 (if both 0.42× and 0.51× prove insufficient), or shorter (C7, C12) if both are already excessive. BPR 7 bar provides the highest O2 pressure head among FEP-20 mL candidates, minimising one source of Stage 2 variability. No yield or conversion can be inferred; adequacy of either stage is uncharacterized.

Alternative 1 / Candidate: 6

Alternative 1 / Reason not selected: S1 at 250 min (1.04× batch) in 5 mL PFA 1 mm provides near-batch Stage 1 exposure and is the preferred near-batch S1 candidate (narrow bore, better photon geometry than ETFE 2.4 mm). However, gas-first as-designed startup simulation did not reach BPR opening pressure within the finite window, leaving full steady-state startup behaviour unconfirmed. Under balanced priority with no prior data, spending the first experiment on a near-batch S1 condition without confirming startup operability is less efficient than C1 as an anchor. If C1 data show Stage 1 conversion is limiting, C6 becomes the priority follow-up.

Alternative 2 / Candidate: 7

Alternative 2 / Reason not selected: S1: 100 min (0.42× batch, same as C1); S2: 10 mL PFA 1 mm, τ_inlet=90.9 min (0.25× batch), BPR 7 bar. Provides the short-S2, short-S1 bracketing point. S2 apparent index is the lowest among narrow-bore 7-bar candidates; if oxidation requires closer to batch time, this condition risks insufficient Stage 2 exposure without yielding diagnostic information. C1's longer S2 (181.8 min, 0.51× batch) is more likely to produce detectable product, making C1 more informative as the first experiment. C7 is the recommended short-S2 follow-up after C1.

Alternative 3 / Candidate: 3

Alternative 3 / Reason not selected: S1: 5 mL ETFE 2.4 mm, τ_liq=250 min (1.04× batch); S2: 20 mL FEP, τ_inlet=181.8 min, BPR 7 bar. Near-batch S1 in large-bore ETFE; photon penetration at 2.4 mm ID is uncharacterized relative to the batch narrow-tube geometry. Additionally, gas-first as-designed startup simulation did not reach BPR opening pressure (finite window). The combination of uncharacterized photon delivery and unconfirmed startup makes C3 less interpretable as a first experiment than C1 (PFA 1 mm, simpler startup profile). If S1 is rate-limiting, C6 (same τ, narrow bore, confirmed photon geometry) is preferred over C3.

Shared assessment / Applies to all candidates 1: Unprotected liquid-branch reverse path: gas introduction junction (Stage 2) connects directly to Stage 1 reactor via the liquid branch with no forward-oriented non-return valve on that branch. The CV-3301 check valve is declared only on the MFC (gas) branch and does not protect the liquid branch. This is an unresolved credible risk (BF-REVERSE-PATH:st2_mixer:s3:st1_reactor) across all twelve candidates. Reducing gas flow does not close this path. Branch-specific non-return protection on the liquid line must be reviewed before any candidate is executed.

Shared assessment / Applies to all candidates 2: Pure O2 substitution: The protocol specifies air (ambient atmosphere). Use of pure O2 was not approved in the original design; the prior experiment used O2 without design approval and observed reverse flow. Pure O2 use constitutes an unapproved gas-source change requiring chemist confirmation, oxygen-service review, and materials compatibility assessment before laboratory execution.

Shared assessment / Applies to all candidates 3: CV-3301 pressure rating: Cracking pressure (1.0 bar) is not a pressure rating. Maximum pressure rating of CV-3301 is unconfirmed and must be verified before use at 7–8 bar system pressures. The cracking pressure concerns the differential pressure across that valve, not BPR minus cracking pressure.

Shared assessment / Applies to all candidates 4: Physics tool results: DrFluidics and DrSafety outputs are conditional on illustrative unmeasured parameters (viscosity 1.0 mPa·s, gas plenum 0.2 mL, MFC ΔP 1.0 bar, liquid compliance 0.01 mL/bar, multiphase resistance multiplier 3.0). No liquid displacement predicted does not establish safe operation or exclude backflow under real startup, transient, or shutdown conditions. Tool output is predicted liquid displacement at the junction, not confirmation that oxygen reaches or does not reach Stage 1.

Shared assessment / Applies to all candidates 5: Gas residence-time index: Q_gas_sccm and Q_gas_STP values are gas volume at inlet/STP (273.15 K, 1.01325 bar). The apparent Stage 2 index V/(Q_liquid + Q_gas_STP) is not a measured liquid residence time or photon exposure. Gas equivalents are proposed inlet molar ratios, not measured stoichiometric uptake.

Shared assessment / Applies to all candidates 6: Wavelength: Protocol specifies 452 nm. UV-150/450 nm (Stage 1) and Manual 2/448 nm (Stage 2) are nearest available; photon adequacy for catalyst excitation is not established.

Shared assessment / Applies to all candidates 7: Stage 2 temperature: Not batch-resolved. All candidates assign 25°C; no validated rate correction is available.

Shared assessment / Applies to all candidates 8: Kinetics: Uncharacterized for both stages. No yield, conversion, or rate superiority can be inferred from any hold ratio.

Shared assessment / Applies to all candidates 9: Acrylonitrile: Excluded from RED pump tubing. All candidates assign liquid feed to BLUE pump; consistent with exclusion constraint.

Answer impact 1 / Question: Q-GAS-001

Answer impact 1 / Effect: Pure O2 confirmed by chemist (Q-GAS-001); however, this is an unapproved gas-source change relative to the batch air atmosphere and requires oxygen-service review and laboratory safety confirmation before use. The answer sets the gas identity but does not establish approval for oxygen service.

Answer impact 2 / Question: Q-GAS-003

Answer impact 2 / Effect: Oxygen introduction at Stage 2 inlet only is implemented in C1 (and all candidates). The unprotected liquid-branch reverse path means this constraint cannot be guaranteed without branch-specific non-return protection; this remains unresolved.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: The KHU qualitative backflow observation (15 Sep 2026) is the only measured operability evidence. It directly motivates the unresolved reverse-path finding and confirms that reducing gas flow alone does not resolve the topology risk. It does not quantify backflow rate or conditions.

Answer impact 4 / Question: Q-HYP-001

Answer impact 4 / Effect: Stage 1 oxygen exclusion depends on resolving the liquid-branch reverse path before execution. Stage 2 oxygen availability is determined by BPR (7 bar in C1 maximises O2 partial pressure) and inlet equivalents (2.01 equiv); gas-liquid mass transfer is uncharacterized. The hypothesis that Stage 1 sulfide formation limits Stage 2 oxidation cannot be tested until Stage 1 conversion data are available.

Limitation 1: No kinetic data exist for either stage; stage-hold ratios are exposure indices only, not conversion or yield predictors.

Limitation 2: The unprotected liquid-branch reverse path is unresolved across all candidates; no candidate can be executed safely until branch-specific non-return protection is reviewed.

Limitation 3: Pure O2 is unapproved as a gas source; chemist confirmation, O2-service review, and materials compatibility assessment are required before execution.

Limitation 4: CV-3301 maximum pressure rating is unconfirmed; cracking pressure (1.0 bar) is not a pressure rating.

Limitation 5: Physics tool (DrFluidics/DrSafety) results are conditional on illustrative unmeasured parameters; no reversal predicted does not establish safe operation.

Limitation 6: Stage 2 inlet/STP apparent residence time (181.8 min) is not a measured liquid exposure; liquid τ in Stage 2 is 1000 min (unconstrained by gas index) and actual photon exposure time is also uncharacterized.

Limitation 7: Stage 2 temperature is not batch-resolved; 25°C assignment is unverified.

Limitation 8: Wavelength mismatch (450/448 nm vs 452 nm target) is unconfirmed as adequate for catalyst excitation.

Limitation 9: FEP coil (Stage 2) max temperature 50°C limits future elevated-temperature Stage 2 exploration.

Limitation 10: Candidates 3, 6, and 11 as-designed gas-first startup simulations did not reach BPR opening pressure; startup operability for those designs is unconfirmed.

Suggested measurement 1: Verify and install branch-specific non-return protection on the liquid line between Stage 1 outlet and the Stage 2 gas-introduction junction; confirm device cracking pressure differential, maximum pressure rating, O2 compatibility, and low-flow seating before any execution.

Suggested measurement 2: Confirm CV-3301 maximum pressure rating independent of cracking pressure specification; verify O2 service compatibility.

Suggested measurement 3: Obtain chemist, inventory, and oxygen-service review approval for pure O2 use before laboratory execution.

Suggested measurement 4: Measure Stage 1 outlet conversion (Giese product) from C1 run to determine whether 100 min liquid τ is sufficient; this single data point determines whether longer-S1 candidates (C3, C6) are needed.

Suggested measurement 5: Measure Stage 2 sulfoxide yield from C1 run; compare to C7 (shorter S2) and C6 (longer S1, same S2) to resolve the stage-limiting factor.

Suggested measurement 6: Record startup pressure traces and startup sequence for the connected two-stage system to validate or falsify the unresolved reverse-path risk and simulation assumptions.

Suggested measurement 7: Confirm gas supply pressure and MFC inlet differential pressure under operating conditions (illustrative model assumed 12 bar supply, 1.0 bar ΔP required).

Suggested measurement 8: Verify Stage 2 operating temperature resolves to 25°C as assigned; the batch protocol does not explicitly state Stage 2 temperature.

Suggested measurement 9: Assess photon penetration adequacy at 450 nm and 448 nm relative to the protocol 452 nm lamp for the iridium photocatalyst absorption band.

Suggested measurement 10: Measure acrylonitrile solubility and stability in EtOH:pH 9 buffer at 0.2 M stock concentration to confirm no precipitation or phase separation at 0.02 mL/min flow.

Editorial cross-checks (not model statements)

Time interpretation: some reports call Stage 2 liquid exposure approximately 1000 min, obtained from 20 mL/0.02 mL/min. This is a liquid-only space-time calculation using the entire reactor volume, not measured two-phase liquid residence. Likewise, the 181.82 min inlet/STP index must not be interpreted as a measured fraction of batch reaction exposure.

Numerical transcription: the Figure 5 Set 3 Chief response gives the photocatalyst molar flow as 0.001 mmol/min. The final feed record and concentration-times-flow calculation give 0.0005 mol/L x 0.02 mL/min = 0.000010 mmol/min. The original statement is retained in the transcript as a 100-fold numerical error, not adopted in the figure text or experimental recipe.

Geometry and transport claims: the Set 1 DrFluidics shared assessment groups Candidate 6 with large-bore ETFE options even though Candidate 6 uses 1 mm-ID PFA. The Set 3 Chief attributes a smaller in-channel gas volume fraction to the selected reactor relative to other 7 bar candidates with smaller reactors. Their gas and liquid feed rates, pressure and temperature are the same; changing reactor volume alone does not establish a different gas/liquid volumetric flow ratio or measured gas holdup. A known bore defines geometry but does not establish photon-dose adequacy or a measured transport advantage.

Protocol interpretation: several reports describe the Stage 2 batch temperature as unspecified, whereas the supplied procedure continues under otherwise identical irradiation conditions after Stage 1 at 25 deg C. That

wording does not establish an independently measured Stage 2 temperature, but it does not justify presenting 25 deg C as an unrelated new condition. Similarly, the qualitative backflow incident does not prove that every lower-flow alternative will reverse; the defensible finding is that the unprotected pathway remains.

F5-S2: Figure 5, Set 2
Source: outputs/figure5_pure_oxygen_physics_20260918/figure5_set2/attempt_01. Selected Candidate 6; 12-candidate pool; downstream model claude-sonnet-4-6. The objective policy is yield_priority.

Call 1: DrChemistry, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry / Feed identity: Stream A: substrate 1 at 0.1 M, acrylonitrile 2 at 0.2 M (2.0 equiv), [Ir(dF(CF3)ppy)2(dtbpy)]PF6 at 0.0005 M (0.5 mol%), EtOH:pH 9 buffer 5:1 v/v, argon-degassed offline. Stream B: pure O2 (100 mol%), 0.09 mL/min STP, 2.008 equiv, introduced at Stage 2 inlet only.

Shared assessment / Chemistry / Addition order: All Stage 1 components combined in a single premixed stream; oxygen introduced only at Stage 2. Consistent with protocol.

Shared assessment / Chemistry / Acrylonitrile pump exclusion: Acrylonitrile is excluded from pump_e_series_red. All candidates use pump_e_series_blue for Stream A, which has no acrylonitrile exclusion. Compliant.

Shared assessment / Chemistry / Gas source change: Protocol specifies air; chemist answer Q-GAS-001 mandates pure O2. This gas-source change is chemist-confirmed per Q-GAS-001 and is the active design instruction. No additional chemist-confirmation flag is triggered beyond what is already recorded.

Shared assessment / Chemistry / Oxygen stage 1 exclusion: Stage 1 must be oxygen-free (Q-GAS-003). All candidates introduce O2 only at Stage 2. Consistent with constraint.

Shared assessment / Chemistry / Solubility and compatibility: Ir photocatalyst solubility in EtOH:pH 9 buffer 5:1 at 0.0005 M is not independently verified but is consistent with published Giese photoredox protocols. No incompatibility identified.

Shared assessment / Chemistry / Wavelength: Protocol specifies 452 nm. Available sources are UV-150/450 nm (Stage 1, all candidates) and Manual 2/448 nm (Stage 2, all candidates). Both are within 4 nm of target; photon-dose adequacy is unproven.

Shared assessment / Chemistry / Stage 2 temperature: Batch Step 2 temperature is not stage-resolved. All candidates assign 25 °C. Required time and yield impact remain unknown.

Shared assessment / Chemistry / No intermediate isolation: Continuous two-stage series flow with no isolation step. Consistent with protocol intent.

Shared assessment / Chemistry / Stage 1 exposure concern: Stage 1 liquid residence time is 100 min (candidates 1,2,4,5,7,8,9,10,11) or 250 min (candidates 3,6,12) against a 240 min batch anchor. 100 min is 42% of the batch anchor; photon-dose adequacy for Giese conversion is uncharacterized and unproven at this shorter time. This is a conversion adequacy uncertainty, not a hard violation.

Shared assessment / Chemistry / Stage 2 apparent time concern: Stage 2 inlet/STP apparent residence time index is 181.8 min (candidates 1,2,3,4,6,12; 20 mL FEP reactor) or 90.9 min (candidates 5,7,8,9,10,11; 10 mL reactor) against a 360 min batch anchor. Gas uptake and liquid residence are not equivalent to the batch soak time; adequacy is uncharacterized.

Shared assessment / Chemistry / Degradation risk: Longer Stage 1 holds (250 min) under continuous irradiation with Ir photocatalyst and acrylonitrile present carry an unquantified risk of side-product formation or catalyst fatigue. Shorter holds risk incomplete Giese conversion. Neither risk is resolved without data.

Shared assessment / Chemistry / Pure oxygen hazard: Pure O2 at elevated BPR in organic solvent requires oxygen-service material review, inventory approval, and formal safety assessment before laboratory use. This is flagged as a required pre-execution action for all candidates. It is not an automatic hard violation that vetoes candidates, but no candidate may be executed without this review.

Shared assessment / Operability and backflow / Reverse path status: Unresolved for all candidates. The liquid branch from Stage 2 mixer back to Stage 1 reactor is unprotected (no non-return valve declared on that branch). The MFC (gas) branch carries a declared CV-3301 check valve (cracking pressure 1.0 bar), but this does not protect the liquid branch. KHU observation of reverse gas flow at 0.02 mL/min liquid / 0.43 mL/min gas is qualitative evidence of risk; reduced gas flow to 0.09 mL/min STP does not close the unprotected reverse path. A branch-specific non-return device or pressure-decoupled arrangement on the interstage liquid line is required before execution.

Shared assessment / Operability and backflow / Cv 3301 pressure rating: Maximum pressure rating of CV-3301 is not supplied. Cracking pressure (1.0 bar) is not a pressure rating. Verification required before execution for all candidates.

Shared assessment / Operability and backflow / Mfc check valve cracking: CV-3301 cracking pressure (1.0 bar) concerns the pressure difference across that valve. At BPR 7 bar the MFC outlet must supply above system back-pressure; MFC differential-pressure adequacy at 0.09 sccm against 7-8 bar absolute is unverified.

Shared assessment / Operability and backflow / Startup sequence: No startup sequence data supplied. Transient states (pressurization, gas introduction) may create reverse-flow conditions not captured by steady-state analysis.

Shared assessment / Operability and backflow / Gas volume accounting: Q_gas_sccm and Q_gas_actual_mL_min are inlet/STP index values. Actual gas-liquid ratio, slug flow regime, and O2 mass transfer to liquid are not modeled. Oxygen uptake is unconfirmed.

Shared assessment / Objective policy: Yield priority. Stage 1 conversion adequacy is a primary concern. Stage 2 hold must be sufficient for oxidation without over-oxidation to sulfone. Neither is established experimentally.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA / UV-150 450 nm / 100 min (ratio 0.42 vs 240 min batch). Stage 2: 20 mL FEP / Manual2 448 nm / 181.8 min apparent (ratio 0.51 vs 360 min batch), BPR 7 bar, 2.008 equiv O2. Both stages are sub-batch in exposure. Conversion adequacy is uncharacterized but the design bracket is a legitimate screening point. Reverse-path risk unresolved; oxygen-service review required.

Reviews 1 / Uncertainty 1: Stage 1 at 100 min (42% batch) may be insufficient for Giese conversion; uncharacterized.

Reviews 1 / Uncertainty 2: Stage 2 at 181.8 min apparent index (51% batch) may under-oxidize; uncharacterized.

Reviews 1 / Uncertainty 3: Unprotected reverse path on interstage liquid branch; not resolved.

Reviews 1 / Uncertainty 4: CV-3301 pressure rating unverified; cracking vs. BPR adequacy unconfirmed.

Reviews 1 / Uncertainty 5: Pure O2 oxygen-service and inventory approval required before execution.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical to Candidate 1 in Stage 1 (100 min) and Stage 2 apparent time (181.8 min), differing only in BPR 2.5 bar vs 7 bar. Lower BPR increases in-channel gas volume (0.028 mL/min vs 0.012 mL/min at pressure), potentially improving gas-liquid contact, but this is an untested hypothesis. Both stages sub-batch. Reverse-path unresolved; oxygen-service review required.

Reviews 2 / Uncertainty 1: Stage 1 100 min adequacy uncharacterized.

Reviews 2 / Uncertainty 2: Stage 2 181.8 min apparent index adequacy uncharacterized.

Reviews 2 / Uncertainty 3: Whether lower BPR improves O2 mass transfer is an untested hypothesis.

Reviews 2 / Uncertainty 4: Unprotected reverse path on interstage liquid branch unresolved.

Reviews 2 / Uncertainty 5: CV-3301 cracking pressure vs. 2.5 bar BPR differential adequacy unverified.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 5 mL ETFE / UV-150 450 nm / 250 min (ratio 1.04 vs 240 min batch anchor) — the only Stage 1 hold that meets or exceeds the batch anchor, supporting yield-priority conversion objective. Stage 2: 20 mL FEP / 181.8 min apparent (ratio 0.51). Larger ID (2.4 mm) in Stage 1 may affect light penetration depth; adequacy unproven. Reverse-path and oxygen-service risks unresolved but no hard violation.

Reviews 3 / Uncertainty 1: ETFE 2.4 mm ID may reduce photon penetration vs. 1 mm; effect on conversion unproven.

Reviews 3 / Uncertainty 2: Stage 2 181.8 min apparent index (51% batch) may under-oxidize; uncharacterized.

Reviews 3 / Uncertainty 3: Unprotected reverse path on interstage liquid branch unresolved.

Reviews 3 / Uncertainty 4: CV-3301 pressure rating and service unverified.

Reviews 3 / Uncertainty 5: Pure O2 oxygen-service approval required before execution.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 20 mL FEP / 181.8 min apparent (ratio 0.51), BPR 1.5 bar. Functionally identical to Candidates 1 and 2 in stage exposures; lowest BPR gives largest in-channel gas volume (0.040 mL/min) but mass-transfer benefit is unproven. Stage 1 conversion adequacy concern same as C1/C2. Reverse-path and oxygen-service risks unresolved.

Reviews 4 / Uncertainty 1: Stage 1 100 min adequacy uncharacterized.

Reviews 4 / Uncertainty 2: Stage 2 181.8 min apparent index adequacy uncharacterized.

Reviews 4 / Uncertainty 3: Gas-liquid mass transfer benefit at BPR 1.5 bar is an untested hypothesis.

Reviews 4 / Uncertainty 4: Unprotected reverse path unresolved.

Reviews 4 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL ETFE 2.4 mm / 90.9 min apparent (ratio 0.25), BPR 7 bar. Stage 2 apparent index is the shortest across the set at 25% of batch; adequacy

for oxidation is most uncertain. Stage 1 also sub-batch. Largest-ID Stage 2 coil; light penetration unproven. Acceptable as a short-time bracket but highest oxidation under-exposure risk.

Reviews 5 / Uncertainty 1: Stage 2 90.9 min apparent (25% batch) is the lowest Stage 2 exposure; oxidation adequacy most uncertain.

Reviews 5 / Uncertainty 2: Stage 1 100 min adequacy uncharacterized.

Reviews 5 / Uncertainty 3: ETFE 2.4 mm ID light penetration in Stage 2 unproven.

Reviews 5 / Uncertainty 4: Unprotected reverse path unresolved.

Reviews 5 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: prefer

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA 1 mm / UV-150 450 nm / 250 min (ratio 1.04) — meets batch anchor, supports Giese conversion objective; 1 mm ID consistent with UV-150 photon penetration. Stage 2: 20 mL FEP / 181.8 min apparent (ratio 0.51), BPR 7 bar. Stronger Stage 1 coverage than Candidates 1,2,4,5,7–11 with better light-path geometry than Candidate 3. Reverse-path and oxygen-service risks unresolved.

Reviews 6 / Uncertainty 1: Stage 1 250 min under irradiation with acrylonitrile present: unquantified side-reaction or catalyst-fatigue risk.

Reviews 6 / Uncertainty 2: Stage 2 181.8 min apparent index (51% batch) oxidation adequacy uncharacterized.

Reviews 6 / Uncertainty 3: Unprotected reverse path on interstage liquid branch unresolved.

Reviews 6 / Uncertainty 4: CV-3301 pressure rating unverified.

Reviews 6 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL PFA 1 mm / 90.9 min apparent (ratio 0.25), BPR 7 bar. Both stages sub-batch; Stage 2 at 25% batch is the lowest exposure group. 1 mm ID consistent with Manual2 light source. Acceptable as a short screening bracket; conversion adequacy in both stages most uncertain in this group.

Reviews 7 / Uncertainty 1: Stage 1 100 min adequacy uncharacterized.

Reviews 7 / Uncertainty 2: Stage 2 90.9 min apparent (25% batch) oxidation adequacy most uncertain.

Reviews 7 / Uncertainty 3: Unprotected reverse path unresolved.

Reviews 7 / Uncertainty 4: CV-3301 pressure rating unverified.

Reviews 7 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL FEP 1 mm / 90.9 min apparent (ratio 0.25), BPR 7 bar. Functionally equivalent to Candidate 7 with FEP vs. PFA in Stage 2; FEP max temperature 50 °C limits future elevated-temperature screening but is not a constraint at 25 °C. Both stages sub-batch; same conversion adequacy concerns.

Reviews 8 / Uncertainty 1: Stage 1 100 min adequacy uncharacterized.

Reviews 8 / Uncertainty 2: Stage 2 90.9 min apparent (25% batch) oxidation adequacy most uncertain.

Reviews 8 / Uncertainty 3: FEP Stage 2 coil constrains future temperature elevation to ≤50 °C.

Reviews 8 / Uncertainty 4: Unprotected reverse path unresolved.

Reviews 8 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL ETFE 2.4 mm / 90.9 min apparent (ratio 0.25), BPR 2.5 bar. Same exposure profile as Candidates 5,7,8 with lower BPR and ETFE large-ID Stage 2. Gas volume larger at pressure; mass-transfer benefit unproven. Light penetration in 2.4 mm ID unproven. Both stages sub-batch.

Reviews 9 / Uncertainty 1: Stage 2 90.9 min apparent (25% batch) oxidation adequacy most uncertain.

Reviews 9 / Uncertainty 2: ETFE 2.4 mm ID photon penetration unproven.

Reviews 9 / Uncertainty 3: BPR 2.5 bar mass-transfer benefit is an untested hypothesis.

Reviews 9 / Uncertainty 4: Unprotected reverse path unresolved.

Reviews 9 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL PFA 1 mm / 90.9 min apparent (ratio 0.25), BPR 2.5 bar. Same sub-batch exposures as Candidates 7,9; 1 mm ID consistent with Manual2. Lower BPR than C7 gives larger in-channel gas volume; mass-transfer benefit unproven. Both stages sub-batch; adequacy concerns same.

Reviews 10 / Uncertainty 1: Stage 1 100 min adequacy uncharacterized.

Reviews 10 / Uncertainty 2: Stage 2 90.9 min apparent (25% batch) oxidation adequacy most uncertain.

Reviews 10 / Uncertainty 3: BPR 2.5 bar vs. 7 bar mass-transfer difference is an untested hypothesis.

Reviews 10 / Uncertainty 4: Unprotected reverse path unresolved.

Reviews 10 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL FEP 1 mm / 90.9 min apparent (ratio 0.25), BPR 2.5 bar. Equivalent to Candidate 10 with FEP Stage 2; FEP max 50 °C limits future temperature exploration. Both stages sub-batch; conversion and oxidation adequacy concerns same as C7,C8,C10.

Reviews 11 / Uncertainty 1: Stage 1 100 min adequacy uncharacterized.

Reviews 11 / Uncertainty 2: Stage 2 90.9 min apparent (25% batch) oxidation adequacy most uncertain.

Reviews 11 / Uncertainty 3: FEP Stage 2 limits future temperature screening to ≤50 °C.

Reviews 11 / Uncertainty 4: Unprotected reverse path unresolved.

Reviews 11 / Uncertainty 5: Pure O2 oxygen-service approval required.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: prefer

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 5 mL PFA 0.75 mm / UV-150 450 nm / 250 min (ratio 1.04) — meets batch anchor; narrowest ID (0.75 mm) maximizes photon penetration relative to all Stage 1 options, supporting Giese conversion. Stage 2: 20 mL FEP / 181.8 min apparent (ratio 0.51), BPR 7 bar. Combines strongest Stage 1 light-path geometry with adequate Stage 2 volume. Reverse-path and oxygen-service risks unresolved.

Reviews 12 / Uncertainty 1: 0.75 mm ID increases pressure drop and may affect slug-flow patterns at 0.02 mL/min; operability unverified.

Reviews 12 / Uncertainty 2: Stage 2 181.8 min apparent (51% batch) oxidation adequacy uncharacterized.

Reviews 12 / Uncertainty 3: Unquantified side-reaction or catalyst-fatigue risk at 250 min under continuous irradiation.

Reviews 12 / Uncertainty 4: Unprotected reverse path on interstage liquid branch unresolved.

Reviews 12 / Uncertainty 5: Pure O2 oxygen-service approval required before execution.

Cross cutting notes / Preferred candidates: Candidates 3, 6, and 12 are preferred because Stage 1 holds (250 min, ratio 1.04) meet the batch anchor, which best supports the yield-priority conversion objective. Among these, Candidate 6 (PFA 5 mL, 1 mm ID) and Candidate 12 (PFA 5 mL, 0.75 mm ID) use narrow-bore PFA consistent with UV-150 illumination geometry; Candidate 3 uses ETFE 2.4 mm whose larger ID may reduce photon penetration. No yield or conversion ranking is established.

Cross cutting notes / Stage 2 exposure ranking: Candidates 1,2,3,4,6,12 (20 mL FEP, 181.8 min apparent, ratio 0.51) provide more Stage 2 index time than Candidates 5,7,8,9,10,11 (10 mL, 90.9 min, ratio 0.25). Given yield-priority objective and uncharacterized oxidation kinetics, higher Stage 2 index is less risky for under-oxidation, though over-oxidation to sulfone remains possible at any hold.

Cross cutting notes / Mandatory pre execution actions 1: Resolve unprotected interstage liquid reverse path: install branch-specific non-return protection on the liquid line entering Stage 2 mixer, or use a pressure-decoupled architecture.

Cross cutting notes / Mandatory pre execution actions 2: Verify CV-3301 maximum pressure rating (not cracking pressure) and confirm service compatibility with O2 and EtOH/water mixture.

Cross cutting notes / Mandatory pre execution actions 3: Confirm MFC (mfc_ffc00) cracking-pressure differential is achievable at 0.09 sccm against system back-pressure for each BPR setpoint.

Cross cutting notes / Mandatory pre execution actions 4: Complete oxygen-service material and inventory review for all wetted components before using pure O2.

Cross cutting notes / Mandatory pre execution actions 5: Confirm startup sequence prevents reverse gas flow before steady-state pressurization.

Cross cutting notes / Mandatory pre execution actions 6: Verify Ir photocatalyst solubility and stability in EtOH:pH 9 buffer 5:1 at 0.0005 M under continuous flow irradiation.

Cross cutting notes / Objective fit note: Hold ratios are experimental exposure indices only; physical equivalence between flow and batch exposure is not established. No yield, conversion, or kinetic prediction is made.

Call 2: DrKinetics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / All candidates / Gas source change: Pure O2 was used in the prior experiment instead of the specified air. This change has not received laboratory approval. It must be confirmed by the chemist before any candidate is executed.

Shared assessment / All candidates / Reverse path unresolved: All 12 candidates share an unprotected reverse path from the Stage 2 gas junction back through the liquid line to Stage 1 (st2_mixer → st1_reactor). No forward-oriented non-return device is declared on the liquid branch before Stage 1. A CV-3301 is declared only on the MFC gas branch; a valve on a different branch does not protect the liquid branch. This finding is not resolved in any candidate. Reduced gas flow does not close an unprotected reverse path. Observed backflow in the prior experiment corroborates this structural risk. Engineering review and a branch-specific liquid-side non-return device (or pressure-decoupled arrangement) are required before laboratory execution of any candidate.

Shared assessment / All candidates / Cv 3301 pressure rating: CV-3301 maximum pressure rating is not supplied. Cracking pressure (1.0 bar) is not a pressure rating. Verification required before execution.

Shared assessment / All candidates / Kinetics uncharacterized: No flow kinetic data exist for either stage. Batch times (Stage 1: 240 min, Stage 2: 360 min) are screening anchors, not established flow optima. Hold ratios are indices of experimental exposure, not predicted conversion or yield.

Shared assessment / All candidates / Stage2 temperature: Stage 2 batch temperature is not stage-resolved. All candidates assign 25 °C; no rate correction is validated.

Shared assessment / All candidates / Gas residence time basis: Stage 2 residence times are inlet/STP apparent indices V/(Q_liquid + Q_gas_STP), not measured liquid exposure or photon dose. They are not equivalent to batch hold times.

Shared assessment / All candidates / Wavelength mismatch: Protocol specifies 452 nm. Stage 1 uses UV-150 at 450 nm (acceptable proximity). Stage 2 uses Manual 2 Blue Strip LED at 448 nm. Neither wavelength identity nor irradiance equivalence to the batch 5 W LED pair is established. Photon dose adequacy is unproven in all candidates.

Shared assessment / All candidates / Oxygen service: Pure oxygen use requires oxygen-service review of all wetted components, fittings, and check valves. This is an unapproved proposal pending chemistry, inventory, and oxygen-service review.

Shared assessment / All candidates / Stage1 exposure adequacy candidates 1 2 4: Candidates 1, 2, 4 assign Stage 1 liquid τ = 100 min (ratio 0.42 vs 240 min batch). Whether this is sufficient for the Giese photocatalytic step is uncharacterized. Under yield-priority policy, conversion adequacy at sub-batch exposure is uncertain, not confirmed.

Shared assessment / All candidates / Stage2 exposure adequacy candidates 5 7 8 9 10 11: Candidates 5, 7, 8, 9, 10, 11 assign Stage 2 inlet/STP $\tau \approx 91$ min (ratio 0.25 vs 360 min batch). This is the shortest Stage 2 index in the set. Under yield-priority policy, whether this delivers adequate oxidation is uncharacterized.

Shared assessment / All candidates / Acrylonitrile pump exclusion: Acrylonitrile is excluded from red pump tubing. All candidates assign the E-series blue pump for Stream A. This is consistent with the exclusion list.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1 τ=100 min (ratio 0.42), Stage 2 inlet/STP τ=182 min (ratio 0.51). Stage 1 sub-batch exposure is the primary conversion uncertainty under yield-priority policy. BPR 7 bar increases O2 solubility (hypothesis, not established). Unprotected reverse path requires engineering resolution before execution.

Reviews 1 / Uncertainty 1: Stage 1 at 100 min may be insufficient for Giese conversion; uncharacterized.

Reviews 1 / Uncertainty 2: Stage 2 at 182 min inlet/STP adequacy for oxidation is uncharacterized.

Reviews 1 / Uncertainty 3: Unprotected liquid-branch reverse path to Stage 1 unresolved.

Reviews 1 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 1 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical stage exposure indices to Candidate 1 (S1 τ=100 min, S2 inlet/STP τ=182 min). BPR 2.5 bar lowers O2 in-channel partial pressure vs Candidate 1; O2 solubility difference is a hypothesis. Same unresolved reverse-path and gas-source concerns apply.

Reviews 2 / Uncertainty 1: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 2 / Uncertainty 2: Lower BPR may reduce dissolved O2 availability; untested.

Reviews 2 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 2 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 2 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 uses ETFE 5 mL coil (2.4 mm ID) with τ=250 min (ratio 1.04 vs batch 240 min), providing the most complete Stage 1 exposure anchor among the set. Stage 2 inlet/STP τ=182 min (ratio 0.51). Larger ID may affect light penetration; uncharacterized. Reverse-path unresolved.

Reviews 3 / Uncertainty 1: ETFE 2.4 mm ID light penetration depth in UV-150 module uncharacterized.

Reviews 3 / Uncertainty 2: Stage 2 oxidation adequacy at 182 min inlet/STP uncharacterized.

Reviews 3 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 3 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 3 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1 τ=100 min (ratio 0.42), Stage 2 inlet/STP τ=182 min (ratio 0.51). BPR 1.5 bar is the lowest pressure; lowest O2 partial pressure in Stage 2 among candidates with 20 mL reactor. Same exposure and reverse-path uncertainties as Candidates 1–2.

Reviews 4 / Uncertainty 1: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 4 / Uncertainty 2: BPR 1.5 bar gives lowest in-channel O2 partial pressure; adequacy uncharacterized.

Reviews 4 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 4 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 4 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1 τ=100 min (ratio 0.42); Stage 2 uses ETFE 10 mL 2.4 mm ID coil, inlet/STP $\tau\approx$91 min (ratio 0.25), the shortest Stage 2 index. Under yield-priority policy, this is the highest-risk Stage 2 exposure among the set. Larger ID may affect O2 mass transfer; uncharacterized. Reverse-path unresolved.

Reviews 5 / Uncertainty 1: Stage 2 at 91 min inlet/STP is 25% of batch; oxidation adequacy highly uncertain.

Reviews 5 / Uncertainty 2: ETFE 2.4 mm ID gas-liquid mass transfer and light penetration uncharacterized.

Reviews 5 / Uncertainty 3: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 5 / Uncertainty 4: Unprotected liquid-branch reverse path unresolved.

Reviews 5 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 uses PFA 5 mL 1 mm ID coil, τ=250 min (ratio 1.04 vs batch). Stage 2 inlet/STP τ=182 min (ratio 0.51). Provides near-batch Stage 1 exposure with standard 1 mm geometry, avoiding large-ID light-penetration concern of Candidate 3. BPR 7 bar. Reverse-path unresolved.

Reviews 6 / Uncertainty 1: Stage 2 oxidation adequacy at 182 min inlet/STP uncharacterized.

Reviews 6 / Uncertainty 2: Whether near-batch Stage 1 exposure improves yield over shorter candidates is uncharacterized.

Reviews 6 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 6 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 6 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1 τ=100 min (ratio 0.42); Stage 2 uses PFA 10 mL 1 mm coil, inlet/STP $\tau\approx$91 min (ratio 0.25). Shortest Stage 2 index with standard 1 mm geometry. Under yield-priority policy, 91 min vs 360 min batch is the key conversion-adequacy uncertainty. Reverse-path unresolved.

Reviews 7 / Uncertainty 1: Stage 2 at 91 min inlet/STP is 25% of batch; oxidation adequacy uncertain.

Reviews 7 / Uncertainty 2: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 7 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 7 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 7 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1 τ=100 min (ratio 0.42); Stage 2 uses FEP 10 mL 1 mm coil, inlet/STP $\tau\approx$91 min (ratio 0.25). Functionally identical to Candidate 7 in exposure indices; FEP vs PFA material difference is minor at 25 °C. Same yield-priority Stage 2 adequacy concern applies. Reverse-path unresolved.

Reviews 8 / Uncertainty 1: Stage 2 at 91 min inlet/STP adequacy for oxidation uncharacterized.

Reviews 8 / Uncertainty 2: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 8 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 8 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 8 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 τ=100 min (ratio 0.42); Stage 2 uses ETFE 10 mL 2.4 mm ID coil, inlet/STP $\tau\approx$91 min (ratio 0.25), BPR 2.5 bar. Combines shortest Stage 2 index with largest ID and lowest BPR; O2 mass transfer and partial pressure effects are uncharacterized hypotheses. Reverse-path unresolved.

Reviews 9 / Uncertainty 1: Stage 2 adequacy at 91 min inlet/STP with 2.4 mm ID and 2.5 bar highly uncertain.

Reviews 9 / Uncertainty 2: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 9 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 9 / Uncertainty 4: O2 service and gas-source change unapproved.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1 τ=100 min (ratio 0.42); Stage 2 uses PFA 10 mL 1 mm coil, inlet/STP $\tau\approx$91 min (ratio 0.25), BPR 2.5 bar. Short Stage 2 index with lower O2 partial pressure than Candidate 7; adequacy uncharacterized. Reverse-path unresolved.

Reviews 10 / Uncertainty 1: Stage 2 at 91 min inlet/STP and 2.5 bar: oxidation adequacy uncharacterized.

Reviews 10 / Uncertainty 2: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 10 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 10 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 10 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1 τ=100 min (ratio 0.42); Stage 2 uses FEP 10 mL 1 mm coil, inlet/STP $\tau\approx$91 min (ratio 0.25), BPR 2.5 bar. Functionally identical to Candidate 10 in exposure indices with FEP vs PFA material; difference minor at 25 °C. Same yield-priority Stage 2 adequacy concern. Reverse-path unresolved.

Reviews 11 / Uncertainty 1: Stage 2 at 91 min inlet/STP adequacy for oxidation uncharacterized.

Reviews 11 / Uncertainty 2: Stage 1 conversion adequacy at 100 min uncharacterized.

Reviews 11 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 11 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 11 / Uncertainty 5: O2 service and gas-source change unapproved.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1 uses PFA 5 mL 0.75 mm ID coil, τ=250 min (ratio 1.04). Smaller ID increases pressure drop and may improve plug flow; effect on photon penetration vs standard 1 mm is uncharacterized. Stage 2 inlet/STP τ=182 min (ratio 0.51), BPR 7 bar. Near-batch Stage 1 exposure; reverse-path unresolved.

Reviews 12 / Uncertainty 1: 0.75 mm ID coil: light penetration and flow stability at 0.02 mL/min uncharacterized.

Reviews 12 / Uncertainty 2: Stage 2 oxidation adequacy at 182 min inlet/STP uncharacterized.

Reviews 12 / Uncertainty 3: Unprotected liquid-branch reverse path unresolved.

Reviews 12 / Uncertainty 4: CV-3301 pressure rating unconfirmed.

Reviews 12 / Uncertainty 5: O2 service and gas-source change unapproved.

Call 3: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Tool name: simulate_flow_transients

Candidate ids 1: 1

Candidate ids 2: 2

Candidate ids 3: 3

Candidate ids 4: 4

Candidate ids 5: 5

Candidate ids 6: 6

Candidate ids 7: 7

Candidate ids 8: 8

Candidate ids 9: 9

Candidate ids 10: 10

Candidate ids 11: 11

Candidate ids 12: 12

Call 4: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch), 7 bar BPR, 2.007 equiv O_2 at 0.09 sccm. Light: 450 nm UV-150 (S1) + 448 nm Manual 2 (S2). Both stages below batch anchors; Stage 1 conversion adequacy uncharacterized. Unprotected reverse path st2_mixer→st1_reactor unresolved. Pure O_2 source unapproved; requires chemistry, inventory, and O_2-service review. BPR_opening_pressure_reached; no liquid displacement predicted under entered profile, but unprotected path not closed. Screening candidate subject to backflow review.

Reviews 1 / Uncertainty 1: Stage 1 photon adequacy unconfirmed at 100 min; kinetics uncharacterized.

Reviews 1 / Uncertainty 2: Unprotected liquid reverse path st2_mixer→st1_reactor not mitigated.

Reviews 1 / Uncertainty 3: CV-3301 pressure rating unconfirmed; cracking pressure ≠ pressure rating.

Reviews 1 / Uncertainty 4: Pure O_2 supply not approved; requires chemistry and O_2-service review.

Reviews 1 / Uncertainty 5: Gas uptake, dissolution, and actual O_2 delivery to liquid are not modeled.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch), 2.5 bar BPR, 2.007 equiv O_2 at 0.09 sccm (in-channel 0.0283 mL/min at 3.51 bar abs). Lower BPR → higher in-channel gas volume fraction vs. C1. Same reactor geometry as C1. Unprotected reverse path unresolved. No liquid displacement predicted under entered profile. Pure O_2 unapproved.

Reviews 2 / Uncertainty 1: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 2 / Uncertainty 2: Unprotected liquid reverse path unresolved for all BPR settings.

Reviews 2 / Uncertainty 3: Lower BPR reduces gas compression but does not resolve reverse connectivity.

Reviews 2 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 2 / Uncertainty 5: In-channel gas volume fraction and mass transfer unquantified.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 5 mL ETFE (ID 2.4 mm), 250 min liquid τ (1.04× batch anchor), 7 bar BPR. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch). Larger ID in Stage 1 reduces light path uniformity vs. 1 mm coil; photon delivery adequacy untested. Per tool, as_designed startup scenario did not reach BPR opening pressure in the simulation window—steady-state pressure dynamics not confirmed for this candidate. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 3 / Uncertainty 1: 2.4 mm ID Stage 1 coil: light penetration depth and photon uniformity uncharacterized.

Reviews 3 / Uncertainty 2: Tool reported BPR_opening_pressure_reached=false in as_designed startup window; steady-state behaviour not simulated.

Reviews 3 / Uncertainty 3: Unprotected liquid reverse path st2_mixer→st1_reactor unresolved.

Reviews 3 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 3 / Uncertainty 5: Gas uptake and actual O_2 delivery to liquid not modeled.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch), 1.5 bar BPR. Lowest system pressure of the 20 mL FEP group; highest in-channel gas volume fraction (0.0396 mL/min at 2.51 bar abs). No liquid displacement predicted under entered profile. Peristaltic pump rated to 10 bar; 1.5 bar BPR provides minimal upstream backpressure if transients occur. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 4 / Uncertainty 1: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 4 / Uncertainty 2: 1.5 bar BPR provides minimal transient back-pressure; startup sequence critical.

Reviews 4 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 4 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 4 / Uncertainty 5: In-channel gas fraction highest in group; mass transfer and gas-liquid regime uncharacterized.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 10 mL ETFE (ID 2.4 mm), 90.9 min inlet/STP τ (0.25× batch), 7 bar BPR. Stage 2 inlet/STP τ is the shortest in the candidate set relative to the batch Stage 2 anchor; oxygen exposure adequacy most uncertain here. Larger ID Stage 2 coil has uncharacterized mass transfer. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 5 / Uncertainty 1: Stage 2 inlet/STP τ 90.9 min = 0.25× batch; oxidation adequacy most uncertain.

Reviews 5 / Uncertainty 2: 2.4 mm ID Stage 2 coil: gas-liquid regime and mass transfer uncharacterized.

Reviews 5 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 5 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 5 / Uncertainty 5: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA (ID 1 mm), 250 min liquid τ (1.04× batch), 7 bar BPR. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch). Per tool, as_designed startup window did not reach BPR opening pressure; steady-state not confirmed in that window, though hypothetical_liquid_check steady-state shows no reversal. Unprotected reverse path unresolved. Pure O_2 unapproved. Stage 1 residence time near batch anchor; yield-priority objective better served than candidates with 100 min Stage 1.

Reviews 6 / Uncertainty 1: Tool BPR_opening_pressure_reached=false in as_designed startup window; steady-state unconfirmed.

Reviews 6 / Uncertainty 2: Unprotected liquid reverse path st2_mixer→st1_reactor unresolved.

Reviews 6 / Uncertainty 3: Pure O_2 supply unapproved; O_2-service review required.

Reviews 6 / Uncertainty 4: Stage 2 inlet/STP τ 0.51× batch; oxidation adequacy uncharacterized.

Reviews 6 / Uncertainty 5: Photon adequacy and degradation risk at 250 min Stage 1 not established.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 10 mL PFA (ID 1 mm), 90.9 min inlet/STP τ (0.25× batch), 7 bar BPR. Stage 2 τ is shortest among 1 mm ID coil candidates; oxidation adequacy most uncertain with 1 mm ID Stage 2. BPR_opening_pressure_reached=true; no liquid displacement predicted. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 7 / Uncertainty 1: Stage 2 inlet/STP τ 90.9 min = 0.25× batch; oxidation adequacy uncertain.

Reviews 7 / Uncertainty 2: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 7 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 7 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 7 / Uncertainty 5: No degradation risk data at 25 °C for either stage duration.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 10 mL FEP (ID 1 mm, max 50 °C), 90.9 min inlet/STP τ (0.25× batch), 7 bar BPR, 25 °C—within FEP temperature limit. Identical τ profile to C7; same oxidation adequacy uncertainty. BPR_opening_pressure_reached=true; no displacement predicted. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 8 / Uncertainty 1: Stage 2 inlet/STP τ 90.9 min = 0.25× batch; oxidation adequacy uncertain.

Reviews 8 / Uncertainty 2: FEP Stage 2 coil limited to 50 °C; operating at 25 °C is within limit but constrains future temperature exploration.

Reviews 8 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 8 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 8 / Uncertainty 5: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 10 mL ETFE (ID 2.4 mm), 90.9 min inlet/STP τ (0.25× batch), 2.5 bar BPR. Same τ as C5 but with lower BPR; higher in-channel gas volume fraction vs. 7 bar. 2.4 mm ID Stage 2: gas-liquid regime and mass transfer uncharacterized. No liquid displacement predicted. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 9 / Uncertainty 1: Stage 2 inlet/STP τ 0.25× batch; oxidation adequacy most uncertain in set.

Reviews 9 / Uncertainty 2: 2.4 mm ID + 2.5 bar BPR: highest in-channel gas fraction; regime unknown.

Reviews 9 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 9 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 9 / Uncertainty 5: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 10 mL PFA (ID 1 mm), 90.9 min inlet/STP τ (0.25× batch), 2.5 bar BPR. Same τ as C7 but lower BPR; in-channel gas volume fraction higher than 7 bar equivalent. BPR_opening_pressure_reached=true; no displacement predicted. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 10 / Uncertainty 1: Stage 2 inlet/STP τ 0.25× batch; oxidation adequacy uncertain.

Reviews 10 / Uncertainty 2: Lower BPR increases in-channel gas fraction; mass transfer regime uncharacterized.

Reviews 10 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 10 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 10 / Uncertainty 5: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch). Stage 2: 10 mL FEP (ID 1 mm, max 50 °C), 90.9 min inlet/STP τ (0.25× batch), 2.5 bar BPR, 25 °C—within FEP limit. Identical τ to C10; same oxidation adequacy concern. FEP temperature ceiling constrains future parameter exploration. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 11 / Uncertainty 1: Stage 2 inlet/STP τ 0.25× batch; oxidation adequacy uncertain.

Reviews 11 / Uncertainty 2: FEP 50 °C ceiling constrains temperature screening space.

Reviews 11 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 11 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 11 / Uncertainty 5: Stage 1 photon adequacy at 100 min uncharacterized.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 5 mL PFA (ID 0.75 mm), 250 min liquid τ (1.04× batch), 7 bar BPR. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch). Narrower ID (0.75 mm) in Stage 1 increases pressure drop (0.00486 bar calculated) and may improve photon penetration but reduces throughput window. Per tool, as_designed startup window did not reach BPR opening; steady-state not confirmed in that window. Unprotected reverse path unresolved. Pure O_2 unapproved.

Reviews 12 / Uncertainty 1: Tool BPR_opening_pressure_reached=false in as_designed startup window; steady-state dynamics unconfirmed.

Reviews 12 / Uncertainty 2: 0.75 mm ID Stage 1: plugging risk with any particulates; not characterized.

Reviews 12 / Uncertainty 3: Unprotected liquid reverse path st2_mixer→st1_reactor unresolved.

Reviews 12 / Uncertainty 4: Pure O_2 supply unapproved; O_2-service review required.

Reviews 12 / Uncertainty 5: Stage 2 inlet/STP τ 0.51× batch; oxidation adequacy uncharacterized.

Shared assessment / Issues common to all candidates 1: Unprotected reverse path: st2_mixer→st1_reactor has no declared forward non-return device on the liquid branch. The CV-3301 on the MFC (gas) branch does not protect this separate liquid inlet branch. This is an unresolved credible risk across all 12 candidates and requires engineering review before execution.

Shared assessment / Issues common to all candidates 2: Pure O_2 gas supply: The gas-source change from the batch protocol (air) to pure O_2 was chemist-confirmed (Q-GAS-001) but has not received chemistry, inventory, and O_2-service safety review. It must not be treated as approved for execution.

Shared assessment / Issues common to all candidates 3: CV-3301 pressure rating unconfirmed: cracking pressure (1.0 bar) is not a pressure rating. Maximum pressure rating requires laboratory verification before use.

Shared assessment / Issues common to all candidates 4: KHU qualitative observation (2026-09-15): gas reverse flow was observed experimentally at 0.02 mL/min liquid / 0.43 mL/min O_2 (unverified reference conditions). Reducing gas flow to 0.09 sccm does not close the unprotected liquid reverse path and does not resolve the finding.

Shared assessment / Issues common to all candidates 5: Gas-phase residence time index: Q_gas_sccm = 0.09 mL/min at inlet/STP (273.15 K, 1.01325 bar) is the MFC-reported volumetric flow. The apparent inlet/STP residence time $\tau = V/(Q_{liquid} + Q_{gas_STP})$ is a geometric index only; it is not measured liquid exposure time or photon exposure time. Gas uptake, dissolution, and actual O_2 delivered to the liquid phase are not modeled.

Shared assessment / Issues common to all candidates 6: Wavelength mismatch: protocol specifies 452 nm; UV-150 Stage 1 source is 450 nm (light_uv150_row10) and Manual 2 Stage 2 source is 448 nm (light_manual2_row13). No photon-dose equivalence is established.

Shared assessment / Issues common to all candidates 7: Stage 2 temperature not stage-resolved in batch protocol: screen temperature 25 °C assumed; no validated rate correction available.

Shared assessment / Issues common to all candidates 8: Kinetics uncharacterized: no flow kinetic data exist for this Giese/oxidation sequence. Batch times are screening anchors only; flow conversion and yield are unpredicted.

Shared assessment / Issues common to all candidates 9: No inline degasser available: offline argon degassing of feed required; O_2 contamination of Stage 1 feed is an operational risk requiring procedural control.

Shared assessment / Issues common to all candidates 10: Candidates 3, 6, 12: tool BPR_opening_pressure_reached=false in as_designed startup window—steady-state pressure behaviour not confirmed within the simulation window for these candidates; this is a simulation limitation, not proof of failure.

Shared assessment / Issues common to all candidates 11: Physics tool results are conditional on unmeasured illustrative parameters (viscosity, compliance, gas plenum, MFC differential pressure, pump response). No reversal predicted in these tests does not establish safety. Results are not probabilities.

Physics assessment / Tool result id: c5f4faa53d16cff0ba8ff1194becf8a69effa7c49391d142e9ad66b3a12cbc05

Physics assessment / Evidence ids 1: C1/as_designed/entered_profile/steady

Physics assessment / Evidence ids 2: C1/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 3: C2/as_designed/entered_profile/steady

Physics assessment / Evidence ids 4: C2/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 5: C3/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 6: C3/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 7: C4/as_designed/entered_profile/steady

Physics assessment / Evidence ids 8: C4/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 9: C5/as_designed/entered_profile/steady

Physics assessment / Evidence ids 10: C5/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 11: C6/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 12: C6/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 13: C7/as_designed/entered_profile/steady

Physics assessment / Evidence ids 14: C7/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 15: C8/as_designed/entered_profile/steady

Physics assessment / Evidence ids 16: C8/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 17: C9/as_designed/entered_profile/steady

Physics assessment / Evidence ids 18: C9/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 19: C10/as_designed/entered_profile/steady

Physics assessment / Evidence ids 20: C10/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 21: C11/as_designed/entered_profile/steady

Physics assessment / Evidence ids 22: C11/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 23: C12/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 24: C12/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Proposed alternatives 1: Install a forward-oriented, service-verified check valve on the liquid branch between Stage 1 outlet and the Stage 2 gas-introduction junction. This is an unapproved proposal: inventory item, cracking pressure, pressure rating, O_2-service compatibility, and low-flow sealing must be verified by the laboratory before adoption.

Physics assessment / Proposed alternatives 2: A pressure-decoupled interstage arrangement (e.g., collection and re-injection) could isolate Stage 1 from Stage 2 gas pressure; this is an unapproved architectural proposal requiring full engineering and chemistry review.

Physics assessment / Proposed alternatives 3: Controlled startup sequence (liquid flow established and BPR pressure reached before gas introduction) may reduce transient risk but does not close the unprotected reverse path structurally and requires laboratory-verified procedures.

Physics assessment / Limitation 1: Tool uses illustrative, uncalibrated parameters: viscosity 1.0 mPa·s, liquid compliance 0.01 mL/bar, gas plenum 0.2 mL, MFC required ΔP 1.0 bar, pump response 1.0 s, MFC response 0.5 s. None are measured KHU values.

Physics assessment / Limitation 2: No reversal predicted under entered profile does not establish safety; it reflects model assumptions, not measured pressure traces.

Physics assessment / Limitation 3: Predicted quantity is liquid displacement at the junction, not oxygen reaching Reactor 1; gas-front propagation, dissolution, and slug dynamics are not modeled.

Physics assessment / Limitation 4: The CV-3301 cracking pressure (1.0 bar) concerns the pressure difference across that valve on the gas branch only; it does not protect the liquid branch reverse path.

Physics assessment / Limitation 5: Candidates 3, 6, 12: as_designed startup simulations did not reach BPR opening pressure within the simulation window; steady-state results are available only for the hypothetical_liquid_check alternative, not for the as-designed configuration.

Physics assessment / Limitation 6: Finite scenario window; not a lifetime or reliability assessment. Transient scenarios at startup and shutdown are not fully characterized.

Physics assessment / Limitation 7: Hypothetical liquid check valve is not an inventory item and is not service-verified; its inclusion in simulation does not constitute an approved design change.

Physics assessment / Limitation 8: Gas identity (pure O_2) is chemist-confirmed but not approved for O_2 service; the tool does not evaluate material compatibility, ignition risk, or O_2-enriched atmosphere hazards.

Physics assessment / Limitation 9: Actual supply pressures, MFC differential-pressure requirements, valve characteristics, and startup sequencing behaviour require laboratory measurement.

Call 5: DrSafety, attempt 1
Status: recorded response, parsed successfully.

Tool name: simulate_flow_transients

Candidate ids 1: 1

Candidate ids 2: 2

Candidate ids 3: 3

Candidate ids 4: 4

Candidate ids 5: 5

Candidate ids 6: 6

Candidate ids 7: 7

Candidate ids 8: 8

Candidate ids 9: 9

Candidate ids 10: 10

Candidate ids 11: 11

Candidate ids 12: 12

Call 6: DrSafety, attempt 1
Status: recorded response, parsed successfully.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA, 100 min liquid hold (ratio 0.42 vs batch 240 min); Stage 2: 20 mL FEP, 181.8 min inlet/STP index (ratio 0.51 vs batch 360 min). BPR 7 bar. Pure O2 at 0.09 sccm (2.01 equiv inlet/STP) confirmed chemist-approved change requiring chemistry, inventory and oxygen-service review before use. Unprotected reverse path on liquid branch to Stage 1 persists; reverse path not closed by gas check valve on MFC branch. Tool predicted no liquid displacement in entered-profile steady scenarios (unvalidated, finite window). CV-3301 pressure rating unconfirmed. Acrylonitrile excluded from red pump tubing; blue pump used correctly.

Reviews 1 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved; no liquid-branch non-return valve declared.

Reviews 1 / Uncertainty 2: CV-3301 max pressure rating unconfirmed; cracking pressure is not a pressure rating.

Reviews 1 / Uncertainty 3: Pure O2 gas-source change unapproved; requires chemistry, inventory and oxygen-service review.

Reviews 1 / Uncertainty 4: Tool results are conditional on unmeasured parameters; no reversal predicted is not proof of safety.

Reviews 1 / Uncertainty 5: Stage 1 photon adequacy unconfirmed; 450 nm vs 452 nm target, 2 mL at 0.02 mL/min, no dose model.

Reviews 1 / Uncertainty 6: Stage 2 temperature not stage-resolved in protocol; rate correction unknown.

Reviews 1 / Uncertainty 7: Gas uptake, slug regime and dissolution not modelled; inlet/STP index is not liquid residence time.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Same reactor pair as C1 but BPR 2.5 bar, reducing system pressure and increasing in-channel gas volume (0.028 mL/min at 3.51 bar abs). Stage 2 inlet/STP index 181.8 min unchanged. Lower BPR provides larger MFC margin (7.29 bar above required). All O2, reverse-path and check-valve uncertainties identical to C1. Unprotected liquid reverse path unresolved.

Reviews 2 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 2 / Uncertainty 2: Lower BPR does not close reverse path; gas-to-liquid pressure relationship at junction unverified.

Reviews 2 / Uncertainty 3: CV-3301 max pressure rating unconfirmed.

Reviews 2 / Uncertainty 4: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 2 / Uncertainty 5: Tool results conditional on unmeasured parameters.

Reviews 2 / Uncertainty 6: Stage 2 temperature not stage-resolved; yield unknown.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 5 mL ETFE ID 2.4 mm, 250 min liquid hold (ratio 1.04 vs batch 240 min). Stage 2: 20 mL FEP, 181.8 min inlet/STP index, BPR 7 bar. ETFE large-bore at 2.4 mm ID with UV-150 at 25 °C: light penetration across wider bore is unconfirmed. Tool as_designed startup scenario did not reach BPR operating pressure in simulated window; steady-state behaviour unverified for this candidate. Reverse-path and O2 issues shared with all candidates.

Reviews 3 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 3 / Uncertainty 2: ETFE 2.4 mm ID: light penetration and photocatalyst excitation depth unconfirmed.

Reviews 3 / Uncertainty 3: Tool startup simulation did not reach BPR operating pressure; steady-state not verified.

Reviews 3 / Uncertainty 4: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 3 / Uncertainty 5: CV-3301 max pressure rating unconfirmed.

Reviews 3 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Same reactor pair as C1 and C2 (2 mL PFA Stage 1, 20 mL FEP Stage 2) with BPR 1.5 bar. Lowest system pressure of the 20 mL FEP group; gas in-channel volume largest (0.040 mL/min at 2.51 bar abs). MFC margin 8.28 bar above required. Peristaltic pump max 10 bar; system pressure ~1.54 bar, ample headroom. Reverse-path and O2 issues shared. FEP max 50 °C; screening at 25 °C is within limit.

Reviews 4 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 4 / Uncertainty 2: Lowest BPR (1.5 bar): startup pressure sequencing and gas/liquid phase fraction at low back-pressure uncharacterised.

Reviews 4 / Uncertainty 3: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 4 / Uncertainty 4: CV-3301 max pressure rating unconfirmed.

Reviews 4 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA, 100 min. Stage 2: 10 mL ETFE ID 2.4 mm, inlet/STP index 90.9 min (ratio 0.25 vs batch 360 min), BPR 7 bar. Large-bore ETFE in Stage 2 may impair gas-liquid contact and light penetration; both unconfirmed. Shorter Stage 2 index increases under-oxidation risk for yield priority. Reverse-path and O2 issues shared.

Reviews 5 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 5 / Uncertainty 2: ETFE 2.4 mm bore Stage 2: gas-liquid contacting and light distribution unconfirmed.

Reviews 5 / Uncertainty 3: Stage 2 inlet/STP index 90.9 min is 25% of batch time; oxidation adequacy unknown.

Reviews 5 / Uncertainty 4: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 5 / Uncertainty 5: CV-3301 max pressure rating unconfirmed.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA ID 1 mm, 250 min liquid hold (ratio 1.04), BPR 7 bar. Stage 2: 20 mL FEP, 181.8 min inlet/STP. Tool as_designed startup did not reach BPR operating pressure in simulated window. Consumes one of two available 5 mL PFA coils (coil_row4). Stage 2 same as C1. Reverse-path and O2 issues shared.

Reviews 6 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 6 / Uncertainty 2: Tool startup simulation did not reach BPR operating pressure; steady-state behaviour unverified.

Reviews 6 / Uncertainty 3: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 6 / Uncertainty 4: CV-3301 max pressure rating unconfirmed.

Reviews 6 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA, 100 min. Stage 2: 10 mL PFA ID 1 mm, inlet/STP index 90.9 min (ratio 0.25), BPR 7 bar. PFA 1 mm bore in Stage 2 preserves gas-liquid contacting geometry. Shorter Stage 2 index is a yield risk at this conversion priority. Reverse-path and O2 issues shared.

Reviews 7 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 7 / Uncertainty 2: Stage 2 inlet/STP index 90.9 min is 25% of batch time; oxidation adequacy unknown.

Reviews 7 / Uncertainty 3: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 7 / Uncertainty 4: CV-3301 max pressure rating unconfirmed.

Reviews 7 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Identical flow parameters to C7 but Stage 2 reactor is 10 mL FEP ID 1 mm (coil_row6, max 50 °C). At 25 °C screening temperature FEP limit is respected. FEP gas permeability is higher than PFA; O2 loss through wall is unquantified but unlikely to be controlling at these flow rates. Reverse-path and O2 issues shared.

Reviews 8 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 8 / Uncertainty 2: FEP higher gas permeability: O2 wall loss unquantified.

Reviews 8 / Uncertainty 3: Stage 2 inlet/STP index 90.9 min is 25% of batch time; oxidation adequacy unknown.

Reviews 8 / Uncertainty 4: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 8 / Uncertainty 5: CV-3301 max pressure rating unconfirmed.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 2 mL PFA, 100 min. Stage 2: 10 mL ETFE ID 2.4 mm, inlet/STP index 90.9 min, BPR 2.5 bar. Large-bore ETFE and lower BPR combine two uncertainties (gas-liquid contact, phase fraction). MFC margin large (7.32 bar). Reverse-path and O2 issues shared.

Reviews 9 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 9 / Uncertainty 2: ETFE 2.4 mm bore and low BPR: gas fraction and contacting behaviour uncharacterised.

Reviews 9 / Uncertainty 3: Stage 2 inlet/STP index 90.9 min is 25% of batch time; oxidation adequacy unknown.

Reviews 9 / Uncertainty 4: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 9 / Uncertainty 5: CV-3301 max pressure rating unconfirmed.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA, 100 min. Stage 2: 10 mL PFA ID 1 mm, inlet/STP index 90.9 min, BPR 2.5 bar. 1 mm PFA preserves contacting geometry. Lower BPR vs C7 increases in-channel gas volume. MFC margin 7.30 bar. Reverse-path and O2 issues shared.

Reviews 10 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 10 / Uncertainty 2: Stage 2 inlet/STP index 90.9 min is 25% of batch time; oxidation adequacy unknown.

Reviews 10 / Uncertainty 3: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 10 / Uncertainty 4: CV-3301 max pressure rating unconfirmed.

Reviews 10 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Identical flow parameters to C10 but Stage 2 uses 10 mL FEP ID 1 mm. At 25 °C within FEP 50 °C limit. Same BPR 2.5 bar, same inlet/STP index 90.9 min, same MFC margin. FEP gas permeability caveat applies. Reverse-path and O2 issues shared.

Reviews 11 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 11 / Uncertainty 2: FEP higher gas permeability: O2 wall loss unquantified.

Reviews 11 / Uncertainty 3: Stage 2 inlet/STP index 90.9 min is 25% of batch time; oxidation adequacy unknown.

Reviews 11 / Uncertainty 4: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 11 / Uncertainty 5: CV-3301 max pressure rating unconfirmed.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 5 mL PFA ID 0.75 mm (coil_row8), 250 min liquid hold, BPR 7 bar. Narrow 0.75 mm bore increases pressure drop (0.005 bar, within rating) but may increase wall-proximity photon delivery. Stage 2: 20 mL FEP, 181.8 min inlet/STP. Tool as_designed startup did not reach BPR operating pressure in simulated window. Unique narrow-bore reactor; no sedimentation risk for clear solution. Reverse-path and O2 issues shared.

Reviews 12 / Uncertainty 1: Unprotected reverse path (st2_mixer→Stage 1 reactor) unresolved.

Reviews 12 / Uncertainty 2: Tool startup simulation did not reach BPR operating pressure; steady-state not verified.

Reviews 12 / Uncertainty 3: 0.75 mm ID: plugging risk if any precipitate forms is uncharacterised.

Reviews 12 / Uncertainty 4: Pure O2 gas-source change unapproved; oxygen-service review required.

Reviews 12 / Uncertainty 5: CV-3301 max pressure rating unconfirmed.

Reviews 12 / Uncertainty 6: Stage 2 temperature not stage-resolved.

Shared assessment / Applies to all candidates: true

Shared assessment / Issues 1 / Id: SA-01

Shared assessment / Issues 1 / Title: Unprotected reverse path: Stage 2 mixer to Stage 1 reactor

Shared assessment / Issues 1 / Description: All twelve candidates share the topology where the O2 gas junction (st2_mixer) has a connected reverse path through the liquid branch directly to the Stage 1 reactor. No forward-oriented non-return valve is declared on this liquid branch. The CV-3301 check valve is on the MFC gas branch only and does not protect the liquid branch. The KHU observation of gas reverse flow under similar conditions (0.02 mL/min liquid, 0.43 mL/min O2) is qualitative evidence that this path is credible. Reducing gas flow alone does not close an unprotected reverse path. Branch-specific non-return protection on the liquid branch between Stage 1 and the gas junction must be evaluated and confirmed by the laboratory before execution.

Shared assessment / Issues 1 / Severity: requires_engineering_review

Shared assessment / Issues 1 / Resolved: false

Shared assessment / Issues 2 / Id: SA-02

Shared assessment / Issues 2 / Title: Pure oxygen gas-source change: unapproved, requires chemist confirmation

Shared assessment / Issues 2 / Description: The Bronkhorst FF-C00 MFC is listed as compatible with O2, N2, Air, CO2. Use of pure O2 (100 mol%) was confirmed by the chemist (Q-GAS-001) but constitutes a change from the batch protocol (ambient air). This change requires chemistry review (over-oxidation to sulfone, peroxide formation), inventory review (O2-service materials, fittings, tubing compatibility), and oxygen-service safety review before laboratory use. This is not an approved gas supply for execution; it remains a design requirement pending review.

Shared assessment / Issues 2 / Severity: requires_chemistry_and_safety_review

Shared assessment / Issues 2 / Resolved: false

Shared assessment / Issues 3 / Id: SA-03

Shared assessment / Issues 3 / Title: CV-3301 pressure rating unconfirmed

Shared assessment / Issues 3 / Description: The CV-3301 has a declared cracking pressure of 1.0 bar but no maximum pressure rating is supplied in inventory. The cracking pressure is the differential pressure required to

open the valve, not its pressure rating. Verification of the maximum allowable pressure rating is required before laboratory execution, particularly for candidates with BPR 7 bar where the gas plenum may reach ~8.2 bar g.

Shared assessment / Issues 3 / Severity: requires_verification

Shared assessment / Issues 3 / Resolved: false

Shared assessment / Issues 4 / Id: SA-04

Shared assessment / Issues 4 / Title: Physics tool results are conditional on unmeasured parameters

Shared assessment / Issues 4 / Description: DrSafety and DrFluidics tool results use illustrative assumption profiles (viscosity 1 mPa·s, liquid compliance 0.01 mL/bar, gas plenum 0.2 mL, pump response 1 s, MFC response 0.5 s, etc.) not derived from measured KHU data. No reversal predicted across the eight scenario/sensitivity combinations in the finite simulation window is not proof of safe operation and does not establish backflow probability. Candidates 3, 6 and 12 (gas-first startup scenarios) did not reach BPR operating pressure within the simulated window; steady-state transient behaviour is unverified for those candidates. All results are conditional model outputs requiring laboratory measurement and engineering review.

Shared assessment / Issues 4 / Severity: informational_limitation

Shared assessment / Issues 4 / Resolved: false

Shared assessment / Issues 5 / Id: SA-05

Shared assessment / Issues 5 / Title: Acrylonitrile exclusion from red pump tubing

Shared assessment / Issues 5 / Description: Acrylonitrile is listed in excluded_chemicals for all red pump tubing variants. All candidates correctly assign stream A (containing acrylonitrile) to pump_e_series_blue (blue tubing). This constraint is satisfied across all candidates.

Shared assessment / Issues 5 / Severity: satisfied

Shared assessment / Issues 5 / Resolved: true

Shared assessment / Issues 6 / Id: SA-06

Shared assessment / Issues 6 / Title: Stage 2 protocol temperature not stage-resolved

Shared assessment / Issues 6 / Description: The batch protocol does not provide an explicit temperature for Stage 2; the 25 °C assignment in all candidates is a design assumption. No validated rate correction is available. Yield and required residence time for Stage 2 remain unknown.

Shared assessment / Issues 6 / Severity: unresolved_experimental_uncertainty

Shared assessment / Issues 6 / Resolved: false

Shared assessment / Issues 7 / Id: SA-07

Shared assessment / Issues 7 / Title: No inline degasser; offline preparation required

Shared assessment / Issues 7 / Description: Inline degasser is declared unavailable. Stage 1 must be oxygen-free. All candidates rely on offline argon degassing of feed solution. The adequacy of offline deoxygenation and any dissolved oxygen residual entering Stage 1 are unconfirmed. Startup purge protocol must be defined.

Shared assessment / Issues 7 / Severity: procedural_requirement

Shared assessment / Issues 7 / Resolved: false

Shared assessment / Issues 8 / Id: SA-08

Shared assessment / Issues 8 / Title: Manual 2 light source minimum temperature

Shared assessment / Issues 8 / Description: Manual 2 Blue Strip LED (light_manual2_row13) has min_temperature_C 25.0 °C listed. All Stage 2 screens are at 25 °C. Operation at the minimum rated temperature should be confirmed to be within the module's operating range.

Shared assessment / Issues 8 / Severity: requires_verification

Shared assessment / Issues 8 / Resolved: false

Physics assessment / Tool result id: c5f4faa53d16cff0ba8ff1194becf8a69effa7c49391d142e9ad66b3a12cbc05

Physics assessment / Evidence ids 1: C1/as_designed/entered_profile/steady

Physics assessment / Evidence ids 2: C1/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 3: C2/as_designed/entered_profile/steady

Physics assessment / Evidence ids 4: C2/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 5: C3/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 6: C3/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 7: C4/as_designed/entered_profile/steady

Physics assessment / Evidence ids 8: C4/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 9: C5/as_designed/entered_profile/steady

Physics assessment / Evidence ids 10: C5/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 11: C6/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 12: C6/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 13: C7/as_designed/entered_profile/steady

Physics assessment / Evidence ids 14: C7/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 15: C8/as_designed/entered_profile/steady

Physics assessment / Evidence ids 16: C8/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 17: C9/as_designed/entered_profile/steady

Physics assessment / Evidence ids 18: C9/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 19: C10/as_designed/entered_profile/steady

Physics assessment / Evidence ids 20: C10/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 21: C11/as_designed/entered_profile/steady

Physics assessment / Evidence ids 22: C11/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 23: C12/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 24: C12/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Proposed alternatives 1: Unapproved proposal: install a forward-oriented non-return valve (e.g. CV-3301 or equivalent, service-verified for O2 and EtOH/buffer mixture) on the liquid branch between Stage 1 reactor outlet and the Stage 2 gas junction. This would provide branch-specific reverse-flow protection for Stage 1. Requires laboratory review of cracking pressure compatibility with pump operating pressure, O2 and solvent compatibility, and pressure rating confirmation before any design change is adopted.

Physics assessment / Proposed alternatives 2: Unapproved proposal: pressure-decoupled interstage arrangement (e.g. collection vessel with argon blanket between Stage 1 and Stage 2) to eliminate direct gas reverse connectivity. Requires full chemistry, operability and safety review.

Physics assessment / Limitation 1: Tool is an uncalibrated deterministic screening model; results are conditional on illustrative assumption profiles, not measured KHU device parameters.

Physics assessment / Limitation 2: No reversal predicted in the simulated window is not proof of safe operation and does not establish backflow probability.

Physics assessment / Limitation 3: Negative liquid-branch flow in model output means predicted liquid displacement toward Stage 1 pump, not proof that O2 gas reaches Reactor 1; actual species transport depends on slug dynamics, dissolution and flow regime not modelled.

Physics assessment / Limitation 4: Candidates 3, 6 and 12 as_designed startup simulations did not reach BPR operating pressure within the simulated window; their steady-state transient behaviour under operating pressure is not verified by these results.

Physics assessment / Limitation 5: Hypothetical liquid check valve alternative is not assigned inventory and is not service-verified; zero leakage assumption is an idealisation.

Physics assessment / Limitation 6: Gas plenum volume (0.2 mL illustrative), liquid compliance (0.01 mL/bar), MFC response (0.5 s), pump response (1.0 s) and two-phase resistance multiplier (3.0) are all unmeasured and require laboratory calibration.

Physics assessment / Limitation 7: Cracking pressure of CV-3301 (1.0 bar) concerns the differential pressure across that valve only; it is not a BPR minus cracking pressure calculation.

Physics assessment / Limitation 8: Model does not resolve slug flow, gas dissolution, capillary effects, BPR hysteresis or pressure-wave propagation.

Physics assessment / Limitation 9: Finite scenario window covers eight fixed scenario/sensitivity combinations; this is not a lifetime or probabilistic reliability assessment.

Physics assessment / Limitation 10: All tool outputs require engineering review before laboratory execution; no candidate is approved for execution by these results alone.

Call 7: Skeptic, attempt 1
Status: unsuccessful response; retained for provenance. Unterminated string starting at: line 183 column 28 (char 13447)

```json
{
  "vetoes": [],
  "shared_assessment": {
    "applies_to_all_candidates": true,
    "issues": [
      {
        "id": "SA-1",
        "title": "Unprotected reverse path to Stage 1 — unresolved for all candidates",
        "detail": "An unprotected reverse connectivity exists from the Stage 2 gas-introduction junction back through the interstage liquid branch to Stage 1 (path: st2_mixer → st1_reactor). No forward-oriented non-return valve is declared on this liquid branch in any candidate. The CV-3301 check valve is on the MFC/gas branch only; it does not protect the liquid branch. Qualitative experimental evidence (KHU, 15 Sep 2026) confirmed gas reverse flow under an earlier configuration. The tool predicted no liquid displacement under the entered illustrative profile, but this is a conditional model result based on unmeasured parameters and a finite scenario window — it does not close the structural reverse path and is not proof of safety. Reducing gas flow does not resolve reverse connectivity. This is an unresolved engineering risk requiring laboratory review before execution of any candidate."
      },
      {
```

"id": "SA-2",

"title": "Pure O2 gas source — unapproved change requiring chemist confirmation",

"detail": "The protocol specifies atmospheric air; the chemist has requested pure O2 (100 mol%). The measured experimental event (KHU) used pure O2 without prior design approval. The MFC (Bronkhorst FLEXI-FLOW FF-C00) supports O2, but oxygen-service compatibility of all wetted materials (tubing, mixers, check valves, BPR), ignition risk assessment, and laboratory inventory/safety review have not been confirmed. The gas-source change is unapproved and requires chemist and laboratory review before use. This applies to all candidates equally."

},

{

"id": "SA-3",

"title": "CV-3301 pressure rating unconfirmed",

"detail": "The declared cracking pressure is 1.0 bar. The maximum pressure rating is not supplied in the inventory. The cracking pressure concerns the differential pressure across the gas-branch check valve only; it is not equivalent to BPR setpoint minus cracking pressure. Pressure rating must be verified before laboratory execution."

},

{

"id": "SA-4",

"title": "Stage 2 temperature not stage-resolved in protocol",

"detail": "The batch protocol does not state a temperature for Stage 2 independently. All candidates assign 25 °C to Stage 2. No rate correction or yield prediction is available; this deviation from an unresolved source condition is an unconfirmed assumption."

},

{

"id": "SA-5",

"title": "Gas stage inlet/STP residence time index is not liquid residence time",

"detail": "The Stage 2 residence time is reported as V_R / (Q_liquid + Q_gas_STP), an apparent index only. Gas uptake, dissolution, slug dynamics, and actual photon exposure time are not modelled. This index cannot be compared directly to the batch 360 min hold as an equivalent exposure."

},

{

"id": "SA-6",

"title": "Wavelength mismatch — minor, not a hard violation",

"detail": "The protocol specifies 452 nm. The UV-150 assigned to Stage 1 delivers 450 nm; Manual 2 delivers 448 nm. No photocatalyst absorption data are provided to quantify the impact. This is a screening approximation, not a confirmed hard violation, but photon adequacy remains uncharacterized."

},

{

"id": "SA-7",

"title": "Candidates 3, 6, 12 — as_designed startup simulations did not reach BPR operating pressure",

"detail": "For Candidates 3, 6, and 12, the DrFluidics and DrSafety tool reported BPR_opening_pressure_reached=false in the as_designed startup scenario. Steady-state pressure dynamics are not confirmed for the as-designed configuration in these three candidates. The hypothetical_liquid_check alternative reached steady state, but that alternative uses an unapproved, unassigned valve. This is an unresolved uncertainty, not a confirmed hard violation, but it requires attention to startup sequencing for these three candidates."

},

{

"id": "SA-8",

"title": "Stage 1 conversion adequacy at 100 min — the primary yield-priority uncertainty for Candidates 1, 2, 4, 5, 7, 8, 9, 10, 11",

"detail": "Nine candidates use a 2 mL PFA Stage 1 reactor yielding a liquid residence time of 100 min (ratio 0.42 vs the 240 min batch anchor). Under the yield-priority objective, inadequate Stage 1 Giese conversion directly limits the maximum achievable Stage 2 yield. This is an uncharacterized kinetic risk, not a hard violation, but it is the dominant yield-limiting uncertainty for the majority of the candidate set."

}

]

},

"candidate_justifications": [

{

"candidate_id": 1,

"recommendation": "acceptable",

"hard_violation": false,

"justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 20 mL FEP / 181.8 min inlet/STP index (ratio 0.51) / BPR 7 bar / 2.01 equiv O2. Both stages sub-batch. No hard violation identified. Unresolved reverse path (SA-1) and unapproved O2 source (SA-2) apply.",

"uncertainties": [

"Stage 1 Giese conversion adequacy at 100 min uncharacterized (SA-8).",

"Unprotected reverse path to Stage 1 unresolved (SA-1).",

"Pure O2 unapproved; oxygen-service review required (SA-2)."

]

},

{

"candidate_id": 2,

"recommendation": "acceptable",

"hard_violation": false,

"justification": "Identical stage geometry and exposure indices to Candidate 1; BPR 2.5 bar gives larger in-channel gas volume (0.028 vs 0.012 mL/min). Whether lower BPR improves O2 mass transfer is an untested hypothesis. Same conversion adequacy and reverse-path concerns as C1.",

"uncertainties": [

```
      "Stage 1 Giese conversion adequacy at 100 min uncharacterized.",
      "BPR 2.5 bar mass-transfer benefit is an untested hypothesis.",
      "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
  },
  {
    "candidate_id": 3,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 5 mL ETFE 2.4 mm ID / 250 min (ratio 1.04) — meets batch Stage 1 anchor. Stage 2: 20 mL FEP / 181.8 min inlet/STP (ratio 0.51) / BPR 7 bar. ETFE 2.4 mm ID light penetration adequacy in UV-150 uncharacterized. As_designed startup simulation did not reach BPR pressure (SA-7). Reverse-path and O2 concerns shared.",
    "uncertainties": [
      "ETFE 2.4 mm ID: photon penetration depth in UV-150 uncharacterized.",
      "As_designed startup did not reach BPR operating pressure (SA-7).",
      "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
  },
  {
    "candidate_id": 4,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 20 mL FEP / 181.8 min inlet/STP / BPR 1.5 bar. Lowest system pressure; highest in-channel gas fraction (0.040 mL/min). Gas-liquid regime at BPR 1.5 bar uncharacterized. Stage 1 adequacy concern same as C1. Reverse-path and O2 concerns shared.",
    "uncertainties": [
      "Stage 1 Giese conversion adequacy at 100 min uncharacterized.",
      "BPR 1.5 bar: gas-liquid phase fraction and mass transfer at minimal back-pressure uncharacterized.",
      "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
  },
  {
    "candidate_id": 5,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL ETFE 2.4 mm / 90.9 min inlet/STP (ratio 0.25) / BPR 7 bar. Stage 2 index is the lowest in the set at 25% of batch; under yield-priority
```

policy this carries the highest oxidation-adequacy uncertainty. Large-bore and short Stage 2 combine two risks.",
"uncertainties": [
"Stage 2 inlet/STP 90.9 min (25% batch) — highest oxidation under-exposure risk in the set.",
"ETFE 2.4 mm ID Stage 2: gas-liquid contact and photon distribution uncharacterized.",
"Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
]
},
{
"candidate_id": 6,
"recommendation": "acceptable",
"hard_violation": false,
"justification": "Stage 1: 5 mL PFA 1 mm / 250 min (ratio 1.04) — meets batch Stage 1 anchor with preferred 1 mm ID geometry for UV-150. Stage 2: 20 mL FEP / 181.8 min inlet/STP / BPR 7 bar. As_designed startup did not reach BPR pressure (SA-7). Preferred over Candidate 3 for Stage 1 light-path geometry. Reverse-path and O2 concerns shared.",
"uncertainties": [
"As_designed startup simulation did not reach BPR operating pressure (SA-7).",
"Stage 2 oxidation adequacy at 181.8 min inlet/STP uncharacterized.",
"Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
]
},
{
"candidate_id": 7,
"recommendation": "acceptable",
"hard_violation": false,
"justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL PFA 1 mm / 90.9 min inlet/STP (ratio 0.25) / BPR 7 bar. Shortest Stage 2 index with standard 1 mm ID; 1 mm geometry suitable for Manual2 but 25% batch exposure is the dominant yield-priority uncertainty. Both stages sub-batch.",
"uncertainties": [
"Stage 2 inlet/STP 90.9 min (25% batch) — oxidation adequacy most uncertain.",
"Stage 1 Giese conversion at 100 min uncharacterized.",
"Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
]
},
{
"candidate_id": 8,
"recommendation": "acceptable",
"hard_violation": false,

```
    "justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL FEP 1 mm / 90.9 min inlet/STP (ratio 0.25) / BPR 7 bar / 25 °C (within FEP 50 °C limit). Functionally equivalent to Candidate 7; FEP Stage 2 limits future temperature exploration to ≤50 °C but is not a violation at 25 °C. Same exposure adequacy concerns.",
    "uncertainties": [
     "Stage 2 inlet/STP 90.9 min (25% batch) — oxidation adequacy uncertain.",
     "FEP Stage 2 max 50 °C constrains future temperature parameter space.",
     "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
   },
   {
    "candidate_id": 9,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL ETFE 2.4 mm / 90.9 min inlet/STP (ratio 0.25) / BPR 2.5 bar. Combines the shortest Stage 2 index with large bore and lowest BPR; gas-liquid regime and mass transfer carry the most combined uncertainty of the 10 mL group. Reverse-path and O2 concerns shared.",
    "uncertainties": [
     "Stage 2 at 90.9 min inlet/STP, 2.4 mm ID, 2.5 bar: highest combined Stage 2 uncertainty in set.",
     "Stage 1 Giese conversion at 100 min uncharacterized.",
     "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
   },
   {
    "candidate_id": 10,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL PFA 1 mm / 90.9 min inlet/STP (ratio 0.25) / BPR 2.5 bar. Standard 1 mm ID in Stage 2; lower BPR than C7 increases in-channel gas volume. Mass-transfer benefit of lower BPR is an untested hypothesis. Both stages sub-batch.",
    "uncertainties": [
     "Stage 2 inlet/STP 90.9 min (25% batch) — oxidation adequacy uncertain.",
     "BPR 2.5 bar mass-transfer benefit relative to 7 bar is an untested hypothesis.",
     "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
   },
   {
    "candidate_id": 11,
```

```
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA / 100 min (ratio 0.42). Stage 2: 10 mL FEP 1 mm / 90.9 min inlet/STP (ratio 0.25) / BPR 2.5 bar / 25 °C (within FEP limit). Equivalent to Candidate 10 with FEP Stage 2; FEP 50 °C ceiling limits future temperature exploration. Same exposure adequacy concerns as C10.",
    "uncertainties": [
      "Stage 2 inlet/STP 90.9 min (25% batch) — oxidation adequacy uncertain.",
      "FEP Stage 2 max 50 °C constrains future temperature screening.",
      "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
  },
  {
    "candidate_id": 12,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 5 mL PFA 0.75 mm ID / 250 min (ratio 1.04) — meets batch Stage 1 anchor; narrowest ID maximizes wall-proximity photon delivery but increases pressure drop and plugging risk if any precipitate forms. Stage 2: 20 mL FEP / 181.8 min inlet/STP / BPR 7 bar. As_designed startup did not reach BPR pressure (SA-7). Reverse-path and O2 concerns shared.",
    "uncertainties": [
      "0.75 mm ID Stage 1: plugging risk with any precipitate; flow stability at 0.02 mL/min unverified.",
      "As_designed startup simulation did not reach BPR operating pressure (SA-7).",
      "Unprotected reverse path unresolved (SA-1); O2 source unapproved (SA-2)."
    ]
  }
 ],
 "screening_observations": {
  "yield_priority_note": "Under the yield-priority objective, Stage 1 conversion adequacy is the primary determinant of overall yield. Nine of twelve candidates deliver a Stage 1 liquid residence time of 100 min (ratio 0.42 vs the 240 min batch anchor). Candidates 3, 6, and 12 deliver 250 min (ratio 1.04) and are more aligned with the batch Stage 1 anchor; this is an exposure argument, not a kinetic or yield claim. Whether 100 min Stage 1 provides adequate Giese conversion in flow is uncharacterized. Similarly, Stage 2 inlet/STP indices of 91 min (ratio 0.25) in Candidates 5, 7, 8, 9, 10, 11 represent the shortest oxidation exposures and carry the highest adequacy uncertainty under yield priority. Candidates 1, 2, 3, 4, 6, 12 deliver Stage 2 apparent indices of 182
```


Call 8: Skeptic, attempt 2
Status: recorded response, parsed successfully.

Vetoes: []

Assessment / Shared issues / Reverse path: All 12 candidates carry an unresolved unprotected reverse path on the liquid branch (st2_mixer → Stage 1 reactor). No forward-oriented non-return valve is declared on that branch. The CV-3301 check valve sits on the MFC/gas branch only and does not protect the liquid branch. The KHU observation of gas reverse flow at 0.02 mL/min liquid / 0.43 mL/min oxygen confirms the credibility of

this risk. Reducing gas flow does not close the unprotected path. This is a persistent engineering review requirement for all candidates, not a hard veto, because it is an unresolved experimental outcome requiring mitigation review rather than a confirmed design-rules violation.

Assessment / Shared issues / Tool status: DrFluidics and DrSafety tool results are conditional on unmeasured parameters (illustrative viscosity 1.0 mPa·s, gas plenum 0.2 mL, MFC ΔP 1.0 bar, pump response 1.0 s). No reversal predicted in the finite scenario window is not proof of safety. Predicted quantity is liquid displacement at the junction, not oxygen reaching Reactor 1. Results for Candidates 3, 6, and 12 (as_designed) did not reach BPR opening pressure in the startup simulation window; their steady-state transient behaviour under operating conditions is not confirmed by these results.

Assessment / Shared issues / Gas source: Pure oxygen (O2) is chemist-confirmed (Q-GAS-001) but this constitutes a gas-source change from the air specified in the earlier design that was actually used in the KHU experiment. This change is unapproved for oxygen service; chemistry, inventory, and O2-service compatibility review are required before laboratory use across all candidates.

Assessment / Shared issues / Cv3301: CV-3301 maximum pressure rating is not supplied. Cracking pressure (1.0 bar) is not a pressure rating. Verification required before execution for all candidates.

Assessment / Shared issues / Stage1 conversion: All candidates that use the 2 mL PFA coil at 0.02 mL/min (Candidates 1, 2, 4, 5, 7, 8, 9, 10, 11) deliver a Stage 1 liquid residence time of 100 min (ratio 0.42 vs the 240 min batch anchor). Whether this is adequate for Giese conversion is uncharacterized; kinetics are unknown.

Assessment / Shared issues / Stage2 apparent index: The Stage 2 inlet/STP apparent residence-time index is V/(Q_liquid + Q_gas_STP) and is not equivalent to liquid residence time or photon exposure time. Gas uptake, dissolution, slug regime, and actual O2 delivery to the liquid phase are not modeled in any candidate.

Assessment / Shared issues / Stage2 temperature: Stage 2 batch temperature is not stage-resolved in the protocol. All candidates assign 25 °C to Stage 2; no validated rate correction is available.

Assessment / Shared issues / Wavelength: Target wavelength is 452 nm. Stage 1 uses UV-150 450 nm (2 nm deviation); Stage 2 uses Manual 2 Blue Strip LED at 448 nm (4 nm deviation). No photon-dose adequacy or intensification is established for either source.

Assessment / Shared issues / Acrylonitrile pump: Acrylonitrile is excluded from red pump tubing. All candidates correctly assign Stream A (containing acrylonitrile) to pump_e_series_blue (blue tubing). No violation.

Assessment / Shared issues / Objective policy: Yield-priority policy requires investigation of conversion adequacy at all stage holds. Longer hold times are not inherently better; degradation risks at extended irradiation in the presence of acrylonitrile and Ir photocatalyst are unquantified for any candidate. Neither shorter nor longer holds are proven superior.

Assessment / Candidate specific notes / Candidates 1 2 4: Stage 1 at 100 min (0.42× batch) is the primary yield-priority concern. These candidates provide the shortest Stage 1 exposure in the set and represent legitimate low-end screening brackets, but conversion adequacy is the most uncertain dimension under the yield-priority objective.

Assessment / Candidate specific notes / Candidate 3: Stage 1 ETFE 2.4 mm ID at 250 min meets the batch anchor but introduces uncharacterized light-penetration depth concern in the wider bore. As-designed startup simulation did not reach BPR operating pressure in the simulation window; steady-state behaviour unverified for as-designed configuration.

Assessment / Candidate specific notes / Candidate 6: Stage 1 PFA 5 mL 1 mm ID at 250 min meets the batch anchor with standard bore geometry, avoiding the large-ID photon penetration concern. Preferred over Candidate 3 on geometry grounds under yield priority, subject to unresolved reverse-path and O2-service review. As-designed startup simulation did not reach BPR operating pressure in simulation window.

Assessment / Candidate specific notes / Candidate 12: Stage 1 PFA 5 mL 0.75 mm ID at 250 min meets the batch anchor with the narrowest bore, potentially maximizing wall-proximity photon delivery. Pressure drop is within rating (0.005 bar calculated). As-designed startup simulation did not reach BPR operating pressure in the simulation window. Plugging risk at 0.75 mm if any precipitate forms is uncharacterized.

Assessment / Candidate specific notes / Candidates 5 7 8 9 10 11: Stage 2 inlet/STP apparent index of 90.9 min (0.25× batch 360 min anchor) is the shortest Stage 2 exposure in the set. Under yield-priority policy this is the highest-risk Stage 2 bracket for oxidation adequacy. These are legitimate short-time screening points but carry the greatest under-oxidation uncertainty.

Assessment / Candidate specific notes / Candidates 8 11: FEP Stage 2 coil (max 50 °C) is within the 25 °C screening temperature but constrains future temperature-elevation experiments. Not a violation at current conditions.

Assessment / Candidate specific notes / Candidate 4: BPR 1.5 bar provides the lowest system back-pressure; startup pressure sequencing is most critical here to prevent gas intrusion during transients. Unresolved reverse path concern is highest-consequence at lowest back-pressure, though no reversal was predicted in the tool's entered-profile steady scenarios.

Assessment / Preferred candidates for yield priority: Candidates 6 and 12 are most aligned with the yield-priority objective: both provide Stage 1 liquid residence time of 250 min (1.04× batch anchor) with standard or narrow 1 mm bore PFA geometry in the UV-150 module, and Stage 2 at 181.8 min inlet/STP index (0.51× batch) in the 20 mL FEP coil at 7 bar. Candidate 6 uses the 1 mm PFA coil (coil_row4); Candidate 12 uses the 0.75 mm PFA coil (coil_row8) with higher potential photon-wall proximity but uncharacterized plugging risk. Candidate 3 is also near-batch in Stage 1 but the 2.4 mm ETFE bore introduces a light-penetration uncertainty that makes Candidates 6 and 12 preferable under yield priority. All three remain subject to the unresolved reverse-path and O2-service review requirements.

Assessment / No hard violations found: No candidate is vetoed. The unresolved reverse path is a credible engineering risk requiring mitigation before execution but is not a confirmed design-rules hard violation that eliminates any candidate from screening consideration. The gas-source change to pure O2 requires approval but is chemist-confirmed as the intended reagent. No cross-domain scientific claim conflicts were identified that constitute hard violations.

Requested measurement 1: Branch-specific non-return valve (liquid branch, st2_mixer to Stage 1 outlet): inventory item, cracking pressure compatible with pump operating pressure, O2 and EtOH/buffer service compatibility, and pressure rating—verified before laboratory execution

Requested measurement 2: CV-3301 maximum pressure rating (distinct from 1.0 bar cracking pressure)—verified before laboratory execution

Requested measurement 3: Pure O2 gas-source approval: chemistry review, O2-service compatibility of all wetted materials and tubing, and laboratory safety review before use

Requested measurement 4: Actual gas supply pressure and MFC differential-pressure requirement at operating conditions—measured, not assumed

Requested measurement 5: Startup pressure sequence verification: liquid flow and BPR back-pressure established before gas introduction—procedurally confirmed in laboratory

Requested measurement 6: Stage 2 batch temperature (not stage-resolved in protocol)—confirmed for rate and yield comparison

Requested measurement 7: Stage 1 kinetics under flow conditions: qualitative conversion check at 100 min and 250 min liquid residence times to confirm adequacy for Giese reaction

Requested measurement 8: Stage 2 oxidation kinetics: qualitative conversion check at 90.9 min and 181.8 min inlet/STP apparent index to confirm adequacy for sulfoxide formation

Requested measurement 9: Light penetration adequacy: photon dose delivered by UV-150 450 nm to 2 mL PFA (1 mm ID) at 0.02 mL/min, and by Manual 2 448 nm to Stage 2 coil geometry—not established by equipment listing alone

Requested measurement 10: ETFE 2.4 mm ID coil (Candidates 3, 5, 9): light penetration depth and gas-liquid contacting regime characterization before comparing to 1 mm ID results

Requested measurement 11: PFA 0.75 mm ID coil (Candidate 12): plugging risk assessment with the EtOH/pH 9 buffer/Ir-catalyst solution at 0.02 mL/min

Requested measurement 12: Pressure traces during startup and steady-state operation to validate or replace illustrative tool parameters (liquid compliance, gas plenum volume, MFC response time, pump droop)

Call 9: Chief, attempt 1
Status: recorded response, parsed successfully.

Candidate: 6

Justification: Stage 1: 5 mL PFA 1 mm ID / UV-150 450 nm / 250 min liquid τ (ratio 1.04 vs 240 min batch anchor) — the only 1 mm-bore Stage 1 option that meets the batch anchor, preserving standard photon-penetration geometry. Stage 2: 20 mL FEP / Manual 2 448 nm / 181.8 min inlet/STP apparent index (ratio 0.51 vs 360 min batch), BPR 7 bar, 2.008 equiv O2 at 0.09 sccm. Under yield-priority policy, Stage 1 conversion adequacy is the primary risk; 250 min at 1 mm bore is the most defensible starting point without inventing kinetics. Stage 2 at 0.51× batch is a sub-batch but reasonable first screening exposure at 7 bar. DrFluidics/DrSafety as-designed startup simulation did not reach BPR opening pressure in the simulation window — this is an unresolved limitation, not a confirmed violation. All shared risks apply (reverse path, O2 service, CV-3301 rating).

Objective alignment: Yield-priority requires adequate Stage 1 Giese conversion before Stage 2 oxidation can produce sulfoxide 4a. The 1 mm bore PFA coil at 250 min provides the batch-anchored Stage 1 exposure without the light-penetration uncertainty introduced by the 2.4 mm ETFE bore (Candidate 3) or the plugging risk of the 0.75 mm bore (Candidate 12). Stage 2 at 181.8 min inlet/STP is a sub-batch starting bracket; the objective policy does not require matching the full 360 min batch Stage 2 time, only a reasonably justified exposure. No degradation or kinetic superiority is claimed.

Alternative 1 / Candidate: 3

Alternative 1 / Reason not selected: Stage 1 uses ETFE 5 mL 2.4 mm ID at 250 min — same batch-anchor exposure as Candidate 6, but the wider bore introduces uncharacterized light-penetration depth uncertainty in the UV-150 module. Under yield priority, uncharacterized photon delivery is a weaker basis than the established 1 mm geometry. As-designed startup simulation also did not reach BPR operating pressure in the simulation window. No hard violation, but geometry uncertainty tips the trade-off to Candidate 6.

Alternative 2 / Candidate: 12

Alternative 2 / Reason not selected: Stage 1 uses PFA 5 mL 0.75 mm ID at 250 min — meets batch anchor and may improve wall-proximity photon delivery, but the narrower bore carries an uncharacterized plugging risk with the EtOH/buffer/Ir-catalyst solution at 0.02 mL/min. As-designed startup simulation did not reach BPR operating pressure in the simulation window. Stage 2 is identical to Candidate 6. Under yield priority, the plugging risk at 0.75 mm is an additional unresolved experimental uncertainty not present in Candidate 6.

Alternative 3 / Candidate: 1

Alternative 3 / Reason not selected: Stage 1 uses 2 mL PFA at 100 min (ratio 0.42 vs 240 min batch anchor) — a legitimate short-time screening bracket but the lowest Stage 1 exposure. Under yield-priority policy, starting with sub-batch Stage 1 exposure creates the highest risk of insufficient Giese conversion entering Stage 2, undermining the stated objective. Stage 2 is identical to Candidate 6. If Stage 1 kinetics prove fast, Candidate 1 becomes a valid follow-on point.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: Yield-priority drives selection of Candidate 6 over shorter Stage 1 candidates (1, 2, 4, 5, 7–11). The objective policy does not guarantee higher yield; it selects the most defensible starting exposure for conversion-focused screening.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: The chemist hypothesis that Stage 1 requires sufficient photon delivery and residence time supports choosing 250 min over 100 min for Stage 1. Stage 2 time is treated as a first screening bracket, not an optimized value.

Answer impact 3 / Question: Q-GAS-001

Answer impact 3 / Effect: Pure O2 confirmed by chemist but unapproved for oxygen service. This requirement applies identically to Candidate 6 and all other candidates; it does not change the relative selection but is a hard prerequisite for execution.

Answer impact 4 / Question: Q-GAS-003

Answer impact 4 / Effect: O2 introduction at Stage 2 inlet only is satisfied by Candidate 6 as designed. The unprotected liquid reverse path means Stage 1 oxygen-free requirement is not structurally guaranteed and requires engineering mitigation before execution.

Answer impact 5 / Question: Q-HIST-001

Answer impact 5 / Effect: The KHU observation of gas reverse flow at 0.02 mL/min liquid / 0.43 mL/min O2 confirms the credibility of the unprotected reverse-path risk. Candidate 6 at 0.09 sccm O2 is a lower gas flow, but reducing gas flow does not close the unprotected liquid-branch path; the risk is not resolved by this change.

Answer impact 6 / Question: Q-PREF-001

Answer impact 6 / Effect: Stream A: substrate 0.1 M, 0.002 mmol/min (1.0 equiv); acrylonitrile 0.2 M, 0.004 mmol/min (2.0 equiv); Ir catalyst 0.0005 M, 0.000001 mmol/min (0.5 mol%); solvent EtOH:pH 9 buffer 5:1 v/v, degassed offline; pump_e_series_blue, 0.02 mL/min. Stream B: O2 pure, 0.09 sccm inlet/STP (0.012422 mL/min pressure-corrected at 8.01 bar abs), 2.008 equiv inlet/STP; mfc_ffc00. Stage 1: 25 °C, 5 mL PFA 1 mm, 250 min liquid τ. Stage 2: 25 °C, 20 mL FEP 1 mm, 181.8 min inlet/STP apparent index. BPR 7 bar. Unprotected reverse path and O2-service approval are unresolved prerequisites.

Limitation 1: No kinetic data exist for this Giese/oxidation sequence in flow; 250 min Stage 1 and 181.8 min Stage 2 inlet/STP index are batch-anchored screening starting points, not proven adequate exposures.

Limitation 2: Stage 2 inlet/STP apparent residence-time index is V/(Q_liquid + Q_gas_STP) and is not equivalent to liquid residence time or photon exposure time; gas uptake, dissolution, and slug dynamics are not modeled.

Limitation 3: UV-150 450 nm vs. target 452 nm (2 nm deviation) and Manual 2 448 nm (4 nm deviation): photon-dose adequacy is not established for either source.

Limitation 4: Stage 2 batch temperature is not stage-resolved in the protocol; the 25 °C assignment carries no validated rate correction.

Limitation 5: As-designed startup simulation for Candidate 6 did not reach BPR opening pressure in the DrFluidics/DrSafety simulation window; steady-state transient behavior under operating conditions is not confirmed by these tool results.

Limitation 6: Unprotected reverse path (st2_mixer → Stage 1 reactor) is unresolved; no liquid-branch non-return valve is declared. The CV-3301 on the MFC branch does not protect this path.

Limitation 7: Pure O2 is chemist-confirmed but not approved for oxygen service; chemistry, inventory, and O2-service compatibility review are required before execution.

Limitation 8: CV-3301 maximum pressure rating is unconfirmed; cracking pressure is not a pressure rating.

Limitation 9: Whether 250 min Stage 1 irradiation with acrylonitrile present leads to side reactions or Ir catalyst fatigue is unquantified.

Limitation 10: Tool results are conditional on unmeasured illustrative parameters; no reversal predicted is not proof of safety.

Suggested measurement 1: Verify and install a forward-oriented, service-verified (O2 and EtOH/buffer compatible) non-return valve on the liquid branch between Stage 1 reactor outlet and the Stage 2 gas-introduction junction before any execution — this is an unapproved proposal requiring laboratory and safety review.

Suggested measurement 2: Confirm CV-3301 maximum pressure rating (distinct from 1.0 bar cracking pressure) before execution.

Suggested measurement 3: Complete chemistry, inventory, and O2-service compatibility review for pure O2 across all wetted materials before execution.

Suggested measurement 4: Measure or verify actual gas supply pressure and MFC differential-pressure requirement at operating conditions to replace illustrative tool assumptions.

Suggested measurement 5: Establish and verify a startup pressure sequence: confirm liquid flow and BPR back-pressure are reached before gas introduction to reduce transient reverse-flow risk.

Suggested measurement 6: Confirm Stage 2 batch temperature from the protocol to allow rate comparison.

Suggested measurement 7: Qualitative Stage 1 conversion check at 100 min and 250 min liquid τ to determine whether 100 min is sufficient (which would make Candidates 1, 2, 4 viable alternatives and reduce reactor volume requirements).

Suggested measurement 8: Qualitative Stage 2 oxidation check at 90.9 min and 181.8 min inlet/STP apparent index to bound the minimum required Stage 2 exposure.

Suggested measurement 9: Measure pressure traces during startup and steady-state to validate or replace illustrative DrFluidics/DrSafety parameters and confirm BPR opening-pressure behavior for Candidate 6 as-designed.

Editorial cross-checks (not model statements)
Time interpretation: some reports call Stage 2 liquid exposure approximately 1000 min, obtained from 20 mL/0.02 mL/min. This is a liquid-only space-time calculation using the entire reactor volume, not measured two-phase liquid residence. Likewise, the 181.82 min inlet/STP index must not be interpreted as a measured fraction of batch reaction exposure.

Numerical transcription: the Figure 5 Set 3 Chief response gives the photocatalyst molar flow as 0.001 mmol/min. The final feed record and concentration-times-flow calculation give 0.0005 mol/L x 0.02 mL/min = 0.000010 mmol/min. The original statement is retained in the transcript as a 100-fold numerical error, not adopted in the figure text or experimental recipe.

Geometry and transport claims: the Set 1 DrFluidics shared assessment groups Candidate 6 with large-bore ETFE options even though Candidate 6 uses 1 mm-ID PFA. The Set 3 Chief attributes a smaller in-channel gas volume fraction to the selected reactor relative to other 7 bar candidates with smaller reactors. Their gas and liquid feed rates, pressure and temperature are the same; changing reactor volume alone does not establish a different gas/liquid volumetric flow ratio or measured gas holdup. A known bore defines geometry but does not establish photon-dose adequacy or a measured transport advantage.

Protocol interpretation: several reports describe the Stage 2 batch temperature as unspecified, whereas the supplied procedure continues under otherwise identical irradiation conditions after Stage 1 at 25 deg C. That wording does not establish an independently measured Stage 2 temperature, but it does not justify presenting 25

deg C as an unrelated new condition. Similarly, the qualitative backflow incident does not prove that every lower-flow alternative will reverse; the defensible finding is that the unprotected pathway remains.

F5-S3: Figure 5, Set 3
Source: outputs/figure5_pure_oxygen_physics_20260918/figure5_set3/attempt_01. Selected Candidate 1; 12-candidate pool; downstream model claude-sonnet-4-6. The objective policy is balanced.

Call 1: DrChemistry, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry / Stage 1 atmosphere: All candidates correctly maintain an oxygen-free (Ar-degassed) Stage 1 for the Giese radical addition.

Shared assessment / Chemistry / Stage 2 gas: Pure O2 introduced at Stage 2 inlet only, per chemist instruction. Gas source change from batch air to pure O2 is a confirmed chemist instruction; oxygen-service compatibility of all wetted materials, fittings, and the MFC requires independent laboratory review before execution.

Shared assessment / Chemistry / Gas delivery: All candidates: O2 at 0.09 sccm inlet/STP, delivering 2.0078 equiv. MFC is Bronkhorst FF-C00 rated for O2. CV-3301 check valve max pressure rating unconfirmed; cracking pressure (1.0 bar) concerns pressure difference across that valve, not BPR minus cracking pressure.

Shared assessment / Chemistry / Wavelength: Protocol specifies 452 nm. UV-150/450 nm (Stage 1) and Manual 2/448 nm (Stage 2) are nearest available; wavelength offset is unresolved—photocatalytic efficiency at these wavelengths is not established.

Shared assessment / Chemistry / Acrylonitrile pump: Acrylonitrile is excluded from red pump tubing; all candidates route feed via blue E-series pump. Solubility of all components in EtOH:pH9 buffer (5:1) under flow is not independently verified.

Shared assessment / Chemistry / Stage 2 temperature: Stage 2 batch temperature is not stage-resolved in the protocol. All candidates assign 25 °C; no rate correction is validated.

Shared assessment / Chemistry / Flow rate: All candidates: Q_liquid = 0.02 mL/min (minimum E-series setting). Feed throughput = 0.002 mmol/min substrate. This is feed throughput; product throughput is unmeasured.

Shared assessment / Chemistry / Stage 1 residence time note: Stage 1 liquid residence time (V/Q_liquid) is the correct basis for a single-phase photochemical stage. Stage 2 apparent residence time uses V/(Q_liquid + Q_gas_STP) index; actual liquid contact time and gas-liquid mass transfer are unmeasured.

Shared assessment / Chemistry / Etfe photoreactor: Candidates 3, 9, 11 use ETFE coil (ID 2.4 mm) in Stage 1 with UV-150. ETFE is listed as compatible with UV-150 in the photoreactor_module_ids. However, the larger ID (2.4 mm vs 1.0 mm) gives lower path-length/volume ratio; optical path and photon delivery efficiency relative to 1 mm coils are unverified.

Shared assessment / Chemistry / Offline deoxygenation: No inline degasser available; feed must be degassed offline. Adequacy of offline degassing under flow conditions is unconfirmed.

Shared assessment / Chemistry / Pure oxygen safety: Pure oxygen use requires oxygen-service review of all wetted components, fittings, tubing, and the MFC. This is an unapproved proposal pending chemistry, inventory, and oxygen-service review.

Shared assessment / Backflow / Status: Unresolved across all candidates. The reverse path st2_mixer → st1_reactor (liquid branch) has no declared non-return protection. The MFC branch carries a CV-3301 check valve (cracking pressure 1.0 bar, max pressure rating unconfirmed, service unverified); this valve protects only the MFC branch, not the liquid inlet branch to Stage 1. Qualitative backflow was experimentally observed (KHU, 2026-09-15) at 0.02 mL/min liquid / 0.43 mL/min O2; reducing gas flow does not close the unprotected

reverse path. A branch-specific non-return device or pressure-decoupled arrangement on the liquid outlet of Stage 1 is required before execution. BPR setpoint does not determine cracking pressure adequacy; the cracking pressure concerns differential pressure across the check valve itself.

Shared assessment / Backflow / Predicted consequence: Upstream liquid displacement by gas is the predicted consequence of reverse flow, not oxygen reaching Reactor 1 directly; however, upstream gas intrusion would violate the Stage 1 oxygen-free requirement.

Shared assessment / Backflow / Resolution required: Engineering review of branch-specific protection mandatory for all candidates before laboratory execution.

Shared assessment / Objective policy: Balanced screen: batch times are anchors, not targets. Stage hold ratios are indices of experimental exposure; no kinetic, yield, or intensification claims are made. All candidates run at minimum pump flow (0.02 mL/min); feed throughput is identical across all. No candidate has demonstrated superior product throughput.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: prefer

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA in UV-150/450 nm, τ_liq=100 min (0.42× batch). Stage 2: 20 mL FEP in Manual2/448 nm, τ_inlet=181.8 min (0.51× batch), BPR 7 bar. Compact baseline with intermediate stage-time ratios for both steps; interpretable anchor for the screen. Backflow unresolved; oxygen-service review required.

Reviews 1 / Uncertainty 1: Unprotected reverse path to Stage 1 unresolved; engineering review required before execution.

Reviews 1 / Uncertainty 2: Stage 2 batch temperature not stage-resolved; 25 °C assumed.

Reviews 1 / Uncertainty 3: Wavelength offset (450/448 vs 452 nm) photocatalytic efficiency unverified.

Reviews 1 / Uncertainty 4: Actual liquid exposure and gas-liquid mass transfer in Stage 2 unmeasured.

Reviews 1 / Uncertainty 5: Offline deoxygenation adequacy under flow unconfirmed.

Reviews 1 / Uncertainty 6: Pure O2 oxygen-service compatibility review required.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical reactor geometry to Candidate 1 but BPR reduced to 2.5 bar, increasing in-channel gas volume (0.028 mL/min vs 0.012 mL/min). Lower operating pressure; gas fraction per unit volume is higher, which may reduce liquid contact time. Useful BPR contrast versus Candidate 1 but gas-liquid contact effects are unmeasured.

Reviews 2 / Uncertainty 1: Unprotected reverse path to Stage 1 unresolved; lower BPR increases backflow risk window at startup/shutdown.

Reviews 2 / Uncertainty 2: Higher gas volume fraction at lower pressure; actual liquid residence and mass transfer unmeasured.

Reviews 2 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 2 / Uncertainty 4: Oxygen-service and CV-3301 pressure rating unconfirmed.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 5 mL ETFE (ID 2.4 mm) in UV-150/450 nm, τ_liq=250 min (1.04× batch). Stage 2: 20 mL FEP, τ_inlet=181.8 min, BPR 7 bar. Near-batch Stage 1 exposure is useful for yield-priority comparison. ETFE ID 2.4 mm photon delivery unverified vs 1 mm PFA. Longer total time (431.8 min) reduces comparative throughput.

Reviews 3 / Uncertainty 1: ETFE 2.4 mm ID optical path efficiency in UV-150 unverified vs 1 mm coils.

Reviews 3 / Uncertainty 2: Unprotected reverse path to Stage 1 unresolved.

Reviews 3 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 3 / Uncertainty 4: Oxygen-service review required.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 2 mL PFA, τ_liq=100 min; Stage 2: 20 mL FEP, τ_inlet=181.8 min, BPR 1.5 bar (lowest available). Highest in-channel gas volume fraction (0.040 mL/min). Provides lowest-pressure BPR data point. Gas fraction effects on liquid contact and selectivity unmeasured; useful contrast but lower pressure increases startup backflow risk window.

Reviews 4 / Uncertainty 1: Unprotected reverse path unresolved; 1.5 bar BPR is lowest setpoint, potentially narrowest margin against backflow at startup.

Reviews 4 / Uncertainty 2: Highest gas fraction; actual liquid exposure unmeasured.

Reviews 4 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 4 / Uncertainty 4: Oxygen-service review required.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA, τ_liq=100 min; Stage 2: 10 mL ETFE (ID 2.4 mm), τ_inlet=90.9 min (0.25× batch), BPR 7 bar. Shortest Stage 2 index among 7-bar candidates; shortest total inlet time (190.9 min). Explores whether 90.9 min apparent Stage 2 is sufficient. ETFE 2.4 mm gas-liquid contact characteristics unverified.

Reviews 5 / Uncertainty 1: Stage 2 τ_inlet=90.9 min is 0.25× batch; adequacy of oxidation exposure is unknown.

Reviews 5 / Uncertainty 2: ETFE 2.4 mm ID gas-liquid contact and photon delivery unverified.

Reviews 5 / Uncertainty 3: Unprotected reverse path unresolved.

Reviews 5 / Uncertainty 4: Oxygen-service review required.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA (ID 1 mm), τ_liq=250 min (1.04× batch), UV-150/450 nm. Stage 2: 20 mL FEP, τ_inlet=181.8 min, BPR 7 bar. Near-batch Stage 1 in 1 mm PFA (better-defined photon path than ETFE); longer total time (431.8 min). Preferred over Candidate 3 for Stage 1 optical path confidence, otherwise equivalent.

Reviews 6 / Uncertainty 1: Unprotected reverse path unresolved.

Reviews 6 / Uncertainty 2: Stage 2 temperature unresolved.

Reviews 6 / Uncertainty 3: 250 min Stage 1 exposure: over-irradiation risk for photoredox is uncharacterized.

Reviews 6 / Uncertainty 4: Oxygen-service review required.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA, τ_liq=100 min; Stage 2: 10 mL PFA (ID 1 mm), τ_inlet=90.9 min, BPR 7 bar. Compact design; shorter Stage 2 with 1 mm PFA (better-defined path than ETFE). Distinguishable from Candidate 5 by Stage 2 reactor material; otherwise similar exposure profile. Total inlet time 190.9 min.

Reviews 7 / Uncertainty 1: Stage 2 τ_inlet=90.9 min adequacy unknown.

Reviews 7 / Uncertainty 2: Unprotected reverse path unresolved.

Reviews 7 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 7 / Uncertainty 4: Oxygen-service review required.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 2 mL PFA, τ_liq=100 min; Stage 2: 10 mL FEP (ID 1 mm, max 50 °C), τ_inlet=90.9 min, BPR 7 bar. Structurally identical timing to Candidates 5 and 7; only material difference is FEP vs PFA/ETFE. FEP max temperature 50 °C is not a constraint at 25 °C. No distinguishing advantage over Candidate 7; informative as material control.

Reviews 8 / Uncertainty 1: Stage 2 τ_inlet=90.9 min adequacy unknown.

Reviews 8 / Uncertainty 2: Unprotected reverse path unresolved.

Reviews 8 / Uncertainty 3: FEP max 50 °C; not violated at 25 °C but confirmed operating margin is small.

Reviews 8 / Uncertainty 4: Oxygen-service review required.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 5 mL ETFE (2.4 mm), τ_liq=250 min; Stage 2: 10 mL ETFE (2.4 mm), τ_inlet=90.9 min, BPR 7 bar. Both stages use wide-bore ETFE; total inlet time 340.9 min. ETFE optical path unverified for both stages. Offers a wide-bore all-ETFE comparison but photon delivery uncertainty is doubled.

Reviews 9 / Uncertainty 1: ETFE 2.4 mm ID photon delivery unverified in both stages.

Reviews 9 / Uncertainty 2: Unprotected reverse path unresolved.

Reviews 9 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 9 / Uncertainty 4: Oxygen-service review required.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA, τ_liq=100 min; Stage 2: 10 mL ETFE (2.4 mm), τ_inlet=90.9 min, BPR 2.5 bar. Lower BPR increases gas volume fraction in Stage 2 (0.028 mL/min). Provides BPR×reactor-material interaction data point vs Candidates 5 and 12. ETFE 2.4 mm and lower pressure both add uncertainty.

Reviews 10 / Uncertainty 1: ETFE 2.4 mm ID gas-liquid contact unverified; lower BPR compounds uncertainty.

Reviews 10 / Uncertainty 2: Unprotected reverse path unresolved; lower BPR narrows startup margin.

Reviews 10 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 10 / Uncertainty 4: Oxygen-service review required.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 5 mL ETFE (2.4 mm), τ_liq=250 min; Stage 2: 10 mL PFA (1 mm), τ_inlet=90.9 min, BPR 7 bar. Near-batch Stage 1 in wide-bore ETFE with narrower-bore PFA Stage 2. Contrasts Candidate 9 (all-ETFE) and Candidate 6 (larger Stage 2). ETFE Stage 1 photon delivery unverified.

Reviews 11 / Uncertainty 1: ETFE 2.4 mm Stage 1 photon delivery unverified.

Reviews 11 / Uncertainty 2: Unprotected reverse path unresolved.

Reviews 11 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 11 / Uncertainty 4: Oxygen-service review required.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 2 mL PFA, τ_liq=100 min; Stage 2: 10 mL PFA (1 mm), τ_inlet=90.9 min, BPR 2.5 bar. Lower BPR contrast to Candidate 7 (same geometry, BPR 7 bar). Useful BPR sensitivity data point; lower pressure increases gas fraction (0.028 mL/min). Gas-liquid contact and backflow margin both less favorable than Candidate 7.

Reviews 12 / Uncertainty 1: Unprotected reverse path unresolved; lower BPR reduces startup margin.

Reviews 12 / Uncertainty 2: Stage 2 τ_inlet=90.9 min adequacy unknown.

Reviews 12 / Uncertainty 3: Stage 2 temperature unresolved.

Reviews 12 / Uncertainty 4: Oxygen-service review required.

Screening priority note: Candidate 1 is preferred as the interpretable balanced baseline (compact reactor pair, intermediate stage ratios, highest BPR, 1 mm PFA in both stages). Candidates 6 and 3 are preferred alternatives for near-batch Stage 1 exposure; Candidate 6 is preferred over Candidate 3 due to 1 mm PFA optical path. Candidates 7 and 8 are preferred for the short-Stage-2 group at 7 bar. All candidates require resolution of the unprotected Stage 1 reverse path and oxygen-service review before laboratory execution. Pure O2 use is an unapproved proposal pending chemistry, inventory, and oxygen-service confirmation.

Call 2: DrKinetics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Gas source: Pure O2 confirmed by chemist (Q-GAS-001). The earlier experiment used O2 despite air being specified; the gas-source change is noted as a protocol deviation. Pure O2 at Stage 2 is the approved design basis; oxygen-service review and inventory verification remain required before laboratory execution.

Shared assessment / Backflow finding: All twelve candidates carry two unresolved backflow findings: (1) BF-REVERSE-PATH — the liquid branch from Stage-2 mixer back through Stage-1 reactor has no declared forward-oriented non-return protection; O2 can propagate upstream, violating the hard constraint that Stage 1 must remain oxygen-free. (2) BF-VERIFY-PROTECTION — the CV-3301 on the gas branch has unconfirmed

pressure rating, direction, leakage, and low-flow service suitability. A check valve on the MFC branch does not protect the liquid (Stage-1) branch. Reducing gas flow does not close this reverse path. These findings are unresolved and require engineering review before any execution.

Shared assessment / Gas flow basis: Q_gas_sccm = 0.09 mL/min at inlet/STP (273.15 K, 1.01325 bar) for all candidates. Stage-2 apparent residence-time index = V_R / (Q_liquid + Q_gas_STP) = not a measured liquid exposure or photon dose. Gas equivalents (2.0078 equiv) are proposed delivered molar ratios at inlet; actual uptake is unmodeled.

Shared assessment / Wavelength: Stage 1 uses UV-150 / 450 nm (target 452 nm; 2 nm offset, minor). Stage 2 uses Manual 2 / Blue Strip LED at 448 nm. Neither wavelength match nor light-dose adequacy is proven for flow conditions. UV-150 is restricted to E-series or R-series platforms; this assignment is respected in all candidates.

Shared assessment / Kinetics: No flow kinetic data exist. Batch times (Stage 1: 240 min; Stage 2: 360 min) are screening anchors only. All hold ratios are unvalidated exposure indices, not yield predictors. Stage-2 batch temperature is not stage-resolved.

Shared assessment / Liquid flow: All candidates: Q_liquid = 0.02 mL/min (E-series minimum), substrate 0.1 M, 0.002 mmol/min molar throughput. Feed throughput is identical across all candidates; product throughput is unmeasured.

Shared assessment / Acrylonitrile pump: Acrylonitrile is excluded from pump_e_series_red and pump_sf10_red. All candidates route acrylonitrile through pump_e_series_blue (BLUE tubing), which has no such exclusion. This is correct.

Shared assessment / Cv3301 rating: CV-3301 maximum pressure rating is not supplied. Cracking pressure (1.0 bar) is not a pressure rating. Verification required before execution.

Shared assessment / Pure o2 service: Pure oxygen requires dedicated oxygen-service verification of all wetted components, fittings, and lubricants. This is an unapproved proposal pending chemistry, inventory, and oxygen-service review.

Shared assessment / Etfe uv150 compatibility: Candidates 3, 9, 11 assign ETFE coil_row11 to Stage 1 under UV-150. ETFE coils (coil_row11, coil_row12) are listed with photoreactor_module_ids including uv150; assignment is within declared inventory.

Shared assessment / Fep temperature: FEP coils (coil_row6, coil_row7) have max_temperature_C = 50 °C; all candidates using them operate at 25 °C, within limit.

Shared assessment / Stage2 temperature: Stage-2 batch temperature is not stage-resolved. Screening at 25 °C is a design choice, not a validated rate-compensated equivalence.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA / UV-150 450 nm / tau_liquid=100 min (ratio 0.42 vs 240 min batch). Stage 2: 20 mL FEP / Manual-2 448 nm / tau_inlet=181.8 min (ratio 0.51 vs 360 min batch). BPR 7 bar; O2 0.09 sccm / 2.0078 equiv. Compact baseline with intermediate Stage-1 exposure and moderate Stage-2 apparent index. No hard violation; unresolved backflow and CV-3301 verification are shared limitations. Suitable as interpretable screening baseline per objective.

Reviews 1 / Uncertainty 1: Unresolved reverse path: Stage-1 reactor unprotected against upstream O2 intrusion; engineering review required before execution.

Reviews 1 / Uncertainty 2: CV-3301 pressure rating and low-flow gas-branch service unconfirmed.

Reviews 1 / Uncertainty 3: Stage-1 liquid tau 100 min is 0.42× batch; adequacy for Giese step unvalidated.

Reviews 1 / Uncertainty 4: Stage-2 tau_inlet 181.8 min is apparent index only; actual liquid exposure and O2 uptake unmodeled.

Reviews 1 / Uncertainty 5: Pure O2 service review pending for all wetted components.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical reactor geometry to C1 (2 mL / 20 mL) but BPR 2.5 bar; gas in-channel volume larger (0.028 mL/min vs 0.012 mL/min) due to lower pressure. Tau_inlet identical (181.8 min) because it is STP-based. Lower BPR reduces dissolved-O2 partial pressure; effect on oxidation unvalidated. Useful pressure-variable companion to C1.

Reviews 2 / Uncertainty 1: Unresolved reverse path: identical topology to C1; Stage-1 unprotected.

Reviews 2 / Uncertainty 2: Lower BPR reduces O2 partial pressure in Stage 2; oxidation sufficiency unvalidated.

Reviews 2 / Uncertainty 3: CV-3301 service on gas branch unconfirmed.

Reviews 2 / Uncertainty 4: Stage-2 temperature not stage-resolved from batch protocol.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 5 mL ETFE coil_row11 / UV-150 450 nm / tau_liquid=250 min (ratio 1.04, near-batch). Stage 2: 20 mL FEP / Manual-2 / tau_inlet=181.8 min (ratio 0.51). Provides near-batch Stage-1 exposure; informative for yield-floor assessment. ETFE with ID 2.4 mm in UV-150 module: photon penetration depth unvalidated for wider bore. BPR 7 bar.

Reviews 3 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected against O2; shared finding.

Reviews 3 / Uncertainty 2: ETFE 2.4 mm ID under UV-150: photon penetration adequacy unvalidated for this geometry.

Reviews 3 / Uncertainty 3: Near-batch Stage-1 time does not guarantee equivalent conversion; flow kinetics uncharacterized.

Reviews 3 / Uncertainty 4: CV-3301 gas-branch service unconfirmed.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 2 mL PFA / UV-150 / tau_liquid=100 min. Stage 2: 20 mL FEP / Manual-2 / tau_inlet=181.8 min. BPR 1.5 bar (lowest in set); lowest O2 partial pressure in Stage 2. Companion to C1 and C2 for pressure-effect comparison. Shorter Stage-1 exposure risk is shared with C1/C2.

Reviews 4 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 4 / Uncertainty 2: 1.5 bar BPR: lowest O2 partial pressure; adequacy for sulfoxide oxidation unvalidated.

Reviews 4 / Uncertainty 3: Stage-1 tau 100 min may be insufficient; unvalidated.

Reviews 4 / Uncertainty 4: CV-3301 gas branch unconfirmed.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA / UV-150 / tau_liquid=100 min (ratio 0.42). Stage 2: 10 mL ETFE coil_row12 / Manual-2 / tau_inlet=90.9 min (ratio 0.25 vs 360 min batch). Short Stage-2 apparent index; O2 delivery and oxidation time adequacy at this exposure are unvalidated. BPR 7 bar. Compact design; screening value is in establishing whether short Stage-2 time is informative.

Reviews 5 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 5 / Uncertainty 2: Stage-2 tau_inlet 90.9 min is 0.25× batch; oxidation sufficiency unvalidated.

Reviews 5 / Uncertainty 3: ETFE 2.4 mm ID: photon penetration for Stage-2 light unvalidated.

Reviews 5 / Uncertainty 4: Short Stage-1 and Stage-2 holds reduce information on each step independently.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA coil_row4 / UV-150 / tau_liquid=250 min (ratio 1.04). Stage 2: 20 mL FEP / Manual-2 / tau_inlet=181.8 min (ratio 0.51). BPR 7 bar. Near-batch Stage-1 with moderate Stage-2 index; informative for Giese-step adequacy. Shares topology with C3 but uses 1 mm PFA for Stage 1 (better photon penetration hypothesis, unvalidated).

Reviews 6 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 6 / Uncertainty 2: PFA 1 mm vs ETFE 2.4 mm photon penetration comparison is a hypothesis; unvalidated.

Reviews 6 / Uncertainty 3: Near-batch Stage-1 time does not guarantee equivalent conversion.

Reviews 6 / Uncertainty 4: CV-3301 gas-branch service unconfirmed.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA / UV-150 / tau_liquid=100 min (ratio 0.42). Stage 2: 10 mL PFA coil_row5 / Manual-2 / tau_inlet=90.9 min (ratio 0.25). BPR 7 bar. Short Stage-2 index with 1 mm PFA in Stage 2; informative companion to C5 (ETFE vs PFA geometry at same volume). Both Stage-1 and Stage-2 holds are well below batch.

Reviews 7 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 7 / Uncertainty 2: Stage-2 tau_inlet 90.9 min may be insufficient for oxidation; unvalidated.

Reviews 7 / Uncertainty 3: Stage-1 tau 100 min may be insufficient for Giese step; unvalidated.

Reviews 7 / Uncertainty 4: CV-3301 gas-branch service unconfirmed.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 2 mL PFA / UV-150 / tau_liquid=100 min. Stage 2: 10 mL FEP coil_row6 / Manual-2 / tau_inlet=90.9 min. BPR 7 bar. FEP in Stage 2 (max 50 °C, within limit at 25 °C). Identical apparent times to C7 but FEP vs PFA material; marginal screening variable. Shared short-hold concerns.

Reviews 8 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 8 / Uncertainty 2: Stage-2 tau_inlet 90.9 min adequacy unvalidated.

Reviews 8 / Uncertainty 3: FEP vs PFA material difference has no proven chemical consequence at 25 °C.

Reviews 8 / Uncertainty 4: CV-3301 service unconfirmed.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 5 mL ETFE coil_row11 / UV-150 / tau_liquid=250 min (ratio 1.04). Stage 2: 10 mL ETFE coil_row12 / Manual-2 / tau_inlet=90.9 min (ratio 0.25). BPR 7 bar. Near-batch Stage-1 with short Stage-2; informative for decoupling step requirements. Both ETFE wide-bore: photon penetration unvalidated in both stages.

Reviews 9 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 9 / Uncertainty 2: ETFE 2.4 mm ID photon penetration unvalidated in both stages.

Reviews 9 / Uncertainty 3: Stage-2 tau_inlet 90.9 min adequacy for oxidation unvalidated.

Reviews 9 / Uncertainty 4: CV-3301 service unconfirmed.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA / UV-150 / tau_liquid=100 min. Stage 2: 10 mL ETFE coil_row12 / Manual-2 / tau_inlet=90.9 min. BPR 2.5 bar. Lower pressure variant of C5; lower O2 partial pressure. Short both stages, low BPR: most uncertain candidate for oxidation adequacy. Screening value limited unless shorter holds are specifically targeted.

Reviews 10 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 10 / Uncertainty 2: Lowest apparent O2 partial pressure among short-Stage-2 candidates (2.5 bar); oxidation adequacy most uncertain here.

Reviews 10 / Uncertainty 3: Stage-2 tau_inlet 90.9 min at lower O2 partial pressure: double risk for oxidation.

Reviews 10 / Uncertainty 4: CV-3301 service unconfirmed.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 5 mL ETFE coil_row11 / UV-150 / tau_liquid=250 min (ratio 1.04). Stage 2: 10 mL PFA coil_row5 / Manual-2 / tau_inlet=90.9 min (ratio 0.25). BPR 7 bar. Near-batch Stage-1 with short Stage-2; companion to C9 but PFA 1 mm in Stage 2. ETFE wide-bore Stage-1 photon penetration unvalidated.

Reviews 11 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 11 / Uncertainty 2: ETFE 2.4 mm ID in Stage 1: photon penetration unvalidated.

Reviews 11 / Uncertainty 3: Stage-2 tau_inlet 90.9 min adequacy unvalidated.

Reviews 11 / Uncertainty 4: CV-3301 service unconfirmed.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 2 mL PFA / UV-150 / tau_liquid=100 min. Stage 2: 10 mL PFA coil_row5 / Manual-2 / tau_inlet=90.9 min. BPR 2.5 bar. Lower-pressure variant of C7; both stages short, lower O2 partial pressure. Most compact design but screening informativeness for oxidation is lowest among 2.5-bar candidates. Companion to C7 and C10 for pressure screening.

Reviews 12 / Uncertainty 1: Unresolved reverse path: Stage-1 unprotected; shared finding.

Reviews 12 / Uncertainty 2: 2.5 bar with 90.9 min Stage-2 apparent index: O2 partial pressure and contact time both reduced; oxidation adequacy most uncertain.

Reviews 12 / Uncertainty 3: Stage-1 tau 100 min adequacy unvalidated.

Reviews 12 / Uncertainty 4: CV-3301 service unconfirmed.

Priority recommendation / Preferred starting candidate: 1

Priority recommendation / Rationale: Candidate 1 (2 mL PFA Stage 1, 20 mL FEP Stage 2, BPR 7 bar) provides the most interpretable baseline: intermediate Stage-1 liquid exposure (tau=100 min, ratio 0.42), moderate Stage-2 apparent index (tau_inlet=181.8 min, ratio 0.51), highest O2 partial pressure in the dominant reactor volume, and compact geometry. It is the most informative single starting point under the balanced objective before kinetics are established. Candidates 3 and 6 (near-batch Stage-1, same Stage-2) are recommended as immediate second experiments to bracket Stage-1 adequacy. Candidates 5, 7, 8 (short Stage-2) should be run after Stage-2 adequacy is assessed. All candidates require resolution of the unprotected Stage-1 reverse-path finding and CV-3301 verification before any execution.

Priority recommendation / Prerequisite for any execution: Engineering review and resolution of BF-REVERSE-PATH (Stage-1 branch unprotected against O2 intrusion) and BF-VERIFY-PROTECTION (CV-3301 gas-branch confirmation), plus oxygen-service review of all wetted components.

Call 3: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Tool name: simulate_flow_transients

Candidate ids 1: 1

Candidate ids 2: 2

Candidate ids 3: 3

Candidate ids 4: 4

Candidate ids 5: 5

Candidate ids 6: 6

Candidate ids 7: 7

Candidate ids 8: 8

Candidate ids 9: 9

Candidate ids 10: 10

Candidate ids 11: 11

Candidate ids 12: 12

Call 4: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: prefer

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ (0.42× batch), UV-150 450 nm, 25°C. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ (0.51× batch), 0.09 sccm O2 (2.01 equiv), 7 bar BPR. Compact balanced baseline matching objective. Unprotected liquid reverse path and unverified check valve require engineering resolution before execution.

Reviews 1 / Uncertainty 1: Unprotected reverse path from Stage 2 junction to Stage 1 reactor; no non-return device declared on liquid branch.

Reviews 1 / Uncertainty 2: CV-3301 cracking pressure, max pressure rating, and service compatibility unverified.

Reviews 1 / Uncertainty 3: Adequacy of 100 min photon exposure for Giese reaction unestablished; kinetics uncharacterized.

Reviews 1 / Uncertainty 4: Gas-source change from batch air to pure O2 requires chemist and safety confirmation.

Reviews 1 / Uncertainty 5: Stage 2 temperature not stage-resolved in batch protocol; 25°C assumption unvalidated.

Reviews 1 / Uncertainty 6: Inlet/STP apparent τ is a hydraulic index; actual liquid exposure and O2 uptake are not modeled.

Reviews 1 / Uncertainty 7: Offline Ar degassing is the only deoxygenation route; effectiveness in flow not confirmed.

Reviews 1 / Uncertainty 8: FEP coil max 50°C; no thermal deviation here but no active temperature control confirmed.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Identical reactor volumes and flow rates to C1; only BPR differs (2.5 bar vs 7 bar). Lower system pressure increases in-channel gas volume (0.028 mL/min vs 0.012 mL/min) and reduces O2 dissolved-phase driving force. Inlet/STP τ identical to C1 (181.8 min). Unprotected liquid reverse path unresolved. Lower BPR reduces headroom against gas back-pressure transients.

Reviews 2 / Uncertainty 1: Unprotected reverse path from Stage 2 junction to Stage 1 reactor unresolved.

Reviews 2 / Uncertainty 2: Lower BPR (2.5 bar) reduces suppression of gas back-pressure; operability at low liquid flow unverified.

Reviews 2 / Uncertainty 3: CV-3301 rating and service compatibility unverified.

Reviews 2 / Uncertainty 4: Whether lower dissolved-O2 driving force at 2.5 bar affects oxidation selectivity is uncharacterized.

Reviews 2 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 2 / Uncertainty 6: Stage 2 temperature not stage-resolved in source protocol.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 5 mL ETFE coil, 250 min liquid τ (1.04× batch anchor), UV-150 450 nm. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ, 7 bar BPR. Longer Stage 1 exceeds batch hold as a reference;

photon coupling of 2.4 mm ID ETFE in UV-150 unverified. Tool reports BPR not reached in gas_first_start scenario; startup sequence requires laboratory protocol. Reverse path unresolved.

Reviews 3 / Uncertainty 1: UV-150 photon penetration into 2.4 mm ID ETFE coil uncharacterized; light coupling not established.

Reviews 3 / Uncertainty 2: Gas-first startup: BPR not reached in simulation window; startup sequence unconfirmed.

Reviews 3 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 3 / Uncertainty 4: CV-3301 rating and service compatibility unverified.

Reviews 3 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 3 / Uncertainty 6: 250 min Stage 1 exceeds batch anchor; longer irradiation degradation risk is hypothesis only.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ, UV-150 450 nm. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ, 1.5 bar BPR. Lower BPR further expands in-channel gas volume (0.040 mL/min) and reduces dissolved-O2 driving force relative to C1/C2. Tool shows no predicted reversal; unprotected liquid reverse path structurally identical to all candidates. Operability at 1.5 bar with gas feed at minimum liquid flow warrants extra caution.

Reviews 4 / Uncertainty 1: 1.5 bar BPR provides least back-pressure suppression; gas-liquid stability at 0.02 mL/min liquid flow unverified.

Reviews 4 / Uncertainty 2: Unprotected liquid reverse path unresolved.

Reviews 4 / Uncertainty 3: CV-3301 rating and service compatibility unverified.

Reviews 4 / Uncertainty 4: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 4 / Uncertainty 5: Dissolved-O2 availability at low pressure uncharacterized for this reaction.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ. Stage 2: 10 mL ETFE (2.4 mm ID), 90.9 min inlet/STP τ (0.25× batch), 7 bar BPR. Most compact Stage 2; shortest combined inlet/STP τ (190.9 min). Whether 90.9 min inlet/STP τ provides sufficient oxidation exposure is uncharacterized. Large-bore ETFE in Stage 2 gas-liquid mixing behavior unverified.

Reviews 5 / Uncertainty 1: 90.9 min Stage 2 inlet/STP τ is 0.25× batch hold; adequacy for oxidation unestablished.

Reviews 5 / Uncertainty 2: Unprotected liquid reverse path unresolved.

Reviews 5 / Uncertainty 3: Gas-liquid mixing and slug formation in 2.4 mm ID coil uncharacterized.

Reviews 5 / Uncertainty 4: CV-3301 rating and service compatibility unverified.

Reviews 5 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 5 mL PFA (1 mm ID), 250 min liquid τ, UV-150 450 nm. Stage 2: 20 mL FEP, 181.8 min inlet/STP τ, 7 bar BPR. Gas-first startup: BPR not reached in tool window; startup sequence requires laboratory protocol. Stage 1 τ exceeds batch anchor; degradation risk from extended irradiation is an untested hypothesis. Reverse path unresolved.

Reviews 6 / Uncertainty 1: Gas-first startup: BPR not reached in simulation window; startup procedure unconfirmed.

Reviews 6 / Uncertainty 2: Unprotected liquid reverse path unresolved.

Reviews 6 / Uncertainty 3: CV-3301 rating and service compatibility unverified.

Reviews 6 / Uncertainty 4: Stage 1 at 250 min exceeds batch hold; over-irradiation degradation is a hypothesis.

Reviews 6 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ. Stage 2: 10 mL PFA (1 mm ID), 90.9 min inlet/STP τ, 7 bar BPR. Combined inlet/STP τ 190.9 min. Narrower bore Stage 2 vs C5 may improve gas-liquid contact; uptake unmodeled. Adequacy of 90.9 min Stage 2 uncharacterized. Reverse path unresolved.

Reviews 7 / Uncertainty 1: 90.9 min Stage 2 τ adequacy for oxidation unestablished.

Reviews 7 / Uncertainty 2: Unprotected liquid reverse path unresolved.

Reviews 7 / Uncertainty 3: CV-3301 rating and service compatibility unverified.

Reviews 7 / Uncertainty 4: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 7 / Uncertainty 5: Gas uptake and liquid-phase O2 distribution in 1 mm PFA coil at these flow rates uncharacterized.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ. Stage 2: 10 mL FEP (1 mm ID), 90.9 min inlet/STP τ, 7 bar BPR. Functionally identical to C7 except FEP material in Stage 2; FEP max 50°C respected at 25°C. Combined inlet/STP τ identical to C7 (190.9 min). Reverse path unresolved.

Reviews 8 / Uncertainty 1: 90.9 min Stage 2 τ adequacy unestablished.

Reviews 8 / Uncertainty 2: Unprotected liquid reverse path unresolved.

Reviews 8 / Uncertainty 3: FEP max 50°C limits future temperature excursions in Stage 2.

Reviews 8 / Uncertainty 4: CV-3301 rating and service compatibility unverified.

Reviews 8 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 5 mL ETFE (2.4 mm ID), 250 min liquid τ. Stage 2: 10 mL ETFE (2.4 mm ID), 90.9 min inlet/STP τ, 7 bar BPR. Both stages large-bore; UV-150 coupling to 2.4 mm ID ETFE in Stage 1

unverified. Tool pump_interruption scenario: negligible predicted displacement (~0.5 pL), but this is model-conditional on unmeasured parameters. Startup sequence unconfirmed. Reverse path unresolved.

Reviews 9 / Uncertainty 1: UV-150 photon penetration into 2.4 mm ID ETFE Stage 1 coil uncharacterized.

Reviews 9 / Uncertainty 2: Unprotected liquid reverse path unresolved.

Reviews 9 / Uncertainty 3: Both stage residence times differ from batch anchors; yield implications uncharacterized.

Reviews 9 / Uncertainty 4: CV-3301 rating and service compatibility unverified.

Reviews 9 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 9 / Uncertainty 6: Large-bore gas-liquid mixing and slug behavior in both stages unverified.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ. Stage 2: 10 mL ETFE (2.4 mm ID), 90.9 min inlet/STP τ, 2.5 bar BPR. Lower BPR expands gas volume (0.028 mL/min in-channel). Large-bore coil with lower pressure: gas-liquid behavior least characterized. Combined inlet/STP τ 190.9 min. Reverse path unresolved.

Reviews 10 / Uncertainty 1: 2.5 bar BPR with 2.4 mm ID Stage 2: gas-liquid stability and mixing unverified.

Reviews 10 / Uncertainty 2: 90.9 min Stage 2 τ adequacy unestablished.

Reviews 10 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 10 / Uncertainty 4: CV-3301 rating and service compatibility unverified.

Reviews 10 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 5 mL ETFE (2.4 mm ID), 250 min liquid τ. Stage 2: 10 mL PFA (1 mm ID), 90.9 min inlet/STP τ, 7 bar BPR. Gas-first startup in tool: BPR not reached in window; startup sequence unconfirmed. UV-150 coupling to large-bore Stage 1 unverified. Combined inlet/STP τ 340.9 min. Reverse path unresolved.

Reviews 11 / Uncertainty 1: Gas-first startup: BPR not reached in simulation window; startup procedure unconfirmed.

Reviews 11 / Uncertainty 2: UV-150 photon penetration into 2.4 mm ID ETFE Stage 1 uncharacterized.

Reviews 11 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 11 / Uncertainty 4: CV-3301 rating and service compatibility unverified.

Reviews 11 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 2 mL PFA, 100 min liquid τ. Stage 2: 10 mL PFA (1 mm ID), 90.9 min inlet/STP τ, 2.5 bar BPR. Lower BPR expands in-channel gas volume (0.028 mL/min). Combined inlet/STP τ 190.9 min. Functionally similar to C7 but lower pressure. Reverse path unresolved.

Reviews 12 / Uncertainty 1: 2.5 bar BPR reduces dissolved-O2 driving force; effect on oxidation selectivity uncharacterized.

Reviews 12 / Uncertainty 2: 90.9 min Stage 2 τ adequacy unestablished.

Reviews 12 / Uncertainty 3: Unprotected liquid reverse path unresolved.

Reviews 12 / Uncertainty 4: CV-3301 rating and service compatibility unverified.

Reviews 12 / Uncertainty 5: Gas-source change to pure O2 requires chemist and safety confirmation.

Shared assessment / Issues common to all candidates 1: UNPROTECTED REVERSE PATH (BF-REVERSE-PATH:st2_mixer:s3:st1_reactor): All 12 candidates share an identical topology. The liquid branch from the Stage 2 mixer back to the Stage 1 reactor has no declared forward-oriented non-return device. This is a credible operability risk under gas introduction, startup, pump interruption, or pressure transient scenarios. Stage 1 must remain oxygen-free; liquid-branch protection (e.g., a verified check valve on the liquid branch between Stage 1 outlet and Stage 2 mixer) is an unapproved proposal requiring engineering, chemistry, and inventory review before use.

Shared assessment / Issues common to all candidates 2: CV-3301 CHECK VALVE (BF-VERIFY-PROTECTION:st2_mixer:s2:pump_c): Declared on the MFC/gas branch only. Its cracking pressure (1.0 bar) concerns the pressure difference across that valve, not BPR setpoint minus cracking pressure. Maximum pressure rating not supplied; service compatibility with pure O2 unverified. A valve on the MFC branch does not protect the liquid branch leading to Stage 1.

Shared assessment / Issues common to all candidates 3: GAS SOURCE CHANGE: All candidates use pure O2 (100 mol%) at Stage 2. The historical experiment also used pure oxygen instead of the originally specified air. This change from the batch protocol (ambient air) is a chemist-confirmed requirement but has NOT received independent oxygen-service, materials compatibility, inventory, or safety review. Pure oxygen use requires laboratory review before execution; no candidate is approved for pure O2 operation without that review.

Shared assessment / Issues common to all candidates 4: PHYSICS TOOL LIMITATIONS: Tool results (simulate_flow_transients, result_id 2c7417b2d6e18efec57d2ac24fb50b4b956bb08bcbc7e87758333f011f32bf59) are conditional on illustrative, unmeasured parameters (viscosity, compliance, plenum volume, MFC response, pump droop, etc.). No predicted liquid displacement in these tests does not establish safe operation or confirm absence of reverse flow. These are screening results from an unvalidated model, not measured evidence.

Shared assessment / Issues common to all candidates 5: STARTUP SCENARIOS (C3, C6, C11): Tool reports BPR not reached within the simulated gas-first startup window. This is a model-conditional result; it does not confirm that these candidates fail, but the startup sequence requires explicit laboratory protocol definition before execution.

Shared assessment / Issues common to all candidates 6: STAGE 2 TEMPERATURE: Not stage-resolved in source batch protocol. All candidates assign 25°C to Stage 2; this is unvalidated. No yield or rate correction is established.

Shared assessment / Issues common to all candidates 7: KINETICS: Uncharacterized for both stages. Residence time ratios relative to batch holds (Stage 1: 0.42× or 1.04×; Stage 2: 0.25× or 0.51×) are hydraulic screening indices, not equivalent chemical exposures. No conversion, yield, or selectivity predictions are made.

Shared assessment / Issues common to all candidates 8: OFFLINE DEGASSING: No inline degasser available. Feed must be degassed offline with Ar; effectiveness in continuous flow is unconfirmed.

Shared assessment / Issues common to all candidates 9: LIGHT SOURCE WAVELENGTH: UV-150 450 nm (actual 450 nm vs target 452 nm) assigned to Stage 1 for all candidates; Manual 2 Blue Strip LED (448 nm vs target 452 nm) assigned to Stage 2. Both are 2 nm deviations from protocol. Light-dose adequacy and photon penetration are not proven for any candidate.

Shared assessment / Issues common to all candidates 10: ACRYLONITRILE PUMP EXCLUSION: pump_e_series_red excluded for acrylonitrile; all candidates correctly assign pump_e_series_blue. Confirmed.

Shared assessment / Issues common to all candidates 11: CV-3301 MAX PRESSURE RATING: Not supplied in inventory. Must be verified before laboratory execution. Cracking pressure (1.0 bar) is not a pressure rating.

Shared assessment / Throughput note: All candidates operate at 0.02 mL/min liquid flow (minimum pump setting, 0.1 M substrate), giving 0.002 mmol/min limiting feed throughput. This is feed throughput only; product throughput is unmeasured. No candidate claims higher product throughput than another.

Shared assessment / Preferred baseline: Candidate 1 is identified as the preferred starting point under the balanced objective: smallest footprint with intermediate Stage 2 hold (181.8 min inlet/STP τ), 7 bar BPR providing best gas-liquid suppression, and inventory-efficient reactor pairing. Resolution of the liquid reverse path and gas-source review are prerequisites for execution of any candidate.

Physics assessment / Tool result id:
2c7417b2d6e18efec57d2ac24fb50b4b956bb08bcbc7e87758333f011f32bf59

Physics assessment / Evidence ids 1: C1/as_designed/entered_profile/steady

Physics assessment / Evidence ids 2: C1/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 3: C2/as_designed/entered_profile/steady

Physics assessment / Evidence ids 4: C2/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 5: C3/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 6: C3/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 7: C4/as_designed/entered_profile/steady

Physics assessment / Evidence ids 8: C4/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 9: C5/as_designed/entered_profile/steady

Physics assessment / Evidence ids 10: C5/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 11: C6/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 12: C6/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 13: C7/as_designed/entered_profile/steady

Physics assessment / Evidence ids 14: C7/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 15: C8/as_designed/entered_profile/steady

Physics assessment / Evidence ids 16: C8/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 17: C9/as_designed/entered_profile/pump_interruption

Physics assessment / Evidence ids 18: C9/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 19: C10/as_designed/entered_profile/steady

Physics assessment / Evidence ids 20: C10/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 21: C11/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 22: C11/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 23: C12/as_designed/entered_profile/steady

Physics assessment / Evidence ids 24: C12/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Proposed alternatives 1: Install a verified forward-oriented non-return valve (e.g., a checked CV-3301 or equivalent) on the liquid branch between Stage 1 outlet and the Stage 2 T-mixer inlet. This

is an unapproved proposal: requires engineering design review, chemistry compatibility assessment (EtOH/buffer/photocatalyst), O2-service suitability confirmation, and inventory verification before use.

Physics assessment / Proposed alternatives 2: Define an explicit startup sequence (liquid-first pressurization to BPR setpoint before gas introduction) to address the gas-first startup gap in C3, C6, and C11; this remains an unverified laboratory procedure.

Physics assessment / Proposed alternatives 3: Verify CV-3301 maximum pressure rating and O2-service compatibility in laboratory before any execution.

Physics assessment / Limitation 1: Tool uses illustrative, uncalibrated assumption profile: viscosity 1.0 mPa·s, liquid compliance 0.01 mL/bar, gas plenum 0.2 mL, pump response 1.0 s, MFC response 0.5 s. None of these are measured KHU laboratory values.

Physics assessment / Limitation 2: No predicted liquid displacement in the simulated scenarios does not establish that reverse flow cannot occur; the unprotected liquid reverse path remains a structural risk independent of tool output.

Physics assessment / Limitation 3: Negative liquid-branch flow in the model represents predicted liquid displacement, not proof that oxygen reaches Reactor 1; the model does not simulate reactive transport.

Physics assessment / Limitation 4: The CV-3301 cracking pressure (1.0 bar) concerns the pressure difference across that specific valve; it is not compared to BPR setpoint minus cracking pressure.

Physics assessment / Limitation 5: Gas-first startup scenarios for C3, C6, C11 show BPR not reached within the simulation window only; this is a model artifact of the finite scenario window, not confirmation of a hard failure.

Physics assessment / Limitation 6: Model is isothermal ideal gas with lumped parameters; slug flow, dissolution, photochemical kinetics, and gas-front propagation are not represented.

Physics assessment / Limitation 7: Hypothetical liquid-branch check valve results use an unassigned, service-unverified hypothetical device with assumed zero leakage; these results do not approve or confirm any hardware change.

Physics assessment / Limitation 8: Backflow probability is null; no probabilistic or reliability assessment is available.

Physics assessment / Limitation 9: Sensitivity bracket (entered gas plenum and one-tenth value) is illustrative only, not a calibrated uncertainty range.

Physics assessment / Limitation 10: All simulation results are conditional on unmeasured parameters and do not constitute safety certification or proof of safe operation.

Call 5: DrSafety, attempt 1
Status: recorded response, parsed successfully.

Tool name: simulate_flow_transients

Candidate ids 1: 1

Candidate ids 2: 2

Candidate ids 3: 3

Candidate ids 4: 4

Candidate ids 5: 5

Candidate ids 6: 6

Candidate ids 7: 7

Candidate ids 8: 8

Candidate ids 9: 9

Candidate ids 10: 10

Candidate ids 11: 11

Candidate ids 12: 12

Call 6: DrSafety, attempt 1
Status: unsuccessful response; retained for provenance. Unterminated string starting at: line 256 column 7 (char 19401)

```json
{
  "reviews": [
    {
      "candidate_id": 1,
      "recommendation": "acceptable",
      "hard_violation": false,
      "justification": "Stage 1: 2 mL PFA coil, τ_liquid=100 min (0.42× batch), UV-150 450 nm, 0.02 mL/min, 25°C, O2-free. Stage 2: 20 mL FEP coil, τ_inlet=181.8 min (0.51× batch apparent index), 0.09 sccm O2 (2.01 equiv inlet/STP), BPR 7 bar, Manual-2 448 nm. Unprotected reverse path (st2_mixer→st1_reactor) is a credible risk unresolved by topology. Tool predicts no liquid displacement (illustrative assumptions, finite window, unvalidated). Pure O2 use confirmed by chemist; gas_delivery.requires_chemist_confirmation not flagged as unapproved. CV-3301 pressure rating unconfirmed; cracking pressure (1 bar) is not a pressure rating. Peristaltic pump at 0.02 mL/min is minimum-flow operation; pulsation risk unconfirmed.",
      "uncertainties": [
        "Unprotected reverse path to Stage 1 unresolved; no branch-specific non-return valve on liquid branch.",
        "CV-3301 maximum pressure rating not verified for O2 service.",
        "Tool predictions conditional on illustrative, unmeasured parameters; not safety certification.",
        "Stage 1 τ_liquid=100 min is 0.42× batch anchor; photon adequacy and conversion uncharacterized.",
        "Stage 2 τ_inlet=181.8 min is apparent index only; actual liquid exposure and O2 uptake not modeled.",
        "Peristaltic pump at minimum flow (0.02 mL/min) pulsation impact unconfirmed.",
        "Acrylonitrile compatibility with RED tubing excluded; BLUE tubing used—verify solvent/reagent compatibility."
      ]
    },
    {
      "candidate_id": 2,
      "recommendation": "acceptable",
      "hard_violation": false,
      "justification": "Identical to C1 except BPR=2.5 bar (reduced system pressure). Stage 2 in-channel gas volume larger (0.028 mL/min vs 0.012 mL/min at 2.5 bar). Higher gas-phase fraction at lower pressure may improve gas-liquid contact (untested hypothesis). Lower BPR reduces MFC differential pressure margin (7.29
```

bar margin vs 2.80 bar in C1)—both within rating. Unprotected reverse path and CV-3301 uncertainties persist. Tool predicts no liquid displacement (same caveats).",

"uncertainties": [

"Unprotected reverse path to Stage 1 unresolved; no branch-specific non-return valve on liquid branch.",

"CV-3301 maximum pressure rating not verified for O2 service.",

"Higher gas fraction at lower pressure is an untested hypothesis; gas-liquid contact not characterized.",

"Tool predictions conditional on illustrative, unmeasured parameters.",

"Stage 1 and Stage 2 apparent residence times identical to C1; same adequacy uncertainty applies."

]

},

{

"candidate_id": 3,

"recommendation": "acceptable",

"hard_violation": false,

"justification": "Stage 1: 5 mL ETFE coil (ID 2.4 mm), τ_liquid=250 min (1.04× batch), UV-150 450 nm. Wider bore reduces linear velocity and Re further (Re=0.177); lamp-to-fluid optical path concerns for 2.4 mm ID coil in UV-150 are unconfirmed. Stage 2: 20 mL FEP, BPR 7 bar, identical to C1. Startup simulation only reached 0.027 bar_g (BPR not opened); full-pressure steady-state dynamics unvalidated. Unprotected reverse path persists. ETFE coil in UV-150: photoreactor_module_ids confirms compatibility.",

"uncertainties": [

"Unprotected reverse path to Stage 1 unresolved.",

"UV-150 with 5 mL ETFE coil (ID 2.4 mm): light penetration at wider bore uncharacterized.",

"Startup simulation did not reach operating pressure; steady-state tool result is hypothetical_liquid_check only.",

"CV-3301 pressure rating unconfirmed for O2.",

"Stage 1 τ_liquid=250 min exceeds batch anchor but photon delivery adequacy is unproven at 2.4 mm ID."

]

},

{

"candidate_id": 4,

"recommendation": "acceptable",

"hard_violation": false,

"justification": "Stage 1: 2 mL PFA, τ_liquid=100 min, identical to C1. Stage 2: 20 mL FEP, BPR=1.5 bar; in-channel gas volume largest of 7-bar group alternatives (0.040 mL/min). At 1.5 bar, O2 solubility is lowest; high gas fraction may affect flow stability. Tool predicts no displacement (8.28 bar MFC margin, wide headroom). Unprotected reverse path and CV-3301 issues persist. Low BPR increases gas fraction and possible slug/flow regime uncertainty.",

"uncertainties": [

"Unprotected reverse path to Stage 1 unresolved.",

"BPR 1.5 bar: highest gas fraction; slug/annular flow regime and stability uncharacterized.",

```
      "O2 solubility lowest at 1.5 bar; impact on oxidation step uncharacterized.",
      "CV-3301 pressure rating unconfirmed for O2 service.",
      "Tool predictions conditional on illustrative assumptions; finite window."
    ]
  },
  {
    "candidate_id": 5,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA, τ_liquid=100 min (0.42× batch), UV-150 450 nm. Stage 2: 10 mL ETFE coil (ID 2.4 mm), τ_inlet=90.9 min (0.25× batch apparent index), BPR 7 bar, Manual-2 448 nm. Shorter Stage 2 apparent index is the largest departure from batch anchor; adequacy uncharacterized. Wider bore (2.4 mm) in Stage 2 with gas: gas distribution and residence uncharacterized. Tool predicts no displacement. Unprotected reverse path persists.",
    "uncertainties": [
      "Unprotected reverse path to Stage 1 unresolved.",
      "Stage 2 τ_inlet=90.9 min is 0.25× batch anchor; oxidation adequacy uncharacterized.",
      "ETFE 2.4 mm ID in Stage 2 gas-liquid: flow regime, gas distribution, and light path uncharacterized.",
      "CV-3301 pressure rating unconfirmed.",
      "Tool predictions conditional on illustrative assumptions."
    ]
  },
  {
    "candidate_id": 6,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 5 mL PFA coil (ID 1.0 mm), τ_liquid=250 min (1.04× batch), UV-150 450 nm. Stage 2: 20 mL FEP, BPR 7 bar, Manual-2 448 nm, τ_inlet=181.8 min. Startup simulation did not reach operating pressure (BPR not opened); full-pressure dynamics unvalidated. Stage 1 longer exposure exceeds batch anchor; extended O2-free irradiation risk with acrylonitrile not characterized. Unprotected reverse path and CV-3301 issues persist.",
    "uncertainties": [
      "Unprotected reverse path to Stage 1 unresolved.",
      "Startup simulation only: operating pressure not reached in modeled window.",
      "Extended Stage 1 irradiation (250 min, 2.5× C1) with acrylonitrile under O2-free conditions: side-reaction risk uncharacterized.",
      "CV-3301 pressure rating unconfirmed.",
      "Tool predictions conditional on illustrative assumptions."
    ]
```

```
  },
  {
   "candidate_id": 7,
   "recommendation": "acceptable",
   "hard_violation": false,
   "justification": "Stage 1: 2 mL PFA, τ_liquid=100 min, UV-150 450 nm. Stage 2: 10 mL PFA coil (ID 1.0 mm), τ_inlet=90.9 min (0.25× batch apparent index), BPR 7 bar, Manual-2 448 nm. Shorter Stage 2 exposure is largest uncertainty. Tool predicts no displacement (steady-state, BPR reached). PFA compatible with O2 at 7 bar. Unprotected reverse path persists; CV-3301 unconfirmed.",
   "uncertainties": [
    "Unprotected reverse path to Stage 1 unresolved.",
    "Stage 2 τ_inlet=90.9 min is 0.25× batch anchor; oxidation adequacy uncharacterized.",
    "CV-3301 pressure rating unconfirmed for O2 service.",
    "Tool predictions conditional on illustrative assumptions; finite window."
   ]
  },
  {
   "candidate_id": 8,
   "recommendation": "acceptable",
   "hard_violation": false,
   "justification": "Stage 1: 2 mL PFA, τ_liquid=100 min, UV-150 450 nm. Stage 2: 10 mL FEP coil (ID 1.0 mm, max 50°C), τ_inlet=90.9 min, BPR 7 bar, Manual-2 448 nm, 25°C — within FEP temperature limit. Identical hydraulics to C7; same concerns apply. FEP max pressure 34 bar, max temperature 50°C; both satisfied at 25°C and 7 bar. Unprotected reverse path and CV-3301 issues unchanged.",
   "uncertainties": [
    "Unprotected reverse path to Stage 1 unresolved.",
    "Stage 2 τ_inlet=90.9 min is 0.25× batch anchor; oxidation adequacy uncharacterized.",
    "FEP max temperature 50°C: any exotherm at Stage 2 must remain below limit (unconfirmed).",
    "CV-3301 pressure rating unconfirmed.",
    "Tool predictions conditional on illustrative assumptions."
   ]
  },
  {
   "candidate_id": 9,
   "recommendation": "acceptable",
   "hard_violation": false,
   "justification": "Stage 1: 5 mL ETFE coil (ID 2.4 mm), τ_liquid=250 min, UV-150 450 nm. Stage 2: 10 mL ETFE coil (ID 2.4 mm), τ_inlet=90.9 min, BPR 7 bar. Both stages use wide-bore ETFE; light penetration and
```

gas-liquid flow regime at 2.4 mm uncharacterized in both stages. Pump interruption scenario modeled; predicted no displacement (illustrative assumptions). Unprotected reverse path persists.",
"uncertainties": [
"Unprotected reverse path to Stage 1 unresolved.",
"2.4 mm ID ETFE in both stages: optical path adequacy and gas-liquid regime uncharacterized.",
"Stage 2 τ_inlet=90.9 min is 0.25× batch anchor.",
"CV-3301 pressure rating unconfirmed for O2.",
"Pump interruption simulation result is illustrative only; unmeasured parameters."
]
},
{
"candidate_id": 10,
"recommendation": "acceptable",
"hard_violation": false,
"justification": "Stage 1: 2 mL PFA, τ_liquid=100 min, UV-150 450 nm. Stage 2: 10 mL ETFE coil (ID 2.4 mm), τ_inlet=90.9 min, BPR 2.5 bar, Manual-2 448 nm. Largest in-channel gas volume in Stage 2 among 10 mL coil candidates (0.028 mL/min at 2.5 bar vs 0.012 at 7 bar). Wide bore with high gas fraction at low pressure: slug/flow regime uncharacterized. Tool predicts no displacement (7.32 bar MFC margin). Unprotected reverse path persists.",
"uncertainties": [
"Unprotected reverse path to Stage 1 unresolved.",
"2.4 mm ID at 2.5 bar BPR: high gas fraction, flow regime uncharacterized.",
"Stage 2 τ_inlet=90.9 min is 0.25× batch anchor; adequacy uncharacterized.",
"CV-3301 pressure rating unconfirmed.",
"Tool predictions conditional on illustrative assumptions."
]
},
{
"candidate_id": 11,
"recommendation": "acceptable",
"hard_violation": false,
"justification": "Stage 1: 5 mL ETFE coil (ID 2.4 mm), τ_liquid=250 min, UV-150 450 nm. Stage 2: 10 mL PFA coil (ID 1.0 mm), τ_inlet=90.9 min, BPR 7 bar, Manual-2 448 nm. Startup simulation did not reach operating pressure; steady-state behavior unvalidated as_designed. Combination of wide-bore Stage 1 (optical uncertainty) and shorter Stage 2 (adequacy uncertainty) gives compound unresolved concerns. Unprotected reverse path persists.",
"uncertainties": [
"Unprotected reverse path to Stage 1 unresolved.",
"Startup simulation did not reach operating pressure; as_designed steady-state not validated.",

```
      "Stage 1 ETFE 2.4 mm ID: optical path adequacy uncharacterized.",
      "Stage 2 τ_inlet=90.9 min is 0.25× batch anchor.",
      "CV-3301 pressure rating unconfirmed."
    ]
   },
   {
    "candidate_id": 12,
    "recommendation": "acceptable",
    "hard_violation": false,
    "justification": "Stage 1: 2 mL PFA, τ_liquid=100 min, UV-150 450 nm. Stage 2: 10 mL PFA coil (ID 1.0 mm), τ_inlet=90.9 min, BPR 2.5 bar, Manual-2 448 nm. Lower BPR increases gas fraction (0.028 mL/min in-channel at 2.5 bar); gas-liquid regime and O2 solubility impact uncharacterized. Tool predicts no displacement (7.30 bar MFC margin). Unprotected reverse path and CV-3301 issues persist. Stage 2 adequacy uncertainty same as C7/C8.",
    "uncertainties": [
      "Unprotected reverse path to Stage 1 unresolved.",
      "BPR 2.5 bar: increased gas fraction vs 7 bar candidates; flow regime uncharacterized.",
      "Stage 2 τ_inlet=90.9 min is 0.25× batch anchor; oxidation adequacy uncharacterized.",
      "CV-3301 pressure rating unconfirmed for O2 service.",
      "Tool predictions conditional on illustrative assumptions."
    ]
   }
  ],
  "shared_assessment": {
   "applies_to_all_candidates": true,
   "issues": [
    {
     "id": "SA-1",
     "topic": "Unprotected reverse path — liquid branch to Stage 1",
     "detail": "All 12 candidates share topology finding BF-REVERSE-PATH: the liquid branch from the Stage 2 T-mixer to Stage 1 reactor has no declared forward-oriented non-return protection. This is distinct from the CV-3301 check valve on the MFC/gas branch, which protects only that branch. Qualitative reverse flow was observed experimentally (KHU, 2026-09-15). Tool predictions of no liquid displacement are conditional on illustrative, uncalibrated assumptions and a finite scenario window; they do not establish safety or resolve the connectivity risk. Stage 1 must remain oxygen-free. A branch-specific non-return valve on the liquid line upstream of the Stage 2 junction, or a pressure-decoupled interstage arrangement, is a required engineering review item before any candidate proceeds to laboratory execution. This is proposed as an unapproved mitigation pending chemistry, inventory, and O2-service review.",
     "status": "unresolved — blocks laboratory execution until engineering review complete"
    },
```

```
{
 "id": "SA-2",
 "topic": "CV-3301 gas check valve — pressure rating unconfirmed",
 "detail": "The CV-3301 cracking pressure is 1.0 bar. Its maximum pressure rating is not supplied in inventory. The cracking pressure is not a pressure rating. Pure O2 service suitability requires laboratory verification before use. This applies to all candidates.",
 "status": "unresolved — verify before laboratory execution"
},
{
 "id": "SA-3",
 "topic": "Pure O2 gas source — chemist confirmation and O2 service",
 "detail": "The earlier design specified air; the experiment used pure O2 (chemist-confirmed per Q-GAS-001). The MFC (Bronkhorst FF-C00) lists O2 as a supported gas. All tubing (PFA, FEP, ETFE) and mixer materials (PEEK, ETFE) require verification of O2 service compatibility at the operating pressure for each candidate. No inventory or final design change is made by this review.",
 "status": "unresolved — O2 service and material compatibility require laboratory confirmation"
},
{
 "id": "SA-4",
 "topic": "Minimum-flow peristaltic pump operation",
 "detail": "All candidates run pump_e_series_blue at 0.02 mL/min, which is the declared minimum flow. Pulsation, flow stability, and residence-time distribution at minimum flow are uncharacterized. This may affect Stage 1 photocatalytic conversion and Stage 2 gas-liquid contact.",
 "status": "unresolved — experimental verification required"
},
{
 "id": "SA-5",
 "topic": "Wavelength mismatch",
 "detail": "Protocol specifies 452 nm. UV-150 450 nm (C1,2,4–12 Stage 1) and Manual-2 448 nm (all Stage 2) are the closest available. No yield or photocatalyst excitation equivalence is established for these wavelengths. Light dose adequacy is not proven.",
 "status": "acknowledged uncertainty — not a hard violation"
},
{
 "id": "SA-6",
 "topic": "Acrylonitrile — RED pump tubing exclusion",
 "detail": "Acrylonitrile is excluded from RED pump tubing. All candidates correctly assign stream A to pump_e_series_blue (BLUE tubing). Confirmed compliant.",
 "status": "resolved as designed"
},
```

```
    {
     "id": "SA-7",
     "topic": "Startup sequence and backflow risk at startup",
     "detail": "For candidates 3, 6, and 11 (large-bore ETFE Stage 1, BPR 7 bar), the as_designed simulation only covered the startup interval and did not reach operating pressure. This does not establish or rule out backflow at startup. The unprotected reverse path (SA-1) remains the controlling concern across all scenarios.",
     "status": "unresolved — startup sequence must be defined and reviewed"
    },
    {
     "id": "SA-8",
     "topic": "No inline degasser available",
     "detail": "Inline degasser is unavailable. All candidates rely on offline Ar degassing of feed. Residual dissolved gas and O2 contamination of feed entering Stage 1 is uncontrolled after offline preparation. Quality and method of offline degassing require laboratory definition.",
     "status": "acknowledged limitation — no mitigation available within current inventory"
    },
    {
     "id": "SA-9",
     "topic": "Stage 2 temperature — not stage-resolved in batch protocol",
     "detail": "Batch Stage 2 temperature is not independently specified; 25°C is assumed. No validated rate correction is available. Any exotherm from the oxidation step is uncharacterized. FEP coils used in Stage 2 have a 50°C maximum; this must not be exceeded.",
     "status": "unresolved — temperature monitoring required during Stage 2 operation"
    },
    {
     "id": "SA-10",
     "topic": "Gas equivalents are proposed delivered molar ratios at inlet STP",
     "detail": "2.01 equiv O2 is calculated from inlet STP flow vs liquid molar flow. Actual O2 consumption, dissolution, and uptake in Stage 2 are not modeled. Delivered equiv is not consumed equiv.",
     "status": "acknowledged — not a hard violation"
    }
   ]
  },
  "physics_assessment": {
   "tool_result_id": "2c7417b2d6e18efec57d2ac24fb50b4b956bb08bcbc7e87758333f011f32bf59",
   "evidence_ids": [
    "C1/as_designed/entered_profile/steady",
    "C1/hypothetical_liquid_check/entered_profile/steady",
    "C2/as_designed/entered_profile/steady",
```

"C2/hypothetical_liquid_check/entered_profile/steady",
"C3/as_designed/entered_profile/gas_first_start",
"C3/hypothetical_liquid_check/entered_profile/steady",
"C4/as_designed/entered_profile/steady",
"C4/hypothetical_liquid_check/entered_profile/steady",
"C5/as_designed/entered_profile/steady",
"C5/hypothetical_liquid_check/entered_profile/steady",
"C6/as_designed/entered_profile/gas_first_start",
"C6/hypothetical_liquid_check/entered_profile/steady",
"C7/as_designed/entered_profile/steady",
"C7/hypothetical_liquid_check/entered_profile/steady",
"C8/as_designed/entered_profile/steady",
"C8/hypothetical_liquid_check/entered_profile/steady",
"C9/as_designed/entered_profile/pump_interruption",
"C9/hypothetical_liquid_check/entered_profile/steady",
"C10/as_designed/entered_profile/steady",
"C10/hypothetical_liquid_check/entered_profile/steady",
"C11/as_designed/entered_profile/gas_first_start",
"C11/hypothetical_liquid_check/entered_profile/steady",
"C12/as_designed/entered_profile/steady",
"C12/hypothetical_liquid_check/entered_profile/steady"
],
"proposed_alternatives": [
"Install a verified, O2-service-rated check valve on the liquid branch (upstream of Stage 2 junction) to address the unprotected reverse path. This is an unapproved proposal pending chemistry, inventory, O2-service, and engineering review; it does not change any current design.",
"Adopt a pressure-decoupled interstage arrangement (

Call 7: DrSafety, attempt 2
Status: recorded response, parsed successfully.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1 liquid τ=100 min (ratio 0.42 vs batch 240 min); Stage 2 inlet/STP apparent τ=181.8 min (ratio 0.51 vs batch 360 min). BPR 7 bar; pure O2 0.09 sccm, 2.01 equiv confirmed. Reverse path to Stage 1 unprotected on liquid branch; no liquid-branch check valve declared. Tool predicted no liquid displacement under entered illustrative profile only—this does not establish safety. CV-3301 pressure rating unverified; pure O2 service requires chemist and oxygen-service review. Acrylonitrile excluded from red tubing; blue peristaltic confirmed. Stage 2 temperature not stage-resolved in protocol.

Reviews 1 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved; no liquid-branch check valve assigned.

Reviews 1 / Uncertainty 2: CV-3301 max pressure rating unverified; cracking pressure ≠ pressure rating.

Reviews 1 / Uncertainty 3: Pure O2 service compatibility with all wetted materials requires laboratory confirmation.

Reviews 1 / Uncertainty 4: Tool results conditional on illustrative, uncalibrated assumptions—no measured pressure traces.

Reviews 1 / Uncertainty 5: Stage 2 batch temperature not stage-resolved; flow temperature deviation consequence unknown.

Reviews 1 / Uncertainty 6: Stage 1 photon dose adequacy at 450 nm vs 452 nm protocol unconfirmed; light-dose sufficiency unproven.

Reviews 1 / Uncertainty 7: Actual gas uptake, slug behaviour, and O2 dissolution not modelled.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1 τ=100 min; Stage 2 inlet/STP τ=181.8 min. BPR 2.5 bar lowers system pressure, increasing in-channel gas volume (0.028 mL/min vs 0.012 mL/min at 7 bar). Same unresolved reverse path and unprotected liquid branch as C1. Lower BPR may reduce oxygen dissolution driving force—untested hypothesis. Pure O2 and CV-3301 rating require confirmation.

Reviews 2 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 2 / Uncertainty 2: Lower BPR effect on O2 dissolution and slug stability is an untested hypothesis.

Reviews 2 / Uncertainty 3: CV-3301 pressure rating unverified.

Reviews 2 / Uncertainty 4: Pure O2 service compatibility requires chemist confirmation.

Reviews 2 / Uncertainty 5: Tool results conditional on illustrative assumptions only.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 uses ETFE 5 mL coil (ID 2.4 mm) giving τ=250 min (ratio 1.04 vs batch 240 min); Stage 2 τ=181.8 min inlet/STP. BPR 7 bar. The 2.4 mm ID at 0.02 mL/min gives very low Re (0.177); photon penetration depth and flow pattern in wide-bore coil under UV-150 are unconfirmed. Tool only simulated gas_first_start startup interval; BPR opening pressure not reached—steady-state reverse path behaviour in this candidate remains uncharacterised by the tool. Reverse liquid branch still unprotected.

Reviews 3 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 3 / Uncertainty 2: Tool only completed startup interval; steady-state not simulated for as_designed profile.

Reviews 3 / Uncertainty 3: Photon penetration and irradiation uniformity in 2.4 mm ID coil under UV-150 unconfirmed.

Reviews 3 / Uncertainty 4: CV-3301 pressure rating unverified; pure O2 service requires confirmation.

Reviews 3 / Uncertainty 5: Stage 2 temperature not stage-resolved in protocol.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1 τ=100 min (ratio 0.42); Stage 2 τ=181.8 min inlet/STP. BPR 1.5 bar—lowest pressure among candidates, highest in-channel gas volume fraction (0.040 mL/min). MFC supply margin remains within rating. Reverse path unprotected on liquid branch. At 1.5 bar, gas fraction is largest; slug flow pattern and O2 contact quality are most uncertain here. Pure O2 service and CV-3301 rating require confirmation.

Reviews 4 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 4 / Uncertainty 2: Highest in-channel gas fraction at 1.5 bar BPR; slug regime and contact quality most uncertain.

Reviews 4 / Uncertainty 3: CV-3301 pressure rating unverified; pure O2 service requires confirmation.

Reviews 4 / Uncertainty 4: Tool results conditional on illustrative assumptions; no measured pressure traces.

Reviews 4 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1 τ=100 min; Stage 2 uses ETFE 10 mL coil (ID 2.4 mm), inlet/STP τ=90.9 min (ratio 0.25 vs batch 360 min). Shortest Stage 2 apparent index among candidates with 7 bar BPR. Whether 90.9 min provides adequate oxidation is experimentally unresolved. Reverse path unprotected. 2.4 mm ID photon penetration for Stage 2 irradiation unconfirmed.

Reviews 5 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 5 / Uncertainty 2: Stage 2 apparent τ 90.9 min is substantially below batch 360 min; oxidation adequacy unresolved.

Reviews 5 / Uncertainty 3: Photon penetration in 2.4 mm ID Stage 2 coil unconfirmed.

Reviews 5 / Uncertainty 4: CV-3301 rating unverified; pure O2 service requires chemist confirmation.

Reviews 5 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 uses PFA 5 mL coil (ID 1 mm), τ=250 min (ratio 1.04); Stage 2 FEP 20 mL, τ=181.8 min inlet/STP, BPR 7 bar. Near-batch Stage 1 exposure. Tool only simulated startup interval for as_designed (BPR not reached); steady-state reverse behaviour uncharacterised by tool. Reverse liquid branch unprotected. Pure O2 and CV-3301 require confirmation.

Reviews 6 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 6 / Uncertainty 2: Tool only completed startup interval; steady-state as_designed not simulated.

Reviews 6 / Uncertainty 3: CV-3301 pressure rating unverified; pure O2 service requires chemist confirmation.

Reviews 6 / Uncertainty 4: Stage 2 temperature not stage-resolved in protocol.

Reviews 6 / Uncertainty 5: Near-batch Stage 1 time does not guarantee equivalent conversion in flow.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1 PFA 2 mL τ=100 min; Stage 2 PFA 10 mL (ID 1 mm) τ=90.9 min inlet/STP, BPR 7 bar. Stage 2 apparent index is 25% of batch 360 min; oxidation adequacy is an open experimental question. Reverse path unprotected on liquid branch. Tool predicted no displacement under entered illustrative assumptions. CV-3301 and pure O2 require confirmation.

Reviews 7 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 7 / Uncertainty 2: Stage 2 τ 90.9 min well below batch 360 min; oxidation sufficiency unknown.

Reviews 7 / Uncertainty 3: CV-3301 pressure rating unverified; pure O2 service requires confirmation.

Reviews 7 / Uncertainty 4: Tool results conditional on illustrative assumptions only.

Reviews 7 / Uncertainty 5: Stage 2 temperature not stage-resolved.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1 PFA 2 mL τ=100 min; Stage 2 FEP 10 mL (ID 1 mm, max 50°C) τ=90.9 min inlet/STP, BPR 7 bar. FEP coil max temperature 50°C is compatible with 25°C operation but leaves no thermal margin if exothermic events or LED heating occur. Reverse path unprotected. CV-3301 and pure O2 require confirmation.

Reviews 8 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 8 / Uncertainty 2: FEP coil max 50°C; thermal margin under exotherm or LED heating not assessed.

Reviews 8 / Uncertainty 3: Stage 2 τ 90.9 min well below batch 360 min; oxidation sufficiency unknown.

Reviews 8 / Uncertainty 4: CV-3301 pressure rating unverified; pure O2 service requires confirmation.

Reviews 8 / Uncertainty 5: Stage 2 temperature not stage-resolved in protocol.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 ETFE 5 mL (ID 2.4 mm) τ=250 min; Stage 2 ETFE 10 mL (ID 2.4 mm) τ=90.9 min inlet/STP, BPR 7 bar. Both stages use wide-bore coils; photon penetration at 2.4 mm ID is unconfirmed for both stages. Tool reported negligible displacement under pump_interruption scenario (illustrative only). Reverse path unprotected. CV-3301 and pure O2 require confirmation.

Reviews 9 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 9 / Uncertainty 2: Photon penetration and irradiation uniformity in 2.4 mm ID coils at both stages unconfirmed.

Reviews 9 / Uncertainty 3: Stage 2 τ 90.9 min well below batch 360 min; oxidation sufficiency unknown.

Reviews 9 / Uncertainty 4: CV-3301 pressure rating unverified; pure O2 service requires confirmation.

Reviews 9 / Uncertainty 5: Tool results conditional on illustrative assumptions; pump_interruption scenario does not cover all credible transients.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1 PFA 2 mL τ=100 min; Stage 2 ETFE 10 mL (ID 2.4 mm) τ=90.9 min inlet/STP, BPR 2.5 bar. Wide-bore Stage 2 coil with lower pressure: largest in-channel gas fraction after C4;

photon penetration in 2.4 mm ID unconfirmed. Reverse path unprotected. CV-3301 and pure O2 require confirmation.

Reviews 10 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 10 / Uncertainty 2: Photon penetration in 2.4 mm ID Stage 2 coil unconfirmed.

Reviews 10 / Uncertainty 3: Lower BPR (2.5 bar) increases gas fraction; O2 contact and slug behaviour uncertain.

Reviews 10 / Uncertainty 4: Stage 2 τ 90.9 min well below batch 360 min; oxidation sufficiency unknown.

Reviews 10 / Uncertainty 5: CV-3301 pressure rating unverified; pure O2 service requires chemist confirmation.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1 ETFE 5 mL (ID 2.4 mm) τ=250 min; Stage 2 PFA 10 mL (ID 1 mm) τ=90.9 min inlet/STP, BPR 7 bar. Tool only simulated startup interval for as_designed (BPR not reached). Stage 1 wide-bore photon penetration unconfirmed. Reverse path unprotected. CV-3301 and pure O2 require confirmation.

Reviews 11 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 11 / Uncertainty 2: Tool only completed startup interval for as_designed; steady-state not simulated.

Reviews 11 / Uncertainty 3: Photon penetration in 2.4 mm ID Stage 1 coil unconfirmed.

Reviews 11 / Uncertainty 4: Stage 2 τ 90.9 min well below batch 360 min; oxidation sufficiency unknown.

Reviews 11 / Uncertainty 5: CV-3301 pressure rating unverified; pure O2 service requires confirmation.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1 PFA 2 mL τ=100 min; Stage 2 PFA 10 mL (ID 1 mm) τ=90.9 min inlet/STP, BPR 2.5 bar. Lowest total reactor volume (12 mL) and inlet/STP total time (190.9 min). Stage 2 apparent index most compressed vs batch; oxidation adequacy most uncertain. Reverse path unprotected. CV-3301 and pure O2 require confirmation.

Reviews 12 / Uncertainty 1: Unprotected reverse path (st2_mixer→st1_reactor) unresolved.

Reviews 12 / Uncertainty 2: Stage 2 τ 90.9 min at 2.5 bar BPR; O2 dissolution and oxidation adequacy most uncertain in this group.

Reviews 12 / Uncertainty 3: CV-3301 pressure rating unverified; pure O2 service requires chemist confirmation.

Reviews 12 / Uncertainty 4: Tool results conditional on illustrative assumptions only.

Reviews 12 / Uncertainty 5: Stage 2 temperature not stage-resolved in protocol.

Shared assessment / Applies to all candidates: All twelve candidates share the following unresolved issues that must be addressed before laboratory execution.

Shared assessment / Reverse path: An unprotected reverse path exists on the liquid branch (st2_mixer→st1_reactor) in every candidate. The gas-branch CV-3301 check valve protects only the MFC branch and does not protect the liquid branch. The KHU observation of observed gas reverse flow toward Stage 1 is qualitative evidence of credible risk. Reducing gas flow does not close an unprotected reverse path. A

branch-specific non-return device on the liquid branch between Stage 1 and the Stage 2 mixer junction is an unapproved proposed mitigation requiring engineering review, inventory confirmation, and service verification before use.

Shared assessment / Gas source confirmation: Pure O2 (100 mol%) is confirmed by chemist answer (Q-GAS-001). This constitutes a gas-source change from the earlier design that specified air. Pure oxygen service requires independent chemist confirmation, oxygen-compatibility review of all wetted materials, and inventory/safety review before laboratory use. This is not yet approved.

Shared assessment / Cv 3301 pressure rating: CV-3301 cracking pressure is 1.0 bar. Maximum pressure rating is not supplied and must be verified before execution. Cracking pressure is not a pressure rating.

Shared assessment / Tool basis: Physics tool results are conditional on illustrative, uncalibrated assumptions (no measured KHU pressure traces, device dynamics, or compliance values). No reversal predicted in these tests does not establish safety. Predicted liquid displacement is not proof that oxygen reaches Reactor 1. Results are not probabilities and do not constitute safety approval.

Shared assessment / Gas stage residence time: Q_gas_sccm and gas_flow_sccm are mL/min at inlet/STP (273.15 K, 1.01325 bar). Stage 2 apparent residence time is V/(Q_liquid + Q_gas_STP), not measured liquid exposure or photon dose. Gas equivalents are proposed delivered molar ratios at inlet STP, not established stoichiometric uptake.

Shared assessment / Wavelength: Available UV-150 450 nm (24 W) and Manual 2 448 nm strip LEDs are assigned. Protocol specifies 452 nm. The 2 nm offset is noted; photon dose adequacy in flow is unproven.

Shared assessment / Acrylonitrile pump exclusion: Acrylonitrile is excluded from red pump tubing. All candidates use blue E-series peristaltic pump for stream A, which is not subject to this exclusion.

Shared assessment / Stage2 temperature: Stage 2 batch temperature is not stage-resolved in the protocol. All candidates assign 25°C. Temperature deviation consequence on oxidation rate and selectivity is unknown.

Shared assessment / Kinetics: Reaction kinetics are uncharacterised. Batch times are screening anchors only. No yield, conversion, or kinetic superiority is established for any candidate.

Physics assessment / Tool result id:
2c7417b2d6e18efec57d2ac24fb50b4b956bb08bcbc7e87758333f011f32bf59

Physics assessment / Evidence ids 1: C1/as_designed/entered_profile/steady

Physics assessment / Evidence ids 2: C1/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 3: C2/as_designed/entered_profile/steady

Physics assessment / Evidence ids 4: C2/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 5: C3/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 6: C3/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 7: C4/as_designed/entered_profile/steady

Physics assessment / Evidence ids 8: C4/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 9: C5/as_designed/entered_profile/steady

Physics assessment / Evidence ids 10: C5/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 11: C6/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 12: C6/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 13: C7/as_designed/entered_profile/steady

Physics assessment / Evidence ids 14: C7/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 15: C8/as_designed/entered_profile/steady

Physics assessment / Evidence ids 16: C8/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 17: C9/as_designed/entered_profile/pump_interruption

Physics assessment / Evidence ids 18: C9/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 19: C10/as_designed/entered_profile/steady

Physics assessment / Evidence ids 20: C10/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 21: C11/as_designed/entered_profile/gas_first_start

Physics assessment / Evidence ids 22: C11/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Evidence ids 23: C12/as_designed/entered_profile/steady

Physics assessment / Evidence ids 24: C12/hypothetical_liquid_check/entered_profile/steady

Physics assessment / Proposed alternatives 1: Install a confirmed, service-verified non-return (check) valve on the liquid branch between Stage 1 reactor outlet and the Stage 2 gas-injection mixer junction. This is an unapproved proposal requiring engineering review, inventory confirmation, oxygen-service compatibility review, and pressure-rating verification before any laboratory use.

Physics assessment / Proposed alternatives 2: Implement a defined startup sequence (liquid flow established and pressure confirmed before gas introduction) to reduce transient reverse-path risk. This is an unapproved operational procedure requiring chemist and safety review.

Physics assessment / Proposed alternatives 3: Verify CV-3301 pressure rating against maximum operating pressure of each candidate before execution. Cracking pressure alone does not confirm suitability.

Physics assessment / Limitation 1: Tool uses illustrative, uncalibrated parameters: no measured KHU pressure traces, compliance values, pump response, MFC dynamics, or multiphase resistance are available.

Physics assessment / Limitation 2: Model is isothermal, single-junction, single liquid branch and single gas branch; does not resolve slug flow, gas dissolution, reaction heat, capillary effects, or gas-front propagation.

Physics assessment / Limitation 3: Negative liquid-branch flow in the model means predicted liquid displacement at the junction, not proof that oxygen reaches Reactor 1.

Physics assessment / Limitation 4: No reversal predicted in these finite-window tests does not establish safety or exclude backflow under all credible transients.

Physics assessment / Limitation 5: Candidates 3, 6, and 11 (large-bore Stage 1 coils) only had the startup interval simulated for the as_designed profile; BPR opening pressure was not reached and steady-state reverse behaviour is uncharacterised by the tool.

Physics assessment / Limitation 6: Hypothetical liquid-branch check valve is not an inventory item, is not service-verified, and zero leakage is an idealisation; its inclusion in the tool does not constitute design approval.

Physics assessment / Limitation 7: BPR cracking pressure concerns the pressure difference across the gas check valve (CV-3301), not the difference between BPR setpoint and cracking pressure.

Physics assessment / Limitation 8: All tool results are conditional on entered illustrative assumptions and do not constitute safety certification or probability estimates.

Call 8: Skeptic, attempt 1
Status: unsuccessful response; retained for provenance. Unterminated string starting at: line 163 column 5 (char 14668)

```json
{
  "vetoes": [],
```

```
"shared_assessment": {
  "backflow_risk_unresolved_all_candidates": "Every candidate carries an unresolved structural risk: the liquid branch from Stage 1 to the Stage 2 mixer has no declared forward-oriented non-return valve. The qualitative KHU observation of gas reverse flow toward Stage 1 remains unrebutted. The DrFluidics/DrSafety tool results (no predicted liquid displacement under entered illustrative assumptions) are conditional on unmeasured parameters and finite scenario windows only; they do not establish safety or close the finding. Reducing gas flow does not protect an unprotected reverse path. A verified, O2-service-compatible check valve on the liquid branch between Stage 1 outlet and the Stage 2 mixer inlet is an unapproved proposal requiring engineering design review, chemistry compatibility assessment, and inventory verification before any execution.",
  "gas_source_status": "All candidates specify pure O2 (100 mol%) at Stage 2. The prior experiment used pure O2 when the design specified air; this change has been confirmed by the chemist (Q-GAS-001) and is reflected in all candidates. Pure O2 oxygen-service suitability for all wetted components (tubing, fittings, mixers, BPR, CV-3301) requires laboratory review before execution.",
  "cv3301_status": "CV-3301 cracking pressure is 1.0 bar; maximum pressure rating is not supplied. Cracking pressure is not a pressure rating. Service compatibility with pure O2 at the declared operating pressures must be verified before use.",
  "gas_stage_temperature": "Stage 2 batch temperature is not stage-resolved in the source protocol; all candidates assume 25 °C. This is an unvalidated assumption; yield and selectivity consequences of any temperature deviation are unknown.",
  "physics_tool_authority": "DrFluidics and DrSafety results are conditional on illustrative, uncalibrated parameters (viscosity 1.0 mPa·s, gas plenum 0.2 mL, liquid compliance 0.01 mL/bar, pump response 1.0 s, MFC response 0.5 s). No measured KHU pressure traces are available. Absence of predicted liquid displacement does not establish that reverse flow cannot occur. The tool reports predicted liquid displacement at the junction, not oxygen reaching Reactor 1.",
  "candidates_3_6_11_startup_gap": "For Candidates 3, 6, and 11 the as-designed gas-first-start simulation did not reach BPR opening pressure within the simulation window. This is a model artifact of a finite scenario window, not a confirmed hard violation. However, steady-state reverse behaviour under as-designed conditions is uncharacterised by the tool for these three candidates, and startup sequence (liquid-first pressurisation to BPR setpoint before gas introduction) must be defined as an explicit laboratory procedure before execution. This remains an unresolved experimental uncertainty, not a vetoed condition.",
  "wavelength_offset": "All Stage 1 assignments use UV-150/450 nm or Manual/448 nm; protocol specifies 452 nm. Photocatalytic efficiency at this offset is unverified.",
  "residence_time_basis": "Stage 1 times are liquid V/Q (no gas). Stage 2 times are inlet/STP apparent index V/(Q_liquid + Q_gas_STP); they are not measured liquid residence times, photon exposure equivalents, or confirmed gas uptake times. Physical exposure equivalence is not established for any candidate.",
  "etfe_24mm_photon_delivery": "Candidates using ETFE 2.4 mm ID coils in Stage 1 (Candidates 3, 9, 11) have unconfirmed photon penetration depth in UV-150. This is an unresolved experimental uncertainty, not a hard violation.",
  "throughput_note": "All candidates share Q_liquid = 0.02 mL/min and Q_gas_STP = 0.09 sccm. Feed throughput is identical across all candidates; total inlet/STP time differs only by Stage 1 reactor volume selection. Computed feed throughput is not measured product throughput.",
  "analogy_records": "All three retrieved analogy records are from different reaction classes or light regimes and are not usable as kinetic evidence for this Giese/oxidation sequence.",
  "offline_deoxygenation": "No inline degasser is available. Offline Ar degassing is the only deoxygenation route; effectiveness under continuous flow has not been confirmed."
},
```

```
"candidate_justifications": [
  {
    "candidate_id": 1,
    "reviewer_consensus": "prefer",
    "justification": "2 mL PFA / UV-150 450 nm, τ_liq=100 min (0.42× batch); 20 mL FEP / Manual-2 448 nm, τ_inlet=181.8 min (0.51× batch); BPR 7 bar; O2 0.09 sccm / 2.01 equiv. Compact interpretable baseline at intermediate stage-time ratios for both steps; 7 bar BPR provides highest dissolved-O2 driving force and gas-check headroom among candidates. Suitable starting design per objective.",
    "retained_uncertainties": [
      "Unprotected liquid reverse path to Stage 1 requires engineering resolution before execution.",
      "CV-3301 pressure rating and O2-service compatibility unconfirmed.",
      "Stage 1 τ_liq=100 min adequacy for Giese step unvalidated; kinetics uncharacterised.",
      "Stage 2 τ_inlet=181.8 min is apparent index only; actual liquid exposure and O2 uptake unmodelled.",
      "Pure O2 wetted-material service review pending.",
      "Stage 2 temperature unresolved from batch protocol."
    ]
  },
  {
    "candidate_id": 2,
    "reviewer_consensus": "acceptable",
    "justification": "Identical reactor volumes and flows to C1; BPR 2.5 bar increases in-channel gas volume (0.028 vs 0.012 mL/min in-channel) and reduces dissolved-O2 driving force hypothesis. Inlet/STP τ identical to C1. Useful BPR-effect data point versus C1; lower pressure reduces suppression of backpressure transients.",
    "retained_uncertainties": [
      "Unprotected liquid reverse path unresolved; lower BPR reduces startup headroom.",
      "Lower dissolved-O2 partial pressure effect on oxidation unvalidated.",
      "CV-3301 rating and O2-service compatibility unconfirmed.",
      "Stage 2 temperature unresolved."
    ]
  },
  {
    "candidate_id": 3,
    "reviewer_consensus": "acceptable",
    "justification": "5 mL ETFE (2.4 mm ID) Stage 1, τ_liq=250 min (1.04× batch), UV-150 450 nm; 20 mL FEP Stage 2, τ_inlet=181.8 min, BPR 7 bar. Near-batch Stage 1 exposure useful for yield-floor reference. ETFE 2.4 mm ID photon penetration in UV-150 unconfirmed. Startup sequence must be defined (gas-first-start tool window did not reach BPR).",
    "retained_uncertainties": [
      "Unprotected liquid reverse path unresolved.",
```

```
      "ETFE 2.4 mm ID photon penetration depth in UV-150 uncharacterised.",
      "Gas-first-start: BPR not reached in tool window; startup protocol must be defined before execution.",
      "CV-3301 rating and O2-service compatibility unconfirmed.",
      "Stage 2 temperature unresolved."
    ]
  },
  {
    "candidate_id": 4,
    "reviewer_consensus": "acceptable",
    "justification": "2 mL PFA Stage 1, τ_liq=100 min; 20 mL FEP Stage 2, τ_inlet=181.8 min, BPR 1.5 bar. Lowest BPR: highest in-channel gas fraction (0.040 mL/min), lowest dissolved-O2 driving force. Provides BPR-sensitivity data point vs C1/C2. Slug stability at 0.02 mL/min liquid with large gas fraction least characterised in this group.",
    "retained_uncertainties": [
      "Unprotected liquid reverse path unresolved; 1.5 bar provides least back-pressure suppression.",
      "Highest gas fraction; slug regime and O2 contact quality most uncertain among 20 mL FEP group.",
      "CV-3301 rating and O2-service compatibility unconfirmed.",
      "Stage 2 temperature unresolved."
    ]
  },
  {
    "candidate_id": 5,
    "reviewer_consensus": "acceptable",
    "justification": "2 mL PFA Stage 1, τ_liq=100 min; 10 mL ETFE (2.4 mm ID) Stage 2, τ_inlet=90.9 min (0.25× batch), BPR 7 bar. Shortest Stage 2 apparent index; total inlet/STP time 190.9 min. Whether 90.9 min provides sufficient oxidation is experimentally unresolved. ETFE 2.4 mm gas-liquid contact and photon delivery in Stage 2 unconfirmed.",
    "retained_uncertainties": [
      "Unprotected liquid reverse path unresolved.",
      "Stage 2 τ_inlet=90.9 min is 0.25× batch; oxidation adequacy unvalidated.",
      "ETFE 2.4 mm ID gas-liquid mixing and photon penetration in Stage 2 uncharacterised.",
      "CV-3301 rating and O2-service compatibility unconfirmed."
    ]
  },
  {
    "candidate_id": 6,
    "reviewer_consensus": "acceptable",
    "justification": "5 mL PFA (1 mm ID) Stage 1, τ_liq=250 min (1.04× batch), UV-150 450 nm; 20 mL FEP Stage 2, τ_inlet=181.8 min, BPR 7 bar. Near-batch Stage 1 in 1 mm PFA (better-defined optical path than C3
```

ETFE). Startup sequence must be defined (gas-first-start tool window did not reach BPR). Longer total time than C1.",

"retained_uncertainties": [

"Unprotected liquid reverse path unresolved.",

"Gas-first-start: BPR not reached in tool window; startup protocol must be defined before execution.",

"Stage 1 at 250 min exceeds batch anchor; over-irradiation degradation risk is an untested hypothesis.",

"CV-3301 rating and O2-service compatibility unconfirmed.",

"Stage 2 temperature unresolved."

]

},

{

"candidate_id": 7,

"reviewer_consensus": "acceptable",

"justification": "2 mL PFA Stage 1, τ_liq=100 min; 10 mL PFA (1 mm ID) Stage 2, τ_inlet=90.9 min (0.25× batch), BPR 7 bar. Compact; 1 mm PFA Stage 2 may offer more defined gas-liquid contact than ETFE 2.4 mm (C5); this is a hypothesis. Total inlet/STP time 190.9 min. Stage 2 adequacy unresolved.",

"retained_uncertainties": [

"Unprotected liquid reverse path unresolved.",

"Stage 2 τ_inlet=90.9 min adequacy for oxidation unvalidated.",

"CV-3301 rating and O2-service compatibility unconfirmed.",

"Stage 2 temperature unresolved."

]

},

{

"candidate_id": 8,

"reviewer_consensus": "acceptable",

"justification": "2 mL PFA Stage 1, τ_liq=100 min; 10 mL FEP (1 mm ID) Stage 2, τ_inlet=90.9 min, BPR 7 bar. Functionally identical timing to C7; FEP vs PFA is the sole differentiator at Stage 2. FEP max 50 °C is satisfied at 25 °C but limits future temperature excursions. No distinguishing chemical advantage over C7.",

"retained_uncertainties": [

"Unprotected liquid reverse path unresolved.",

"Stage 2 τ_inlet=90.9 min adequacy unvalidated.",

"FEP max 50 °C limits upward temperature flexibility in Stage 2.",

"CV-3301 rating and O2-service compatibility unconfirmed."

]

},

{

"candidate_id": 9,

```
    "reviewer_consensus": "acceptable",
    "justification": "5 mL ETFE (2.4 mm ID) Stage 1, τ_liq=250 min; 10 mL ETFE (2.4 mm ID) Stage 2, τ_inlet=90.9 min, BPR 7 bar. Both stages wide-bore; photon penetration unconfirmed for both. Offers all-ETFE wide-bore comparison but compounds optical and gas-liquid uncertainty across stages.",
    "retained_uncertainties": [
      "Unprotected liquid reverse path unresolved.",
      "ETFE 2.4 mm ID photon penetration uncharacterised in both stages.",
      "Stage 2 τ_inlet=90.9 min adequacy unvalidated.",
      "CV-3301 rating and O2-service compatibility unconfirmed."
    ]
  },
  {
    "candidate_id": 10,
    "reviewer_consensus": "acceptable",
    "justification": "2 mL PFA Stage 1, τ_liq=100 min; 10 mL ETFE (2.4 mm ID) Stage 2, τ_inlet=90.9 min, BPR 2.5 bar. Combines wide-bore Stage 2 with lower pressure; highest gas fraction uncertainty among short-Stage-2 group. Most uncertain candidate for oxidation adequacy in this subset; most useful if lower-pressure wide-bore data are specifically targeted.",
    "retained_uncertainties": [
      "Unprotected liquid reverse path unresolved.",
      "ETFE 2.4 mm ID with 2.5 bar BPR: gas-liquid stability and O2 contact most uncertain here.",
      "Stage 2 τ_inlet=90.9 min adequacy unvalidated.",
      "CV-3301 rating and O2-service compatibility unconfirmed."
    ]
  },
  {
    "candidate_id": 11,
    "reviewer_consensus": "acceptable",
    "justification": "5 mL ETFE (2.4 mm ID) Stage 1, τ_liq=250 min; 10 mL PFA (1 mm ID) Stage 2, τ_inlet=90.9 min, BPR 7 bar. Near-batch Stage 1 with compact Stage 2 in narrower bore. Startup sequence must be defined (gas-first-start window did not reach BPR). Stage 1 wide-bore photon delivery unconfirmed.",
    "retained_uncertainties": [
      "Unprotected liquid reverse path unresolved.",
      "Gas-first-start: BPR not reached in tool window; startup protocol must be defined before execution.",
      "ETFE 2.4 mm ID Stage 1 photon penetration uncharacterised.",
      "Stage 2 τ_inlet=90.9 min adequacy unvalidated.",
      "CV-3301 rating and O2-service compatibility unconfirmed."
    ]
  },
```

```
    {
      "candidate_id": 12,
      "reviewer_consensus": "acceptable",
      "justification": "2 mL PFA Stage 1, τ_liq=100 min; 10 mL PFA (1 mm ID) Stage 2, τ_inlet=90.9 min, BPR 2.5 bar. Lower-pressure companion to C7. Reduced dissolved-O2 driving force at 2.5 bar with 90.9 min apparent Stage 2 index makes oxidation adequacy the most uncertain among all-PFA candidates. Useful BPR sensitivity contrast to C7.",
      "retained_uncertainties": [
        "Unprotected liquid reverse path unresolved; 2.5 bar reduces startup headroom.",
        "Stage 2 τ_inlet=90.9 min at lower O2 partial pressure; oxidation adequacy most uncertain in PFA group.",
        "CV-3301 rating and O2-service compatibility unconfirmed.",
        "Stage 2 temperature unresolved."
      ]
    }
  ],
  "required_measurements": [
    "Confirm CV-3301 maximum pressure rating and oxygen-service compatibility before any candidate is executed.",
    "Install and verify a forward-oriented, O2-service-compatible non-return valve on the liquid branch between Stage 1 reactor outlet and Stage 2 mixer junction (unapproved proposal requiring engineering and chemistry review).",
    "Define and validate an explicit startup sequence (liquid flow established and BPR pressure confirmed before gas introduction) for all candidates, and especially for Candidates 3, 6, and 11.",
    "Measure actual pressure at Stage 2 mixer junction and Stage 1 outlet during startup and steady state to confirm reverse-path suppression.",
    "Verify photon penetration depth and irradiation uniformity in ETFE 2.4 mm ID coils under UV-150 module for Candidates 3, 5, 9, 10, 11.",
    "Confirm Stage 2 operating temperature and its effect on oxidation rate and selectivity (batch protocol does not stage-resolve this).",
    "Assess pure O2 service compatibility for all wetted materials: PFA, FEP, ETFE, PEEK, ETFE tubing fitt
```


Call 9: Skeptic, attempt 2
Status: recorded response, parsed successfully.

Vetoes: []

Assessment: No hard violations found across all twelve candidates. All share an identical shared topology finding: the liquid branch from Stage 1 to the Stage 2 mixer junction carries no declared forward-oriented non-return valve, leaving an unprotected reverse path to Stage 1. This is a credible engineering risk requiring resolution before any execution, but it is a shared unresolved uncertainty, not a confirmed hard violation that can be attributed to any individual candidate. The gas branch check valve (CV-3301) requires pressure-rating and O2-service verification in all candidates. Pure O2 was confirmed by the chemist (Q-GAS-001) but no gas-source change approval or oxygen-service review has been completed; this must be documented as unapproved pending laboratory review. Tool results (DrFluidics/DrSafety) are conditional on illustrative, uncalibrated assumptions; no reversal predicted in the finite scenario window does not establish safety or close the unprotected reverse

path. Candidates 3, 6, and 11 (large-bore ETFE Stage 1 coils assigned to UV-150) have only startup-interval tool results; BPR operating pressure was not reached in those windows, meaning steady-state reverse-path behaviour is uncharacterised by the tool for those three. This is a model limitation, not a confirmed hard failure, and does not constitute a veto. The ETFE 2.4 mm ID coils used in UV-150 have unconfirmed photon penetration and coupling efficiency relative to 1 mm PFA or FEP coils; this is an unresolved experimental uncertainty, not a hard constraint violation. FEP coil max temperature 50 °C is respected at 25 °C for Candidate 8 but leaves no margin for LED heating or exothermic events. Stage 2 temperature is not stage-resolved in the source batch protocol for any candidate; 25 °C is an assumption. Acrylonitrile is correctly excluded from red pump tubing in all candidates. The MFC (mfc_ffc00) is a gas MFC and is not used as a liquid pump. All hold ratios are inlet/STP apparent indices or liquid V/Q; none are equivalent to measured liquid exposure or batch kinetic equivalence. No candidate claims yield superiority; none have been rewritten. All candidates carry SCREEN_OR_EXECUTE_AFTER_REVIEW disposition. Candidate 1 (2 mL PFA / 20 mL FEP, BPR 7 bar) is identified as the preferred interpretable baseline per the objective policy: it provides intermediate stage-time ratios for both steps (0.42× and 0.51× batch), uses 1 mm PFA in Stage 1 for better-defined photon path, and is the most compact design with moderate O2 partial pressure. Preference is not a yield claim; it is a screening starting point subject to all shared unresolved requirements.

Requested measurement 1: Confirm liquid-branch non-return protection (e.g. forward-oriented check valve) between Stage 1 outlet and Stage 2 mixer junction before any execution; verify device pressure rating, O2-service compatibility, and low-flow operability

Requested measurement 2: Verify CV-3301 maximum pressure rating against each candidate BPR setpoint plus coil pressure drop; confirm O2-service compatibility

Requested measurement 3: Complete chemist and oxygen-service review for pure O2 use across all wetted materials; document as approved or rejected before laboratory use

Requested measurement 4: Measure actual gas supply pressure and confirm MFC differential pressure requirement under operating conditions

Requested measurement 5: Record pressure traces at Stage 2 inlet, junction, and BPR outlet during startup, steady state, and shutdown to characterise reverse-path risk

Requested measurement 6: Verify UV-150 photon coupling and irradiation uniformity for ETFE 2.4 mm ID coils (Candidates 3, 5, 6, 9, 10, 11) relative to 1 mm PFA reference

Requested measurement 7: Define and document startup sequence (liquid-first pressurisation to BPR setpoint before gas introduction) and validate in laboratory, particularly for Candidates 3, 6, and 11 where tool only completed startup interval

Requested measurement 8: Measure liquid residence time distribution and gas-liquid mass transfer in Stage 2 coil geometries to validate inlet/STP apparent residence time as an exposure index

Requested measurement 9: Confirm offline Ar degassing adequacy and residual O2 in feed solution under flow conditions

Requested measurement 10: Confirm Stage 2 operating temperature in flow and assess LED heating contribution in both stages

Requested measurement 11: Characterise FEP coil thermal margin under flow conditions with LED irradiation for Candidate 8

Requested measurement 12: Obtain measured yield and conversion data at the selected baseline condition (Candidate 1) before comparing stage-time variants

Call 10: Chief, attempt 1
Status: recorded response, parsed successfully.

Candidate: 1

Justification: Stage 1: 2 mL PFA / UV-150 450 nm / Q_liquid=0.02 mL/min / τ_liquid=100 min (0.42× batch 240 min). Stage 2: 20 mL FEP / Manual-2 448 nm / τ_inlet=181.8 min inlet/STP (0.51× batch 360 min) / O2 0.09 sccm at inlet/STP (2.01 equiv) / BPR 7 bar. Compact interpretable baseline with intermediate stage-time ratios for both steps, 1 mm PFA Stage 1 for better-defined photon path, and highest operating pressure among candidates (7 bar), providing greatest O2 partial-pressure driving force and narrowest in-channel gas volume fraction (0.012 mL/min) relative to other 7-bar candidates with smaller reactors. Feed: substrate 0.1 M / 0.002 mmol/min, acrylonitrile 0.2 M / 0.004 mmol/min, Ir-photocatalyst 0.0005 M / 0.001 mmol/min in EtOH:pH9 buffer 5:1 v/v, degassed offline with Ar. Pure O2 via mfc_ffc00 at 0.09 sccm (inlet/STP); chemist confirmed (Q-GAS-001) but oxygen-service and gas-source change require unapproved laboratory review before use. Unprotected reverse path (st2_mixer → st1_reactor) remains unresolved for all candidates; installing a forward-oriented, service-verified non-return valve on the liquid branch between Stage 1 outlet and Stage 2 mixer is an unapproved proposal requiring engineering, chemistry compatibility, inventory, and oxygen-service review. CV-3301 cracking pressure (1.0 bar) is not a pressure rating; maximum pressure rating unconfirmed. Tool results (DrFluidics/DrSafety) predict no liquid displacement under entered illustrative, uncalibrated assumptions within finite scenario windows only; this does not establish safety or close the unprotected reverse path. No yield is claimed.

Objective alignment: Balanced priority requires balancing yield/selectivity potential, substrate throughput, total processing time, oxygen delivery, and stable operation. Candidate 1 provides an interpretable two-point baseline (0.42× Stage 1, 0.51× Stage 2 vs batch anchors) without collapsing either stage to the minimum or extending either beyond the batch anchor, keeping total inlet/STP time at 281.8 min. The 7 bar BPR maximises dissolved-O2 driving force among candidates with the same reactor geometry and delivers the minimum in-channel gas fraction, which is operationally simpler at 0.02 mL/min liquid flow. The 1 mm PFA coil in Stage 1 provides a more predictable optical path than 2.4 mm ETFE alternatives. Feed throughput is identical across all candidates (0.002 mmol/min substrate, 0.02 mL/min liquid); no candidate has higher measured product throughput. Selection as starting design is chemically justified by optical path confidence and O2 pressure margin; it is not a yield claim.

Alternative 1 / Candidate: 6

Alternative 1 / Reason not selected: Stage 1: 5 mL PFA / τ_liquid=250 min (1.04× batch), Stage 2: 20 mL FEP / τ_inlet=181.8 min, BPR 7 bar. Extends Stage 1 to near-batch for yield-floor assessment with 1 mm PFA (better optical path than ETFE). Total inlet/STP time 431.8 min is 53% longer than Candidate 1, reducing comparative throughput. Gas-first startup: tool only completed startup interval; BPR not reached, so steady-state reverse-path behaviour is uncharacterised by the tool for this candidate. Over-irradiation degradation at 250 min is an untested hypothesis. Useful as a near-batch Stage 1 companion after Candidate 1 baseline is established, not as the starting design under the balanced objective.

Alternative 2 / Candidate: 7

Alternative 2 / Reason not selected: Stage 1: 2 mL PFA / τ_liquid=100 min, Stage 2: 10 mL PFA / τ_inlet=90.9 min (0.25× batch 360 min), BPR 7 bar. Total inlet/STP time 190.9 min. Compact design with 1 mm PFA in both stages. Stage 2 apparent index is the shortest at 7 bar; whether 90.9 min provides adequate oxidation exposure is experimentally unresolved and represents a greater yield risk than the more moderate 181.8 min of Candidate 1. Stage 2 gas-liquid uptake and actual liquid contact at this short index are unmodelled. Useful as a short-Stage-2 data point after baseline but not preferred as the starting design when oxidation adequacy is unknown.

Alternative 3 / Candidate: 2

Alternative 3 / Reason not selected: Identical reactor geometry to Candidate 1 (2 mL PFA / 20 mL FEP) but BPR 2.5 bar. In-channel gas volume increases to 0.028 mL/min vs 0.012 mL/min in Candidate 1; lower O2 partial pressure reduces dissolved-O2 driving force. Inlet/STP τ identical (181.8 min) because it is STP-based, not pressure-corrected. Effect of lower pressure on oxidation selectivity is uncharacterised. Unprotected reverse path identical topology; lower BPR provides a narrower pressure margin at startup. Useful as a BPR-contrast

companion to Candidate 1 but not preferred as the starting design when higher O2 partial pressure is operationally simpler and more informative.

Answer impact 1 / Question: Q-GAS-001

Answer impact 1 / Effect: Pure O2 confirmed by chemist; mfc_ffc00 delivers 0.09 sccm (inlet/STP) = 2.01 equiv. Gas-source change from batch air to pure O2 is not approved; oxygen-service compatibility review for all wetted materials, inventory items, and CV-3301 pressure rating required before laboratory execution. This requirement applies equally to all twelve candidates.

Answer impact 2 / Question: Q-GAS-002

Answer impact 2 / Effect: Minimum 2.0 equiv O2 met at 0.09 sccm (2.01 equiv at inlet/STP); grid rounding accounts for 0.01 excess. Applies to all candidates identically; equivalents are inlet/STP delivered molar ratios, not confirmed uptake.

Answer impact 3 / Question: Q-GAS-003

Answer impact 3 / Effect: O2 introduced only at Stage 2 inlet in all candidates; Stage 1 is gas-free. Unprotected reverse path on liquid branch from Stage 2 junction to Stage 1 reactor remains unresolved and requires engineering review before execution.

Answer impact 4 / Question: Q-HYP-001

Answer impact 4 / Effect: Chemist hypothesis that Stage 1 may maintain sufficient sulfide formation at higher throughput and that batch time is a reference not a fixed target is respected: 100 min Stage 1 (0.42× batch) is an exploration of shorter exposure, not a guaranteed equivalent. Yield and kinetics uncharacterised; validation required experimentally.

Answer impact 5 / Question: Q-OBJ-001

Answer impact 5 / Effect: Balanced objective implemented: Candidate 1 selected as compact, higher-throughput starting design with intermediate stage-time ratios, 1 mm PFA optical path, and 7 bar BPR for O2 delivery. Preference is a screening starting point, not a yield or selectivity claim.

Answer impact 6 / Question: Q-HIST-001

Answer impact 6 / Effect: Qualitative observation of gas reverse flow in previous experiment is acknowledged; the unprotected liquid-branch reverse path is flagged as unresolved for all candidates. Reducing gas flow alone (0.09 sccm vs previously reported 0.43 mL/min) does not close the unprotected reverse path. No measured pressure traces or yield from the historical experiment inform candidate selection.

Answer impact 7 / Question: Q-PREF-001

Answer impact 7 / Effect: Reported: substrate 0.1 M / 0.002 mmol/min (1.0 equiv), acrylonitrile 0.2 M / 0.004 mmol/min (2.0 equiv), Ir-catalyst 0.0005 M / 0.001 mmol/min (0.5 mol%), O2 0.09 sccm at inlet/STP (2.01 equiv, 0.012 mL/min pressure-corrected at 8.01 bar absolute). Stage 1: 25°C, 2 mL PFA, τ_liquid=100 min. Stage 2: 25°C (assumed, not stage-resolved), 20 mL FEP, τ_inlet=181.8 min inlet/STP. Conditional backflow: unprotected liquid reverse path unresolved; CV-3301 pressure rating unconfirmed. Missing: measured pressure traces, startup sequence, gas supply pressure, yield data.

Shared assessment / Unprotected reverse path: All twelve candidates share an identical topological finding: no forward-oriented non-return valve is declared on the liquid branch between Stage 1 reactor outlet and the Stage 2 mixer junction. This is a credible engineering risk under gas-introduction, startup, pump-interruption, and shutdown scenarios. The reverse path to Stage 1 is structurally open regardless of BPR setpoint, gas flow rate, or reactor geometry. Installing a confirmed, service-verified, O2-compatible non-return valve on the liquid branch is an unapproved proposal requiring engineering design review, chemistry compatibility assessment, inventory verification, and oxygen-service confirmation before any laboratory use. A valve on the MFC gas branch (CV-3301) does not protect the liquid branch.

Shared assessment / Gas check valve: CV-3301 cracking pressure is 1.0 bar; this concerns the pressure difference across the valve, not BPR setpoint minus cracking pressure. Maximum pressure rating is unconfirmed; verification required before execution.

Shared assessment / Pure o2 approval: Pure O2 use confirmed by chemist (Q-GAS-001) but oxygen-service compatibility with all wetted materials, tubing, fittings, and safety accessories is not approved. This requires laboratory review and documented confirmation before any execution. The gas-source change from batch air to pure O2 is unapproved.

Shared assessment / Physics tool limitations: DrFluidics and DrSafety results are conditional on illustrative, uncalibrated parameters (viscosity 1.0 mPa·s, compliance 0.01 mL/bar, gas plenum 0.2 mL, etc.; no measured KHU values). No reversal predicted in finite scenario windows does not establish safety, close the unprotected reverse path, or constitute proof of safe operation. Predicted liquid displacement means displacement at the junction model node, not proof that O2 reaches Reactor 1.

Shared assessment / Startup interval only: Candidates 3, 6, and 11 (ETFE 2.4 mm ID Stage 1 coils): the tool only completed the startup interval for the as_designed profile; BPR opening pressure was not reached. Steady-state reverse-path behaviour is uncharacterised by the tool for these three candidates. This is a model limitation, not a confirmed hard failure or veto.

Shared assessment / Etfe wide bore photon path: Candidates using ETFE 2.4 mm ID coils (3, 5, 9, 10, 11) have unconfirmed photon penetration and irradiation uniformity in UV-150 relative to 1 mm PFA/FEP coils. This is an unresolved experimental uncertainty.

Shared assessment / Gas stage residence time: τ_inlet/STP is V_reactor / (Q_liquid + Q_gas_STP) and is an apparent hydraulic index only. It is not equivalent to liquid residence time or to batch hold time. Actual liquid exposure, gas-liquid mass transfer, O2 uptake, and slug-flow behaviour are not modelled.

Shared assessment / Temperature: Stage 2 batch temperature is not stage-resolved in the source protocol. 25°C is an assumed screening value for all candidates. No temperature compensation, rate correction, or yield consequence is validated.

Shared assessment / Wavelength: UV-150 / 450 nm and Manual-2 / 448 nm differ from protocol 452 nm. Photocatalytic efficiency at these offsets is unconfirmed; no optical intensification or light-dose adequacy is established.

Shared assessment / Throughput: All candidates have identical feed throughput: 0.02 mL/min liquid, 0.002 mmol/min substrate. Product throughput is unmeasured and cannot be compared.

Shared assessment / Yield: No yield, conversion, or kinetic data exist for any candidate. No kinetic ranking, yield claim, or conversion floor is made or implied.

Limitation 1: Unprotected reverse path from Stage 2 junction to Stage 1 reactor is unresolved for all candidates; no execution before engineering review.

Limitation 2: CV-3301 maximum pressure rating unconfirmed; cracking pressure is not a pressure rating.

Limitation 3: Pure O2 oxygen-service and gas-source change approval pending; all candidates require laboratory review.

Limitation 4: Tool results are conditional on illustrative uncalibrated assumptions; no reversal in finite windows does not establish safety.

Limitation 5: τ_inlet/STP is a hydraulic index; actual liquid exposure, O2 uptake, and gas-liquid mass transfer are unmodelled.

Limitation 6: Stage 2 temperature not stage-resolved; 25°C is an assumption with unknown yield consequence.

Limitation 7: Wavelength offsets (450/448 nm vs 452 nm) photocatalytic efficiency unconfirmed.

Limitation 8: Offline Ar degassing adequacy under flow conditions unconfirmed.

Limitation 9: Feed throughput is identical across all candidates; product throughput is unmeasured.

Limitation 10: No yield, conversion, or kinetic data support selection; preference is for screening starting point only.

Limitation 11: ETFE 2.4 mm ID coil photon penetration in UV-150 uncharacterised.

Limitation 12: Batch hold times are screening anchors, not guaranteed flow equivalents.

Suggested measurement 1: Confirm and install a forward-oriented, service-verified, O2-compatible non-return valve on the liquid branch between Stage 1 outlet and Stage 2 mixer junction; verify cracking pressure, pressure rating, and low-flow operability before execution.

Suggested measurement 2: Verify CV-3301 maximum pressure rating against candidate BPR setpoint plus coil pressure drop; confirm O2-service and chemical compatibility.

Suggested measurement 3: Complete chemist and oxygen-service review for pure O2 across all wetted materials; document as approved before laboratory use.

Suggested measurement 4: Measure and record supply pressure at MFC inlet and confirm MFC differential pressure requirement under operating conditions.

Suggested measurement 5: Record pressure traces at Stage 2 inlet, mixer junction, and BPR outlet during startup, steady-state, and shutdown to characterise reverse-path behaviour.

Suggested measurement 6: Define and validate a liquid-first startup sequence (liquid flow established and system pressure confirmed at BPR setpoint before gas introduction); validate in laboratory particularly for Candidates 3, 6, and 11.

Suggested measurement 7: Measure actual product yield and Stage 1 intermediate conversion at Candidate 1 baseline conditions before screening stage-time variants.

Suggested measurement 8: Measure liquid residence time distribution and gas-liquid mass transfer coefficient in Stage 2 (20 mL FEP, 1 mm ID, 0.02 mL/min + 0.09 sccm O2, 7 bar) to validate τ_inlet/STP as an exposure index.

Suggested measurement 9: Verify UV-150 irradiance and photon penetration depth for ETFE 2.4 mm ID coils relative to 1 mm PFA reference coil.

Suggested measurement 10: Confirm offline Ar degassing residual O2 concentration in feed solution under flow-startup conditions.

Suggested measurement 11: Measure Stage 2 temperature under LED irradiation and flow to confirm 25°C assumption and assess LED heating contribution.

Suggested measurement 12: Characterise FEP coil temperature under LED irradiation and exothermic conditions for Candidate 8 to assess 50°C margin.

Editorial cross-checks (not model statements)

Time interpretation: some reports call Stage 2 liquid exposure approximately 1000 min, obtained from 20 mL/0.02 mL/min. This is a liquid-only space-time calculation using the entire reactor volume, not measured two-phase liquid residence. Likewise, the 181.82 min inlet/STP index must not be interpreted as a measured fraction of batch reaction exposure.

Numerical transcription: the Figure 5 Set 3 Chief response gives the photocatalyst molar flow as 0.001 mmol/min. The final feed record and concentration-times-flow calculation give 0.0005 mol/L x 0.02 mL/min = 0.000010 mmol/min. The original statement is retained in the transcript as a 100-fold numerical error, not adopted in the figure text or experimental recipe.

Geometry and transport claims: the Set 1 DrFluidics shared assessment groups Candidate 6 with large-bore ETFE options even though Candidate 6 uses 1 mm-ID PFA. The Set 3 Chief attributes a smaller in-channel gas

volume fraction to the selected reactor relative to other 7 bar candidates with smaller reactors. Their gas and liquid feed rates, pressure and temperature are the same; changing reactor volume alone does not establish a different gas/liquid volumetric flow ratio or measured gas holdup. A known bore defines geometry but does not establish photon-dose adequacy or a measured transport advantage.

Protocol interpretation: several reports describe the Stage 2 batch temperature as unspecified, whereas the supplied procedure continues under otherwise identical irradiation conditions after Stage 1 at 25 deg C. That wording does not establish an independently measured Stage 2 temperature, but it does not justify presenting 25 deg C as an unrelated new condition. Similarly, the qualitative backflow incident does not prove that every lower-flow alternative will reverse; the defensible finding is that the unprotected pathway remains.

F6-S1: Figure 6, Set 1
Source: outputs/khu_revised_six_20260915/figure6_set1/attempt_01. Selected Candidate 4; 12-candidate pool; downstream model claude-sonnet-4-6. The objective policy is balanced.

Call 1: DrChemistry, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated thioester formation (Stage 1) followed by aminolysis with benzylamine (Stage 2). Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M / DPDTC 0.525 M / DMAP 0.05 M in 2-MeTHF. Stream B: benzylamine 2.1 M in 2-MeTHF. All candidates use E-series blue pump (2-MeTHF compatible), BPR 7 bar, 95 °C, no gas, no light. Stoichiometric flow closure confirmed for all candidates.

Shared assessment / Topology note: A T-mixer or Y-mixer is required at the Stage 2 inlet to introduce Stream B into the Stage 1 effluent. This physical mixing point is not explicitly enumerated in any candidate's equipment list but inventory contains PEEK/ETFE T-mixers and Y-mixers; laboratory must confirm mixer is installed.

Shared assessment / Solubility note: DPDTC solubility in 2-MeTHF at 0.525 M and 95 °C is unconfirmed. Precipitation in Stage 1 coil would be a hard failure; chemist should verify before execution.

Shared assessment / Benzylamine concentration note: Stream B is listed at 2.1 M benzylamine in 2-MeTHF. This is concentrated but feasible; confirm miscibility and no precipitation on mixing with Stage 1 effluent.

Shared assessment / Batch anchor note: Batch holds are 30 min each stage at 95 °C (total reactive hold 60 min), used as screening anchors only. Achieved flow residence times are V/Q ratios, not measured exposure equivalents.

Shared assessment / Stage2 hold asymmetry: Due to Q dilution by Stream B, Stage 2 volumetric flow is always higher than Stage 1. For a fixed Stage 2 reactor volume the Stage 2 residence time is always shorter than Stage 1 at the same reactor volume. Candidates with short Stage 2 holds may not provide adequate aminolysis exposure; this is an experimental uncertainty, not a proven failure.

Shared assessment / Degradation risk note: Extended residence times at 95 °C for thioester intermediate or product amide carry unknown degradation risk. This is an untested hypothesis, not established.

Shared assessment / Pump flow minimum note: E-series minimum is 0.02 mL/min. Stream B at 0.02 mL/min (candidates 5, 9, 10, 11, 12) is at the minimum operating limit; peristaltic pulse behaviour and reproducibility at minimum flow require laboratory evaluation.

Shared assessment / Candidate 11 note: Candidate 11 target label is 120/30 min but achieved Stage 1 hold is 62.5 min, not 120 min. The target label is aspirational; the actual achieved ratio is 2.08 for Stage 1, identical to candidates 6, 7, 9 at the same flow scale.

Shared assessment / Candidate 12 note: Candidate 12 target label is 120/120 min but achieved Stage 1 hold is 62.5 min and Stage 2 is 50 min; the 120 min targets are not achieved. Identical flow parameters to candidate 7 Stage 1; Stage 2 uses coil_row11 vs coil_row12.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Both stages below batch anchor (τ1=17.9 min, τ2=14.3 min; ratios 0.60/0.48). Provides the shortest-time exploratory point. Uses two ETFE 5 mL coils at highest throughput (0.14 mmol/min). Adequate as a sub-batch baseline screening point; Stage 1 thioester formation and Stage 2 aminolysis may both be incomplete at these holds.

Reviews 1 / Uncertainty 1: Stage 1 thioester conversion unknown at 17.9 min; sub-batch hold may leave unreacted acid.

Reviews 1 / Uncertainty 2: Stage 2 aminolysis at 14.3 min may be insufficient; untested.

Reviews 1 / Uncertainty 3: DPDTC solubility in 2-MeTHF at 0.525 M unconfirmed.

Reviews 1 / Uncertainty 4: T-mixer for Stage 2 Stream B introduction not enumerated in candidate equipment list.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: prefer

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1 sub-batch (τ1=16.7 min, ratio 0.56) in compact 2 mL PFA coil; Stage 2 above batch anchor (τ2=33.3 min, ratio 1.11) in ETFE 5 mL coil. This asymmetry (shorter thioester formation, longer aminolysis) matches the chemist hypothesis that Stage 2 may need adequate exposure. Differentiates Stage 1 speed from Stage 2 completeness. Throughput 0.06 mmol/min.

Reviews 2 / Uncertainty 1: Stage 1 thioester completeness at 16.7 min is unknown; incomplete Stage 1 will reduce Stage 2 yield.

Reviews 2 / Uncertainty 2: Whether extra Stage 2 time compensates incomplete Stage 1 is untested.

Reviews 2 / Uncertainty 3: Stream B at 0.03 mL/min is near E-series minimum; pulsation may affect reproducibility.

Reviews 2 / Uncertainty 4: DPDTC solubility unconfirmed.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 above batch anchor (τ1=35.7 min, ratio 1.19) in ETFE 10 mL coil; Stage 2 sub-batch (τ2=14.3 min, ratio 0.48) in ETFE 5 mL coil. Tests whether complete thioester formation in Stage 1 compensates short Stage 2. Complementary asymmetry to candidate 2. Throughput 0.14 mmol/min.

Reviews 3 / Uncertainty 1: Stage 2 aminolysis at 14.3 min may be insufficient for full amide yield; unknown.

Reviews 3 / Uncertainty 2: Whether Stage 1 is complete at 35.7 min is uncharacterized.

Reviews 3 / Uncertainty 3: T-mixer for Stream B introduction not enumerated.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Both stages near batch anchor (τ1=31.3 min ratio 1.04, τ2=25.0 min ratio 0.83); closest symmetric match to batch 30/30 min protocol. Both ETFE 5 mL coils. Natural baseline reference candidate. Throughput 0.08 mmol/min.

Reviews 4 / Uncertainty 1: Stage 2 ratio 0.83 is below batch; aminolysis completeness at 25 min unknown.

Reviews 4 / Uncertainty 2: Equivalence between batch stirred and flow plug-flow exposure not established.

Reviews 4 / Uncertainty 3: DPDTC solubility unconfirmed.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1 sub-batch ($\tau1$=25 min, ratio 0.83) in 2 mL PFA; Stage 2 extended ($\tau2$=50 min, ratio 1.67) in ETFE 5 mL. Explores whether long aminolysis exposure compensates moderate Stage 1 hold. Lowest throughput group (0.04 mmol/min). Stream B at minimum pump flow 0.02 mL/min.

Reviews 5 / Uncertainty 1: Stream B at 0.02 mL/min is at E-series minimum; pulsation risk.

Reviews 5 / Uncertainty 2: Extended Stage 2 at 95 °C: amide or thioester degradation risk is untested hypothesis.

Reviews 5 / Uncertainty 3: Stage 1 thioester completeness at 25 min unknown.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 extended ($\tau1$=62.5 min, ratio 2.08) in ETFE 10 mL; Stage 2 near-batch ($\tau2$=25 min, ratio 0.83). Tests highly extended thioester formation with moderate aminolysis. Total time 87.5 min. Throughput 0.08 mmol/min.

Reviews 6 / Uncertainty 1: 62.5 min Stage 1 at 95 °C: DPDTC or thioester degradation is untested hypothesis.

Reviews 6 / Uncertainty 2: Stage 2 at 25 min may not complete aminolysis; unknown.

Reviews 6 / Uncertainty 3: Long total time conflicts with objective to avoid unnecessarily long residence times.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Both stages extended ($\tau1$=62.5 min ratio 2.08, $\tau2$=50 min ratio 1.67) using two ETFE 10 mL coils. Longest practical screen at 95 °C within inventory. Serves as upper-bound reference. Total time 112.5 min. Throughput 0.08 mmol/min.

Reviews 7 / Uncertainty 1: Extended holds at 95 °C: degradation of thioester intermediate or amide product is untested.

Reviews 7 / Uncertainty 2: Whether additional time beyond batch improves yield is speculative.

Reviews 7 / Uncertainty 3: Conflicts with objective to avoid unnecessarily long residence times; warranted only as upper bound.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1 sub-batch ($\tau1$=16.7 min, ratio 0.56) in 2 mL PFA; Stage 2 highly extended ($\tau2$=66.7 min, ratio 2.22) in ETFE 10 mL. Most extreme short-Stage-1 / long-Stage-2 asymmetry. Stream B at 0.03 mL/min near minimum.

Reviews 8 / Uncertainty 1: Short Stage 1 may leave significant unreacted acid; long Stage 2 cannot recover unconverted thioester.

Reviews 8 / Uncertainty 2: 66.7 min Stage 2 at 95 °C: amide degradation risk untested.

Reviews 8 / Uncertainty 3: Stream B near minimum pump flow; reproducibility uncertain.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 extended ($\tau1$=62.5 min, ratio 2.08) in ETFE 5 mL; Stage 2 sub-batch ($\tau2$=20 min, ratio 0.67) in PFA 2 mL. Identical hold ratios to candidate 11 but different Stage 1 reactor geometry (ETFE 2.4 mm vs PFA 1 mm). Throughput 0.04 mmol/min; Stream B at minimum 0.02 mL/min.

Reviews 9 / Uncertainty 1: Stage 2 at 20 min may be insufficient for complete aminolysis.

Reviews 9 / Uncertainty 2: Stream B at 0.02 mL/min at E-series minimum; pulsation risk.

Reviews 9 / Uncertainty 3: Extended Stage 1 degradation risk untested.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1 sub-batch ($\tau1$=25 min, ratio 0.83) in 2 mL PFA; Stage 2 highly extended ($\tau2$=100 min, ratio 3.33) in ETFE 10 mL. Most extreme long-Stage-2 candidate. Total time 125 min is longest in set. Stream B at minimum 0.02 mL/min.

Reviews 10 / Uncertainty 1: 100 min Stage 2 at 95 °C: amide degradation is untested; conflicts with no-unnecessary-hold objective.

Reviews 10 / Uncertainty 2: Stream B at 0.02 mL/min minimum flow; reproducibility uncertain.

Reviews 10 / Uncertainty 3: Stage 1 thioester completeness at 25 min unknown.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Achieved $\tau1$=62.5 min (ratio 2.08) in PFA 5 mL / $\tau2$=20 min (ratio 0.67) in PFA 2 mL. Note: target label 120/30 min is not achieved; actual holds are same as candidate 9 but Stage 1 uses PFA 1 mm coil vs ETFE 2.4 mm. Differentiates coil geometry effect on Stage 1 at same hold. Stream B at minimum 0.02 mL/min.

Reviews 11 / Uncertainty 1: Target 120 min Stage 1 is not achieved; actual is 62.5 min — label mismatch should be noted.

Reviews 11 / Uncertainty 2: Stage 2 at 20 min aminolysis completeness unknown.

Reviews 11 / Uncertainty 3: Stream B at minimum flow; pulsation risk.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Achieved $\tau1$=62.5 min (ratio 2.08) and $\tau2$=50 min (ratio 1.67) both in ETFE 5 mL coils. Note: target label 120/120 min is not achieved; actual holds are same flow parameters as candidate 7 Stage

1 but Stage 2 uses coil_row11 (5 mL) rather than coil_row12 (10 mL). Label mismatch. Stream B at minimum 0.02 mL/min.

Reviews 12 / Uncertainty 1: Target 120/120 min not achieved; actual 62.5/50 min — label mismatch should be noted.

Reviews 12 / Uncertainty 2: Extended holds at 95 °C: degradation risk untested.

Reviews 12 / Uncertainty 3: Stream B at minimum flow; reproducibility uncertain.

Call 2: DrKinetics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated thioester formation (Stage 1) then aminolysis (Stage 2) at 95 °C in 2-MeTHF. No measured flow kinetics, yield, or conversion data; batch 30+30 min at 95 °C are screening anchors only.

Shared assessment / Stream notes: Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M (1.0 equiv), DPDTC 0.525 M (1.05 equiv), DMAP 0.05 M (0.1 equiv) in 2-MeTHF. Stream B: benzylamine 2.1 M (1.05 equiv) in 2-MeTHF, introduced at Stage 2 mixer. Stoichiometry and concentrations are consistent across all candidates; molar flow scales with the selected scale factor.

Shared assessment / Pump compatibility: 2-MeTHF is excluded from all RED pump tubing (E-series and SF-10). All candidates correctly assign pump_e_series_blue (BLUE tubing). A second BLUE pump channel is required for Stream B; both streams share the same pump_equipment_id label 'pump_e_series_blue' — laboratory must confirm two independent BLUE channels are allocated, one per stream.

Shared assessment / Mixer topology: A T-mixer or Y-mixer must be installed between Stage 1 outlet and the Stage 2 reactor to introduce Stream B. No mixer is listed in any candidate's equipment set; laboratory must confirm a PEEK or ETFE T/Y-mixer is included in the physical assembly.

Shared assessment / Temperature limit: All reactors operate at 95 °C. PFA and ETFE coils are rated to 100 °C — 5 °C headroom only; FEP coils (max 50 °C) are not used in any candidate. Thermal headroom is adequate but narrow.

Shared assessment / Bpr: 7-bar fixed BPR selected for all candidates. 2-MeTHF bp ~80 °C at 1 bar; 7 bar suppresses boiling at 95 °C. Adequate.

Shared assessment / Kinetics: Uncharacterized. Stage 1 conversion is unknown in flow; it directly determines thioester available for Stage 2. No degradation kinetics available for any stage. Prolonged Stage 2 exposure risk (aminolysis side-reactions, thioester hydrolysis) is an untested hypothesis.

Shared assessment / Screening coverage: The 12 candidates span Stage 1 holds of ~17–63 min and Stage 2 holds of ~14–100 min, providing informative coverage of sub-batch, near-batch, and super-batch exposures for both stages.

Shared assessment / Objective policy: Balanced screen: establish interpretable baseline (near-batch), compare shorter and longer holds. No candidate is vetoed for being too short or too long absent a hard constraint violation.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 17.9 min (0.60× batch), Stage 2: 14.3 min (0.48× batch). Both sub-batch; tests whether reaction is faster than batch implies. Total 32 min. Highest limiting-feed throughput (0.14 mmol/min). Valid short-short baseline. Stage 1 thioester completeness unproven at this exposure.

Reviews 1 / Uncertainty 1: Stage 1 thioester formation may be incomplete at 17.9 min — unknown in flow.

Reviews 1 / Uncertainty 2: Stage 2 aminolysis adequacy at 14.3 min uncharacterized.

Reviews 1 / Uncertainty 3: Two independent BLUE pump channels required; laboratory must confirm allocation.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: prefer

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1: 16.7 min (0.56× batch), Stage 2: 33.3 min (1.11× batch). Asymmetric: shorter thioester formation, longer aminolysis. Informative if Stage 1 is fast and Stage 2 is rate-limiting. Mixed reactor geometry (PFA 2 mL + ETFE 5 mL) is valid. Good screening contrast with candidate 3.

Reviews 2 / Uncertainty 1: Stage 1 thioester completeness at 16.7 min unknown.

Reviews 2 / Uncertainty 2: Longer Stage 2 aminolysis side-reaction risk is an untested hypothesis.

Reviews 2 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 35.7 min (1.19× batch), Stage 2: 14.3 min (0.48× batch). Longer thioester formation, shorter aminolysis. Directly contrasts candidate 2; together they bracket the rate-limiting stage hypothesis. Same total time (50 min). Good informative pairing.

Reviews 3 / Uncertainty 1: Stage 2 aminolysis adequacy at 14.3 min uncharacterized.

Reviews 3 / Uncertainty 2: Prolonged Stage 1 reagent exposure risk (DPDTC degradation) is untested.

Reviews 3 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 31.3 min (1.04× batch), Stage 2: 25.0 min (0.83× batch). Closest to the batch protocol in both stages; serves as the primary flow baseline. Both ETFE 5 mL coils, uniform geometry. Essential reference point for the screen.

Reviews 4 / Uncertainty 1: Stage 2 at 25 min is 0.83× batch; adequacy for full aminolysis uncharacterized.

Reviews 4 / Uncertainty 2: Two independent BLUE pump channels required.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 25.0 min (0.83× batch), Stage 2: 50.0 min (1.67× batch). Sub-batch Stage 1 with extended Stage 2. Tests whether excess aminolysis time compensates for potentially incomplete thioester. Informative but Stage 2 prolongation risk is untested.

Reviews 5 / Uncertainty 1: Stage 1 thioester completeness at 25 min unknown.

Reviews 5 / Uncertainty 2: Aminolysis side-reactions at 50 min Stage 2 are an untested hypothesis.

Reviews 5 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 62.5 min (2.08× batch), Stage 2: 25.0 min (0.83× batch). Extended thioester formation with near-batch aminolysis. Informative for Stage 1 ceiling; Stage 1 DPDTC over-exposure risk untested. Complementary to candidate 9.

Reviews 6 / Uncertainty 1: DPDTC/thioester stability at 62.5 min Stage 1 uncharacterized.

Reviews 6 / Uncertainty 2: Stage 2 adequacy at 25 min uncharacterized.

Reviews 6 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 62.5 min (2.08× batch), Stage 2: 50.0 min (1.67× batch). Both stages well above batch; uses two ETFE 10 mL coils (both available). Tests degradation versus completion at long total exposure (112.5 min). Useful outer bound but extended exposure risks are untested.

Reviews 7 / Uncertainty 1: Degradation of thioester or amide product at combined 112.5 min exposure is untested.

Reviews 7 / Uncertainty 2: Two independent BLUE pump channels required.

Reviews 7 / Uncertainty 3: Two coil_row12 units consumed; laboratory must confirm both are available simultaneously.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 16.7 min (0.56× batch), Stage 2: 66.7 min (2.22× batch). Extreme asymmetry: very short Stage 1 with very long Stage 2. Informative only if Stage 1 is fast; otherwise incomplete thioester will limit amide yield regardless of Stage 2 length.

Reviews 8 / Uncertainty 1: Stage 1 thioester completeness at 16.7 min is the dominant unknown.

Reviews 8 / Uncertainty 2: 66.7 min Stage 2 aminolysis side-reaction risk is untested.

Reviews 8 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 62.5 min (2.08× batch), Stage 2: 20.0 min (0.67× batch). Extended Stage 1 with short Stage 2; tests whether thioester formation is the bottleneck. Achieved Stage 2 hold (20 min) is below batch; aminolysis adequacy unknown. Informative pairing with candidate 11.

Reviews 9 / Uncertainty 1: Stage 2 aminolysis at 20 min may be insufficient — uncharacterized.

Reviews 9 / Uncertainty 2: DPDTC/thioester stability at 62.5 min Stage 1 untested.

Reviews 9 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 25.0 min (0.83× batch), Stage 2: 100.0 min (3.33× batch). Extreme Stage 2 extension. Useful as an outer-bound degradation probe for aminolysis stage; however, 100 min Stage 2 exposure risk is entirely untested and this point adds limited yield information relative to candidates 5 and 8.

Reviews 10 / Uncertainty 1: Aminolysis side-reactions or product degradation at 100 min Stage 2 are untested.

Reviews 10 / Uncertainty 2: Stage 1 thioester completeness at 25 min unknown.

Reviews 10 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 62.5 min (2.08× batch) in PFA 5 mL (ID 1 mm), Stage 2: 20.0 min (0.67× batch) in PFA 2 mL. Identical achieved times to candidate 9 but different reactor geometry (narrower ID). Provides a geometry comparison at the same time coordinates; informative for axial dispersion differences.

Reviews 11 / Uncertainty 1: Stage 2 aminolysis at 20 min adequacy uncharacterized.

Reviews 11 / Uncertainty 2: Narrower ID (1 mm) vs. 2.4 mm may affect mixing; effect is untested.

Reviews 11 / Uncertainty 3: Two independent BLUE pump channels required.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 62.5 min (2.08× batch), Stage 2: 50.0 min (1.67× batch). Both stages extended; both use ETFE 5 mL coils (coil_row11 used twice — quantity is 2, confirmed available). Longest informative outer bound alongside candidate 7. Degradation risk at 112.5 min total is untested.

Reviews 12 / Uncertainty 1: Both stages extended well above batch; degradation risk at combined 112.5 min is untested.

Reviews 12 / Uncertainty 2: Two coil_row11 units required simultaneously; laboratory must confirm both available.

Reviews 12 / Uncertainty 3: Two independent BLUE pump channels required.

Call 3: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage telescoped DPDTC-mediated amide formation. Stage 1 thioester formation feeds Stage 2 amidation directly; thioester intermediate concentration entering Stage 2 is theoretical. Kinetics uncharacterized; batch 30+30 min at 95°C are screening anchors only.

Shared assessment / Streams: Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M (1.0 equiv), DPDTC 0.525 M (1.05 equiv), DMAP 0.05 M (0.1 equiv) in 2-MeTHF. Stream B: benzylamine 2.1 M in 2-MeTHF (1.05 equiv relative to substrate), introduced at Stage 2 T-mixer. Molar flows scale with Q_A × 0.5 mmol/mL.

Shared assessment / Pump solvent constraint: 2-MeTHF is excluded from RED pump tubing on all Vapourtec platforms. All candidates correctly use BLUE pump tubing (pump_e_series_blue). Verified compliant.

Shared assessment / Temperature limit: All reactors operate at 95°C. PFA and ETFE coils rated ≤100°C; FEP coils rated ≤50°C. No FEP coils appear in any candidate. All compliant.

Shared assessment / Bpr: 7 bar BPR applied across all candidates. 2-MeTHF bp ~80°C at 1 atm; elevated pressure suppresses boiling at 95°C. Appropriate.

Shared assessment / Mixer topology: Each candidate requires a T- or Y-mixer between Stage 1 outlet and Stream B inlet. Mixers are available (P-632 ETFE, P-713 PEEK, P-513 PEEK). No candidate explicitly names the mixer in equipment lists; this is an unresolved topology assignment requiring laboratory confirmation of connector selection before execution.

Shared assessment / Pump assignment note: Both streams assigned to pump_e_series_blue (a single pump ID). In practice two separate pump channels are required — one for Stream A, one for Stream B. Lab must verify two independent E-series blue channels are allocated.

Shared assessment / Concentration note: Stream B benzylamine at 2.1 M in 2-MeTHF: solubility of benzylamine in 2-MeTHF at this concentration has not been confirmed. Chemist verification required.

Shared assessment / Degradation risk: Whether prolonged Stage 1 exposure (>30 min) risks thioester decomposition or side-product formation at 95°C is uncharacterized. Extended Stage 2 exposure risk for amide product is similarly unknown. These are hypotheses, not established facts.

Shared assessment / Stage2 hold note: Stage 2 residence times below ~14 min (candidates 1, 3) provide less exposure than the batch anchor; whether Stage 2 kinetics allow adequate amide formation at these shorter times is unknown.

Shared assessment / No yield prediction: No yield, conversion, or throughput ranking is made. Computed comparisons provide hold ratios and limiting molar flows only.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1 τ 17.9 min (0.60× batch), Stage 2 τ 14.3 min (0.48× batch). Both stages below batch anchor; provides short-exposure baseline. Equipment valid: two ETFE 5 mL coils at 95°C, BLUE pumps, 7 bar BPR. Mixer topology unconfirmed.

Reviews 1 / Uncertainty 1: Stage 2 at 14.3 min may be insufficient for amide formation; uncharacterized kinetics.

Reviews 1 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 1 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: prefer

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1 τ 16.7 min (0.56× batch, PFA 2 mL), Stage 2 τ 33.3 min (1.11× batch, ETFE 5 mL). Near-batch Stage 2 exposure after shorter Stage 1; balanced screen point. Different reactor geometry per stage adds a variable but uses available inventory.

Reviews 2 / Uncertainty 1: Stage 1 thioester completeness at 16.7 min unknown; directly limits Stage 2 substrate.

Reviews 2 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 2 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 τ 35.7 min (1.19× batch, ETFE 10 mL), Stage 2 τ 14.3 min (0.48× batch, ETFE 5 mL). Long Stage 1 / short Stage 2 asymmetry tests thioester formation priority. Stage 2 below batch anchor is an unresolved kinetic risk.

Reviews 3 / Uncertainty 1: Stage 2 at 14.3 min may be insufficient for amidation; uncharacterized.

Reviews 3 / Uncertainty 2: Extended Stage 1 degradation risk at 95°C is uncharacterized.

Reviews 3 / Uncertainty 3: Mixer connector selection not explicitly assigned.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1 τ 31.3 min (1.04×), Stage 2 τ 25.0 min (0.83×); both stages near batch anchor. Single reactor type (ETFE 5 mL × 2) simplifies setup. Serves as the primary near-batch reference condition for a balanced screen.

Reviews 4 / Uncertainty 1: Stage 2 at 25 min is slightly below batch anchor; amide yield adequacy uncharacterized.

Reviews 4 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 4 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1 τ 25 min (0.83×, PFA 2 mL), Stage 2 τ 50 min (1.67×, ETFE 5 mL). Extended Stage 2 probes whether longer amidation exposure improves yield; degradation risk at extended hold is uncharacterized.

Reviews 5 / Uncertainty 1: Extended Stage 2 amide degradation risk at 95°C is uncharacterized.

Reviews 5 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 5 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 τ 62.5 min (2.08×, ETFE 10 mL), Stage 2 τ 25 min (0.83×, ETFE 5 mL). Tests whether extended thioester formation improves Stage 2 yield; extended Stage 1 degradation risk is uncharacterized. Mixer not assigned.

Reviews 6 / Uncertainty 1: Stage 1 thioester stability at 62.5 min / 95°C uncharacterized; degradation risk.

Reviews 6 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 6 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1 τ 62.5 min (2.08×, ETFE 10 mL), Stage 2 τ 50 min (1.67×, ETFE 10 mL). Both stages well above batch anchor; uses maximum available ETFE volume. Extended exposure degradation risk for both stages uncharacterized.

Reviews 7 / Uncertainty 1: Extended hold degradation risk for thioester (Stage 1) and amide product (Stage 2) uncharacterized.

Reviews 7 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 7 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1 τ 16.7 min (0.56×, PFA 2 mL), Stage 2 τ 66.7 min (2.22×, ETFE 10 mL). Short Stage 1 / very long Stage 2; tests amidation with potentially incomplete thioester feed. Stage 2 degradation risk elevated.

Reviews 8 / Uncertainty 1: Short Stage 1 may limit thioester available; incomplete conversion propagates to Stage 2.

Reviews 8 / Uncertainty 2: Stage 2 at 66.7 min extended degradation risk uncharacterized.

Reviews 8 / Uncertainty 3: Mixer connector selection not explicitly assigned.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 τ 62.5 min (2.08×, ETFE 5 mL), Stage 2 τ 20 min (0.67×, PFA 2 mL). Long Stage 1 ensures thioester formation; short Stage 2 tests minimum amidation time. Stage 2 below batch anchor is unresolved.

Reviews 9 / Uncertainty 1: Stage 2 at 20 min may be insufficient; uncharacterized amidation kinetics.

Reviews 9 / Uncertainty 2: Extended Stage 1 thioester stability at 95°C uncharacterized.

Reviews 9 / Uncertainty 3: Mixer connector selection not explicitly assigned.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1 τ 25 min (0.83×, PFA 2 mL), Stage 2 τ 100 min (3.33×, ETFE 10 mL). Very extended Stage 2 probes upper amidation limit; prolonged 95°C exposure degradation risk is the primary concern. Useful as an outer-bound data point.

Reviews 10 / Uncertainty 1: Stage 2 at 100 min / 95°C: amide degradation risk substantially elevated and uncharacterized.

Reviews 10 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 10 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1 τ 62.5 min (2.08×, PFA 5 mL), Stage 2 τ 20 min (0.67×, PFA 2 mL). Same hold profile as Candidate 9 but uses PFA coils for both stages; directly comparable geometry variant. Stage 2 below batch anchor.

Reviews 11 / Uncertainty 1: Stage 2 at 20 min may be insufficient; uncharacterized kinetics.

Reviews 11 / Uncertainty 2: Stage 1 thioester stability at 62.5 min uncharacterized.

Reviews 11 / Uncertainty 3: Mixer connector selection not explicitly assigned.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1 τ 62.5 min (2.08×), Stage 2 τ 50 min (1.67×), both ETFE 5 mL. Same total time as Candidate 7 but uses smaller reactors and lower flow scale; provides a throughput-geometry comparison. Extended degradation risk applies.

Reviews 12 / Uncertainty 1: Extended hold degradation risk for thioester and amide at 95°C uncharacterized.

Reviews 12 / Uncertainty 2: Mixer connector selection not explicitly assigned.

Reviews 12 / Uncertainty 3: Two independent pump channels required under single pump_e_series_blue ID.

Call 4: DrSafety, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated thioester activation (Stage 1) followed by benzylamine amidation (Stage 2). Both stages at 95 °C; BPR 7 bar suppresses 2-MeTHF boiling (bp ~80 °C at 1 bar). PFA/ETFE materials are compatible with 2-MeTHF.

Shared assessment / Pressure adequacy: 7 bar BPR provides adequate headroom above 2-MeTHF vapour pressure at 95 °C; all reactor max pressures ≥34 bar; pump max 10 bar with negligible pressure drops (<0.003 bar) across all candidates — no pressure hard violation in any candidate.

Shared assessment / Temperature limit: All reactors used at 95 °C are PFA (max 100 °C) or ETFE (max 100 °C). FEP coils (max 50 °C) are not used. Coil_row3 PFA 2 mL (max 100 °C) is used at 95 °C — within limit but only 5 °C margin; chemist should confirm stable thermostat control at this limit.

Shared assessment / Red pump exclusion: 2-MeTHF is excluded from RED pump tubing. All feeds assigned to BLUE E-series pump — no violation.

Shared assessment / Stream b concentration: Benzylamine Stream B is specified at 2.1 M in 2-MeTHF at 0.02–0.07 mL/min. Benzylamine neat density ~0.98 g/mL (MW 107.15) gives ~9.1 M neat; 2.1 M requires ~23% v/v. Solubility and solution stability in 2-MeTHF not confirmed — chemist verification required.

Shared assessment / Interstage mixing: Benzylamine (Stream B) must be introduced via a mixer (T or Y) between Stage 1 and Stage 2 reactors. Mixer type and availability are inventory-confirmed; connection is not detailed in designs but standard connectors are available.

Shared assessment / Dpdtc solubility: DPDTC at 0.525 M in 2-MeTHF at room temperature — solubility at flow startup (ambient) not confirmed; could precipitate before reaching 95 °C. Chemist confirmation required.

Shared assessment / No kinetic evidence: No measured flow kinetics, yield data, or conversion data available. Hold-time ratios are experimental screen coordinates only; no yield or conversion ranking is possible.

Shared assessment / Degradation risk: Extended hold times at 95 °C (Candidates 7, 8, 10, 12 with Stage 2 ≥50 min) carry an untested risk of thioester or amide degradation — this is a chemist-level hypothesis, not an established fact.

Shared assessment / Gas source: No gas stage in any candidate. MFC hardware not invoked.

Shared assessment / Cooling step: Batch protocol includes a 5–10 min cooling step before benzylamine addition. In flow, interstage cooling is not implemented; elimination of this step is an untested hypothesis and requires experimental confirmation that direct hot transfer to Stage 2 does not cause side reactions.

Shared assessment / Peristaltic pump flow accuracy: E-series peristaltic pumps at 0.02–0.08 mL/min (Stream B range) operate near minimum flow (0.02 mL/min) in several candidates; pulsation and accuracy at minimum settings require laboratory verification.

Shared assessment / Two pumps one label: Both streams are assigned to 'pump_e_series_blue' — in practice two separate pump heads/channels are needed. Three E-series BLUE pumps are available; this is not a hard violation but must be confirmed in physical setup.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Short sub-batch holds (18 min S1, 14 min S2) at 95 °C with BPR 7 bar. Equipment within limits. Serves as short-time screen anchor. Stream B at 0.07 mL/min is above minimum. No hard violations.

Reviews 1 / Uncertainty 1: S1 thioester activation may be incomplete at 18 min — uncharacterized kinetics.

Reviews 1 / Uncertainty 2: S2 amidation at 14 min sub-batch — yield impact unknown.

Reviews 1 / Uncertainty 3: Interstage cooling elimination untested.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: prefer

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: S1 ~17 min (sub-batch), S2 ~33 min (above batch). Balanced asymmetric screen: tests whether extended S2 compensates for shorter S1. Stream B at 0.03 mL/min — near minimum, needs verification. Equipment within limits.

Reviews 2 / Uncertainty 1: S2 33 min at 95 °C degradation risk uncharacterized.

Reviews 2 / Uncertainty 2: Stream B 0.03 mL/min peristaltic accuracy near minimum.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: S1 ~36 min (above batch), S2 ~14 min. Tests whether complete S1 thioester formation supports fast S2 amidation. Highest feed throughput among 50-min total candidates (0.14 mmol/min). Equipment within limits.

Reviews 3 / Uncertainty 1: S2 14 min may be insufficient for amidation — uncharacterized.

Reviews 3 / Uncertainty 2: S1 over-exposure degradation risk at 36 min untested.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Near-batch holds both stages (31 min S1, 25 min S2) at 95 °C. Single reactor type (coil_row11 ×2). Serves as primary batch-reference screen. Stream B 0.04 mL/min within range. No hard violations.

Reviews 4 / Uncertainty 1: S2 25 min below 30 min batch — amidation completeness uncertain.

Reviews 4 / Uncertainty 2: Peristaltic accuracy at 0.04 mL/min requires verification.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: S1 25 min, S2 50 min. Extended S2 tests whether longer amidation exposure improves yield. Stream B at 0.02 mL/min is at peristaltic minimum — accuracy risk. PFA coil_row3 at 95 °C near 100 °C limit.

Reviews 5 / Uncertainty 1: Stream B 0.02 mL/min is at E-series minimum — pulsation and accuracy unconfirmed.

Reviews 5 / Uncertainty 2: S2 50 min degradation risk at 95 °C untested.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: S1 62.5 min (2× batch), S2 25 min. Tests whether extended thioester activation at 95 °C is stable before amidation. Degradation risk of DPDTC/thioester at 62.5 min is untested but flagged. No hard violations.

Reviews 6 / Uncertainty 1: Thioester stability over 62.5 min at 95 °C unconfirmed — degradation hypothesis only.

Reviews 6 / Uncertainty 2: Stream B 0.04 mL/min peristaltic accuracy needs verification.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: S1 62.5 min, S2 50 min — both well above batch. Long total hold (112.5 min) at 95 °C increases degradation risk for both intermediate and product. Useful only if shorter candidates show incomplete reaction. No hard violations.

Reviews 7 / Uncertainty 1: Extended 95 °C exposure both stages — degradation risk for thioester and amide uncharacterized.

Reviews 7 / Uncertainty 2: Stream B 0.04 mL/min minimum accuracy concern.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: S1 ~17 min, S2 ~67 min. Very long S2 amidation tests upper exposure limit. Stream B at 0.03 mL/min near minimum. S2 67 min at 95 °C carries elevated degradation risk — untested hypothesis.

Reviews 8 / Uncertainty 1: S2 67 min at 95 °C — amide degradation risk uncharacterized.

Reviews 8 / Uncertainty 2: Stream B 0.03 mL/min near peristaltic minimum.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: S1 62.5 min (ETFE 5 mL), S2 20 min (PFA 2 mL at 95 °C, near 100 °C limit). Tests extended activation with short amidation. PFA coil_row3 at 95 °C: only 5 °C margin. Stream B 0.02 mL/min at minimum.

Reviews 9 / Uncertainty 1: PFA coil_row3 at 95 °C — only 5 °C below max; thermostat stability required.

Reviews 9 / Uncertainty 2: Stream B 0.02 mL/min at peristaltic minimum — accuracy risk.

Reviews 9 / Uncertainty 3: Thioester stability at 62.5 min untested.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: S1 25 min, S2 100 min. Very long S2 at 95 °C is an extreme upper bound; degradation risk is an untested concern. Stream B at 0.02 mL/min at minimum. Low throughput (0.04 mmol/min). Informative only if all shorter S2 candidates underperform.

Reviews 10 / Uncertainty 1: S2 100 min at 95 °C — product degradation risk uncharacterized.

Reviews 10 / Uncertainty 2: Stream B 0.02 mL/min at peristaltic minimum — pulsation risk.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: S1 62.5 min (PFA 5 mL), S2 20 min (PFA 2 mL at 95 °C). Functionally similar to Candidate 9; uses PFA coils only. PFA 2 mL at 95 °C near temperature limit. Stream B at 0.02 mL/min at minimum. Partially redundant with C9.

Reviews 11 / Uncertainty 1: PFA coil_row3 at 95 °C — 5 °C margin; thermostat control required.

Reviews 11 / Uncertainty 2: Stream B 0.02 mL/min at peristaltic minimum.

Reviews 11 / Uncertainty 3: Partially redundant with Candidate 9 — consider prioritising one.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: S1 62.5 min, S2 50 min, both ETFE 5 mL coils at 95 °C. Functionally similar to Candidate 7 (different reactor pairing). Long dual-stage exposure at 95 °C — degradation risk untested. Low throughput (0.04 mmol/min). Largely redundant with C7.

Reviews 12 / Uncertainty 1: Extended 95 °C in both stages — degradation risk uncharacterized.

Reviews 12 / Uncertainty 2: Functionally near-redundant with Candidate 7.

Reviews 12 / Uncertainty 3: Stream B 0.02 mL/min at peristaltic minimum.

Call 5: Skeptic, attempt 1
Status: recorded response, parsed successfully.

Vetoes: []

Shared assessment / Common issues 1: All candidates assign both Stream A and Stream B to 'pump_e_series_blue' as a single equipment_id, but two independent peristaltic channels are physically required. Laboratory must confirm two separate BLUE pump tubing channels are allocated before any candidate is executed.

Shared assessment / Common issues 2: No candidate explicitly assigns a T-mixer or Y-mixer for Stream B introduction at Stage 2. A union fitting is not a three-port mixer. Mixer selection must be confirmed from available inventory (P-632, P-713, P-513) before execution.

Shared assessment / Common issues 3: 2-MeTHF is listed in excluded_chemicals for all RED pump tubing. All candidates correctly route 2-MeTHF through BLUE pump tubing; this is consistent and not a violation.

Shared assessment / Common issues 4: PFA coils (coil_row3, coil_row4) have max_temperature_C of 100 °C; operating at 95 °C leaves only 5 °C margin. Thermostat stability at this margin requires laboratory verification for candidates 2, 5, 8, 9, 10, 11.

Shared assessment / Common issues 5: Benzylamine Stream B concentration of 2.1 M in 2-MeTHF is not confirmed as a soluble, stable, homogeneous solution; laboratory must verify before use.

Shared assessment / Common issues 6: DPDTC solubility at 0.525 M in 2-MeTHF at the operating temperature is unconfirmed across all candidates.

Shared assessment / Common issues 7: All achieved stage hold ratios are V/Q liquid indices, not proven measures of chemical exposure equivalence to batch; kinetics are entirely uncharacterized.

Shared assessment / Common issues 8: No BPR_available list items are present in the hard constraints; BPR assignments (7 bar) use bpr_row6 (fixed, 1 unit available). Laboratory must confirm this unit is unallocated.

Shared assessment / Common issues 9: Candidates 11 and 12 carry label mismatches: target stage times of 120/30 and 120/120 min respectively are not achieved; actual holds are 62.5/20 and 62.5/50 min. These are not hard violations but must be corrected in experimental records to avoid confusion.

Shared assessment / Common issues 10: Extended Stage 1 and/or Stage 2 holds at 95 °C introduce a plausible but entirely untested degradation risk for DPDTC, the thioester intermediate, and the amide product. This is a hypothesis, not an established constraint, and does not veto any candidate.

Shared assessment / Common issues 11: No gas stages, photochemistry, or analogy kinetic data are applicable to this chemistry. No yield, conversion, or residence-time superiority can be claimed for any candidate.

Candidate reviews 1 / Candidate: 1

Candidate reviews 1 / Recommendation: acceptable

Candidate reviews 1 / Hard violation reported: false

Candidate reviews 1 / Justification: Both stages sub-batch ($\tau1$=17.9 min, $\tau2$=14.3 min). Valid short-time exploratory baseline. Highest limiting-feed throughput (0.14 mmol/min). No hard violations; mixer and dual-channel pump allocation unconfirmed.

Candidate reviews 1 / Uncertainty 1: Dual BLUE channel allocation unconfirmed.

Candidate reviews 1 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 2 / Candidate: 2

Candidate reviews 2 / Recommendation: prefer

Candidate reviews 2 / Hard violation reported: false

Candidate reviews 2 / Justification: $\tau1$=16.7 min (0.56×), $\tau2$=33.3 min (1.11×). Asymmetric design with extended Stage 2 provides informative contrast against C3. PFA coil at 95 °C within 5 °C of limit; thermostat verification required.

Candidate reviews 2 / Uncertainty 1: Dual BLUE channel allocation unconfirmed.

Candidate reviews 2 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 3 / Candidate: 3

Candidate reviews 3 / Recommendation: prefer

Candidate reviews 3 / Hard violation reported: false

Candidate reviews 3 / Justification: $\tau1$=35.7 min (1.19×), $\tau2$=14.3 min (0.48×). Complementary asymmetry to C2; tests Stage 1 as rate-limiting step. Highest throughput among 50-min total candidates. No hard violations.

Candidate reviews 3 / Uncertainty 1: Stage 2 aminolysis adequacy at 14.3 min uncharacterized.

Candidate reviews 3 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 4 / Candidate: 4

Candidate reviews 4 / Recommendation: prefer

Candidate reviews 4 / Hard violation reported: false

Candidate reviews 4 / Justification: τ1=31.3 min (1.04×), τ2=25.0 min (0.83×). Nearest match to batch protocol; natural baseline. Uniform reactor geometry (ETFE 5 mL ×2). No hard violations.

Candidate reviews 4 / Uncertainty 1: Stage 2 at 0.83× batch anchor; amidation adequacy uncharacterized.

Candidate reviews 4 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 5 / Candidate: 5

Candidate reviews 5 / Recommendation: acceptable

Candidate reviews 5 / Hard violation reported: false

Candidate reviews 5 / Justification: τ1=25 min (0.83×), τ2=50 min (1.67×). Extended Stage 2 probes amidation upper limit. Stream B at 0.02 mL/min is E-series peristaltic minimum; pulsation accuracy requires verification.

Candidate reviews 5 / Uncertainty 1: Stream B at peristaltic minimum; pulsation risk.

Candidate reviews 5 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 6 / Candidate: 6

Candidate reviews 6 / Recommendation: acceptable

Candidate reviews 6 / Hard violation reported: false

Candidate reviews 6 / Justification: τ1=62.5 min (2.08×), τ2=25 min (0.83×). Extended Stage 1 tests thioester formation ceiling. Degradation of DPDTC/thioester at 62.5 min is an untested hypothesis, not a veto.

Candidate reviews 6 / Uncertainty 1: Thioester stability at 62.5 min/95 °C uncharacterized.

Candidate reviews 6 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 7 / Candidate: 7

Candidate reviews 7 / Recommendation: acceptable

Candidate reviews 7 / Hard violation reported: false

Candidate reviews 7 / Justification: τ1=62.5 min (2.08×), τ2=50 min (1.67×). Upper-bound reference using both ETFE 10 mL coils. Two coil_row12 units required; inventory shows quantity 2 — laboratory must confirm simultaneous availability.

Candidate reviews 7 / Uncertainty 1: Both coil_row12 units required simultaneously; laboratory must confirm.

Candidate reviews 7 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 8 / Candidate: 8

Candidate reviews 8 / Recommendation: acceptable

Candidate reviews 8 / Hard violation reported: false

Candidate reviews 8 / Justification: τ1=16.7 min (0.56×), τ2=66.7 min (2.22×). Most extreme short-S1/long-S2 design. Informative only if Stage 1 conversion is adequate; incomplete thioester cannot be remedied by extended Stage 2.

Candidate reviews 8 / Uncertainty 1: Short Stage 1 may propagate incomplete thioester into Stage 2.

Candidate reviews 8 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 9 / Candidate: 9

Candidate reviews 9 / Recommendation: acceptable

Candidate reviews 9 / Hard violation reported: false

Candidate reviews 9 / Justification: τ1=62.5 min (2.08×, ETFE 5 mL), τ2=20 min (0.67×, PFA 2 mL at 95 °C). PFA coil operates 5 °C below max; thermostat stability must be verified. Stream B at 0.02 mL/min at minimum.

Candidate reviews 9 / Uncertainty 1: PFA coil_row3 at 95 °C — 5 °C margin; thermostat verification required.

Candidate reviews 9 / Uncertainty 2: Stream B at peristaltic minimum.

Candidate reviews 10 / Candidate: 10

Candidate reviews 10 / Recommendation: acceptable

Candidate reviews 10 / Hard violation reported: false

Candidate reviews 10 / Justification: τ1=25 min (0.83×), τ2=100 min (3.33×). Extreme outer-bound for Stage 2. Adds limited incremental information beyond C8 and C5; useful only if all shorter Stage 2 candidates are incomplete. Degradation risk untested.

Candidate reviews 10 / Uncertainty 1: Stage 2 at 100 min/95 °C degradation risk uncharacterized.

Candidate reviews 10 / Uncertainty 2: Stage 2 mixer not assigned.

Candidate reviews 11 / Candidate: 11

Candidate reviews 11 / Recommendation: acceptable

Candidate reviews 11 / Hard violation reported: false

Candidate reviews 11 / Justification: Achieved τ1=62.5 min, τ2=20 min in PFA coils (both ID 1 mm). Label mismatch: target 120/30 min not achieved. Same hold profile as C9 but narrower-ID geometry; geometry comparison is scientifically valid.

Candidate reviews 11 / Uncertainty 1: Label mismatch (target 120/30 min vs achieved 62.5/20 min) must be corrected in records.

Candidate reviews 11 / Uncertainty 2: PFA coil_row3 at 95 °C — 5 °C margin.

Candidate reviews 12 / Candidate: 12

Candidate reviews 12 / Recommendation: acceptable

Candidate reviews 12 / Hard violation reported: false

Candidate reviews 12 / Justification: Achieved τ1=62.5 min, τ2=50 min, both ETFE 5 mL. Label mismatch: target 120/120 min not achieved. Near-redundant with C7 but uses smaller reactors at lower throughput — provides a throughput-geometry contrast.

Candidate reviews 12 / Uncertainty 1: Label mismatch (target 120/120 min vs achieved 62.5/50 min) must be corrected in records.

Candidate reviews 12 / Uncertainty 2: Stage 2 mixer not assigned.

Requested measurement 1: Confirm two independent BLUE peristaltic pump channels are available and allocated before any candidate is executed.

Requested measurement 2: Confirm BPR 7 bar (bpr_row6, quantity 1) is unallocated for the planned screen runs.

Requested measurement 3: Assign and install an explicit T-mixer or Y-mixer from inventory (P-632, P-713, or P-513) for Stream B introduction at the Stage 1/Stage 2 junction in every candidate.

Requested measurement 4: Verify thermostat stability at 95 °C for PFA coils (coil_row3, coil_row4) operating within 5 °C of their 100 °C maximum limit (affects candidates 2, 5, 8, 9, 10, 11).

Requested measurement 5: Confirm DPDTC solubility and homogeneity at 0.525 M in 2-MeTHF at ambient and 95 °C.

Requested measurement 6: Confirm benzylamine at 2.1 M in 2-MeTHF forms a stable, homogeneous solution at operating conditions.

Requested measurement 7: Verify E-series BLUE pump accuracy and pulsation behaviour at 0.02–0.03 mL/min (Stream B in candidates 5, 9, 10, 11, 12).

Requested measurement 8: Confirm simultaneous availability of both coil_row12 units for candidate 7, and both coil_row11 units for candidates 1 and 12.

Requested measurement 9: Correct label mismatches for candidates 11 (target 120/30 min, achieved 62.5/20 min) and 12 (target 120/120 min, achieved 62.5/50 min) in experimental records before execution.

Requested measurement 10: Assess thioester intermediate and amide product stability at 95 °C across the range of Stage 1 hold times (16.7–62.5 min) and Stage 2 hold times (14.3–100 min) through analytical sampling, as no stability data are currently available.

Call 6: Chief, attempt 1
Status: recorded response, parsed successfully.

Candidate: 4

Justification: Stage 1 τ=31.3 min (1.04× batch) and Stage 2 τ=25.0 min (0.83× batch) are the nearest achievable match to the batch 30/30 min protocol using a single uniform reactor type (ETFE 5 mL × 2, coil_row11). This provides the most interpretable baseline against which all other candidates are compared. Both holds are adequate relative to the batch anchor without unnecessarily extending exposure. No hard violations; no label mismatch; Stream B at 0.04 mL/min is above the peristaltic minimum.

Objective alignment: The objective requires maximising amide yield while avoiding unnecessarily long residence times. Candidate 4 anchors the screen at the batch reference condition without extending either stage beyond what the batch protocol demonstrated, satisfying both constraints simultaneously. It is the natural starting point for a balanced screen: shorter candidates (C1, C2, C3) test whether the batch times are unnecessarily long; longer candidates (C6, C7) test whether they are insufficient. Without kinetic data, C4 is the most defensible single selection for a first screen run because any yield gap relative to the batch result directly quantifies the flow penalty or equivalence, rather than conflating a time variable with a geometric variable.

Alternative 1 / Candidate: 3

Alternative 1 / Reason not selected: Stage 1 τ=35.7 min (1.19×) with Stage 2 τ=14.3 min (0.48×). Stage 2 is substantially below the batch anchor; with no kinetic data, incomplete amidation at 14.3 min is a plausible unresolved risk that makes this a higher-uncertainty candidate than C4 as the primary run. It is a scientifically valuable asymmetric contrast run after C4 results are in hand.

Alternative 2 / Candidate: 2

Alternative 2 / Reason not selected: Stage 1 τ=16.7 min (0.56×) and Stage 2 τ=33.3 min (1.11×). Stage 1 is well below the batch anchor; incomplete thioester formation in Stage 1 propagates directly into Stage 2 and cannot be remedied by extended aminolysis time. PFA coil operates within 5 °C of its temperature limit, adding a thermostat-stability requirement. C2 is informative as an asymmetric contrast candidate once C4 establishes the baseline.

Alternative 3 / Candidate: 6

Alternative 3 / Reason not selected: Stage 1 τ=62.5 min (2.08×) is more than double the batch anchor with an untested degradation risk for DPDTC and the thioester intermediate at 95 °C. Stage 2 at 25 min is also below batch. Total time 87.5 min conflicts with the objective to avoid unnecessarily long residence times unless shorter candidates show incomplete Stage 1 conversion.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: C4 directly serves the dual objective: near-batch holds avoid unnecessary extension, and the uniform geometry simplifies interpretation of any yield measurement against the batch result.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: The chemist hypothesis that the cooling step is a handling artefact is addressed by C4 running both stages at 95 °C without interstage cooling; whether this changes yield relative to the batch cooling step cannot be inferred and requires measurement.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: No historical yield data are available; C4 will generate the first flow yield reference point. This does not change the selection rationale but means no yield prediction is possible before execution.

Answer impact 4 / Question: Q-PREF-001

Answer impact 4 / Effect: Reporting requirements are fully met: Stream A at 0.5 M / 0.08 mmol·min^{-1} (1.0 equiv), Stream B at 2.1 M / 0.02 mmol·min^{-1} (1.05 equiv) in 2-MeTHF via BLUE pump; no gas stages. No yield is predicted.

Shared assessment / Common issues all candidates 1: Both Stream A and Stream B are assigned to 'pump_e_series_blue' as a single equipment_id; two independent BLUE peristaltic channels must be physically confirmed and allocated before any candidate is executed.

Shared assessment / Common issues all candidates 2: No candidate explicitly assigns a T- or Y-mixer for Stream B introduction at the Stage 1/Stage 2 junction. A union fitting is not a three-port mixer. An inventory mixer (P-632, P-713, or P-513) must be selected and installed.

Shared assessment / Common issues all candidates 3: BPR 7 bar (bpr_row6, fixed, quantity 1) must be confirmed as unallocated for the planned screen runs.

Shared assessment / Common issues all candidates 4: DPDTC solubility at 0.525 M in 2-MeTHF at ambient and 95 °C is unconfirmed across all candidates.

Shared assessment / Common issues all candidates 5: Benzylamine at 2.1 M in 2-MeTHF: homogeneity and stability at operating conditions unconfirmed.

Shared assessment / Common issues all candidates 6: PFA coils (coil_row3, coil_row4) operating at 95 °C are within 5 °C of their 100 °C maximum; thermostat stability must be verified for candidates 2, 5, 8, 9, 10, 11.

Shared assessment / Common issues all candidates 7: All stage hold times are V/Q liquid indices, not proven measures of chemical exposure equivalence to the batch; kinetics are entirely uncharacterized.

Shared assessment / Common issues all candidates 8: Candidates 11 and 12 carry label mismatches (targets 120/30 and 120/120 min; achieved 62.5/20 and 62.5/50 min); these must be corrected in experimental records before execution.

Shared assessment / Common issues all candidates 9: Extended holds at 95 °C (Stage 1 >35 min or Stage 2 >33 min) carry an untested degradation hypothesis for DPDTC, the thioester intermediate, and the amide product. This is a hypothesis, not a hard constraint, and does not veto any candidate.

Shared assessment / Common issues all candidates 10: No yield, conversion, or residence-time superiority can be claimed for any candidate; no analogy kinetic data are applicable.

Limitation 1: Stage 2 τ=25 min is 0.83× the batch anchor; whether aminolysis is complete at this hold is uncharacterized and cannot be inferred from available data.

Limitation 2: Equivalence between batch stirred-vessel exposure and flow plug-flow exposure at the same nominal time is not established.

Limitation 3: Throughput (0.08 mmol·min^{-1} limiting feed) is a feed throughput; product throughput depends on uncharacterized conversion and yield.

Limitation 4: Elimination of the interstage cooling step relative to the batch procedure is an untested change; its effect on Stage 2 yield is unknown.

Limitation 5: DPDTC and benzylamine solution stability and solubility at operating conditions are unconfirmed.

Suggested measurement 1: Confirm two independent E-series BLUE peristaltic pump channels are available and allocated.

Suggested measurement 2: Select and install a T- or Y-mixer from inventory for Stream B introduction at the Stage 2 inlet.

Suggested measurement 3: Confirm BPR 7 bar (bpr_row6) is unallocated.

Suggested measurement 4: Verify DPDTC solubility at 0.525 M in 2-MeTHF at ambient temperature and at 95 °C.

Suggested measurement 5: Verify benzylamine at 2.1 M in 2-MeTHF is homogeneous and stable at operating conditions.

Suggested measurement 6: After C4 execution, collect inline or offline analytical data (HPLC or UV) on the Stage 1 effluent to quantify thioester formation before committing to the asymmetric candidates (C2, C3) or extended candidates (C6, C7).

Suggested measurement 7: Assess thioester intermediate and amide product stability at 95 °C by sampling at Stage 1 exit and Stage 2 exit across the range of tested hold times.

Suggested measurement 8: Compare C4 amide yield against the batch result to quantify the flow/batch equivalence and determine whether the cooling step omission has any detectable effect.

Editorial cross-checks (not model statements)

Numerical transcription: in the Set 1 Chief answer_impacts entry for Q-PREF-001, benzylamine molar flow is written as 0.02 mmol/min. The final feed and independent arithmetic give 2.1 mol/L x 0.04 mL/min = 0.084 mmol/min, which is 1.05 times the acid feed of 0.08 mmol/min. For Sets 2/3, the corresponding amine rate is 0.147 mmol/min and acid rate is 0.14 mmol/min. The copied council text remains unedited; these corrected values govern the reporting.

Inference limits: wording such as adequate exposure, confirmed photon geometry, or a single run distinguishing the rate-limiting step is not accepted as an experimentally demonstrated conclusion. A nominal hold near the batch duration does not establish equal conversion in flow. In particular, the Set 2 claim that incomplete Stage 1 conversion must have another cause because its hold exceeds batch is not justified. Changes in mixing, temperature distribution, feed state and residence-time distribution remain possible. The selected designs are therefore described as screening hypotheses.

Inventory reconciliation: references to a missing mixer or one pump label belong to the council's view of the candidate record. The final allocation explicitly assigns the interstage P-713 mixer and separate stream-pump operations. This resolves the reported assignment at the software-record level, not by claiming that the council physically installed equipment or verified the laboratory configuration.

F6-S2: Figure 6, Set 2
Source: outputs/khu_revised_six_20260915/figure6_set2/attempt_01. Selected Candidate 3; 12-candidate pool; downstream model claude-sonnet-4-6. The objective policy is yield_priority.

Call 1: DrChemistry, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated thioester formation (S1) followed by aminolysis with benzylamine (S2). Identity confirmed by chemist. No isolation between stages. All candidates use 2-MeTHF with E-series blue pump (2-MeTHF excluded from red pump tubing — correctly observed). Stream B benzylamine at 2.1 M introduced at S2 mixer. Stoichiometry closure verified across all candidates.

Shared assessment / Temperature: All candidates run at 95 °C matching batch protocol. PFA and ETFE coils rated to 100 °C — within limit. FEP coils (max 50 °C) not used here.

Shared assessment / Pressure: BPR 7 bar selected consistently. 2-MeTHF bp ~80 °C at 1 atm; 7 bar back-pressure suppresses vaporisation at 95 °C. Adequate.

Shared assessment / Kinetics: Uncharacterized. Batch anchors (30 min each stage, 95 °C) are reference, not proven flow optima. Hold ratios are nominal V/Q indices, not validated exposure equivalents.

Shared assessment / Shared uncertainties 1: DPDTC and thioester intermediate solubility at 0.5 M in 2-MeTHF under flow at 95 °C unconfirmed — precipitation risk in narrow-bore coils.

Shared assessment / Shared uncertainties 2: Thioester intermediate stability at prolonged 95 °C exposure uncharacterized; longer S1 holds carry untested degradation risk.

Shared assessment / Shared uncertainties 3: Benzylamine at 2.1 M Stream B — concentration and solubility in 2-MeTHF not confirmed experimentally.

Shared assessment / Shared uncertainties 4: No inline analytics; conversion adequacy in either stage is unverified.

Shared assessment / Shared uncertainties 5: Peristaltic pump (E-series) flow accuracy at low rates (0.02–0.04 mL/min) may introduce stoichiometric drift.

Shared assessment / Shared uncertainties 6: Ambient atmosphere protocol (no inert purge) carried into flow — oxygen sensitivity of thioester intermediate uncharacterized.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: reject

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: S1 hold 17.9 min (ratio 0.60) and S2 hold 14.3 min (ratio 0.48) are both well below batch anchors with no kinetic justification. Under yield-priority policy, both stages carry meaningful under-exposure risk for an uncharacterized reaction. Too short to serve as a credible starting point.

Reviews 1 / Uncertainty 1: S1 thioester conversion adequacy at 17.9 min entirely unknown.

Reviews 1 / Uncertainty 2: S2 aminolysis completeness at 14.3 min unknown.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: prefer

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: S1 hold 16.7 min (ratio 0.56) is modestly short but S2 hold 33.3 min (ratio 1.11) exceeds batch anchor — appropriate for yield-priority given S2 aminolysis is the product-forming step. PFA 2 mL coil (S1) and ETFE 5 mL (S2) are correctly rated. Practical flow rates. Well-balanced exploratory starting point per objective.

Reviews 2 / Uncertainty 1: S1 conversion at 16.7 min sub-batch — thioester yield feeding S2 unknown.

Reviews 2 / Uncertainty 2: Stream B 2.1 M benzylamine solubility unconfirmed.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: S1 hold 35.7 min (ratio 1.19) exceeds batch anchor — supports high thioester conversion before amine addition. S2 hold 14.3 min (ratio 0.48) is sub-batch but aminolysis with benzylamine may be fast; this asymmetry is a reasonable exploratory hypothesis under yield priority. Higher throughput than C2.

Reviews 3 / Uncertainty 1: S2 aminolysis completeness at 14.3 min unverified — could limit final amide yield.

Reviews 3 / Uncertainty 2: Extended S1 thioester stability at 95 °C uncharacterized.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: S1 31.3 min (ratio 1.04) and S2 25.0 min (ratio 0.83) bracket the batch anchors closely in both stages. Both ETFE 5 mL coils at 95 °C. Balanced near-batch design is a strong yield-priority anchor; S2 slightly sub-batch is acceptable as a starting screen point.

Reviews 4 / Uncertainty 1: S2 at 25 min sub-batch; aminolysis extent unconfirmed.

Reviews 4 / Uncertainty 2: ETFE 5 mL coil used twice — inventory quantity is 2, permissible.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: S1 25.0 min (ratio 0.83) slightly sub-batch; S2 50.0 min (ratio 1.67) substantially exceeds anchor. Yield-priority rationale for extended S2 is plausible if aminolysis is slow, but prolonged 95 °C exposure in S2 carries untested degradation risk. Informative but not the preferred starting design.

Reviews 5 / Uncertainty 1: S1 thioester conversion adequacy at 25 min unverified.

Reviews 5 / Uncertainty 2: S2 product degradation risk at 50 min/95 °C uncharacterized.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: S1 62.5 min (ratio 2.08) is 2× batch anchor — plausible for ensuring full thioester conversion but carries untested intermediate degradation risk. S2 25.0 min (ratio 0.83) is sub-batch. Useful for mapping S1 long-hold sensitivity but less well-justified than C3/C4 for a starting design.

Reviews 6 / Uncertainty 1: Thioester stability at 62.5 min/95 °C uncharacterized.

Reviews 6 / Uncertainty 2: S2 aminolysis extent at 25 min unconfirmed.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: S1 62.5 min (ratio 2.08) and S2 50.0 min (ratio 1.67) both exceed batch anchors substantially. Uses two ETFE 10 mL coils — inventory has 2 units, permissible. Extended dual-stage hold is

useful for boundary mapping but lengthy total time (112.5 min) and degradation risk in both stages are unresolved.

Reviews 7 / Uncertainty 1: Both stages carry untested intermediate/product degradation risk at 95 °C.

Reviews 7 / Uncertainty 2: Total 112.5 min run time reduces practical screening throughput.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: S1 62.5 min and S2 100.0 min (ratio 3.33) — extreme S2 hold well beyond any kinetic justification available. Total 162.5 min is the longest in the set. Useful only if prior shorter-hold screens show incomplete S2 conversion. Not a preferred starting point under yield-priority/shortest-justified-time objective.

Reviews 8 / Uncertainty 1: S2 product degradation at 100 min/95 °C entirely uncharacterized.

Reviews 8 / Uncertainty 2: Very low flow rates (0.02 mL/min Stream B) may cause peristaltic metering errors.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: S1 62.5 min (ratio 2.08) and S2 50.0 min (ratio 1.67), same time profile as C7 but uses two ETFE 5 mL coils (inventory: 2 units, permissible). Duplicate hold ratios to C7; both provide similar boundary information. Acceptable as an instrument-variant of C7.

Reviews 9 / Uncertainty 1: Same degradation risk profile as C7.

Reviews 9 / Uncertainty 2: Very low flow rates (0.08/0.02 mL/min) near peristaltic accuracy limit.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: S1 ETFE 5 mL, S2 PFA 5 mL — same hold ratios as C9 (62.5/50.0 min). Mixing a wide-bore ETFE S1 with narrow-bore PFA S2 adds no clear chemical benefit and introduces minor hydraulic discontinuity; otherwise equivalent to C9. Informative as a reactor-geometry variant.

Reviews 10 / Uncertainty 1: ETFE-to-PFA transition at S2 mixer — connection compatibility requires lab confirmation.

Reviews 10 / Uncertainty 2: Same degradation risk as C7/C9.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: S1 25.0 min (ratio 0.83) PFA 2 mL, S2 50.0 min (ratio 1.67) PFA 5 mL. Mirrors C5 in time profile but uses all-PFA narrow-bore coils. Extended S2 at 95 °C degrades risk applies; S1 sub-batch. Useful geometry variant for comparison against ETFE wide-bore candidates but not a preferred starting design.

Reviews 11 / Uncertainty 1: DPDTC/thioester precipitation risk may be higher in 1 mm bore at 95 °C.

Reviews 11 / Uncertainty 2: S1 conversion adequacy at 25 min unverified.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: S1 125 min (ratio 4.17) is far beyond any kinetic rationale from batch data. Longest S1 in the set; intermediate degradation risk is the primary concern. S2 50 min (ratio 1.67) is adequate. Useful only as an extreme boundary point; should not be a first-run candidate under the shortest-justified-time objective.

Reviews 12 / Uncertainty 1: Thioester degradation at 125 min/95 °C in S1 uncharacterized — may reduce S2 substrate.

Reviews 12 / Uncertainty 2: Total 175 min run impractical for initial screening.

Call 2: DrKinetics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated thioester-to-amide sequence. No measured flow kinetics, yield, or degradation data. Batch anchors are 30 min each at 95 °C. Kinetics uncharacterized; adequacy of any sub-batch hold is unknown.

Shared assessment / Pump assignment: All candidates use pump_e_series_blue. 2-MeTHF is excluded from RED tubing; BLUE is correct throughout.

Shared assessment / Bpr: 7 bar BPR selected; 2-MeTHF bp ~80 °C, operation at 95 °C requires pressurization to maintain liquid phase. 7 bar provides adequate margin.

Shared assessment / Stoichiometry: Stream A: acid 0.5 M (1.0 equiv), DPDTC 0.525 M (1.05 equiv), DMAP 0.05 M (0.1 equiv) in 2-MeTHF. Stream B: benzylamine 2.1 M (1.05 equiv) in 2-MeTHF, added at Stage 2. Concentration ratio 4:1 (A:B flow) preserves stoichiometry. Intermediate thioester concentration in Stage 2 is diluted upon Stream B addition.

Shared assessment / Yield priority note: Objective requests shortest reasonably justified times for conversion-focused screening. Stage 1 sub-batch times carry uncharacterized risk of incomplete thioester formation; Stage 2 sub-batch times carry uncharacterized risk of incomplete aminolysis. Extended times carry no proven degradation risk but are not required by the objective. No yield or degradation data exist to rank candidates by expected outcome.

Shared assessment / Common uncertainties 1: No measured flow kinetics; sub-batch hold adequacy for Stage 1 thioester conversion is unknown.

Shared assessment / Common uncertainties 2: No measured flow kinetics; sub-batch hold adequacy for Stage 2 amide conversion is unknown.

Shared assessment / Common uncertainties 3: Degradation of thioester or amide at 95 °C over extended holds is uncharacterized.

Shared assessment / Common uncertainties 4: Stream B benzylamine concentration (2.1 M) in 2-MeTHF: solubility not confirmed.

Shared assessment / Common uncertainties 5: Mixing quality at T-junction between stages not characterized for these low-Re laminar flows.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: reject

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1 τ=17.9 min (ratio 0.60) and Stage 2 τ=14.3 min (ratio 0.48) are both substantially sub-batch. For a conversion-priority objective both stages carry unmitigated risk of incomplete reaction. No data justify either short hold; this is the weakest coverage across both stages simultaneously.

Reviews 1 / Uncertainty 1: Stage 1 thioester conversion at 0.60× batch time uncharacterized.

Reviews 1 / Uncertainty 2: Stage 2 amide conversion at 0.48× batch time uncharacterized.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: prefer

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1 τ=16.7 min (ratio 0.56) is sub-batch but Stage 2 τ=33.3 min (ratio 1.11) exceeds batch anchor, providing conservative amide-forming coverage. Asymmetric design tests whether Stage 1 is rate-limiting while protecting Stage 2. Practical volumes (2 mL + 5 mL) and moderate flow rates. Lowest equipment footprint among asymmetric options.

Reviews 2 / Uncertainty 1: Stage 1 thioester conversion at 0.56× batch time uncharacterized.

Reviews 2 / Uncertainty 2: Stage 2 longer hold degradation risk uncharacterized but no evidence of concern.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 τ=35.7 min (ratio 1.19) exceeds batch anchor, supporting thioester conversion. Stage 2 τ=14.3 min (ratio 0.48) is sub-batch; risk of incomplete aminolysis. Complementary to candidate 2; together these bracket stage-time sensitivity. Higher throughput (0.14 mmol/min) than candidate 2.

Reviews 3 / Uncertainty 1: Stage 2 amide conversion at 0.48× batch time uncharacterized.

Reviews 3 / Uncertainty 2: Thioester stability over 35.7 min Stage 1 hold uncharacterized.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1 τ=31.3 min (ratio 1.04) and Stage 2 τ=25.0 min (ratio 0.83) both near batch anchor. Provides closest analog to batch conditions across both stages simultaneously. Good screening anchor for conversion-priority objective with practical volumes (5+5 mL ETFE) and moderate flow.

Reviews 4 / Uncertainty 1: Stage 2 at 0.83× batch time may be insufficient; aminolysis kinetics uncharacterized.

Reviews 4 / Uncertainty 2: Whether near-batch times translate to equivalent conversion in flow is unestablished.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1 τ=25.0 min (ratio 0.83) sub-batch; Stage 2 τ=50.0 min (ratio 1.67) substantially exceeds batch. Protects Stage 2 amide yield at cost of Stage 1 uncertainty. Lowest stream A flow (0.08 mL/min) reduces throughput. Useful if Stage 2 is slow; redundant if candidate 2 screens adequately.

Reviews 5 / Uncertainty 1: Stage 1 thioester conversion at 0.83× batch time uncharacterized.

Reviews 5 / Uncertainty 2: Stage 2 extended hold degradation uncharacterized.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 τ=62.5 min (ratio 2.08) substantially exceeds batch; Stage 2 τ=25.0 min (ratio 0.83) is sub-batch. Over-invests time in Stage 1 relative to objective's shortest-justified guidance. Stage 2 risk of incomplete aminolysis remains. Useful only if Stage 1 is confirmed slow; redundant alongside candidate 3.

Reviews 6 / Uncertainty 1: Stage 1 extended hold: thioester stability at 2.08× batch uncharacterized.

Reviews 6 / Uncertainty 2: Stage 2 amide conversion at 0.83× batch time uncharacterized.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1 τ=62.5 min (ratio 2.08), Stage 2 τ=50.0 min (ratio 1.67); both stages substantially exceed batch. Maximizes exposure coverage but conflicts with objective's 'shortest reasonably justified' guidance. Two 10 mL ETFE coils required; large footprint. Informative only if all shorter candidates show incomplete conversion.

Reviews 7 / Uncertainty 1: Extended Stage 1 thioester stability uncharacterized.

Reviews 7 / Uncertainty 2: Extended Stage 2 degradation risk uncharacterized.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: reject

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 2 τ=100.0 min (ratio 3.33) is extreme overkill relative to batch anchor and directly contradicts the objective's 'shortest reasonably justified' instruction. Total time 162.5 min offers no screening advantage over candidates 7 or 12. Not justified by conversion-priority objective.

Reviews 8 / Uncertainty 1: Stage 2 extended hold degradation at 3.33× batch uncharacterized.

Reviews 8 / Uncertainty 2: Product stability over 100 min at 95 °C unconfirmed.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 τ=62.5 min (ratio 2.08), Stage 2 τ=50.0 min (ratio 1.67); same hold ratios as candidate 7 but uses two 5 mL coils (smaller footprint). Numerically identical in exposure to candidate 7; same concerns apply. Acceptable if extended exploration is warranted.

Reviews 9 / Uncertainty 1: Extended Stage 1 and Stage 2 holds carry uncharacterized degradation risk.

Reviews 9 / Uncertainty 2: No benefit over candidate 7 in screening coverage.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1 τ=62.5 min, Stage 2 τ=50.0 min; same ratios as candidates 7 and 9. Distinguishing feature: Stage 2 uses PFA coil_row4 (1 mm ID) vs ETFE. Useful to detect material-dependent effects. Otherwise redundant; same exposure concerns as 7 and 9.

Reviews 10 / Uncertainty 1: Whether narrower ID affects mixing or residence time distribution at these Re is uncharacterized.

Reviews 10 / Uncertainty 2: Extended hold degradation risk identical to candidates 7 and 9.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1 τ=25.0 min (ratio 0.83), Stage 2 τ=50.0 min (ratio 1.67). Same hold ratios as candidate 5 but uses different reactor combination (PFA 2+5 mL). Distinguishes any material or ID effect vs candidate 5. Marginally acceptable; largely redundant with candidate 5.

Reviews 11 / Uncertainty 1: Stage 1 at 0.83× batch time: thioester conversion uncharacterized.

Reviews 11 / Uncertainty 2: Redundancy with candidate 5 limits screening value.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: reject

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1 τ=125.0 min (ratio 4.17) is far beyond any conversion-justified need and directly contradicts the objective. Lowest throughput (0.04 mmol/min). Total time 175 min. No screening rationale supports this extreme Stage 1 over-exposure without prior evidence of very slow thioester formation.

Reviews 12 / Uncertainty 1: Stage 1 thioester stability at 4.17× batch time uncharacterized.

Reviews 12 / Uncertainty 2: Extremely low throughput reduces screening efficiency.

Call 3: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage thermal amide synthesis via thioester intermediate. Both stages at 95°C, no gas, no light. All candidates use E-series blue pump (2-MeTHF compatible). BPR 7 bar selected throughout. Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M (1.0 equiv), DPDTC 0.525 M (1.05 equiv), DMAP 0.05 M (0.1 equiv) in 2-MeTHF. Stream B: benzylamine 2.1 M (1.05 equiv) in 2-MeTHF, added at Stage 2. Stoichiometric flow closure verified for all candidates. No hard violations identified across the set.

Shared assessment / Common uncertainties 1: Kinetics uncharacterized: flow residence times are not equivalent to batch exposure; hold-ratio comparisons are geometric indices only.

Shared assessment / Common uncertainties 2: Thioester intermediate stability under extended 95°C hold is unknown; degradation risk increases with Stage 1 time but is unquantified.

Shared assessment / Common uncertainties 3: No measured historical yield data; no analogy record is chemically applicable.

Shared assessment / Common uncertainties 4: Stage 2 residence time is calculated from combined Q_liquid after Stream B addition; benzylamine mixing adequacy at T-mixer is unconfirmed.

Shared assessment / Common uncertainties 5: E-series peristaltic pump flow accuracy at minimum scale (0.02 mL/min Stream B) is a practical concern for stoichiometric fidelity.

Shared assessment / Common uncertainties 6: DPDTC solubility at 0.525 M in 2-MeTHF at room temperature (for feed preparation) is unconfirmed.

Shared assessment / Common uncertainties 7: Ambient atmosphere in batch protocol (no inert purge) is replicated in flow; oxidative or moisture sensitivity of thioester intermediate in flow is untested.

Shared assessment / Common uncertainties 8: PFA coil_row3 max temperature listed as 100°C; operation at 95°C is within limit but margin is narrow and requires confirmation of heating module compatibility.

Shared assessment / Common uncertainties 9: FEP coils (coil_row6, coil_row7) max temperature 50°C: excluded from 95°C operation — none assigned to any candidate, confirmed correct.

Shared assessment / Common uncertainties 10: Candidates 7 and 12 use two ETFE coil_row12 units; inventory lists quantity 2, so simultaneous use of both is at capacity limit and requires laboratory confirmation.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 17.9 min (ratio 0.60×batch); Stage 2: 14.3 min (ratio 0.48×batch). Both stages substantially sub-batch. Useful as shortest-time anchor to detect whether batch times are conservative. Higher throughput (0.14 mmol/min limiting feed). Both stages in ETFE coil_row11 (5 mL, 2.4 mm). Conversion adequacy at these reduced times is the primary experimental question.

Reviews 1 / Uncertainty 1: Sub-batch times for both stages; thioester formation and amide coupling may be incomplete.

Reviews 1 / Uncertainty 2: Two uses of coil_row11 (qty 2 available): confirmed within inventory.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: prefer

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1: 16.7 min (ratio 0.56×) in compact 2 mL PFA coil; Stage 2: 33.3 min (ratio 1.11×) in 5 mL ETFE. Asymmetric allocation directs longer exposure to the amide-forming step, consistent with yield-priority objective. Practical reactor volumes; 0.06 mmol/min limiting feed. Reasonable starting design balancing conversion adequacy against excess time.

Reviews 2 / Uncertainty 1: Stage 1 sub-batch; thioester yield feeding Stage 2 uncharacterized.

Reviews 2 / Uncertainty 2: PFA coil_row3 at 95°C: within 100°C limit but heating module compatibility requires confirmation.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 35.7 min (ratio 1.19×) in 10 mL ETFE; Stage 2: 14.3 min (ratio 0.48×). Inverted allocation: extended Stage 1 supports thioester formation but Stage 2 is well sub-batch. Higher throughput (0.14 mmol/min). Useful for mapping Stage 1 sensitivity. Stage 2 conversion adequacy is the key experimental question.

Reviews 3 / Uncertainty 1: Stage 2 at 0.48× batch may be insufficient for full amide conversion.

Reviews 3 / Uncertainty 2: Extended Stage 1 at 95°C raises unquantified thioester degradation risk.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 31.25 min (ratio 1.04×); Stage 2: 25 min (ratio 0.83×). Both stages near-batch. Best candidate for replicating batch conditions in flow as a reference point. 0.08 mmol/min limiting feed. Two uses of coil_row11 (qty 2): at inventory capacity, confirmed available.

Reviews 4 / Uncertainty 1: Stage 2 at 0.83× batch; amide conversion may be marginally sub-complete.

Reviews 4 / Uncertainty 2: Near-batch times do not guarantee equivalent conversion in flow without measured confirmation.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 25 min (0.83×) in 2 mL PFA; Stage 2: 50 min (1.67×) in 5 mL ETFE. Stage 2 extended to 1.67× batch, providing ample amide-forming exposure. Stage 1 slightly sub-batch. 0.04 mmol/min limiting feed is at minimum scale; pump accuracy concern heightened for Stream B at 0.02 mL/min.

Reviews 5 / Uncertainty 1: Stream B at 0.02 mL/min approaches E-series minimum; stoichiometric accuracy uncertain.

Reviews 5 / Uncertainty 2: Stage 1 at 0.83× batch may limit thioester supply for Stage 2.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 62.5 min (2.08×) in 10 mL ETFE; Stage 2: 25 min (0.83×). Long Stage 1 explores upper thioester formation bound; Stage 2 sub-batch. Useful for degradation vs. conversion trade-off study in Stage 1. 0.08 mmol/min limiting feed.

Reviews 6 / Uncertainty 1: 62.5 min Stage 1 at 95°C: thioester degradation risk unquantified and is a hypothesis, not an established concern.

Reviews 6 / Uncertainty 2: Stage 2 at 0.83× batch may be sub-complete for amide coupling.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 62.5 min (2.08×); Stage 2: 50 min (1.67×); both extended above batch. Uses two coil_row12 units (inventory qty 2: at capacity). Total time 112.5 min. Provides upper-bound exposure data for both stages. Degradation risk hypothesis applies to Stage 1.

Reviews 7 / Uncertainty 1: Two coil_row12 simultaneously: at inventory limit, requires laboratory confirmation.

Reviews 7 / Uncertainty 2: Extended Stage 1 at 95°C: thioester degradation is an untested hypothesis.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 62.5 min (2.08×) in 5 mL ETFE; Stage 2: 100 min (3.33×) in 10 mL ETFE. Longest Stage 2 in the set. Most conservative amide exposure but lowest throughput (0.04 mmol/min; Stream B 0.02 mL/min). Marginal pump accuracy concern applies.

Reviews 8 / Uncertainty 1: Stream B at 0.02 mL/min: E-series minimum flow accuracy concern.

Reviews 8 / Uncertainty 2: 100 min Stage 2 at 95°C: prolonged amide exposure degradation risk is a hypothesis.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 62.5 min (2.08×); Stage 2: 50 min (1.67×). Identical hold ratios to Candidate 7 but uses two coil_row11 (qty 2, confirmed) rather than two coil_row12. More confident inventory assignment than Candidate 7. 0.04 mmol/min limiting feed; minimum flow concern for Stream B.

Reviews 9 / Uncertainty 1: Stream B at 0.02 mL/min: peristaltic pump minimum flow accuracy.

Reviews 9 / Uncertainty 2: Extended Stage 1 degradation risk remains a hypothesis.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 62.5 min ETFE coil_row11; Stage 2: 50 min PFA coil_row4 (1 mm ID). Same hold ratios as Candidate 9 but Stage 2 reactor differs (narrower ID, higher pressure drop 0.004 bar). No hard constraint breach. Inventory and flow closure confirmed.

Reviews 10 / Uncertainty 1: PFA coil_row4 at 95°C and 0.1 mL/min: within limits, but narrow ID may influence mixing post-T-merger.

Reviews 10 / Uncertainty 2: Same minimum-flow concern for Stream B at 0.02 mL/min.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 25 min (0.83×) in 2 mL PFA; Stage 2: 50 min (1.67×) in 5 mL PFA. Same hold ratios as Candidate 5 but both reactors PFA coil (narrower ID). Inventory closure confirmed (coil_row3 qty 1, coil_row4 qty 2). Stream B 0.02 mL/min minimum-flow concern applies.

Reviews 11 / Uncertainty 1: Stream B at 0.02 mL/min: minimum flow accuracy concern.

Reviews 11 / Uncertainty 2: PFA coil_row3 at 95°C: within 100°C limit, heating module compatibility unconfirmed.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 125 min (4.17×) in 10 mL ETFE — the longest Stage 1 in the set. Stage 2: 50 min (1.67×). Represents extreme Stage 1 upper bound for degradation/conversion mapping. 0.04 mmol/min limiting feed; Stream B 0.02 mL/min minimum-flow concern.

Reviews 12 / Uncertainty 1: 125 min Stage 1 at 95°C greatly exceeds batch reference; thioester stability over this duration is entirely uncharacterized.

Reviews 12 / Uncertainty 2: Stream B at 0.02 mL/min: stoichiometric accuracy at E-series minimum is uncertain.

Call 4: DrSafety, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage thermal amide formation via DPDTC thioester intermediate in 2-MeTHF at 95 °C. No measured flow kinetics; batch holds are screening anchors only.

Shared assessment / Temperature: 95 °C is at the declared max_temperature_C for all PFA and ETFE coils (100 °C); operation is within limit but leaves only 5 °C headroom. Chemist should confirm thermal stability of DPDTC intermediate during extended holds.

Shared assessment / Pressure: 7 bar BPR selected across all candidates. 2-MeTHF bp ~80 °C; 7 bar is appropriate to suppress boiling at 95 °C. BPR hardware confirmed available (bpr_row6, fixed 7 bar). Pump max pressure (E-series 10 bar) comfortably exceeds system pressure plus negligible pressure drops.

Shared assessment / Solvent pump compatibility: 2-MeTHF is explicitly excluded from RED pump tubing on both E-series and SF-10. All candidates correctly assign Stream A and B to pump_e_series_blue (BLUE tubing). This is a hard constraint correctly honored.

Shared assessment / Benzylamine: Reactive primary amine; volatility and inhalation hazard require closed-system operation, which flow inherently provides. No additional safety flag beyond standard handling.

Shared assessment / Dpdtc: Activating reagent; stability at 95 °C in 2-MeTHF over extended holds is uncharacterized. Prolonged Stage 1 times in candidates 6, 7, 8, 9, 10, 12 introduce an untested degradation risk for the thioester intermediate.

Shared assessment / Stream b concentration: Benzylamine delivered at 2.1 M in 2-MeTHF. Solubility of benzylamine in 2-MeTHF at this concentration is unconfirmed but physically plausible; chemist confirmation recommended.

Shared assessment / Coil temperature limit: All reactors used (ETFE coils 100 °C max, PFA coils 100 °C max) are within limit at 95 °C operating temperature.

Shared assessment / No gas stage: No gas feeds; no MFC, gas-delivery, or gas-bookkeeping issues applicable.

Shared assessment / No light stage: No photochemical stages; no irradiation source assignment required.

Shared assessment / Peristaltic pump low flow: E-series minimum flow 0.02 mL/min; Stream B at 0.02 mL/min (candidates 5, 8, 9, 10, 11, 12) is at the absolute minimum. Peristaltic delivery accuracy at minimum setting should be confirmed.

Shared assessment / Stage2 hold note: Stage 2 residence times are diluted by Stream B addition (Q increases from 0.28 to 0.35 or 0.08 to 0.10 mL/min). Stated residence times are correctly computed on the post-mix flow rate.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Stage 1: 17.9 min (ratio 0.60×), Stage 2: 14.3 min (ratio 0.48×). Both stages substantially sub-batch. Adequate for a short-time anchor point in the screen; conversion adequacy in both stages is uncharacterized. Practical flow rates (0.28/0.07 mL/min).

Reviews 1 / Uncertainty 1: Stage 1 and Stage 2 both below batch reference; thioester and amide conversion adequacy unknown.

Reviews 1 / Uncertainty 2: Whether 14 min Stage 2 is sufficient for amide formation is uncharacterized.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Stage 1: 16.7 min (0.56×), Stage 2: 33.3 min (1.11×). Stage 2 slightly exceeds batch reference, providing conversion margin. Stage 1 sub-batch is a conversion risk. Stream B at 0.03 mL/min is low but above E-series minimum.

Reviews 2 / Uncertainty 1: Stage 1 sub-batch; thioester conversion adequacy uncharacterized.

Reviews 2 / Uncertainty 2: 0.03 mL/min Stream B delivery precision on peristaltic pump unconfirmed.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1: 35.7 min (1.19×) exceeds batch reference, supporting thioester conversion. Stage 2: 14.3 min (0.48×) is sub-batch but amide formation from a pre-formed thioester may be faster. Practical flow rates; single-platform E-series.

Reviews 3 / Uncertainty 1: Stage 2 sub-batch; amide conversion adequacy uncharacterized.

Reviews 3 / Uncertainty 2: Whether thioester is stable during 35.7 min hold is uncharacterized.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1: 31.25 min (1.04×), Stage 2: 25 min (0.83×). Both stages near batch reference. Balanced coverage; two ETFE 5 mL coils, practical flow rates. Reasonable yield-priority anchor.

Reviews 4 / Uncertainty 1: Stage 2 at 0.83× batch; amide conversion completeness uncharacterized.

Reviews 4 / Uncertainty 2: DPDTC intermediate stability over ~30 min hold unconfirmed.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1: 25 min (0.83×), Stage 2: 50 min (1.67×). Stage 2 extended; possible degradation risk of product at 95 °C for 50 min is uncharacterized. Stream B at 0.02 mL/min is at E-series minimum.

Reviews 5 / Uncertainty 1: Stage 2 at 50 min/95 °C; product or intermediate degradation uncharacterized.

Reviews 5 / Uncertainty 2: Stream B at 0.02 mL/min (E-series minimum); peristaltic delivery accuracy unconfirmed.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1: 62.5 min (2.08×), Stage 2: 25 min (0.83×). Stage 1 strongly exceeds batch; extended thioester intermediate exposure at 95 °C is an untested stability risk. Stage 2 slightly sub-batch.

Reviews 6 / Uncertainty 1: 62.5 min Stage 1 at 95 °C; DPDTC thioester intermediate stability uncharacterized.

Reviews 6 / Uncertainty 2: Stage 2 sub-batch; amide conversion completeness uncharacterized.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Stage 1: 62.5 min (2.08×), Stage 2: 50 min (1.67×). Both extended well beyond batch. Uses two ETFE 10 mL coils. Extended 95 °C exposure in both stages raises untested degradation risk for intermediate and product.

Reviews 7 / Uncertainty 1: 62.5+50 min at 95 °C; thioester and amide degradation risk uncharacterized.

Reviews 7 / Uncertainty 2: Two coil_row12 units required; inventory quantity is 2 — confirmed available but fully consumed.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1: 62.5 min, Stage 2: 100 min; both heavily exceed batch. Total 162.5 min at 95 °C. Very extended Stage 2 amide exposure is an untested degradation risk. Stream B at 0.02 mL/min is at E-series minimum.

Reviews 8 / Uncertainty 1: 100 min Stage 2 at 95 °C; amide and byproduct degradation risk uncharacterized.

Reviews 8 / Uncertainty 2: Stream B at minimum E-series flow; delivery accuracy unconfirmed.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1: 62.5 min (2.08×), Stage 2: 50 min (1.67×). Same hold ratios as candidate 7 but uses two ETFE 5 mL coils instead of 10 mL coils. Same extended-hold degradation uncertainty applies.

Reviews 9 / Uncertainty 1: Extended 95 °C exposure in both stages; degradation risk uncharacterized.

Reviews 9 / Uncertainty 2: Stream B at 0.02 mL/min (E-series minimum); delivery accuracy unconfirmed.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 1: 62.5 min (ETFE 5 mL), Stage 2: 50 min (PFA 5 mL). Identical hold ratios to candidate 9; mixed coil materials introduce no constraint violation. Same degradation and minimum-flow uncertainties.

Reviews 10 / Uncertainty 1: Extended 95 °C holds; degradation risk uncharacterized for both stages.

Reviews 10 / Uncertainty 2: Stream B at 0.02 mL/min; peristaltic minimum flow accuracy unconfirmed.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1: 25 min (0.83×, PFA 2 mL), Stage 2: 50 min (1.67×, PFA 5 mL). Stage 1 slightly sub-batch; Stage 2 extended with untested degradation risk at 95 °C. Stream B at minimum flow.

Reviews 11 / Uncertainty 1: Stage 1 sub-batch; thioester conversion adequacy uncharacterized.

Reviews 11 / Uncertainty 2: Stream B at 0.02 mL/min; E-series minimum flow accuracy unconfirmed.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1: 125 min (4.17×), Stage 2: 50 min (1.67×). Stage 1 far exceeds batch; 125 min DPDTC intermediate exposure at 95 °C is the most extreme untested stability condition in this set. Limited yield-priority rationale for this design.

Reviews 12 / Uncertainty 1: 125 min Stage 1 at 95 °C; DPDTC thioester intermediate stability is the most uncertain across all candidates.

Reviews 12 / Uncertainty 2: Stream B at 0.02 mL/min; E-series minimum flow accuracy unconfirmed.

Call 5: Skeptic, attempt 1
Status: recorded response, parsed successfully.

Vetoes: []

Shared assessment / Chemistry: Two-stage DPDTC-mediated amide formation; kinetics uncharacterized; no measured flow yield data. Batch anchor: 30 min each stage at 95 °C. Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M, DPDTC 0.525 M, DMAP 0.05 M in 2-MeTHF. Stream B: benzylamine 2.1 M in 2-MeTHF, introduced at Stage 2 T-mixer. All candidates use E-series BLUE pump (2-MeTHF compatible); RED pump correctly excluded. BPR 7 bar selected; all PFA and ETFE coils rated ≥ 95 °C and ≥ 34 bar — no temperature or pressure hard violations detected across any candidate. No gas stage, no light source, no inline degasser required.

Shared assessment / Stream b minimum flow flag: Candidates 5, 8, 9, 10, 11, 12 set Stream B at 0.02 mL/min, which equals the E-series peristaltic minimum (0.02 mL/min). Stoichiometric delivery accuracy at the absolute minimum is unconfirmed and should be flagged for laboratory verification before executing these candidates.

Shared assessment / Benzylamine solubility: Stream B benzylamine at 2.1 M in 2-MeTHF is unconfirmed; precipitation risk is uncharacterized for all candidates equally.

Shared assessment / Thioester stability: Prolonged Stage 1 holds at 95 °C (candidates 6–12) carry uncharacterized thioester intermediate degradation risk — this is a hypothesis, not an established fact.

Shared assessment / Product degradation: Extended Stage 2 holds at 95 °C (candidates 5, 7, 8, 9, 10, 11, 12) carry uncharacterized amide product degradation risk — hypothesis only.

Shared assessment / Coil row12 dual use: Candidate 7 uses two coil_row12 units simultaneously; inventory quantity is 2 — at capacity, confirmed permissible but requires laboratory confirmation of simultaneous availability.

Shared assessment / Hold ratio note: All hold ratios are V/Q liquid indices, not measured conversion equivalents. Physical exposure equivalence to batch is not established for any candidate.

Shared assessment / No hard violations found: No candidate breaches a hard constraint (temperature rating, pressure rating, pump chemical exclusion, inventory quantity, or minimum flow). No veto is warranted.

Candidate assessments 1 / Candidate: 1

Candidate assessments 1 / Cross domain verdict: no_additional_veto

Candidate assessments 1 / Justification: S1 τ=17.9 min (0.60×), S2 τ=14.3 min (0.48×). Both stages sub-batch; consensus reject from Chemistry and Kinetics reviewers. No hard constraint violated; rejection is objective-policy-based, not a hard veto. Acceptable as extreme short-time anchor if chemist elects.

Candidate assessments 1 / Uncertainty 1: Thioester and amide conversion adequacy at sub-batch times entirely uncharacterized.

Candidate assessments 2 / Candidate: 2

Candidate assessments 2 / Cross domain verdict: no_additional_veto

Candidate assessments 2 / Justification: S1 τ=16.7 min (0.56×, 2 mL PFA coil_row3), S2 τ=33.3 min (1.11×, 5 mL ETFE coil_row11). Asymmetric design favoring S2 amide step is consistent with yield-priority objective. Three-domain prefer consensus. No hard violations.

Candidate assessments 2 / Uncertainty 1: S1 sub-batch thioester conversion uncharacterized.

Candidate assessments 2 / Uncertainty 2: Stream B 2.1 M benzylamine solubility unconfirmed.

Candidate assessments 3 / Candidate: 3

Candidate assessments 3 / Cross domain verdict: no_additional_veto

Candidate assessments 3 / Justification: S1 τ=35.7 min (1.19×, 10 mL ETFE coil_row12), S2 τ=14.3 min (0.48×, 5 mL ETFE coil_row11). Inverted asymmetry; Chemistry/Kinetics prefer, Safety prefer. Higher throughput (0.14 mmol/min). S2 sub-batch amide conversion is the key experimental unknown. No hard violations.

Candidate assessments 3 / Uncertainty 1: S2 amide conversion at 0.48× batch uncharacterized.

Candidate assessments 3 / Uncertainty 2: Thioester stability over 35.7 min at 95 °C uncharacterized.

Candidate assessments 4 / Candidate: 4

Candidate assessments 4 / Cross domain verdict: no_additional_veto

Candidate assessments 4 / Justification: S1 τ=31.25 min (1.04×), S2 τ=25 min (0.83×), both near batch. Two ETFE 5 mL coil_row11 (qty 2 — at inventory capacity, permissible). Three-domain prefer consensus. Closest flow analog to batch conditions. No hard violations.

Candidate assessments 4 / Uncertainty 1: S2 at 0.83× batch; amide completeness uncharacterized in flow.

Candidate assessments 4 / Uncertainty 2: Near-batch time does not guarantee equivalent conversion.

Candidate assessments 5 / Candidate: 5

Candidate assessments 5 / Cross domain verdict: no_additional_veto

Candidate assessments 5 / Justification: S1 τ=25 min (0.83×, 2 mL PFA), S2 τ=50 min (1.67×, 5 mL ETFE). Stream B at 0.02 mL/min equals E-series minimum — accuracy unconfirmed. No hard violation; minimum-flow concern is an operational uncertainty.

Candidate assessments 5 / Uncertainty 1: Stream B 0.02 mL/min at E-series minimum; stoichiometric accuracy unconfirmed.

Candidate assessments 5 / Uncertainty 2: S2 50 min product degradation at 95 °C uncharacterized hypothesis.

Candidate assessments 6 / Candidate: 6

Candidate assessments 6 / Cross domain verdict: no_additional_veto

Candidate assessments 6 / Justification: S1 τ=62.5 min (2.08×, 10 mL ETFE), S2 τ=25 min (0.83×, 5 mL ETFE). Extended S1 explores upper thioester formation bound; S2 sub-batch. Cross-domain consensus: acceptable. No hard violations.

Candidate assessments 6 / Uncertainty 1: S1 thioester stability at 62.5 min/95 °C is hypothesis only.

Candidate assessments 6 / Uncertainty 2: S2 amide conversion at 0.83× batch uncharacterized.

Candidate assessments 7 / Candidate: 7

Candidate assessments 7 / Cross domain verdict: no_additional_veto

Candidate assessments 7 / Justification: S1 τ=62.5 min (2.08×), S2 τ=50 min (1.67×); both extended. Two coil_row12 simultaneously — at inventory limit (qty 2), permissible but requires lab confirmation. Cross-domain: acceptable. No hard violations.

Candidate assessments 7 / Uncertainty 1: Two coil_row12 at inventory capacity — simultaneous availability needs lab confirmation.

Candidate assessments 7 / Uncertainty 2: Extended 95 °C holds in both stages carry uncharacterized degradation risk.

Candidate assessments 8 / Candidate: 8

Candidate assessments 8 / Cross domain verdict: no_additional_veto

Candidate assessments 8 / Justification: S1 τ=62.5 min, S2 τ=100 min (3.33×). Kinetics reviewer rejects on objective grounds (extreme over-exposure vs. 'shortest justified' instruction); no hard constraint violated. Stream B 0.02 mL/min at E-series minimum. Acceptable as an extreme boundary point if chemist elects.

Candidate assessments 8 / Uncertainty 1: S2 100 min at 95 °C: product degradation risk is hypothesis, uncharacterized.

Candidate assessments 8 / Uncertainty 2: Stream B 0.02 mL/min minimum-flow accuracy unconfirmed.

Candidate assessments 9 / Candidate: 9

Candidate assessments 9 / Cross domain verdict: no_additional_veto

Candidate assessments 9 / Justification: S1 τ=62.5 min (5 mL ETFE coil_row11), S2 τ=50 min (5 mL ETFE coil_row11); two coil_row11 used (qty 2 — at capacity, permissible). Numerically identical to candidate 7 in hold ratios. Stream B 0.02 mL/min minimum-flow concern. No hard violations.

Candidate assessments 9 / Uncertainty 1: Two coil_row11 at inventory capacity — simultaneous availability needs lab confirmation.

Candidate assessments 9 / Uncertainty 2: Stream B 0.02 mL/min at E-series minimum; accuracy unconfirmed.

Candidate assessments 10 / Candidate: 10

Candidate assessments 10 / Cross domain verdict: no_additional_veto

Candidate assessments 10 / Justification: S1 τ=62.5 min (5 mL ETFE coil_row11), S2 τ=50 min (5 mL PFA coil_row4, 1 mm ID). Mixed coil materials; no constraint violation. ETFE-to-PFA transition at S2 inlet — connector compatibility is an operational matter, not a hard constraint. Same hold ratios as candidates 7 and 9.

Candidate assessments 10 / Uncertainty 1: PFA coil_row4 narrow ID (1 mm) may affect RTD relative to wide-bore ETFE — uncharacterized.

Candidate assessments 10 / Uncertainty 2: Stream B 0.02 mL/min minimum-flow accuracy unconfirmed.

Candidate assessments 11 / Candidate: 11

Candidate assessments 11 / Cross domain verdict: no_additional_veto

Candidate assessments 11 / Justification: S1 τ=25 min (0.83×, 2 mL PFA coil_row3), S2 τ=50 min (1.67×, 5 mL PFA coil_row4). All-PFA narrow-bore geometry. Inventory confirmed (coil_row3 qty 1, coil_row4 qty 2). Stream B 0.02 mL/min at E-series minimum. Same hold ratios as candidate 5. No hard violations.

Candidate assessments 11 / Uncertainty 1: Stream B 0.02 mL/min at E-series minimum; stoichiometric accuracy unconfirmed.

Candidate assessments 11 / Uncertainty 2: S2 50 min/95 °C product degradation risk is hypothesis only.

Candidate assessments 12 / Candidate: 12

Candidate assessments 12 / Cross domain verdict: no_additional_veto

Candidate assessments 12 / Justification: S1 τ=125 min (4.17×, 10 mL ETFE coil_row12), S2 τ=50 min (5 mL ETFE coil_row11). Chemistry and Kinetics reject on objective grounds (far exceeds 'shortest justified' instruction); Safety flags S1 stability as most extreme in set. No hard constraint violated. Stream B 0.02 mL/min at E-series minimum.

Candidate assessments 12 / Uncertainty 1: S1 125 min/95 °C: thioester stability most uncertain across all candidates — hypothesis only.

Candidate assessments 12 / Uncertainty 2: Stream B 0.02 mL/min minimum-flow accuracy unconfirmed.

Requested measurement 1: Benzylamine solubility at 2.1 M in 2-MeTHF at 95 °C (all candidates)

Requested measurement 2: E-series BLUE peristaltic pump delivery accuracy at 0.02 mL/min (candidates 5, 8, 9, 10, 11, 12)

Requested measurement 3: E-series BLUE peristaltic pump delivery accuracy at 0.03 mL/min (candidate 2)

Requested measurement 4: DPDTC thioester intermediate stability in 2-MeTHF at 95 °C as a function of hold time (relevant to all candidates; critical for candidates 6, 7, 8, 9, 10, 12)

Requested measurement 5: Amide product stability in 2-MeTHF at 95 °C over Stage 2 hold duration (candidates 5, 7, 8, 9, 10, 11, 12)

Requested measurement 6: Stage 1 thioester conversion as a function of residence time at 95 °C to anchor kinetic rationale for stage-time selection

Requested measurement 7: Stage 2 amide conversion as a function of residence time at 95 °C to confirm aminolysis is complete within sub-batch holds

Requested measurement 8: Simultaneous availability of two coil_row11 or two coil_row12 units for candidates 4, 7, 9

Requested measurement 9: Heating module compatibility confirmation for PFA coil_row3 at 95 °C (candidates 2, 5, 11)

Call 6: Chief, attempt 1
Status: recorded response, parsed successfully.

Candidate: 3

Justification: Stage 1 τ=35.7 min (ratio 1.19×) exceeds the batch anchor, supporting adequate thioester formation before amine addition. Stage 2 τ=14.3 min (ratio 0.48×) is sub-batch but serves the objective's 'shortest reasonably justified' instruction by testing whether aminolysis is fast relative to the batch hold. This asymmetry is the most informative first-run design under yield-priority: it protects the upstream intermediate-forming step while probing whether Stage 2 requires the full 30 min. Highest limiting-feed throughput among the preferred candidates (0.14 mmol/min). Equipment assignment is clean: coil_row12 (ETFE 10 mL, 95°C rated) for Stage 1, coil_row11 (ETFE 5 mL, 95°C rated) for Stage 2, single E-series BLUE platform, BPR 7 bar.

Objective alignment: Yield-priority requires adequate Stage 1 conversion to generate thioester for Stage 2. Candidate 3 is the only preferred candidate that exceeds the batch anchor in Stage 1 (1.19×) while applying the shortest-justified-time instruction to Stage 2 (0.48×). If Stage 2 result is low, extending Stage 2 is the obvious next step (Candidate 4 or 2). If Stage 1 result shows incomplete thioester, the Stage 1 hold is already above batch and the problem lies elsewhere. This asymmetry distinguishes the rate-limiting step in a single run.

Alternative 1 / Candidate: 4

Alternative 1 / Reason not selected: S1 31.25 min (1.04×) and S2 25 min (0.83×) both near-batch. Valuable as reference but does not test whether either stage can be shortened, and S2 is still sub-batch without a mechanistic reason. Both ETFE coil_row11 units consumed simultaneously (at inventory capacity). Less informative than Candidate 3 as a first run because it does not probe Stage 2 speed at all.

Alternative 2 / Candidate: 2

Alternative 2 / Reason not selected: S1 16.7 min (0.56×) and S2 33.3 min (1.11×). Complementary asymmetry to Candidate 3 but S1 is sub-batch, risking insufficient thioester yield that would confound interpretation of any S2 result. Better as a second run after Candidate 3 confirms S2 kinetics. Lower throughput (0.06 mmol/min) than Candidate 3.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: Candidate 3 directly operationalizes the 'shortest reasonably justified' instruction for Stage 2 while protecting Stage 1 conversion above the batch anchor. If Stage 2 at 14.3 min is sufficient, the objective is met efficiently; if not, Stage 2 time is the actionable variable.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: Result will test the chemist hypothesis that Stage 1 conversion governs amide yield: if Stage 1 τ > batch and final amide yield is low, Stage 2 kinetics are implicated. If yield is high, Stage 2 can be shortened further in subsequent runs.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: No measured data available; outcome of Candidate 3 will constitute the first quantitative anchor for both stage-time sensitivity and flow yield.

Shared assessment / All candidates: No hard constraint violations found across any candidate. All coils are 95°C and ≥34 bar rated. E-series BLUE pump is 2-MeTHF compatible; RED pump correctly excluded. BPR 7 bar confirmed available. Stream B benzylamine at 2.1 M in 2-MeTHF is unconfirmed for solubility at 95°C across all candidates. All hold ratios are V/Q liquid indices; physical exposure equivalence to batch is not established. No gas stage, no light source, no degasser required.

Shared assessment / Minimum flow flag: Candidates 5, 8, 9, 10, 11, 12: Stream B at 0.02 mL/min equals the declared E-series minimum. Stoichiometric delivery accuracy at this limit is unconfirmed and requires laboratory verification before executing these candidates.

Shared assessment / Degradation hypotheses: Extended Stage 1 holds >30 min at 95°C (Candidates 6, 7, 8, 9, 10, 12) and extended Stage 2 holds >30 min at 95°C (Candidates 5, 7, 8, 9, 10, 11, 12) carry uncharacterized intermediate and product degradation risk. These are untested hypotheses, not established facts.

Shared assessment / Coil inventory at capacity: Candidates 4, 9 use two coil_row11 simultaneously (qty 2, permissible). Candidate 7 uses two coil_row12 simultaneously (qty 2, permissible). Laboratory confirmation of simultaneous availability required.

Candidate justifications / 1: S1 17.9 min (0.60×), S2 14.3 min (0.48×). Both sub-batch; conversion adequacy in both stages uncharacterized. Useful extreme short-time anchor only.

Candidate justifications / 2: S1 16.7 min (0.56×) sub-batch; S2 33.3 min (1.11×) exceeds anchor. Complementary to C3; lower throughput and S1 risk.

Candidate justifications / 3: SELECTED. S1 35.7 min (1.19×) protects thioester step; S2 14.3 min (0.48×) tests shortest-justified amide hold. Highest preferred-candidate throughput.

Candidate justifications / 4: S1 31.3 min (1.04×), S2 25 min (0.83×). Near-batch both stages; strong reference but does not probe Stage 2 minimum time. Two coil_row11 at inventory limit.

Candidate justifications / 5: S1 25 min (0.83×), S2 50 min (1.67×). Extended S2 degrades risk (hypothesis); Stream B 0.02 mL/min at E-series minimum.

Candidate justifications / 6: S1 62.5 min (2.08×), S2 25 min (0.83×). S1 far exceeds batch without kinetic justification; useful boundary only.

Candidate justifications / 7: S1 62.5 min, S2 50 min; both well above batch. Two coil_row12 at inventory limit. Boundary exploration only.

Candidate justifications / 8: S2 100 min (3.33×) extreme overkill; contradicts 'shortest justified' objective. Stream B at E-series minimum.

Candidate justifications / 9: S1 62.5 min, S2 50 min; identical ratios to C7, smaller coils. Stream B at E-series minimum. Redundant with C7.

Candidate justifications / 10: S1 62.5 min ETFE, S2 50 min PFA 1 mm. Same ratios as C9; RTD difference in narrow bore uncharacterized.

Candidate justifications / 11: S1 25 min (0.83×), S2 50 min (1.67×). Same ratios as C5; all-PFA narrow bore. Stream B at E-series minimum.

Candidate justifications / 12: S1 125 min (4.17×) far exceeds batch; directly contradicts objective. Lowest throughput. Extreme boundary only.

Limitation 1: No measured flow kinetics or yield data exist; hold ratios are V/Q indices, not conversion equivalents.

Limitation 2: Thioester stability at 35.7 min/95°C in Stage 1 is uncharacterized; extended hold may or may not be beneficial.

Limitation 3: Stage 2 aminolysis rate in flow at 95°C is unknown; 14.3 min may be insufficient.

Limitation 4: Stream B benzylamine at 2.1 M in 2-MeTHF at 95°C: solubility unconfirmed.

Limitation 5: Peristaltic pump (E-series) delivery accuracy at 0.07 mL/min (Stream B, Candidate 3) is unconfirmed but above minimum.

Limitation 6: No measured product degradation data for either stage; degradation risk at extended holds is hypothesis only.

Suggested measurement 1: Stage 2 amide conversion as a function of residence time at 95°C (10–30 min range) to determine whether 14.3 min Stage 2 is sufficient — this is the primary decision gate for Candidate 3.

Suggested measurement 2: Stage 1 thioester conversion at 35.7 min/95°C to confirm Stage 1 is not rate-limiting at this hold.

Suggested measurement 3: Benzylamine solubility at 2.1 M in 2-MeTHF at 95°C before any run.

Suggested measurement 4: DPDTC thioester intermediate stability at 95°C as a function of hold time (10–120 min) to bound degradation risk for longer-Stage-1 candidates.

Suggested measurement 5: E-series BLUE pump delivery accuracy at 0.02–0.03 mL/min before executing Candidates 2, 5, 8, 9, 10, 11, 12.

Suggested measurement 6: Heating module compatibility confirmation for PFA coil_row3 at 95°C before executing Candidates 2, 5, 11.

Editorial cross-checks (not model statements)
Numerical transcription: in the Set 1 Chief answer_impacts entry for Q-PREF-001, benzylamine molar flow is written as 0.02 mmol/min. The final feed and independent arithmetic give 2.1 mol/L x 0.04 mL/min = 0.084 mmol/min, which is 1.05 times the acid feed of 0.08 mmol/min. For Sets 2/3, the corresponding amine rate is 0.147 mmol/min and acid rate is 0.14 mmol/min. The copied council text remains unedited; these corrected values govern the reporting.

Inference limits: wording such as adequate exposure, confirmed photon geometry, or a single run distinguishing the rate-limiting step is not accepted as an experimentally demonstrated conclusion. A nominal hold near the batch duration does not establish equal conversion in flow. In particular, the Set 2 claim that incomplete Stage 1 conversion must have another cause because its hold exceeds batch is not justified. Changes in mixing, temperature distribution, feed state and residence-time distribution remain possible. The selected designs are therefore described as screening hypotheses.

Inventory reconciliation: references to a missing mixer or one pump label belong to the council's view of the candidate record. The final allocation explicitly assigns the interstage P-713 mixer and separate stream-pump operations. This resolves the reported assignment at the software-record level, not by claiming that the council physically installed equipment or verified the laboratory configuration.

F6-S3: Figure 6, Set 3
Source: outputs/khu_revised_six_20260915/figure6_set3/attempt_01. Selected Candidate 3; 12-candidate pool; downstream model claude-sonnet-4-6. The objective policy is balanced.

Call 1: DrChemistry, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated thioester activation (Stage 1) then aminolysis with benzylamine (Stage 2) in 2-MeTHF at 95 °C. Both stages at 95 °C match batch conditions; no cooling step between stages is proposed, which is an untested hypothesis requiring chemist confirmation. No measured flow kinetics exist; all hold ratios are experimental exposure indices only, not yield predictors.

Shared assessment / Stoichiometry: Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M (1.0 equiv), DPDTC 1.05 equiv, DMAP 0.1 equiv in 2-MeTHF. Stream B: benzylamine 2.1 M (1.05 equiv relative to substrate) in 2-

MeTHF. Molar flow ratio Stream A : Stream B = 4:1 (v/v) across all candidates; stoichiometric closure confirmed.

Shared assessment / Pump assignment: All candidates use pump_e_series_blue (E-series, blue tubing) exclusively. 2-MeTHF is excluded from RED tubing; BLUE tubing is chemically compatible. Single platform, no mixed-platform issues.

Shared assessment / Bpr: 7-bar BPR (bpr_row6, fixed) applied to all candidates. 2-MeTHF bp ≈ 80 °C at 1 atm; operation at 95 °C requires elevated back-pressure to suppress boiling. 7 bar provides adequate suppression margin. All reactor materials (PFA, ETFE) rated ≥ 34 bar; pressure headroom confirmed.

Shared assessment / Temperature: PFA and ETFE coils all rated max 100 °C; 95 °C operation is within limit. FEP coils (coil_row6, coil_row7) rated max 50 °C and are not used in any candidate — correct.

Shared assessment / Mixer topology: Each candidate joins Stream A into Stage 1 reactor and Stream B into Stage 2 reactor via T- or Y-mixer from inventory. No union substituted for a T-mixer.

Shared assessment / Uncertainties shared 1: Direct hot transfer from Stage 1 to benzylamine addition (no cooling) is an untested hypothesis; batch protocol explicitly cools before amine addition — chemist confirmation required.

Shared assessment / Uncertainties shared 2: Thioester intermediate stability at 95 °C in flow (no isolation, direct amine addition) is uncharacterized.

Shared assessment / Uncertainties shared 3: No measured flow kinetics or yield data; batch times are screening anchors only.

Shared assessment / Uncertainties shared 4: Solubility of DPDTC, intermediate thioester, and product at 0.5 M in 2-MeTHF under flow conditions is unconfirmed.

Shared assessment / Uncertainties shared 5: Feed throughput (limiting_feed_mmol_min) is not product throughput; no conversion data available.

Shared assessment / Uncertainties shared 6: Peristaltic pump pulsation at low flow rates (0.02–0.04 mL/min) may affect mixing and steady-state quality.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: prefer

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Highest limiting feed throughput (0.14 mmol/min) with compact 5+5 mL ETFE geometry. Stage 1 τ=17.9 min (ratio 0.60), Stage 2 τ=14.3 min (ratio 0.48); both sub-batch, suitable as a compact starting screen. Single reactor type simplifies setup. Stream B flow 0.07 mL/min is comfortably above peristaltic minimum.

Reviews 1 / Uncertainty 1: Sub-batch Stage 1 exposure may be insufficient for complete thioester formation — experimentally uncharacterized.

Reviews 1 / Uncertainty 2: Sub-batch Stage 2 aminolysis exposure adequacy is unknown.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Mixed reactor geometry (2 mL PFA Stage 1, 5 mL ETFE Stage 2). Stage 1 τ=16.7 min (ratio 0.56) sub-batch; Stage 2 τ=33.3 min (ratio 1.11) exceeds batch anchor. Throughput 0.06 mmol/min, lower than C1. Stream B at 0.03 mL/min approaches low-flow pulsation risk.

Reviews 2 / Uncertainty 1: Stream B at 0.03 mL/min may exhibit peristaltic pulsation; mixing quality uncertain.

Reviews 2 / Uncertainty 2: Whether Stage 2 over-exposure at 95 °C benefits or degrades amide yield is unknown.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: acceptable

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Stage 1 τ=35.7 min (ratio 1.19) exceeds batch anchor; Stage 2 τ=14.3 min (ratio 0.48) sub-batch. Throughput matches C1 at 0.14 mmol/min. Uses 10+5 mL ETFE. Tests whether extended thioester activation improves amide stage performance, with compact Stage 2.

Reviews 3 / Uncertainty 1: Whether prolonged Stage 1 at 95 °C degrades thioester intermediate is uncharacterized.

Reviews 3 / Uncertainty 2: Sub-batch Stage 2 aminolysis adequacy is uncharacterized.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Both stages near-batch (τ1=31.3 min ratio 1.04, τ2=25 min ratio 0.83) using 5+5 mL ETFE. Throughput 0.08 mmol/min. Compact interpretable baseline close to batch anchors for both stages.

Reviews 4 / Uncertainty 1: Stage 2 at 25 min is 17% below batch anchor; adequacy for aminolysis is uncharacterized.

Reviews 4 / Uncertainty 2: Near-batch times are not guaranteed optimal in flow.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Stage 1 τ=25 min (ratio 0.83) sub-batch; Stage 2 τ=50 min (ratio 1.67) well above batch. Mixed 2+5 mL geometry. Throughput 0.04 mmol/min (lowest tier). Stream B at 0.02 mL/min is at E-series peristaltic minimum, raising pulsation risk.

Reviews 5 / Uncertainty 1: Stream B at 0.02 mL/min is at minimum flow; pulsation and stoichiometric accuracy are at risk.

Reviews 5 / Uncertainty 2: Stage 2 over-exposure effect on amide stability is unknown.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 τ=62.5 min (ratio 2.08) strongly exceeds batch; Stage 2 τ=25 min (ratio 0.83) sub-batch. 10+5 mL ETFE. Throughput 0.08 mmol/min. Tests whether very long thioester activation is needed, with shorter aminolysis.

Reviews 6 / Uncertainty 1: Thioester stability under 62.5 min at 95 °C is uncharacterized; degradation risk is a hypothesis.

Reviews 6 / Uncertainty 2: Stage 2 aminolysis adequacy at 25 min sub-batch is unknown.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Both stages well above batch (τ1=62.5, τ2=50 min). Uses two 10 mL ETFE coils (coil_row12 qty=2, confirmed). Total time 112.5 min; throughput 0.08 mmol/min. Long-exposure anchor for screening; tests upper bound of exposure space.

Reviews 7 / Uncertainty 1: Extended exposure of thioester intermediate at 95 °C may risk degradation — uncharacterized.

Reviews 7 / Uncertainty 2: Long total processing time conflicts with throughput objective.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1 τ=16.7 min sub-batch; Stage 2 τ=66.7 min (ratio 2.22) well above batch. Mixed 2+10 mL. Throughput 0.06 mmol/min. Stream B at 0.03 mL/min near low-flow limit. Tests asymmetric short activation / long aminolysis.

Reviews 8 / Uncertainty 1: Stream B at 0.03 mL/min pulsation risk for peristaltic pump.

Reviews 8 / Uncertainty 2: Stage 2 amide product stability under 66.7 min at 95 °C is unknown.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 τ=62.5 min (ratio 2.08); Stage 2 τ=20 min (ratio 0.67) sub-batch using compact 2 mL PFA. Throughput 0.04 mmol/min. Stream B at 0.02 mL/min is at E-series minimum. Tests long activation / short aminolysis geometry inversion.

Reviews 9 / Uncertainty 1: Stream B at 0.02 mL/min is at minimum; stoichiometric and mixing fidelity uncertain.

Reviews 9 / Uncertainty 2: Whether 20 min aminolysis is adequate is experimentally unknown.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: reject

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 2 τ=100 min (ratio 3.33) is the longest Stage 2 in the set; total time 125 min. Throughput 0.04 mmol/min (lowest tier). Stream B at 0.02 mL/min at minimum flow. Strongly imbalanced against balanced-priority objective; extreme over-exposure risk with minimum throughput.

Reviews 10 / Uncertainty 1: Amide product or intermediate degradation under 100 min at 95 °C is uncharacterized.

Reviews 10 / Uncertainty 2: Stream B minimum flow pulsation risk.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Same τ profile as C9 (62.5+20 min) but Stage 1 uses PFA 5 mL (coil_row4, 1 mm ID) giving higher pressure drop; Stage 2 uses PFA 2 mL. Throughput 0.04 mmol/min; Stream B at 0.02 mL/min minimum. Redundant with C9 in exposure space but different reactor geometry.

Reviews 11 / Uncertainty 1: Stream B at minimum flow introduces stoichiometric fidelity risk.

Reviews 11 / Uncertainty 2: Narrower ID (1 mm) Stage 1 increases pressure drop; solids deposition risk if any precipitation occurs.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: reject

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Stage 1 τ=62.5, Stage 2 τ=50 min using two 5 mL ETFE coils (same IDs as C7 but smaller reactors at same flow). Total time 112.5 min; throughput 0.04 mmol/min (lowest). Stream B at 0.02 mL/min minimum. Duplicates long-exposure space of C7 at lower throughput; poor balance for balanced-priority objective.

Reviews 12 / Uncertainty 1: Stream B minimum flow pulsation risk.

Reviews 12 / Uncertainty 2: Extended Stage 2 amide stability at 95 °C is uncharacterized.

Call 2: DrKinetics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated thioester/amide sequence. Stage 1 forms 2-pyridyl thioester; Stage 2 aminolyzes it with benzylamine. Kinetics uncharacterized; batch times (30+30 min, 95°C) are screening anchors only.

Shared assessment / Solvent pump constraint: 2-MeTHF is excluded from all RED pump tubing. All candidates correctly assign Stream A and Stream B to BLUE E-series pumps. No violation.

Shared assessment / Temperature: All stages at 95°C, within PFA and ETFE coil limits (max 100°C). No temperature deviation flagged.

Shared assessment / Pressure: 7-bar BPR applied to all candidates. Calculated pressure drops are negligible (<0.003 bar). BPR is appropriate for maintaining liquid phase at 95°C in 2-MeTHF (bp ~80°C at 1 atm); this is a necessary operating condition, not just a preference.

Shared assessment / Stoichiometry: Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M (1.0 equiv), DPDTC 1.05 equiv, DMAP 0.1 equiv in 2-MeTHF. Stream B: benzylamine 2.1 M in 2-MeTHF (1.05 equiv relative to substrate). Molar ratio maintained by Q_A/Q_B = 4:1 across all candidates. Correct.

Shared assessment / Thioester intermediate dilution: Upon Stream B addition, Stage 2 inlet substrate concentration is diluted by factor Q_A/Q_total = 0.28/0.35 (scale 0.14 candidates) or 0.12/0.15 (scale 0.06) or 0.08/0.10 (scale 0.04) or 0.16/0.20 (scale 0.08). Theoretical intermediate concentration entering Stage 2 ≈ 0.4 M (0.14 scale) or 0.4 M consistently across all scales. Correctly reported as theoretical.

Shared assessment / Throughput ranking: Feed throughput (mmol/min of limiting substrate): C1=C3: 0.14; C4=C6=C7: 0.08; C2=C8: 0.06; C5=C9=C10=C11=C12: 0.04. Highest feed throughput is C1 and C3 (0.14 mmol/min). Product throughput is unmeasured.

Shared assessment / Stage2 sub batch risk: No kinetic data exist. Short Stage 2 holds (C1: 14.3 min ratio 0.48; C3: 14.3 min ratio 0.48; C9: 20 min ratio 0.67; C11: 20 min ratio 0.67) carry unquantified risk of incomplete aminolysis. This is an experimental uncertainty, not a proven failure.

Shared assessment / Stage1 sub batch risk: Short Stage 1 holds (C1: 17.9 min ratio 0.60; C2: 16.7 min ratio 0.56; C8: 16.7 min ratio 0.56) may not complete thioester formation; uncharacterized risk.

Shared assessment / Extended hold risk: Very long Stage 2 holds (C10: 100 min ratio 3.33; C8: 66.7 min ratio 2.22) extend exposure at 95°C with no evidence of stability. Degradation risk is an untested hypothesis.

Shared assessment / Reactor inventory: coil_row11 (ETFE 5 mL, qty 2) and coil_row12 (ETFE 10 mL, qty 2) used in multiple candidates simultaneously within a single-candidate design—no cross-candidate conflict within a single run. C7 uses coil_row12 twice (both Stage 1 and Stage 2); qty=2 available, so no hard violation.

Shared assessment / No gas no light: No gas streams, no photochemistry. Gas and light hardware not applicable.

Shared assessment / Direct interstage transfer: All candidates implement direct in-line Stage 1→Stage 2 transfer without cooling, consistent with chemist hypothesis that cooling may be avoidable. Chemist confirmation of thermal compatibility is still required.

Shared assessment / Bpr placement: BPR placed at system outlet. With 2-MeTHF (bp ~80°C at 1 atm) at 95°C, backpressure is required. 7-bar BPR provides adequate headroom. Correct.

Shared assessment / Balanced objective: Objective: balance yield, throughput, processing time, stable operation. Candidates 3 and 4 best anchor this balance. C1 is highest feed throughput with shortest total time but both stages are sub-batch. C4 is near-batch for Stage 1 with moderate Stage 2 shortening and uses only two identical ETFE 5 mL coils.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: acceptable

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Highest feed throughput (0.14 mmol/min), shortest total time (32.1 min). Both stages sub-batch (S1: 17.9 min/0.60×; S2: 14.3 min/0.48×). Compact identical-coil setup. Good throughput anchor; yield risk unquantified at both shortened stages.

Reviews 1 / Uncertainty 1: Stage 1 thioester formation may be incomplete at 17.9 min (0.60× batch); uncharacterized.

Reviews 1 / Uncertainty 2: Stage 2 aminolysis at 14.3 min (0.48× batch) may give incomplete amide yield; uncharacterized.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Lower feed throughput (0.06 mmol/min). Stage 1 sub-batch (16.7 min, 0.56×); Stage 2 above batch (33.3 min, 1.11×). Asymmetric hold explores whether extended aminolysis compensates for short thioester stage. Mixed reactor types (PFA 2 mL + ETFE 5 mL).

Reviews 2 / Uncertainty 1: Stage 1 thioester completeness at 16.7 min uncharacterized.

Reviews 2 / Uncertainty 2: No evidence extended Stage 2 compensates for sub-batch Stage 1.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Highest feed throughput (0.14 mmol/min) with Stage 1 above batch (35.7 min, 1.19×), giving thioester formation headroom. Stage 2 shortened (14.3 min, 0.48×). Practical starting point balancing throughput and Stage 1 yield adequacy; Stage 2 risk is the key experimental question.

Reviews 3 / Uncertainty 1: Stage 2 aminolysis at 14.3 min (0.48× batch) may be incomplete; requires experimental confirmation.

Reviews 3 / Uncertainty 2: Thioester stability during extended Stage 1 hold is uncharacterized.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: prefer

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Stage 1 near-batch (31.3 min, 1.04×); Stage 2 shortened but substantial (25 min, 0.83×). Moderate feed throughput (0.08 mmol/min). Best interpretable baseline: near-batch Stage 1 minimizes thioester risk while Stage 2 shortening is modest. Compact identical-coil design.

Reviews 4 / Uncertainty 1: Stage 2 at 25 min (0.83×) may not fully complete aminolysis; kinetics uncharacterized.

Reviews 4 / Uncertainty 2: Direct interstage transfer at 95°C without cooling requires chemist confirmation.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Lowest feed throughput tier (0.04 mmol/min). Stage 1 sub-batch (25 min, 0.83×); Stage 2 well above batch (50 min, 1.67×). Explores whether long aminolysis time recovers yield from shorter thioester stage. Limited throughput value for balanced objective.

Reviews 5 / Uncertainty 1: Stage 1 thioester completeness at 25 min uncharacterized.

Reviews 5 / Uncertainty 2: Stage 2 extended hold at 95°C: product stability over 50 min uncharacterized.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Stage 1 greatly extended (62.5 min, 2.08×); Stage 2 sub-batch (25 min, 0.83×). Moderate feed throughput (0.08 mmol/min). Long Stage 1 at 95°C raises thioester/intermediate stability concern. Provides screening data on over-extended Stage 1.

Reviews 6 / Uncertainty 1: Thioester or intermediate stability at 62.5 min / 95°C is uncharacterized; degradation risk is a hypothesis.

Reviews 6 / Uncertainty 2: Stage 2 at 25 min (0.83× batch) may be incomplete.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Both stages well above batch (S1: 62.5 min 2.08×; S2: 50 min 1.67×). Total 112.5 min. Uses two ETFE 10 mL coils. Lowest throughput-per-time ratio in the set. Useful only as an extended-hold upper bound; limited practical value for balanced objective.

Reviews 7 / Uncertainty 1: Extended exposure at 95°C across both stages: intermediate and product stability uncharacterized.

Reviews 7 / Uncertainty 2: No evidence prolonged hold improves yield beyond batch.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Stage 1 sub-batch (16.7 min, 0.56×); Stage 2 greatly extended (66.7 min, 2.22×). Moderate feed throughput (0.06 mmol/min). Explores maximum aminolysis time; however very long Stage 2 at 95°C with uncharacterized product stability is a concern.

Reviews 8 / Uncertainty 1: Stage 1 thioester completeness at 16.7 min uncharacterized.

Reviews 8 / Uncertainty 2: Product degradation at 66.7 min / 95°C is an untested hypothesis.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Stage 1 greatly extended (62.5 min, 2.08×) using ETFE 5 mL; Stage 2 compressed (20 min, 0.67×) in small PFA 2 mL coil. Lowest throughput tier (0.04 mmol/min). Mirrors C3 logic inverted but at lower throughput; Stage 2 shortening risk similar.

Reviews 9 / Uncertainty 1: Stage 2 aminolysis at 20 min (0.67×) may be incomplete; uncharacterized.

Reviews 9 / Uncertainty 2: Stage 1 intermediate stability at 62.5 min / 95°C uncharacterized.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Stage 2 at 100 min (3.33× batch) is the longest Stage 2 in the set. Lowest throughput (0.04 mmol/min), total 125 min. Extreme upper bound for aminolysis hold. Product stability over 100 min at 95°C is an unresolved safety/chemistry concern warranting careful review.

Reviews 10 / Uncertainty 1: Product or byproduct accumulation at 100 min / 95°C: degradation risk uncharacterized.

Reviews 10 / Uncertainty 2: Practical value for balanced objective is limited at 0.04 mmol/min throughput.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Stage 1 at 62.5 min (2.08×) in PFA 5 mL 1 mm ID; Stage 2 at 20 min (0.67×) in PFA 2 mL. Identical throughput to C9 (0.04 mmol/min) with different reactor geometry. Narrow-bore PFA in Stage 1 increases axial dispersion risk vs. ETFE 2.4 mm alternatives at same flow.

Reviews 11 / Uncertainty 1: Stage 2 aminolysis at 20 min (0.67×) may be incomplete; uncharacterized.

Reviews 11 / Uncertainty 2: Axial dispersion behavior in 1 mm ID vs. 2.4 mm ID coils at low Re not characterized.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Both stages substantially above batch (S1: 62.5 min; S2: 50 min) using two ETFE 5 mL coils. Lowest throughput tier (0.04 mmol/min). Symmetrically extended screen; duplicates C7 concept at lower total reactor volume. Limited additional information versus C7.

Reviews 12 / Uncertainty 1: Extended dual-stage hold at 95°C: intermediate and product stability uncharacterized.

Reviews 12 / Uncertainty 2: Screen informativeness overlap with C7 reduces marginal value.

Call 3: DrFluidics, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated amide formation. Stage 1: thioester formation (acid + DPDTC/DMAP, 2-MeTHF, 95°C). Stage 2: aminolysis with benzylamine at junction, same temperature. No batch cooling step in flow — direct interstage transfer is a chemist hypothesis, not established fact.

Shared assessment / Pump assignments: All candidates use pump_e_series_blue (BLUE tubing). Stream A (2-MeTHF solvent) is correctly excluded from RED tubing. Both streams share a single pump_equipment_id label; in practice two separate E-series channels are needed — one per stream. Confirm two BLUE E-series channels are available simultaneously.

Shared assessment / Stream b concentration: Benzylamine stream B at 2.1 M in 2-MeTHF: solubility of neat benzylamine at this concentration is plausible but unconfirmed in 2-MeTHF at operating temperature.

Shared assessment / Stoichiometry closure: Q_B/Q_A = 0.25 for scale=0.14 candidates (1,3), 0.25 for scale=0.08 candidates (4,6,7), 0.25 for scale=0.06 candidates (2,8), 0.25 for scale=0.04 candidates (5,9,10,11,12). Molar ratio benzylamine/substrate = (2.1 × Q_B)/(0.5 × Q_A) = 1.05 in all cases. Stoichiometric closure confirmed.

Shared assessment / Throughput metric: Feed substrate throughput (limiting reagent mmol/min): C1=C3=0.14, C4=C6=C7=0.08, C2=C8=0.06, C5=C9=C10=C11=C12=0.04. Candidates 1 and 3 have highest substrate feed throughput.

Shared assessment / Temperature: All candidates run at 95°C throughout. No temperature deviation from batch. FEP coils (coil_row6, coil_row7) are excluded at 95°C (max 50°C) — none are used. All selected PFA and ETFE reactors are rated to 100°C.

Shared assessment / Bpr: 7-bar BPR selected across all candidates. 2-MeTHF bp ~80°C; 95°C operation above bp requires back-pressure. 7-bar is adequate to suppress solvent boiling — confirmed feasible.

Shared assessment / Direct interstage transfer: Elimination of batch cooling step between stages is an untested hypothesis. No evidence thioester intermediate is stable or amide coupling is equally efficient without cooling.

Shared assessment / Flow equivalence: Hold ratios are V/Q indices, not measures of equivalence to batch exposure. Kinetics are uncharacterized; no yield or conversion prediction is valid.

Shared assessment / Mixer requirement: Stage 2 requires a T- or Y-mixer to introduce Stream B into Stage 1 effluent. Adequate mixers (P-713 PEEK, P-632 ETFE, P-513 PEEK) are available in inventory. No candidates assign a mixer explicitly — chemist must confirm junction hardware before execution.

Shared assessment / Target time vs achieved: Target screen times are design targets; achieved residence times differ from targets due to discrete pump increment constraints. Stated residence times are the actual achieved values.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: prefer

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Highest substrate feed throughput (0.14 mmol/min). Compact 5+5 mL ETFE geometry. Stage 1: 17.9 min (0.60× batch); Stage 2: 14.3 min (0.48× batch). Both stages below batch anchor — thioester completion and amide formation under shorter exposure are unvalidated but acceptable screening entry points per balanced objective. Single reactor type simplifies setup.

Reviews 1 / Uncertainty 1: Both stages below batch anchor; thioester and amide completion at shorter exposure unconfirmed.

Reviews 1 / Uncertainty 2: Direct interstage transfer at 95°C without cooling is an untested hypothesis.

Reviews 1 / Uncertainty 3: T-mixer for Stage 2 Stream B injection not explicitly assigned.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Reduced throughput (0.06 mmol/min). Stage 1 at 16.7 min (0.56× batch) is sub-batch; Stage 2 at 33.3 min (1.11× batch) exceeds batch anchor for amide coupling. Mixed reactor geometry (PFA 2 mL + ETFE 5 mL). Useful for probing whether longer Stage 2 benefits amide yield versus Stage 1 risk.

Reviews 2 / Uncertainty 1: Thioester completion adequacy at 16.7 min Stage 1 is unconfirmed.

Reviews 2 / Uncertainty 2: Whether extended Stage 2 compensates for incomplete Stage 1 is unknown.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Highest substrate feed throughput (0.14 mmol/min), equal to C1. Stage 1 at 35.7 min (1.19× batch) exceeds batch anchor — more conservative thioester formation. Stage 2 at 14.3 min (0.48× batch) is shorter. Balanced design favoring Stage 1 completion. Requires two different ETFE coils (10+5 mL).

Reviews 3 / Uncertainty 1: Stage 2 sub-batch exposure for amide coupling; aminolysis completion at 14.3 min unconfirmed.

Reviews 3 / Uncertainty 2: coil_row12 (10 mL ETFE, qty 2) and coil_row11 (5 mL ETFE, qty 2) inventory must be confirmed as simultaneously available.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Throughput 0.08 mmol/min. Stage 1: 31.25 min (1.04× batch); Stage 2: 25 min (0.83× batch). Both stages near batch anchor — interpretable baseline. Uses two 5 mL ETFE coils. Adequate balance but lower throughput than C1/C3.

Reviews 4 / Uncertainty 1: Stage 2 at 25 min is slightly sub-batch; amide completion not established.

Reviews 4 / Uncertainty 2: Identical reactor assignment (coil_row11 ×2) requires confirmation of dual availability.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Low throughput (0.04 mmol/min). Stage 1: 25 min (0.83× batch); Stage 2: 50 min (1.67× batch). Long Stage 2 at 95°C — whether extended amide exposure risks degradation is uncharacterized but a concern. Useful as yield-bias long-Stage-2 data point.

Reviews 5 / Uncertainty 1: Extended Stage 2 at 95°C degradation risk is uncharacterized.

Reviews 5 / Uncertainty 2: Stage 1 sub-batch exposure may yield incomplete thioester.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Throughput 0.08 mmol/min. Stage 1: 62.5 min (2.08× batch) — substantially extended thioester exposure. Stage 2: 25 min (0.83× batch). Useful for probing whether excess Stage 1 time causes DPDTC-side product buildup. Requires 10+5 mL ETFE.

Reviews 6 / Uncertainty 1: Prolonged Stage 1 at 95°C may generate DPDTC-derived side products; no evidence either way.

Reviews 6 / Uncertainty 2: Stage 2 sub-batch for aminolysis.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Throughput 0.08 mmol/min. Both stages substantially above batch anchor (2.08×, 1.67×). Total time 112.5 min. Longest-total-time candidate at this throughput. Useful as upper-bound exposure reference. Requires two 10 mL ETFE coils simultaneously.

Reviews 7 / Uncertainty 1: Risk of prolonged 95°C exposure in both stages — degradation potential uncharacterized.

Reviews 7 / Uncertainty 2: coil_row12 (qty 2) dual use must be confirmed.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Throughput 0.06 mmol/min. Stage 1: 16.7 min (0.56× batch); Stage 2: 66.7 min (2.22× batch) — extreme Stage 2 extension at 95°C. Tests whether very long amide exposure degrades product. Mixed PFA 2 mL + ETFE 10 mL geometry.

Reviews 8 / Uncertainty 1: 66.7 min Stage 2 at 95°C — product degradation risk is uncharacterized but elevated.

Reviews 8 / Uncertainty 2: Incomplete Stage 1 thioester combined with long Stage 2 complicates interpretation.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Throughput 0.04 mmol/min. Stage 1: 62.5 min in 5 mL ETFE; Stage 2: 20 min in 2 mL PFA. Long Stage 1, short Stage 2. Reactor size inversion (larger to smaller) is hydraulically valid. Useful probe of over-extended Stage 1 with compressed amide stage.

Reviews 9 / Uncertainty 1: Stage 2 at 20 min (0.67× batch) — aminolysis adequacy unconfirmed.

Reviews 9 / Uncertainty 2: Reactor geometry inversion (ETFE→PFA, 2.4→1.0 mm ID) introduces mixing regime change at Stage 2 entry.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Throughput 0.04 mmol/min. Stage 1: 25 min; Stage 2: 100 min (3.33× batch). Longest Stage 2 in set. Useful as extreme upper bound for amide coupling; however, prolonged 95°C exposure risk is uncharacterized. Total time 125 min.

Reviews 10 / Uncertainty 1: 100 min Stage 2 at 95°C — product degradation risk is uncharacterized and a notable concern.

Reviews 10 / Uncertainty 2: Lowest throughput limits utility as baseline.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Throughput 0.04 mmol/min. Stage 1: 62.5 min in PFA 5 mL (1 mm ID); Stage 2: 20 min in PFA 2 mL. Identical to C9 in residence times but uses PFA coil_row4 for Stage 1 instead of ETFE coil_row11. Narrower ID (1 mm vs 2.4 mm) yields higher Re and better radial mixing in Stage 1.

Reviews 11 / Uncertainty 1: Mixing regime difference between Stage 1 (1 mm PFA) and Stage 2 (1 mm PFA) is modest but ID change at junction entry with Stream B needs mixer confirmation.

Reviews 11 / Uncertainty 2: Same aminolysis adequacy concern as C9 at 20 min Stage 2.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Throughput 0.04 mmol/min. Stage 1: 62.5 min; Stage 2: 50 min; total 112.5 min. Both stages well above batch anchor. Lowest throughput with longest combined hold at 95°C. Useful only as extreme degradation-risk reference; not preferred for balanced objective.

Reviews 12 / Uncertainty 1: Combined 112.5 min at 95°C — degradation risk across both stages is uncharacterized.

Reviews 12 / Uncertainty 2: Two 5 mL ETFE coils (coil_row11, qty 2) simultaneously required — confirm dual availability.

Call 4: DrSafety, attempt 1
Status: recorded response, parsed successfully.

Shared assessment / Chemistry: Two-stage DPDTC-mediated amide formation. Stage 1: thioester formation at 95°C. Stage 2: benzylamine aminolysis at 95°C. No cooling step between stages—direct inline transfer is assumed; chemically and operationally unvalidated in flow.

Shared assessment / Solvent pump constraint: 2-MeTHF is excluded from all RED pump tubing (E-series and SF-10). All candidates correctly use pump_e_series_blue (BLUE tubing). Compliant.

Shared assessment / Temperature constraint: All reactors used (PFA coil_row3/4/5, ETFE coil_row11/12) rated to 100°C. Operating at 95°C is within limits. FEP coils (coil_row6/7) rated only to 50°C—none used. Compliant.

Shared assessment / Pressure: BPR 7 bar selected across all candidates. 2-MeTHF bp ~80°C at 1 atm; at 95°C with 7 bar backpressure, liquid phase maintained. All reactor max pressures ≥34 bar. Chemyx syringe pumps limited to ~3 bar—not used here; E-series peristaltic max 10 bar. Backpressure adequate for single-phase operation.

Shared assessment / Volatility control: 2-MeTHF is volatile (bp ~80°C). Operation at 95°C requires backpressure to suppress vapour formation. 7 bar BPR is appropriate; single-phase flow expected but not confirmed experimentally.

Shared assessment / Benzylamine addition: Stream B is benzylamine (2.1 M in 2-MeTHF) introduced at Stage 2 via T-mixer into hot Stage 1 effluent at 95°C. Benzylamine bp ~185°C; no volatility risk. Exotherm on mixing with reactive thioester intermediate is uncharacterized—rapid aminolysis is plausible but heat release is unknown.

Shared assessment / Dpdtc safety: DPDTC is a reactive acylating/thioesterification reagent. SDS review and compatibility with PFA/ETFE tubing and PEEK/ETFE mixers recommended before execution.

Shared assessment / Nitro group: 3-Methyl-4-nitrobenzoic acid contains a nitro group. No reduction conditions present; no specific pressure/temperature safety flag beyond standard nitro-compound handling.

Shared assessment / Interstage transfer: Batch protocol requires cooling before benzylamine addition. Flow design eliminates cooling step—chemical acceptability (thioester stability, exotherm on hot mixing) is unvalidated and requires chemist confirmation.

Shared assessment / Stream b concentration: Benzylamine stream B is 2.1 M to achieve 1.05 equiv at low flow ratio (Q_B/Q_A = 0.25). This high concentration in 2-MeTHF requires solubility confirmation.

Shared assessment / Throughput note: Limiting feed throughput (mmol/min of substrate) is computed from Stream A flow and concentration. Candidates 1 and 3 share the highest substrate feed rate (0.14 mmol/min); throughput ranking is based solely on feed rate, not measured product output.

Shared assessment / Peristaltic pump note: E-series peristaltic pumps have inherent pulsation. At very low flow rates (0.02–0.04 mL/min), flow stability and stoichiometric accuracy should be verified experimentally.

Shared assessment / Gas stage: No gas stages present in any candidate. No gas bookkeeping issues.

Shared assessment / No kinetic evidence: Kinetics uncharacterized. No yield, conversion, or residence-time superiority can be claimed. Batch times (30 min each stage) are anchors only. All hold ratios describe nominal hydraulic exposure, not validated reaction completeness.

Reviews 1 / Candidate: 1

Reviews 1 / Recommendation: prefer

Reviews 1 / Hard violation reported: false

Reviews 1 / Justification: Highest substrate throughput (0.14 mmol/min). Compact dual ETFE 5 mL coil design. Stage 1 τ=17.9 min (ratio 0.60×batch), Stage 2 τ=14.3 min (0.48×batch)—both sub-batch, consistent with balanced screening intent. Single platform, no equipment conflicts.

Reviews 1 / Uncertainty 1: Sub-batch holds for both stages: thioester completeness and amide yield at shorter exposures unvalidated.

Reviews 1 / Uncertainty 2: Stream B at 2.1 M benzylamine in 2-MeTHF: solubility unconfirmed.

Reviews 1 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 1 / Uncertainty 4: E-series peristaltic accuracy at 0.07 mL/min (Stream B) requires verification.

Reviews 2 / Candidate: 2

Reviews 2 / Recommendation: acceptable

Reviews 2 / Hard violation reported: false

Reviews 2 / Justification: Moderate throughput (0.06 mmol/min). Stage 1 τ=16.7 min (0.56×batch) in compact 2 mL PFA coil; Stage 2 τ=33.3 min (1.11×batch) in 5 mL ETFE coil—asymmetric design providing extended Stage 2 exposure. Mixed reactor geometry is operationally straightforward.

Reviews 2 / Uncertainty 1: Stage 1 sub-batch: thioester conversion at 16.7 min unvalidated.

Reviews 2 / Uncertainty 2: Stream B at 0.03 mL/min: peristaltic accuracy at minimum-range flow.

Reviews 2 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 3 / Candidate: 3

Reviews 3 / Recommendation: prefer

Reviews 3 / Hard violation reported: false

Reviews 3 / Justification: Highest substrate throughput (0.14 mmol/min), equal to Candidate 1. Stage 1 τ=35.7 min (1.19×batch) in 10 mL ETFE coil—exceeds batch anchor, supporting thioester completeness. Stage 2 τ=14.3 min (0.48×batch)—sub-batch for aminolysis. Good screening complement to Candidate 1.

Reviews 3 / Uncertainty 1: Stage 2 sub-batch aminolysis: amide yield at 14.3 min unvalidated.

Reviews 3 / Uncertainty 2: Stream B at 2.1 M benzylamine in 2-MeTHF: solubility unconfirmed.

Reviews 3 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 3 / Uncertainty 4: Uses two ETFE coils of different volumes; both within inventory.

Reviews 4 / Candidate: 4

Reviews 4 / Recommendation: acceptable

Reviews 4 / Hard violation reported: false

Reviews 4 / Justification: Moderate throughput (0.08 mmol/min). Stage 1 τ=31.3 min (1.04×batch) and Stage 2 τ=25.0 min (0.83×batch)—both near-batch. Interpretable near-baseline screen using compact dual ETFE 5 mL coils. Reasonable balanced starting point.

Reviews 4 / Uncertainty 1: Stage 2 τ slightly sub-batch; aminolysis completeness unvalidated.

Reviews 4 / Uncertainty 2: Stream B at 2.1 M benzylamine in 2-MeTHF: solubility unconfirmed.

Reviews 4 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 5 / Candidate: 5

Reviews 5 / Recommendation: acceptable

Reviews 5 / Hard violation reported: false

Reviews 5 / Justification: Lower throughput (0.04 mmol/min). Stage 1 τ=25.0 min (0.83×batch) in 2 mL PFA; Stage 2 τ=50.0 min (1.67×batch) in 5 mL ETFE—extended Stage 2 explores over-exposure risk for aminolysis. Useful as a long-Stage-2 reference point.

Reviews 5 / Uncertainty 1: Stage 2 at 1.67×batch: extended exposure at 95°C may risk product degradation, uncharacterized.

Reviews 5 / Uncertainty 2: Stream B at 0.02 mL/min: at E-series minimum; flow stability uncertain.

Reviews 5 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 6 / Candidate: 6

Reviews 6 / Recommendation: acceptable

Reviews 6 / Hard violation reported: false

Reviews 6 / Justification: Moderate throughput (0.08 mmol/min). Stage 1 τ=62.5 min (2.08×batch) provides extended thioester formation; Stage 2 τ=25.0 min (0.83×batch) is near-batch. Long total time (87.5 min) reduces practical value as a compact design but is a useful long-Stage-1 reference.

Reviews 6 / Uncertainty 1: Stage 1 at 2.08×batch: extended exposure of acid/DPDTC at 95°C; side-product formation uncharacterized.

Reviews 6 / Uncertainty 2: Stream B at 2.1 M benzylamine in 2-MeTHF: solubility unconfirmed.

Reviews 6 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 7 / Candidate: 7

Reviews 7 / Recommendation: acceptable

Reviews 7 / Hard violation reported: false

Reviews 7 / Justification: Moderate throughput (0.08 mmol/min). Both stages significantly exceed batch time (Stage 1 2.08×, Stage 2 1.67×); total 112.5 min. Useful upper-bound screen for extended exposure, but long total time is inconsistent with compact preference. Low practical priority.

Reviews 7 / Uncertainty 1: Both stages >1×batch at 95°C: risk of thioester or amide degradation over prolonged heating is uncharacterized.

Reviews 7 / Uncertainty 2: Stream B at 2.1 M benzylamine in 2-MeTHF: solubility unconfirmed.

Reviews 7 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 8 / Candidate: 8

Reviews 8 / Recommendation: acceptable

Reviews 8 / Hard violation reported: false

Reviews 8 / Justification: Moderate throughput (0.06 mmol/min). Stage 1 τ=16.7 min (0.56×batch); Stage 2 τ=66.7 min (2.22×batch) in 10 mL ETFE—strongly asymmetric. Extended Stage 2 at 95°C explores over-exposure. Less practically attractive; useful only as extreme Stage 2 reference.

Reviews 8 / Uncertainty 1: Stage 2 at 2.22×batch: extended aminolysis at 95°C; degradation risk uncharacterized.

Reviews 8 / Uncertainty 2: Stream B at 0.03 mL/min: low-flow peristaltic accuracy requires verification.

Reviews 8 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 9 / Candidate: 9

Reviews 9 / Recommendation: acceptable

Reviews 9 / Hard violation reported: false

Reviews 9 / Justification: Lower throughput (0.04 mmol/min). Stage 1 τ=62.5 min (2.08×batch) in ETFE 5 mL; Stage 2 τ=20.0 min (0.67×batch) in PFA 2 mL. Inverted relative to objective preference (compact higher-throughput). Long Stage 1, short Stage 2 is an informative design complement.

Reviews 9 / Uncertainty 1: Stage 1 at 2.08×batch: extended exposure of reactive intermediates at 95°C; side reactions uncharacterized.

Reviews 9 / Uncertainty 2: Stream B at 0.02 mL/min: at E-series minimum; flow stability uncertain.

Reviews 9 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 10 / Candidate: 10

Reviews 10 / Recommendation: acceptable

Reviews 10 / Hard violation reported: false

Reviews 10 / Justification: Lowest throughput group (0.04 mmol/min). Stage 2 τ=100 min (3.33×batch) is extreme; total 125 min. Provides maximum Stage 2 exposure data point but is highly impractical for a compact balanced design. Low screening priority.

Reviews 10 / Uncertainty 1: Stage 2 at 3.33×batch (100 min, 95°C): product degradation risk is uncharacterized and elevated.

Reviews 10 / Uncertainty 2: Stream B at 0.02 mL/min: minimum-range peristaltic flow; accuracy requires verification.

Reviews 10 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 11 / Candidate: 11

Reviews 11 / Recommendation: acceptable

Reviews 11 / Hard violation reported: false

Reviews 11 / Justification: Lower throughput (0.04 mmol/min). Stage 1 τ=62.5 min in PFA 5 mL (1 mm ID); Stage 2 τ=20.0 min in PFA 2 mL. PFA 1 mm ID yields higher pressure drop than ETFE 2.4 mm at same flow; still negligible at 0.08 mL/min. Informative but not a preferred compact design.

Reviews 11 / Uncertainty 1: Stage 1 at 2.08×batch: prolonged thioester exposure at 95°C; side reactions uncharacterized.

Reviews 11 / Uncertainty 2: Stream B at 0.02 mL/min: minimum-range peristaltic flow; accuracy requires verification.

Reviews 11 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Reviews 12 / Candidate: 12

Reviews 12 / Recommendation: acceptable

Reviews 12 / Hard violation reported: false

Reviews 12 / Justification: Lowest throughput (0.04 mmol/min). Both stages well above batch time (Stage 1 2.08×, Stage 2 1.67×); total 112.5 min. Least consistent with compact higher-throughput objective. Useful only as maximum-exposure reference; low practical priority.

Reviews 12 / Uncertainty 1: Both stages >1×batch at 95°C: cumulative degradation risk at 112.5 min total is uncharacterized.

Reviews 12 / Uncertainty 2: Stream B at 0.02 mL/min: minimum-range peristaltic flow; accuracy requires verification.

Reviews 12 / Uncertainty 3: Direct hot interstage transfer without cooling: chemical acceptability unvalidated.

Call 5: Skeptic, attempt 1
Status: recorded response, parsed successfully.

Vetoes: []

Shared assessment / Common issues 1 / Issue: Direct hot interstage transfer at 95°C without batch cooling step

Shared assessment / Common issues 1 / Scope: all candidates

Shared assessment / Common issues 1 / Status: unresolved chemist hypothesis — not a hard constraint; requires chemist confirmation before execution

Shared assessment / Common issues 2 / Issue: Stream B benzylamine concentration 2.1 M in 2-MeTHF

Shared assessment / Common issues 2 / Scope: all candidates

Shared assessment / Common issues 2 / Status: solubility unconfirmed; no measured evidence; requires laboratory verification before use

Shared assessment / Common issues 3 / Issue: 2-MeTHF excluded from E-series RED pump tubing; all candidates correctly assign Stream A and B to BLUE pump tubing only

Shared assessment / Common issues 3 / Scope: all candidates

Shared assessment / Common issues 3 / Status: no violation detected

Shared assessment / Common issues 4 / Issue: Interstage T-mixer for Stream B injection not explicitly enumerated in design realizations

Shared assessment / Common issues 4 / Scope: all candidates

Shared assessment / Common issues 4 / Status: PEEK or ETFE T-mixers are available in inventory; assignment must be confirmed at setup; not a hard violation

Shared assessment / Common issues 5 / Issue: Kinetics entirely uncharacterized; no measured flow yield, conversion, or residence-time superiority data exist

Shared assessment / Common issues 5 / Scope: all candidates

Shared assessment / Common issues 5 / Status: all stage hold ratios are screening anchors only; no yield prediction or ranking is supportable

Shared assessment / Common issues 6 / Issue: Feed throughput (limiting reagent mmol/min) is computed from stream A flow × concentration; product throughput is unmeasured and cannot be compared

Shared assessment / Common issues 6 / Scope: all candidates

Shared assessment / Common issues 6 / Status: throughput comparisons are feed-side only

Shared assessment / Common issues 7 / Issue: BPR 7 bar selected across all candidates; 2-MeTHF bp ~80°C at 1 atm, operation at 95°C requires sufficient back-pressure to suppress boiling; 7 bar BPR provides adequate headroom

Shared assessment / Common issues 7 / Scope: all candidates

Shared assessment / Common issues 7 / Status: no hard violation; confirm BPR setting before operation

Shared assessment / Common issues 8 / Issue: Analogy records retrieved are from different reaction classes and are not usable as kinetic evidence for this DPDTC-mediated amide formation

Shared assessment / Common issues 8 / Scope: all candidates

Shared assessment / Common issues 8 / Status: confirmed non-applicable; no kinetic inference drawn

Candidate reviews 1 / Candidate: 1

Candidate reviews 1 / Council recommendation: prefer

Candidate reviews 1 / Hard violation reported: false

Candidate reviews 1 / Justification: Highest feed throughput (0.14 mmol/min); compact identical 5+5 mL ETFE geometry. S1 τ=17.9 min (0.60×), S2 τ=14.3 min (0.48×); both sub-batch. Consistent with compact higher-throughput starting screen per objective. Stream B at 0.07 mL/min is above E-series minimum.

Candidate reviews 1 / Uncertainty 1: Both stages sub-batch: thioester and amide completion at shortened exposures uncharacterized.

Candidate reviews 1 / Uncertainty 2: Benzylamine 2.1 M in 2-MeTHF solubility unconfirmed.

Candidate reviews 2 / Candidate: 2

Candidate reviews 2 / Council recommendation: acceptable

Candidate reviews 2 / Hard violation reported: false

Candidate reviews 2 / Justification: Throughput 0.06 mmol/min; mixed PFA 2 mL + ETFE 5 mL geometry. S1 sub-batch (16.7 min, 0.56×); S2 above batch (33.3 min, 1.11×). Useful asymmetric probe but lower throughput than C1/C3. Stream B at 0.03 mL/min approaches low-flow pulsation threshold for peristaltic pump.

Candidate reviews 2 / Uncertainty 1: Stream B at 0.03 mL/min: peristaltic pulsation and stoichiometric accuracy require verification.

Candidate reviews 2 / Uncertainty 2: S1 thioester completeness at 16.7 min uncharacterized.

Candidate reviews 3 / Candidate: 3

Candidate reviews 3 / Council recommendation: prefer

Candidate reviews 3 / Hard violation reported: false

Candidate reviews 3 / Justification: Highest feed throughput (0.14 mmol/min), equal to C1. S1 τ=35.7 min (1.19×) exceeds batch anchor, providing thioester formation headroom. S2 τ=14.3 min (0.48×) is sub-batch. Requires 10 mL ETFE (coil_row12) + 5 mL ETFE (coil_row11); both available. Key experimental question is S2 aminolysis adequacy.

Candidate reviews 3 / Uncertainty 1: S2 aminolysis at 14.3 min (0.48× batch) may be incomplete; uncharacterized.

Candidate reviews 3 / Uncertainty 2: Thioester stability during extended S1 at 95°C is an untested hypothesis.

Candidate reviews 4 / Candidate: 4

Candidate reviews 4 / Council recommendation: prefer

Candidate reviews 4 / Hard violation reported: false

Candidate reviews 4 / Justification: Throughput 0.08 mmol/min; compact dual 5 mL ETFE coils. S1 τ=31.3 min (1.04×) near-batch; S2 τ=25.0 min (0.83×) modestly sub-batch. Most interpretable near-batch baseline for both stages; minimizes thioester risk while S2 shortening is modest. Consistent with balanced objective.

Candidate reviews 4 / Uncertainty 1: S2 at 25 min (0.83× batch): aminolysis completeness uncharacterized.

Candidate reviews 4 / Uncertainty 2: Dual coil_row11 (qty=2) simultaneous availability requires confirmation.

Candidate reviews 5 / Candidate: 5

Candidate reviews 5 / Council recommendation: acceptable

Candidate reviews 5 / Hard violation reported: false

Candidate reviews 5 / Justification: Lowest throughput tier (0.04 mmol/min). S1 τ=25 min (0.83×); S2 τ=50 min (1.67×). Stream B at 0.02 mL/min is at E-series peristaltic minimum — flow stability and stoichiometric accuracy are at risk and must be verified. Useful as long-S2 reference only.

Candidate reviews 5 / Uncertainty 1: Stream B at 0.02 mL/min at E-series minimum: pulsation and dosing fidelity unverified.

Candidate reviews 5 / Uncertainty 2: S2 product stability at 50 min / 95°C uncharacterized.

Candidate reviews 6 / Candidate: 6

Candidate reviews 6 / Council recommendation: acceptable

Candidate reviews 6 / Hard violation reported: false

Candidate reviews 6 / Justification: Throughput 0.08 mmol/min. S1 τ=62.5 min (2.08×) substantially exceeds batch; S2 τ=25 min (0.83×) sub-batch. Useful for probing whether over-extended S1 generates DPDTC-derived side products. Total 87.5 min reduces practical compactness. Uses 10+5 mL ETFE.

Candidate reviews 6 / Uncertainty 1: DPDTC intermediate stability at 62.5 min / 95°C: side-product formation is an untested hypothesis.

Candidate reviews 6 / Uncertainty 2: S2 aminolysis adequacy at 25 min sub-batch uncharacterized.

Candidate reviews 7 / Candidate: 7

Candidate reviews 7 / Council recommendation: acceptable

Candidate reviews 7 / Hard violation reported: false

Candidate reviews 7 / Justification: Throughput 0.08 mmol/min; both stages well above batch (2.08×, 1.67×). Total 112.5 min. Requires two 10 mL ETFE coils (coil_row12, qty=2 confirmed). Useful upper-bound reference only; long total time conflicts with compact higher-throughput objective.

Candidate reviews 7 / Uncertainty 1: Cumulative degradation risk across both stages at 95°C (112.5 min total) uncharacterized.

Candidate reviews 7 / Uncertainty 2: Marginal practical value for balanced objective given total processing time.

Candidate reviews 8 / Candidate: 8

Candidate reviews 8 / Council recommendation: acceptable

Candidate reviews 8 / Hard violation reported: false

Candidate reviews 8 / Justification: Throughput 0.06 mmol/min. S1 τ=16.7 min (0.56×) sub-batch; S2 τ=66.7 min (2.22×) strongly above batch. Stream B at 0.03 mL/min near low-flow limit. Extreme asymmetric design; S2 over-exposure at 95°C raises product stability concern. Limited practical priority.

Candidate reviews 8 / Uncertainty 1: S2 product degradation at 66.7 min / 95°C uncharacterized; elevated concern.

Candidate reviews 8 / Uncertainty 2: Stream B at 0.03 mL/min peristaltic pulsation risk requires verification.

Candidate reviews 9 / Candidate: 9

Candidate reviews 9 / Council recommendation: acceptable

Candidate reviews 9 / Hard violation reported: false

Candidate reviews 9 / Justification: Throughput 0.04 mmol/min. S1 τ=62.5 min (2.08×) in ETFE 5 mL; S2 τ=20 min (0.67×) in PFA 2 mL. Stream B at 0.02 mL/min at E-series minimum — flow stability must be verified. Reactor geometry inversion (2.4→1.0 mm ID) introduces mixing-regime change at S2 entry.

Candidate reviews 9 / Uncertainty 1: Stream B at 0.02 mL/min at minimum: stoichiometric accuracy uncertain.

Candidate reviews 9 / Uncertainty 2: S2 aminolysis at 20 min (0.67× batch) may be incomplete; uncharacterized.

Candidate reviews 10 / Candidate: 10

Candidate reviews 10 / Council recommendation: low_priority

Candidate reviews 10 / Hard violation reported: false

Candidate reviews 10 / Justification: Throughput 0.04 mmol/min (lowest tier). S2 τ=100 min (3.33×) is the longest Stage 2 in the set; total 125 min. Stream B at 0.02 mL/min at E-series minimum. Strongly imbalanced against the balanced objective; extreme over-exposure risk with minimum throughput and longest total time.

Candidate reviews 10 / Uncertainty 1: S2 product or byproduct accumulation at 100 min / 95°C: degradation risk elevated and uncharacterized.

Candidate reviews 10 / Uncertainty 2: Stream B at 0.02 mL/min minimum-range flow: stoichiometric fidelity uncertain.

Candidate reviews 11 / Candidate: 11

Candidate reviews 11 / Council recommendation: acceptable

Candidate reviews 11 / Hard violation reported: false

Candidate reviews 11 / Justification: Throughput 0.04 mmol/min. S1 τ=62.5 min in PFA 5 mL (1 mm ID); S2 τ=20 min in PFA 2 mL. Exposure profile identical to C9 but different reactor geometry. Stream B at 0.02 mL/min at E-series minimum. Narrower 1 mm ID in S1 increases pressure drop; negligible at this flow but solids risk noted.

Candidate reviews 11 / Uncertainty 1: Stream B at 0.02 mL/min at minimum: stoichiometric fidelity uncertain.

Candidate reviews 11 / Uncertainty 2: S2 aminolysis adequacy at 20 min (0.67× batch) uncharacterized; same concern as C9.

Candidate reviews 12 / Candidate: 12

Candidate reviews 12 / Council recommendation: low_priority

Candidate reviews 12 / Hard violation reported: false

Candidate reviews 12 / Justification: Throughput 0.04 mmol/min (lowest tier). Both stages substantially above batch (62.5+50 min, total 112.5 min). Duplicates long-exposure concept of C7 at lower throughput. Stream B at 0.02 mL/min at minimum. Least consistent with compact higher-throughput objective; minimal marginal information over C7.

Candidate reviews 12 / Uncertainty 1: Combined 112.5 min at 95°C: cumulative degradation risk across both stages uncharacterized.

Candidate reviews 12 / Uncertainty 2: Stream B at 0.02 mL/min minimum-range flow: dosing accuracy uncertain.

Priority recommendations / First screen 1: 3

Priority recommendations / First screen 2: 1

Priority recommendations / Rationale: C3 and C1 share the highest feed throughput (0.14 mmol/min) and compact ETFE geometry. C3 provides conservative S1 exposure (1.19× batch) to support thioester formation while probing compressed S2; C1 is the aggressive short-S1/short-S2 complement. Together they bracket the high-throughput region most consistent with the balanced objective.

Priority recommendations / Baseline reference: 4

Priority recommendations / Baseline rationale: C4 provides the most interpretable near-batch baseline (S1 1.04×, S2 0.83×) at 0.08 mmol/min if either C1 or C3 shows yield deficiency attributable to sub-batch exposure.

Requested measurement 1: Benzylamine solubility in 2-MeTHF at 2.1 M and at 95°C before pressurized operation

Requested measurement 2: E-series peristaltic pump delivery accuracy and pulsation characterization at 0.02–0.07 mL/min for 2-MeTHF streams

Requested measurement 3: Chemist confirmation that direct hot interstage transfer at 95°C without intermediate cooling is chemically and operationally acceptable

Requested measurement 4: T-mixer assignment for Stream B injection at Stage 2 inlet confirmed from inventory before setup

Requested measurement 5: Thioester intermediate stability at 95°C as a function of hold time (to assess C6, C7, C9, C11 extended S1 risk)

Requested measurement 6: Amide product stability at 95°C as a function of hold time (to assess C5, C7, C8, C10, C12 extended S2 risk)

Requested measurement 7: Confirmation that coil_row11 (qty=2) and coil_row12 (qty=2) are simultaneously available for candidates requiring two coils of the same type

Requested measurement 8: BPR back-pressure setting verified sufficient to suppress 2-MeTHF boiling at 95°C before operation

Call 6: Chief, attempt 1
Status: recorded response, parsed successfully.

Candidate: 3

Justification: C3 delivers the highest feed throughput (0.14 mmol/min), equal to C1, while providing Stage 1 τ=35.7 min (1.19× batch anchor) that offers conservative thioester formation headroom before the amide-coupling stage. Stage 2 τ=14.3 min (0.48× batch) is the key experimental question. Compact 10+5 mL ETFE geometry; both reactors within inventory. No hard violations. Stream B at 0.07 mL/min is above E-series minimum, reducing pulsation risk versus lower-throughput candidates.

Objective alignment: Balanced priority requires maximum feed throughput, stable operation, and adequate stage exposure. C3 meets throughput parity with C1 while reducing the Stage 1 thioester-completion risk that C1 accepts by holding both stages sub-batch. The trade-off is accepting a sub-batch Stage 2 (0.48×) in exchange for S1 adequacy at maximum throughput; this is chemically justified as a starting screen because aminolysis of activated thioesters is generally faster than thioester formation, but that hypothesis is uncharacterized and

requires experimental confirmation. Total time 50 min versus C1's 32 min reflects the S1 extension, not an inherent optimum.

Alternative 1 / Candidate: 1

Alternative 1 / Reason not selected: C1 shares the highest feed throughput (0.14 mmol/min) and is the more aggressive compact screen (S1 τ=17.9 min, 0.60×; S2 τ=14.3 min, 0.48×). Both stages are sub-batch, leaving thioester formation adequacy uncharacterized at S1. C3 is preferred as the first screen because it retains throughput parity while reducing S1 risk. C1 is the recommended co-first screen if C3 shows adequate S2 yield, to test whether S1 can be further compressed.

Alternative 2 / Candidate: 4

Alternative 2 / Reason not selected: C4 provides the most interpretable near-batch baseline (S1 τ=31.3 min, 1.04×; S2 τ=25 min, 0.83×) at 0.08 mmol/min, which is 43% lower feed throughput than C3. The balanced objective prefers the higher-throughput C3 as a starting screen; C4 is designated the baseline reference if C3 or C1 show yield deficiency attributable to sub-batch S2 exposure. Near-batch times are not inherently optimal in flow; C4 does not supersede C3 on yield grounds absent measured evidence.

Answer impact 1 / Question: Q-OBJ-001

Answer impact 1 / Effect: C3's maximum-throughput, conservative-S1 design directly addresses the balance of yield adequacy and throughput; sub-batch S2 is the unresolved element that experimental data will resolve.

Answer impact 2 / Question: Q-HYP-001

Answer impact 2 / Effect: C3 operationalizes the chemist hypothesis that direct hot interstage transfer is acceptable and that S1 completeness is the governing constraint; both remain unvalidated. S1 extended beyond batch anchor partially hedges the thioester-formation risk.

Answer impact 3 / Question: Q-HIST-001

Answer impact 3 / Effect: No historical yield data are available; candidate ranking is based solely on throughput, stage exposure ratios, and equipment feasibility. Measured yield from C3 would become the primary evidence for subsequent stage-time decisions.

Answer impact 4 / Question: Q-PREF-001

Answer impact 4 / Effect: Stream A: 3-methyl-4-nitrobenzoic acid 0.5 M / DPDTC 1.05 equiv / DMAP 0.1 equiv in 2-MeTHF; molar flow 0.14 mmol/min substrate; Q=0.28 mL/min, pump_e_series_blue. Stream B: benzylamine 2.1 M in 2-MeTHF; 1.05 equiv relative to substrate; molar flow 0.147 mmol/min; Q=0.07 mL/min, pump_e_series_blue. No gas stages; Q_gas_sccm=0 for all stages. BPR 7 bar.

Shared assessment / Common uncertainties 1: Direct hot interstage transfer at 95°C without batch cooling: chemist confirmation required before execution for all candidates.

Shared assessment / Common uncertainties 2: Benzylamine 2.1 M in 2-MeTHF solubility unconfirmed; laboratory verification required before pressurized operation.

Shared assessment / Common uncertainties 3: 2-MeTHF excluded from E-series RED pump tubing; all candidates correctly use BLUE pump tubing only — no violation.

Shared assessment / Common uncertainties 4: Interstage T-mixer for Stream B injection not explicitly assigned in design realizations; PEEK/ETFE T-mixers are available in inventory and must be confirmed at setup.

Shared assessment / Common uncertainties 5: Kinetics entirely uncharacterized; no yield, conversion, or residence-time superiority can be inferred from stage hold ratios.

Shared assessment / Common uncertainties 6: Feed throughput (mmol/min) is computed from Stream A flow × concentration; product throughput is unmeasured.

Shared assessment / Common uncertainties 7: BPR 7 bar selected for all candidates; 2-MeTHF bp ~80°C at 1 atm — 7 bar provides adequate headroom to suppress boiling at 95°C; confirm before operation.

Shared assessment / Common uncertainties 8: Analogy records retrieved are from different reaction classes and provide no kinetic evidence for this DPDTC-mediated amide formation.

Limitation 1: No measured flow yield, conversion rate, or kinetic data exist; C3 preference is based on throughput parity and stage-exposure reasoning, not proven superiority.

Limitation 2: Stage 2 τ=14.3 min (0.48× batch) for aminolysis is sub-batch; whether this is sufficient cannot be inferred without measurement.

Limitation 3: Thioester intermediate stability at 95°C for 35.7 min is an untested hypothesis; degradation risk cannot be excluded.

Limitation 4: Direct hot interstage transfer replacing batch cooling is a chemist hypothesis, not an established fact.

Limitation 5: Benzylamine 2.1 M concentration in 2-MeTHF is unconfirmed; solubility failure would require stream reformulation across all candidates.

Limitation 6: Feed throughput is not product throughput; measured yield from C3 may differ from batch yield at any stage-time combination.

Suggested measurement 1: Confirm benzylamine solubility in 2-MeTHF at 2.1 M and at 95°C before pressurized operation.

Suggested measurement 2: Chemist must confirm that direct hot interstage transfer at 95°C without intermediate cooling is chemically and operationally acceptable before any candidate is executed.

Suggested measurement 3: Verify E-series peristaltic pump delivery accuracy and pulsation at 0.07 mL/min (Stream B, C3) and at 0.02–0.03 mL/min (lower-throughput candidates) in 2-MeTHF.

Suggested measurement 4: Assign and confirm T-mixer from inventory for Stream B injection at Stage 2 inlet before setup.

Suggested measurement 5: Measure amide yield from C3 at Stage 2 τ=14.3 min; if sub-batch S2 yield is deficient, execute C4 as the near-batch baseline reference before lengthening S2.

Suggested measurement 6: If C3 yield is adequate, execute C1 to determine whether S1 can be further compressed to τ=17.9 min at maximum throughput.

Suggested measurement 7: Characterize thioester intermediate stability at 95°C as a function of hold time before executing C6, C7, C9, or C11 with extended S1 >35 min.

Suggested measurement 8: Confirm simultaneous availability of coil_row12 (10 mL ETFE, qty=2) and coil_row11 (5 mL ETFE, qty=2) for candidates requiring both coils of the same type.

Editorial cross-checks (not model statements)

Numerical transcription: in the Set 1 Chief answer_impacts entry for Q-PREF-001, benzylamine molar flow is written as 0.02 mmol/min. The final feed and independent arithmetic give 2.1 mol/L x 0.04 mL/min = 0.084 mmol/min, which is 1.05 times the acid feed of 0.08 mmol/min. For Sets 2/3, the corresponding amine rate is 0.147 mmol/min and acid rate is 0.14 mmol/min. The copied council text remains unedited; these corrected values govern the reporting.

Inference limits: wording such as adequate exposure, confirmed photon geometry, or a single run distinguishing the rate-limiting step is not accepted as an experimentally demonstrated conclusion. A nominal hold near the batch duration does not establish equal conversion in flow. In particular, the Set 2 claim that incomplete Stage 1 conversion must have another cause because its hold exceeds batch is not justified. Changes in mixing, temperature distribution, feed state and residence-time distribution remain possible. The selected designs are therefore described as screening hypotheses.

Inventory reconciliation: references to a missing mixer or one pump label belong to the council's view of the candidate record. The final allocation explicitly assigns the interstage P-713 mixer and separate stream-pump operations. This resolves the reported assignment at the software-record level, not by claiming that the council physically installed equipment or verified the laboratory configuration.